ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Naval Warfare Officer (NWO)
(00207.1, 00207.2, 00207.3)

Minor War Vessel Navigating Officer (MWVNO) - AEFF
Fleet Navigating Officer (FNO) - AEEA

Version 1.0 (2022)

MITE Course IDs:
Pre-Instruction Package (PIP): TBD
Residential/Sea Phase MWVNO (AEFF): 120308
Residential/Sea Phase FNO (AEEA): 109067

MARPAC RDIMS #
Chap, 1, provisionally approved 10 Jun 2020
Chap 2, 3 approved DD MM 2022

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RECORD OF AMENDMENTS

The "Authority" reference indicates approval of this QSP document. Authority for chapters 1 and 2 of this QSP reside with the Commander Naval Personnel and Training Group (CNPTG). The authority for chapters 3 and 4 of this QSP Commanding Officer Naval Training Development Centre (Pacific) (CO NTDC (P)), who is both the Training Authority (TA) and the Curriculum Control Authority (CCA). The approval date appears in the bottom right corner of every page (the change date shall be updated in the footer). The Revision column is intended to provide a short indication of where the change(s) was/were made. This new QSP was created from a QSP Writing Board held 08-26 June 2020. The history of amendments to previous QSPs is detailed in Annex J "Course History" and shall henceforth be updated with any changes to this version 2.4 of the QSP. Detailed changes and decisions are amplified in Annex C.

Table 1 – Record of Amendments

REVISION	APPROVAL DATE	AUTHORITY
QSPWB Jun 2020	26 Jun 20	NPTG HQ
V 1.0 – Post Pilot Course Update		NPTG HQ

FOREWORD

1. This training document is developed under the authority of the Chief Maritime Staff Commander Naval Personnel and Training Group (CNPTG) and in accordance with the following references:

    1. A-P9-050 Series, Canadian Forces Manual of Individual Training and Education System (CFITES);
    2. A-PD-055-002/PP-003 Canadian Forces Officer General Specification dated 23 April 2014
    3. A-PD-055-003/PQ-001 Occupational Specialty Specification – Minor War Vessel Navigating Officer (MWVNO) –AEFF 31 Jul 90;
    4. A-PD-055-003/PQ-001 Occupation Specialty Specification – Fleet Navigating Officer (FNO).- AEEA 08 Feb 17;
    5. NPTORD 4500-22 Naval Personnel Training Order – Qualification Standard & Plan.

2. This course was developed during the Occupational Analysis (OA) (not complete at the time of this QSP finalization), and the OSSs had not yet been updated. The Task List at Annex B reflects the proposed changes that were to be implemented by the OA into the new OSS for each national qualification (NQual).

3. This qualification standard and plan (QSP) publication is effective on receipt and supersedes FNO QSP revision 2.3, dated Sep 2018 (RDIMS #392011) as captured in Annex J Course History.

4. Pre-Instructional Package (PIP): is located in Annex N.

5. Performance Objectives (POs) and Enabling Objectives (EOs): POs and EOs, which specify the standards that students must achieve to become qualified, are listed in the Table of Contents and are explained fully in Chapter 3 and Annex K Part 2 Activity and Assessment Plan (AAP).

6. Assessment Details: Assessment details for all Enabling Checks (ECs), Critical Enabling Checks (CECs) and Performance Checks (PCs) can be found in Annex K Part 1 - Assessments Overview, Part 2 - Activity and Assessment Plan (AAP) and Part 3 - Assessment Checklists.

7. Activity and Assessment Plan: Each PO's Activity and Assessment Plans (AAP) are located in Annex K Part 2 – AAP.

8. Assessment Checklists: Checklists for all assessments are located in Annex K Part 3

9. Amendments: Suggestions for amendments may be initiated by anyone and must be forwarded through the normal chain of command summarizing all requested amendments in writing to the applicable approving authority. IAW NPTORD 4500-22 Table 6.1, minor amendments to Chapters 2 and 3 are to be forwarded to course control document managers (CCDM) at NTDC (P). Major amendments and final approval for Chapter 1 amendments are required to be submitted to the CNPTG.

10. Description of Need: To ensure the professional navigation of RCN vessels, a requirement exists for a specialist officer to carry out the duties of a Navigation Officer for HMC Ships.

11. Scope: The major areas covered by this course are:

PO 001 – Navigate a Vessel of Any Class;

PO 002 – Navigate a Formation of Vessels of Any Class;

PO 003 – Lead the Navigation Division and Bridge Team of a Vessel of Any Class;

PO 004 – Provide Advice to Command on Navigation Matters;

PO 005 – Manage Navigation Systems and Equipment;

PO 006 – Manage Navigation Aspects of the Maritime Battlespace.

12. Method of Achieving Objectives: The course's objectives are met by a combination of theoretical and experiential training and performance-based assessments in the classroom, in simulators, and at sea.

13. This QSP shall be used by NFS(P) as the primary authority governing development, conduct, program evaluation and validation of the Minor War Vessel Navigating Officer (MWVNO) and Fleet Navigating Officer (FNO) courses.

14. This QSP must be used in conjunction with Occupation Specialty Specifications AEFF and AEEA in A-PD-055-003/PQ-001.

15. The designated TEs for this training are indicated below:

Table 2 - Training Establishments

TE	Phase/Module	MITE Course ID	NQual
NFS(P)	PIP	TBD	N/A
NFS (P)	MWVNO	120308	AEFF
NFS (P)	FNO	109067	AEEA

CHAPTER 1 - **TRAINING MANAGEMENT DETAILS**

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

AIM

1. To ensure that a professional manner of maritime navigation is conducted throughout the Canadian Armed Forces, a requirement exists for a specialist officer to execute the navigation of Royal Canadian Naval Vessels globally. The Qualification Standard and Plan (QSP) establishes the requirements and training for the Fleet Navigating Officer (FNO). This QSP is designed to assist the training system in developing, conducting, evaluating and validating the training. This course is designed to expand upon the knowledge and experience previously acquired by a Bridge Watch Keeper (BWK).

COURSE DURATION

2. Course duration is calculated in accordance with NPTORD 4500-22 Annex B. Detailed sequencing and timings are found at Annex G of this QSP. Table 3 below details the estimated and proposed timing allocation for this course. The estimation is based on the NPTORD 4500-22 direction for physical training and administrative time allocation, and the proposed allocation is based on FNI experience from past course schedules.

Table 3 - Course Duration

	Time in Minutes	
Allocation of ATT	Estimated	Proposed
Lesson Objectives (LOs)/Teaching Points	26775	26775
Enabling Checks (ECs)	3090	3090
Critical Enabling Checks (CECs)	10020	10020
Performance Checks (PCs)	4020	4020
ATT Sub-total	**43905**	**43905**
Allocation of BAT, TAT, TTT, PT (Course > 31 days)		
Base Administrative Time (BAT)		120
Training Administrative Time (TAT)		1195
Training Travel Time (TTT)	4074.3	540
Physical *Fitness* Training (PT)	3951.45	1740
Sub-total	**8025.75**	**3595**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT)	51930.75	47500
Total mins/400 = # days (rounded up)	129.826875	111.76471
Total Training Days	**130**	**112**

RESOURCES

3. Required Course Staffing: Table 4 below details key personnel required to conduct the training:

Table 4 - Required Course Staffing

Required Course Staffing					
Position	Minimum Rank	Qualifications Required	Qty	Remarks	PC/EC, PO/EO
Sr Evaluator Venture Div Cdr (preferred)	LCdr	NWO (MOSID 00207), AEEA, AEEG preferred	1	Required for final evaluation. Require 2 if more than 8 students on MWVNO. For PC 001A consideration may be given to FNIs to conduct evaluations if needed.	PC 001A PC 001B EC 001.05
Fleet Navigation Instructor (FNI)	Lt (N)	NWO (MOSID 00207), AEEA, ALPQ	2	Preferred 1 from each coast East & West; preferred recommendation for instruction	All
Met Instructor	Sgt	MetTech (MOSID 00100) – with sea time	1	Guest Lecturer, Preferred Port Met Inspector	PO 004
AJAG Officer	Lt (N)/Capt	Legal Officer (MOSID 00204)	1	Guest Lecturer	PO 004
Engineering Officer	Lt (N)	Preferred Naval Architecture experience, UEnvO course (MOSID 00346)	2	Guest Lecturer	PO 004
SAR Instructor	Civ	CCG Officer with SAR Ops Officer Course	1	Guest Lecturer	PO 004
JSS Instructor	Lt (N) or equivalen t (OF2)	NavO with Berlin Class/Protecteur Class experience	1	Guest Lecturer JSS Specific	PO 001, PO 004
Senior Mentor	Cdr/Civ	Former Commanding Officer (MOSID 00207)	1	Guest Lecturer	PO 001 PO 003 PO 004
Nav Systems Instructor	Civ	Senior Sonar Underwater Systems	1	Guest Lecturer	PO 005
Class Specific Instructor	Lt (N)/Civ	Shiphandling experience (MOSID 00207)	1	Guest Lecturer – supplement FNI Shiphandling experience	PO 001 PO 002 PO 004

Required Course Staffing					
Position	Minimum Rank	Qualifications Required	Qty	Remarks	PC/EC, PO/EO
Hydrographic Services Expert	Civ	CHS or HSO preferred	1	Guest Lecturer	PO001 PO006, EO006.02
OROCO	LCdr	AEEU, ALPY (MOSID 00207)	1	Guest Lecturer	PO006 EO006.01
Sea Training (Pacific)	LCdr	OpsO, WpnsO AEEU, ALPY (MOSID 00207)	1	Guest Lecturer	PO006 EO006.01
NABS Bridge Team	NCdt	NWO III min (MOSID 00207)	4	OOW, FixO	PO 001 PO 002
Helmsman	OS	NETP (ATR)	2	Nil	PO 001 PO 002
NavComm	LS	RQ-LS (MOSID 00299)	3	1 NavComm per cubicle and 1 for control	PO 002
NavComm Supervisor	PO2	RQ-PO2 (MOSID 00299)	1	Supervise all signals	PO 002

4. Critical Training Resources: Table 5 below, details the critical training resources to conduct the training. It should be read in conjunction with the recommended course timetable in Annex G and the provision of additional training resources required in Annex H of this QSP.

Table 5 - Critical Training Resources

Critical Training Resources	Training Days	PO/EO
DWAN Laptop (each student) which includes: -WECDIS, NavPac, ADP, Total Tides, ADC, MS Office Suite	FC	All
FNO Classroom	FC	All
Sensitive Discussion Area (SDA) N92A	3	PO 006
CSNI Computer N92A	3	PO 006
Smart Board	FC	All
DWAN Desktop connected to Smart Board	FC	All
DWAN drops (each student)	FC	All
Colour Printer	FC	All
Navigation and Bridge Simulator (NABS)	FC	All
Naval Training Vessel (Orca Class preferred)	See Table 6	All
Major Warship (Halifax Class Preferred)	See Table 6	All except PO 002
Celestial Navigation Software (NavPac preferred)	FC	PO 001
Chart and Publication Outfit	FC	All
USB Thumb Drive (Protected A) – (each trainee)	FC	All
Functional Sextant	FC	PO001
Synthetic Environment Advanced Combat Operator Trainer (SEACOT)	A/R	PO006

5. Critical Training Resources – Afloat: Tables 6 and 7 below detail critical training requirements during the sea phases for this course. See notes following the table for amplification and clarification

Table 6 - Critical Training Resources MWVNO Sea Phase

MWVNO Course Sea Days						
Event	**Platform**	**Days at Sea**	**Resources Based On:**			
			4 Students	**6 Students**	**8 Students**	**9 - 10 Students**
Developmental	Orca[1]	5[2]	1 Orca 2 CTOs Minimum[4] 5 passages per student x 4 students = 20 hours	1 Orca 2 CTOs Minimum[4] 5 passages per student x 6 students = 30 hours	1 or 2 Orca [3] 2 CTOs Minimum[4] 5 passages per student x 8 students = 40 hours	2 Orca [4] 2 CTOs Minimum[4] 5 passages per student x 5 students(max/orca) = 25 hours
Assessment	Orca[1]	5[2]	1 Orca 2 CTOs 1Sr Evaluator [5] 4 Passages per student x 4 = 16 hours	1 Orca 2 CTOs 1 Sr Evaluator [5] 4 Passages per student x 6 = 24 hours	1 or 2 Orca [3] 2 CTOs 1 Sr Evaluator [5] 4 Passages per student x 8 = 32 hours	2 Orca 2 CTOs 2 Sr Evaluator [5] 4 Passages per student x 5 (max/orca) = 20 hours

Table 7 - Critical Training Resources FNO Sea Phase

FNO Course Sea Days					
Event	**Platform**	**Days at sea**	**Resources Based On:**		
			4 Students	**6 Students**	**8 Students**
Developmental	FFH or Orca[7]	5[2]	1 FFH CO[8] 2 CTOs Minimum[6] 5 passages per student x 4 = 20 hours	1 FFH CO[8] 2 CTOs Minimum[6] 5 passages per student x 6 = 30 hours	1 FFH CO[8] 2 CTOs Minimum[6] 5 passages per student x 8 = 40 hours
Assessment	FFH[9]	5[2]	1 FFH 2 CTOs 1 Sr Evaluator 4 passages per student x 4 = 16 hours	1 FFH 2 CTOs 1 Sr Evaluator 4 passages per student x 6 = 24 hours	1 FFH 2 CTOs 1 Sr Evaluator 4 passages per student x 8 = 32 hours

Notes: 1. Orca platform preferred. KIN or HDW Class acceptable for MWVNO standard.
2. 5 days at sea is the required time for each phase; however, running phases consecutively without a minimum a 1 day break is unachievable. Ideally 5 day sea phases are split up with a 2 day weekend in port or "off" days at sea.
3. Based on previous course serials, 1 orca with 8 students is achievable but any delays due to engineering or weather, etc risk severely impacting the course standards. It is therefore recommended that if 2 orcas and sufficient crewing/staff can be put to sea for 8 students – this is ideal.

4. In the event that 2 Orcas cannot be put to sea when there are 8-10 students, it is recommended that one of the NABS developmental weeks be conducted at sea, extending the orca sea phase to 3 weeks and allowing greater flexibility to begin assessments earlier to meet minimum run standards. This would also be the recommended COA in the event that additional CTOs could not be sourced to adequately crew each orca.
5. Senior Evaluator qualification covered under key personnel in Chapter 1 para 3.
6. Minimum runs are the absolute requirement to meet the course standard. Vessels at sea should attempt to give increased training opportunities to the students. At least 1 developmental run during the FNO Sea Phase must be conducted in an FFH
7. FFH preferred to give students increased exposure to major warship navigation.
8. The FFH CO who is supporting the final sea-phase is the preferred CO for this phase.
9. Historically, Fleet staff attempt to layer other exercises and serials onto a ship conducting FNO. As the training is critical for the success of the course, any interference to the required runs will result in additional sea days being required to meet the minimum standard. If, due to scheduling concerns, this is deemed a difficulty, the situation will be referred to CNPTG on the risk of the programme interruption.

COURSE CAPACITY

6. Planning for the conduct of training is based on the guidelines in CFITES Vol 5 Annex D (Instructor to Learner Ratios). Justification can be found in Annex C – Record of Proceedings (ROP).

Course Load	MWVNO	FNO
Maximum	10	8
Minimum	4	4
Optimum	6	6

*Notes:

1. Exceeding the maximum number of students will require an extended warship sea phase and extend the course duration.
2. Failing to meet the minimum course loads places significant resource allocation challenges to support the course and is not recommended.

COURSE LOADING

7. Loading: The authority to load members onto this course lies with the NFS (P). NPTG is to be included as an INFO addressee on all loading messages.

PREREQUISITES

8. The loading authority must ensure that members have the associated prerequisites listed in table 8 below before being selected for this course:

Table 8 - Prerequisites for Loading on Course

MWVNO Reservist	**MWVNO Submariner**	**FNO**
Medically fit sea	Medically fit sea	Medically fit sea
Qualified NWO IV 00207	Qualified NWO IV 00207	Qualified NWO IV 00207
Qualified BWK 108983 (legacy) OR BWK Class Endorsement from: KIN Class – 123007 HAL Class – 123003 HDW Class – 123004	Qualified BWK 108983 (legacy) OR BWK Class Endorsement from: KIN Class – 123007 HAL Class – 123003 HDW Class – 123004	Qualified BWK 109113 (legacy) OR Qualified NWOPQ Board 123005 which requires either *Harry DeWolf*-class endorsement 123004 or *Halifax*-class endorsement 123003
PIP completed (MITE TBD)	PIP completed (MITE TBD)	PIP completed (MITE TBD)
Enhanced Reliability Status	Enhanced Reliability Status	Secret Security Clearance
CO Recommendation	CO Recommendation	CO Recommendation
3 months sea time with BWK Class Endorsement	Submariner NOPQ or NOPQ-02	NWOPQ + 6 months consolidation time[1]
	Submarine 2nd OOW ALDE	MWVNO (AEFF) completed (MITE 120308)
	3 months sea time as BWK	

Notes: 1. Consolidation time is necessary to support PIP completion and provide requisite mariner experience prior to course and follow-on employment as a ship's senior watchkeeper at sea. Therefore, the 6 month consolidation time should be value-added consolidation, on a sea going vessel where candiates can hone and expand their mariner skills. In addition pre MWVNO/FNO candidates should ideally be posted to a ship with a qualified FNO who can provide mentorship and oversight of their PIP progression and consolidation. Posting to *Kingston*-class as an OpsO is not considered value-added consolidation in preparation of this course. In the event where 6 months consolidation cannot be achieved, or there is a request to waive this requirement, FNIs and the Venture Div Commander should be contacted to provide guidance on a case-by-case basis. No candidate shall be loaded on course in such a circumstance without express approval from the Venture Div Commander. It must be noted that historical data shows candidates who attempt the MWVNO and FNO courses without adequate consolidation and/or an incomplete PIP are more likely to be unsuccessful on course.

SCHEDULING

9. Scheduling: Each course session shall be scheduled in MITE by NFS(P) (conducting the training) and NPTG through the Quantity Control section. The

course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays.

MITE

10. MITE Qualification: Two navigation qualifications are available throughout this course depending on the needs of the service and the student's capabilities, including Successful completion of this course constitutes the primary means of attaining the MWVNO (AEFF) and FNO (AEEA). Below is the breakdown that indicates the MITE codes associated with each training phase.:

    a. Pre-Instruction Package (PIP) –**MITE TBD** is awarded to students upon the completion of the package, which can be found at Annex N. Specifically, Part 1 Checklist and any additional items assigned by the FNIs via email instructor prior to the course start must be complete and submitted to NFS (P) FNIs along with proof of prerequisites listed in Table 8 of this QSP.

    b. Minor War Vessel Navigating Officer (MWVNO) – AEFF/**MITE Code 120308**: awarded to students upon successfully completing the learning objectives for AEFF (IAW Annex K). Submarine students will challenge the MWVNO portion of the FNO course. Chapter 2 Table 9 lists the Performance Objectives (POs) and specific Performance Checks (PCs) that the MWVNO must achieve IOT be granted the qualification of AEFF

    c. Fleet Navigating Officer (FNO) – AEEA/**MITE Code 109067** is awarded upon successfully completing the full course, including all PCs listed in Chapter 2 Table 9.

COURSE REPORTING

11. Successful trainees shall receive a Course Report (CF 377) and Certificate of Military Achievement (CF 289) completed IAW DAOD 5031-9 "Course Reporting and Certificates" and NPTORD 4500-51 "Course Reports and Certification."

12. Unsuccessful trainees shall receive a Course Report (CF377) only.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

13. PLARs shall be conducted in accordance with NAVORD 4500-2 "RCN Prior Learning Assessment and Recognition," DAOD 5031-1 "Canadian Forces Military Equivalencies Program," and CFITES Vol 12. "Canadian Forces Military Equivalencies Programme: Prior Learning Assessment."

SPECIAL INSTRUCTIONS

14. Foreign students that are attending the course will have a leave period while the other students are receiving Canadian-specific warfare instruction during PO 006.

15. The CF377 for foreign students will indicate course completion of the FNO course less the Canadian-specific warfare instruction.

16. For PO 006, instruction will take place in a sensitive discussion area (SDA) facility that requires Visitor Clearance Requests (VCRs) for students and instructors. Course staff must submit VCRs in a timely fashion IOT meet NFS(P) requirements. A security brief will also need to be scheduled before SDA is used.

17. This course supersedes and incorporates all previous RCN Navigating Officer Course content, including Destroyer Navigating Officer, Deep Draught Navigating Officer, Master Navigating Officer, Fleet Navigating Officer, and Minor War Vessel Navigating Officer. This course is to be considered equivalent to all previous courses with respect to Transport Canada TP2293E (2016).

CHAPTER 2 - **ASSESSMENT**

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE NAVAL TRAINING DEVELOPMENT CENTRE (PACIFIC)

GENERAL

1. Philosophy: Unlike initial NWO training, after which successful students progress to the fleet to gain a BWK and NOPQ qualification under supervision, graduates of this course are expected to be capable of operational pilotage and managing the navigation of an RCN ship from the moment they are posted to a sea-going unit. It is expected that Navigation Officers be able to provide safe, honest and timely navigation advice to their Commanding Officers. They are expected to have good knowledge of shiphandling, bridge resource management and Rules of the Road. Therefore, the examiners must ensure that the student is capable of bearing the ultimate responsibility of the safe navigation of a warship.

2. This is a pass/fail course. Students must successfully complete all POs to pass this course. Performance Checks (PCs), Critical Enabling Checks (CECs) and Enabling Checks (ECs) are based on the Performance Objectives (POs) and Enabling Objectives (EOs) in Chapter 3 of this QSP. All PCs must be successfully completed to achieve the qualification. The following table lists the requirements for MWVNO (AEFF) and FNO (AEEA):

Table 9 - Requirements for Qualification

Performance Objective	MWVNO (AEFF)	FNO (AEEA)
PO 001	PC 001A	PC 001A, PC 001B
PO 002	PC 002	PC 002
PO 003	PC 003	PC 003
PO 004	PC 004	PC 004
PO 005	PC 005A	PC 005A, PC 005B
PO 006	PC 006	PC 006

3. To be successful, students must consistently conduct navigation in all environmental conditions and traffic density levels, adjust course and/or speed as required, assess positional accuracy, and provide Command with accurate and timely navigation and shiphandling advice. It is crucial that navigators consistently maintain their integrity and honesty at all times with the CO/OIC.

4. Students will be developed and assessed using a combination of formative and summative assessment events. At a minimum, these will involve PCs, ECs, end of lesson confirmation, homework, assignments, and/or quizzes. See Annex K for details.

5. Dress and deportment, ethics, conduct and behaviour demonstrated by the trainee shall be evaluated and assessed based on the standards laid out under Operation Honour and the LRH program. A trainee who displays inappropriate dress and deportment, ethics, and/or behaviour is considered to have demonstrated unsatisfactory conduct, which may result in the convening of a Training Review Board (TRB).

ENABLING AND PERFORMANCE CHECKS

6. Enabling Checks (ECs): ECs are formative assessments that help the trainee, and the instructors recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning to improve retention of course content. Formative assessments are designed to provide feedback to

both instructional staff and trainees. Each EO must be formatively assessed prior to summative assessments.

7. In the event of an EC failure, the trainee will be counselled and given remedial training and/or additional homework, as appropriate. In the event of a consecutive EC failure, the trainee will be counselled IAW CFITES Vol 6 with the unsatisfactory progress documented in their Student File.

8. Critical Enabling Checks (CECs): CECs are summative assessments at the end of an EO which are considered critical because trainees cannot progress in their training without first successfully completing them. They are related to elements such as safety and security, where harm could result from poor performance. CECs are treated the same as PCs.

9. Performance Checks (PCs): PCs are summative tests used to determine if the trainee has met the standard under the PO's conditions (where possible) and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC. IAW CFITES Vol 7 a flexible policy for the action to be taken upon learner failure of a PC or CEC will depend on many variables. For any assessment that would necessitate additional study time or preparation by the student to be best situated to succeed during the supplemental PC, appropriate time should be given to allow for this.

10. PCs will be conducted after a formative assessment has occurred and the trainee has had the opportunity to develop skills and apply knowledge related to the PO.

PHASES

11. NABS Developmental Phase– General Concept: Within the overall course philosophy, the simulator phase remains a developmental element in preparing the student for the sea phase and does not represent the final standard. The student must demonstrate a firm knowledge of navigational principles and be considered safe to be given the responsibility of navigating a warship at sea under supervision. All students will undertake MWVNO training (MWVNO NABS Developmental Phase) prior to proceeding to sea for the MWVNO Orca Sea Phase. FNO students will embark upon an additional three-week developmental NABS phase (FNO NABS Developmental), followed immediately by a one-week Final FNO NABS Assessment Phase (CEC 001.05). Students will then proceed to sea for the FNO Developmental Sea Phase before commencing the FNO Assessment Phase in a major warship.

12. Orca Phases: This intermediate level of practical training significantly increases the realism presented to the students, enabling a training environment with real consequences for action. Bridge Resource Management, time management and interaction with other vessels all are experienced in the real world, giving students a real opportunity to gain experience. All students will undertake the MWVNO Orca Sea Phase, with students completing assessments in the final week of this phase. FNO students will continue embarking on an FNO Developmental Sea Phase (ideally on a major warship).

13. MWVNO Sea Phase: All students must pass the MWVNO phase, including those intending to proceed onto FNO. Input by the ship's Officer-in-Charge will be taken into account during assessments; however, the enforcement of the training standard and awarding of the MWVNO qualification lies with embarked NFS(P) navigation staff. The role of the evaluator during this assessment period is to make a reasonable and objective judgement of whether a student is capable of safely executing navigation passages and reacting to COLREGs situations that consistently meet the standard, IAW PC001A. The aim for the student is to demonstrate the necessary skills to navigate a minor warship in an operational environment, particularly when

tired and under pressure. This phase is 12 days consisting of 10 days underway conducting MWVNO training with no more than 5 consecutive days of training and should incorporate a 2 day training pause after 5 days to allow students adequate recovery and planning time in preparation of assessments (typically alongside or at anchor).

14. At the end of MWVNO assessment period, students will be told whether they have obtained the MWVNO qualification. If a student fails to achieve the standard, a TRB will be convened ashore. Successful Reservists and Submariners will be given the opportunity to continue to challenge the FNO qualification if they have the required prerequisites and approval from their CoC. Students intending to challenge the FNO qualification will be awarded the MWVNO (AEFF) qualification upon the completion of this phase.

15. The FNO Sea Phase To maintain the RCN's high standard of navigation, the training must be carefully managed throughout the FNO Sea Phase by following a model that provides students with progressively challenging navigation and BRM situations. The intent is to have the ship's team from the designated major warship support the FNO Developmental Phase. Ideally, the Commanding Officer (CO) acts as the cubicle CO for CEC 001.05 (Conduct a High-Speed Pilotage Run), as the CO for the FNO Developmental Phase (Orca or major warship (preferred)) and Final Assessment Phase. The major warship is a critical resource in the final assessment, enabling students to gain experience and confidence in a real-world pilotage situation with a ship's team. In total, this phase (Developmental and Assessment) is to be 12 days consisting of 10 days underway conducting FNO training with no more than 5 consecutive days of training and should incorporate a 2-day training pause after 5 days to allow students adequate recovery and planning time in preparation of assessments (typically alongside).

16. The entirety of this course is intended to be delivered on the West Coast (MARPAC). In particular, sea phases conducted outside of the MARPAC AO are strongly discouraged. This is largely due to the availability of effective training areas (pilotage waters/geography) and training platforms (Orcas) which the MARPAC AO provides and elsewhere does not. Any training outside of the MARPAC AO (academic or at sea) would require additional planning time and significantly increase course duration. If the final Frigate sea phase was required to be conducted on the East Coast it is recommended that the developmental and assessment weeks, for a total of 2 weeks, be the minimum length for any east coast sea phase.

17. Table 10 below details the minimum required number of developmental runs per student per phase:

Table 10 – Minimum Number of Runs per Phase

MWVNO NABS Developmental Phase	MWVNO Developmental Sea Phase	FNO NABS Developmental Phase	FNO Developmental Sea Phase
7 runs	5 runs	10 runs	5 runs

(3day/3night/1extra)	(2day/2night/1 extra)	(8 FFH, 2 Multi-Class) (4day/4night/2extra)	(min 1 in FFH) (2day/2night/1extra)

PROGRESS MONITORING

18. Student progress will be monitored using formative assessment tools to help the trainee and instructional staff recognize progress or lapses in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. Formative assessment is used during a phase of instruction to reinforce learning so that content is retained.
19. Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW TE SOs or SOPs and CFITES Vol 6 "Conduct of Instruction Programmes."
20. Continuous monitoring of student progress is required to provide the following:
    a. Early warning of trainee difficulties;
    b. Feedback on the effectiveness of training;
    c. Information for any required TRBs; and
    d. Information to assist in the completion of the CF 377 Course Report.
21. Students experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs or SOPs. A record of trainee counselling is to be maintained by the training establishment.
22. Instructional staff shall maintain individual trainee files, including the results of all:
    a. Student performance documentation (e.g. Weekly write-up, medical chits, dress inspection chits, conduct chits, etc.);
    b. Formative assessment material (e.g. ECs, practice periods, homework, assignments and quizzes, etc.); and
    c. Summative assessment material (PCs, CECs).

TRAINING REVIEW BOARD

23. All CEC and PC failures and supplemental failures are to be handled IAW NPTORD 4500-54 "Training Review Board Process." Supplemental CECs and PCs will normally be scheduled not less than 24 hours post counselling, depending on assessment resource availability.
24. TRBs shall be convened and conducted IAW with NPTORD 4500-54 "Training Review Board Process" with the exception that at any time during the course, the FNIs or Venture Div Commander can recommend a student for TRB for such instances as failing to "maintain the standard," developing an unprofessional attitude, acting in such a way to betray the trust of a ship's commanding officer (dishonesty/lying), or other similar reasons.
25. After passing any particular CEC or PC, students are expected to "maintain that standard" for the duration of the course. This could include failing to demonstrate continued mastery of academic knowledge or deteriorating performance during a practical phase. "Maintaining the standard" does not equate to the student needing to pass all follow on pilotage runs, for example, but they should not regress and fail for safety or other significant concerns (what would be known as "Automatic Safety Failure Criteria" on the Assessment Sheet for PC 001A/B & CEC 001.05). The expectation is that once a skill or knowledge base has been mastered, there will not be significant regression during the remainder of the course.
26. In the event of prolonged trainee absence, TE Staff shall determine, on a case by case basis, the necessity to convene a TRB.

RESPONSIBILITY FOR ASSESSMENT

27. Assessment of trainees is to be done IAW CFITES Vol. 7 "Evaluation of Learners."
28. NFS staff provide quality control by administering and scoring all ECs and PCs IAW the standards laid out in this QSP and the assessments developed by NTDC(P).

COURSE REPORTS

29. Course reports (CF 377) shall be prepared and distributed IAW DAOD 5031-9 "Course Reporting and Certificates" and amplified in NPTORD 4500-51 "Course Reports and Certifications."
30. Students will be graded pass or fail. The narrative will indicate the student's strengths and weaknesses based on PC results.
31. The Course Report narrative comments could include:
    a. Ability to work with others;
    b. Ability to work under pressure;
    c. Written and verbal communication skills;
    d. Organizational skills; and
    e. Leadership ability.
32. Training limitations and/or incomplete training must be included in the course report.

CHAPTER 3 **- PERFORMANCE AND ENABLING OBJECTIVES**

AUTHORIZATION FOR THIS CHAPTER AND/OR AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE NAVAL TRAINING DEVELOPMENT CENTRE (PACIFIC)

PO 001 Navigate a vessel of any class

1. Performance: Navigate a vessel of any class (FT0001, AT0001).
2. Conditions:
    1. Given:
        (1) A vessel at sea;
        (2) Bridge Team;
        (3) Access to DWAN;
        (4) References (See Annex K Part 2);
        (5) Supervision;
    2. Denied:
        (1) Assistance;
    3. Environmental:
        (1) On board vessel at sea;
        (2) Any time of day;
        (3) Any weather conditions.
3. Standard: The navigator shall navigate a vessel (AT0001 & FT0001), IAW refs CFCD 130 & BR45 by:
    1. Performing a navigation appraisal;
    2. Conducting navigation planning; and
    3. Executing the navigation by:
        (1) navigating a minor war vessel (MWV) or submarine at speeds up to 15kts (MWVNO);
        (2) navigating any vessel at any speed (FNO).
4. References: See Annex K Part 2
5. Task Numbers (TKSA):
    1. **AEFF** T0001 (FT0001), S0001, S0002, S0003, S0005, S0006, S0008, K0001, K0004, K0011, K0018, K0025, K0037, K0047, K0048; and
    2. **AEEA** T0001 (AT0001), S0001, S0005, S0006, S0007, S0010, K0001, K0002, K0003, K0008, K0009.
6. Training Limitations:

1. Sea phase will only take place on a naval training vessel (Orca Class preferred) for MVWNO and a Major Warship (Halifax Class Preferred) for FNO.
2. Submarines will not be used for training.

7. Remarks:

a. Assistance is commensurate with what would be available from the bridge team and fellow navigating officers.

b. Refer to Annex F for clarification of terms.

PO 001 ***Navigate a vessel (any class)***

EO 001.01 **Perform a navigation appraisal**

1. Performance: Perform a navigation appraisal.

2. Conditions:

   a. Given:

      (1) References (See Annex K Part 2)

      (2) BR45 Vol 1 – 3 Checklists;

      (3) DWAN Laptop with internet access

      (4) WECDIS;

      (5) Total Tides software;

      (6) Parallel ruler;

      (7) Chart and Publication outfit;

      (8) Cue (commander's intent or run sheet);

      (9) Assistance; and

      (10) Supervision;

   b. Denied:

      (1) Nil;

   c. Environmental:

      (1) Any.

3. Standard: The trainee shall perform a navigation appraisal using the BR45 Vol 1 (3 checklists) as performance support and IAW the refs (listed in Annex K Part 2 AAP) by:

   a. Performing navigation mission analysis by:

      (1) Confirming COs intent;

      (2) Considering the operational situation:

         (a) Displaying AMLs; and

         (b) Reviewing intelligence products including;

            (i) Port information reports from the IntO; and

            (ii) Port recce reports from the FLS; and

   b. Conducting navigation research by:

(1) Gathering relevant navigation information including:

(a) Ordering charts and publications;

(b) Liaising with outside agencies;

(2) Reviewing relevant navigation information;

(3) Plotting initial routing IAW SOA and fuel considerations;

(4) Interpreting meteorological data;

(5) Calculating tides and currents;

(6) Calculating safety and clearances including;

(a) LDL; and

(b) Clearance over/under; and

(7) Confirming the reliability and condition of the vessel's navigational systems and equipment;

(8) Calculating astronomic factors; and

(9) Calculating time zones.

c. Complete a Command Appraisal Brief.

4. Time Required: See Annex K & Annex G

5. Remarks: (limitations, development considerations)

a. Assistance is commensurate with what would be available from the bridge team and fellow navigating officers.

b. Refer to Annex F for clarification of terms.

c. Chart and publication outfit as issued to HMC Ships (as described in Annex F);

d. Run sheet as described in Annex F - Table 2.

6. Lesson Specifications: Learning objective details and associated references are listed in Annex K Part 2 Activity and Assessment Plan.

7. Assessment Details: Assessment details can be found in Annex K Part 1 (Assessments Overview), Part 2 (AAP) and Part 3 (Assessment Checklists).

PO 001 ***Navigate a vessel (any class)***

EO 001.02 **Conduct Navigation Planning**

1. Performance: Conduct navigation planning.

2. Conditions:

   a. Given:

      (1) References (See Annex K Part 2)

      (2) BR45 Vol 1 – 3 Checklists;

      (3) DWAN Laptop with internet access

      (4) WECDIS;

      (5) Total Tides software;

      (6) Navigation Tools;

      (7) Chart and Publication Outfit;

      (8) Cue (commander's intent or run sheet);

      (9) Assistance; and

      (10) Supervision;

   b. Denied:

      (1) Nil;

   c. Environmental:

      (1) Any.

3. Standard: The trainee shall conduct navigation planning using the BR45 Vol 1 (3 checklists) as performance support and IAW the refs by:

   a. Creating a NavPlan IAW the current NavPlan checklist in the QSP at Annex K Part 3 including:

      (1) Ship-handing;

      (2) Speed plan;

      (3) Time gates;

      (4) CDP/PNR;

      (5) CIPs;

      (6) RAS/RVs/SOE(serials)

   b. Incorporating operational factors including:

(1) Risk management including:

(i) Satellite communications;

(ii) Alternate navigation methods in C2D2E;

(2) Force protection;

(3) Water-space management;

c. Conducting internal preparations including:

(1) Liaising with OpsO regarding the ship's plan;

(2) Submitting timings and flex inputs;

(3) Updating charts and permits;

(4) Planning system and equipment checks; and

d. Briefing the navigation IAW with CFCD 130 by:

(1) Creating a navigation brief; and

(2) Scheduling the brief.

4. Time Required: See Annex K & Annex G

5. Remarks: (limitations, development considerations)

a. See Annex F for what is meant by Navigation Tools and Chart & Publication Outfit;

b. Assistance is commensurate with what would be available from the bridge team and fellow navigating officers.

c. Chart and publication outfit as issued to HMC Ships;

d. References are listed in Annex K Part 2;

e. Run sheet is the tool used to detail the commander's intent, weather, and other scene set-up directives

6. Lesson Specifications: Learning objective details and associated references are listed in Annex K Part 2 Activity and Assessment Plan.

7. Assessment Details: Assessment details can be found in Annex K Part 1 (Assessments Overview), Part 2 (AAP) and Part 3 (Assessment Checklists).

PO 001Navigate a vessel (any class)

EO 001.03 Execute the navigation

1. Performance: Execute the navigation of a vessel of any class.

2. Conditions:

   a. Given:

      (1) References (See Annex K Part 2);

      (2) Cue (commander's intent or run sheet);

      (3) Simulator or vessel at sea;

      (4) Navigation Systems and Equipment (see Remark 5.b.);

      (5) Chart and Publication Outfit;

      (6) Assistance (see Remark 5.c.);

      (7) Supervision (see Remark 5.d);

   b. Denied:

      (1) Nil;

   c. Environmental:

      (1) In a simulator or in a vessel at sea;

      (2) Any time of day;

      (3) Any weather conditions.

3. Standard: The trainee shall execute navigation of any class of vessel IAW refs by:

   a. Employing collision avoidance techniques;

   b. Employing bridge resource management (BRM) principles including:

      (1) Consideration of light/shapes; and

      (2) Sound signal requirements; and

   c. Assistance is commensurate with what would be available from the bridge team and fellow navigating officers.

   d. Adapting to navigation in a satellite denied or EMCON environment;

   e. Determining the ship's position;

   f. Conducting a wheel-over routine;

   g. Performing Shiphandling in seamanship evolutions; and

h. Monitoring the navigation passage;

4. Time Required: See Annex K & Annex G

5. Remarks: (limitations, development considerations)

   a. Assistance in the givens refers only to the level of bridge team assistance that can be expected at sea;

   b. Supervision is commensurate with navigation regime, and to the level of a non-informed Captain.

   c. Navigation systems and equipment shall be in various states of functionality;

   d. References are listed in Annex K Part 2;

6. Lesson Specifications: Learning objective details and associated references are listed in Annex K Part 2 Activity and Assessment Plan.

7. Assessment Details: Assessment details can be found in Annex K Part 1 (Assessments Overview), Part 2 (AAP) and Part 3 (Assessment Checklists).

PO 002 Navigate a Formation of Vessels (of any class) (AT0002)

1. Performance: Lead the navigation a formation of vessels (of any class) (AT0002).

2. Conditions:

   a. Given:

      (1) Multiple vessels at sea;

      (2) Bridge Team;

      (3) Access to DWAN;

      (4) References (See Annex K Part 2);

      (5) Supervision;

   b. Denied:

      (1) Assistance;

   c. Environmental:

      (1) On board vessel at sea;

      (2) Any time of day;

      (3) Any weather conditions.

3. Standard: The navigator shall lead the navigation of a formation of vessels of any class at sea (AT0002), IAW CFCD 130 & BR45 by:

   a. Performing a formation navigation appraisal;

   b. Conducting formation navigation planning; and

   c. Executing the formation navigation including:

      (1) Navigating a minor war vessel (MWV) or submarine at speeds up to 15kts (MWV NO);

      (2) Navigating any vessel at any speed (FNO);

      (3) Pilotage; and

      (4) Task Group Operations.

4. References: See Annex K Part 2

5. Task Numbers (TKSA):

   a. **AEFF**: T0001, S0005, S0005, S0006, S0007, S0008, K0016, K0049; and

   b. **AEEA**: T0002 (AT0002) K0004, K0009, K0011, K0013, K0014, K0015, K0018, K0019

6. Training Limitations:

   a. Training for formation navigation will only occur in the simulator;

   b. Task Group staff cannot be replicated in the simulator environment.

PO 002 ***Navigate a formation of vessels (any class)***

EO 002.01 **Perform a formation navigation appraisal**

1. Performance: Perform a formation navigation appraisal.

2. Conditions:

   a. Given:

      (1) References (See Annex K Part 2);

      (2) BR45 Vol 1 – Voyage Planning Checklist;

      (3) DWAN Laptop with internet access

      (4) WECDIS;

      (5) Total Tides software;

      (6) Parallel ruler;

      (7) Chart and Publication Outfit (See Annex F);

      (8) Cue (commander's intent or run sheet);

      (9) Assistance; and

      (10) Supervision;

   b. Denied:

      (1) Nil;

   c. Environmental:

      (1) Any.

3. Standard: The trainee shall perform a formation navigation appraisal using the BR45 Vol 1 Voyage Planning Checklist as performance support and IAW the refs (see Annex K – AAP) by:

   a. Performing the single ship navigation appraisal;

   b. Confirming COs intent WRT the formation;

   c. Conducting formation navigation research;

      (1) Liaising with formation ships' navigating officers;

      (2) Researching all formation ships' capabilities and characteristics including:

         (a) Military or civilian;

         (b) Size;

(c) Maneuvering characteristics; and

(d) Weapons and sensors;

(e) Communications; and

(3) Reviewing formation doctrine;

(4) Researching Q-routes and lead-throughs;

4. Time Required: See Annex K & Annex G

5. Remarks: (limitations, development considerations)

a. Assistance is commensurate with what would be available from the bridge team and fellow navigating officers.

b. This EO builds on the information contained in EO 001.01 with specific considerations for formation navigation;

c. References are listed in Annex K Part 2.

6. Lesson Specifications: Learning objective details and associated references are listed in Annex K Part 2 Activity and Assessment Plan.

7. Assessment Details: Assessment details can be found in Annex K Part 1 (Assessments Overview), Part 2 (AAP) and Part 3 (Assessment Checklists).

PO 002 ***Navigate a formation of vessels (any class)***

EO 002.02 **Plan for formation navigation**

1. Performance: Plan for formation navigation.

2. Conditions:

   a. Given:

      (1) References (See Annex K Part 2);

      (2) BR45 Vol 1 – Voyage Planning Checklist;

      (3) DWAN Laptop with internet access;

      (4) WECDIS;

      (5) Total Tides software;

      (6) Navigation tools;

      (7) Chart and Publication Outfit;

      (8) Cue (commander's intent or run sheet);

      (9) Assistance; and

      (10) Supervision;

   b. Denied:

      (1) Nil;

   c. Environmental:

      (1) Any.

3. Standard: The student shall conduct formation navigation planning using the BR45 Vol 1 (3 checklists) as performance support and IAW the refs by:

   a. Planning multi-ship seamanship and/or Shiphandling evolutions including:

      (1) Formation anchoring;

      (2) Harbour entrance and exit;

      (3) Coastal and open-ocean transits;

      (4) OOW maneuvers;

      (5) Photo exercises; and

      (6) Formation anchoring; and

   b. Communicating with other ships' navigating officers regarding:

(1) Confirming signals;

(2) Determining safety requirements;

(3) Review/draft messages;

(4) Review NavComm language and terminology;

(5) Formations;

(6) Screens; and

(7) Patrol boxes; and

c. Briefing the formation navigation plan to the commanding officer by:

(1) Creating the brief; and

(2) Scheduling the brief.

4. Time Required: See Annex K & Annex G

5. Remarks: (limitations, development considerations)

a. Assistance is commensurate with what would be available from the bridge team and fellow navigating officers.

b. This EO builds on the information contained in EO 001.02 with specific considerations for formation navigation.

c. References are listed in Annex K Part 2 for each LO;

6. Lesson Specifications: Learning objective details and associated references are listed in Annex K Part 2 Activity and Assessment Plan.

7. Assessment Details: Assessment details can be found in Annex K Part 1 (Assessments Overview), Part 2 (AAP) and Part 3 (Assessment Checklists).

PO 002 Navigate a formation of vessels (any class)

EO 002.03 Execute formation navigation

1. Performance: Execute formation navigation.

2. Conditions:

   a. Given:

      (1) References (See Annex K Part 2);

      (2) BR45 Vol 1 – Voyage Planning Checklist

      (3) Assistance (see remark 5.b.);

      (4) Supervision (see remark 5.c.);

   b. Denied:

      (1) Nil;

   c. Environmental:

      (1) Any.

3. Standard: The trainee shall execute the navigation of a formation of any class of vessels IAW refs by:

   a. Overseeing multi-vessel seamanship/Shiphandling evolutions including:

      (1) Formation anchoring;

      (2) Harbour entrance and exit;

      (3) Coastal and open-ocean transits;

      (4) OOW manoeuvres;

      (5) Photo exercises;

   b. Communicating with other ships by:

      (1) Executing signals;

      (2) Supervising NavComms;

   c. Communicating with Officer in Tactical Command (OTC)/Task Group Commander by:

      (1) Reporting progress;

      (2) Recommending changes;

   d. Monitoring formation navigation including:

      (1) Own vessel; and

(2) Other vessels within the formation.

4. Time Required: See Annex K & Annex G

5. Remarks: (limitations, development considerations)

   a. This EO builds on the information contained in EO 001.03 with specific considerations for formation navigation;

   b. Assistance in this EO is commensurate with assistance that would be provided by the bridge team in an operational environment;

6. Lesson Specifications: Learning objective details and associated references are listed in Annex K Part 2 Activity and Assessment Plan.

7. Assessment Details: Assessment details can be found in Annex K Part 1 (Assessments Overview), Part 2 (AAP) and Part 3 (Assessment Checklists).

PO 003 Lead the Navigation Division and Bridge Team on Any Class of Vessel

1. Performance: Lead the navigation division and bridge team on any class of vessel. (FT0002, AT0003).

2. Conditions:

   a. Given:

      (1) Reference (See Annex K Part 2);

      (2) Supervision;

   b. Denied:

      (1) Nil;

   c. Environmental:

      (1) On board vessel at sea or alongside;

      (2) Any time of day;

      (3) Any weather conditions.

3. Standard: The navigator shall lead the navigation division and bridge team on any class of vessel (FT0002, AT0003), IAW refs by:

   a. Performing divisional duties;

   b. Performing administrative duties;

   c. Managing the operational readiness of the division; and

   d. Leading the bridge team, to include:

      (1) Setting and enforcing bridge standards;

      (2) Providing oversight on training; and

      (3) Leading the team using Bridge Resource Management concepts.

4. References: See Annex K Part 2

5. Task Numbers (TKSA):

   a. **AEFF** T0002, K0034, K0035, K0037; and

   b. **AEEA** T0003, K0011, K0013, K0015, K0017, K0021, K0022.

6. Training Limitations: No met techs or members of the division will be available or simulated during the course.

PO 003 ***Lead the Navigation Division and Bridge Team on Any Class of Vessel***

EO 003.01 **Conduct divisional duties**

1. Performance: Conduct divisional duties
2. Conditions:
   a. Given:
      (1) References (See Annex K);
      (2) DWAN Laptop with internet access
      (3) Assistance; and
      (4) Supervision;
   b. Denied:
      (1) Nil;
   c. Environmental:
      (1) Any.
3. Standard: The trainee shall conduct divisional duties IAW refs by:
   a. Conducting general officer duties by:
      (1) Acting as a divisional officer;
      (2) Creating CFPAS products; and
      (3) Administering disciplinary and administrative measures.
   b. Managing subordinates by:
      (1) Controlling records including:
         (a) Personnel; and
         (b) Training;
      (2) Liaising with the NWO BTL Manager;
      (3) Supervising NOPQ training; and
      (4) Directing Meteorological Technicians.
4. Time Required: See Annex K & Annex G
5. Remarks:
   a. Assistance is commensurate with what would be available from the bridge team and fellow navigating officers.

b. The conduct of general officer duties is covered under other coursework and will not be specifically instructed in this QSP;

c. References are listed in Annex K Part 2;

d. Run sheet is the tool used to detail the (limitations, development considerations)

6. Lesson Specifications: Learning objective details and associated references are listed in Annex K Part 2 Activity and Assessment Plan.

7. Assessment Details: Assessment details can be found in Annex K Part 1 (Assessments Overview), Part 2 (AAP) and Part 3 (Assessment Checklists).

PO 003 Lead the Navigation Division and Bridge Team on Any Class of Vessel

EO 003.02 Conduct administrative duties

1. Performance: Conducting administrative duties
2. Conditions:
   a. Given:
      (1) References (See Annex K Part 2);
      (2) DWAN Laptop with internet access;
      (3) Assistance; and
      (4) Supervision;
   b. Denied:
      (1) Nil;
   c. Environmental:
      (2) Any.
3. Standard: The student shall conduct navigation administrative duties IAW refs by:
   a. Maintaining Navigation Records by:
      (1) Logging and maintaining the ship's VDR;
      (2) Controlling the OOW Notebook and Ship's Log;
      (3) Maintaining the NavO's Workbook;
      (4) Maintaining the Navigation Data Book; and
      (5) Maintaining Vol 1 of the Captain's Ship's Book;
   b. Drafting navigation messages including:
      (1) Significant Incident Reports (SIRs);
      (2) Position and Intended Movement (PIM) Messages;
      (3) OOW Designator Messages;
      (4) BWK qualification Messages; and
      (5) Ships Activity Reports.
   c. Managing Bridge Watch Routines;
   d. Drafting Navigation Administration including:

(1) Captain's Standing Orders;

(2) Occasions to Call;

(3) Pro Notes;

(4) Night Orders;

(5) Sailing Orders;

(6) MARPIFs; and

(7) Bridge Packs; and

e. Preparing for a Navigation Administration Inspection IAW CFCD 130;

f. Conducting secondary duty administration including:

(1) Supervising the ship's mail program;

(2) Supervising Food Services Administration;

(3) Administering a SLoC.

4. Time Required: See Annex K & Annex G

5. Remarks:

a. Assistance is commensurate with what would be available from the bridge team and fellow navigating officers.

b. The conduct of general officer duties is covered under other coursework and will not be specifically instructed in this QSP;

c. Run sheet is the tool used to detail the (limitations, development considerations)

6. Lesson Specifications: Learning objective details and associated references are listed in Annex K Part 2 Activity and Assessment Plan.

7. Assessment Details: Assessment details can be found in Annex K Part 1 (Assessment Overview), Part 2 (AAP) and Part 3 (Assessment Checklists)

PO 003 ***Lead the Navigation Division and Bridge Team on Any Class of Vessel***

EO 003.03 **Manage the bridge team**

1. Performance: Manage the bridge team
2. Conditions:
   a. Given:
      (1) A bridge team;
      (2) References (See Annex K Part 2);
      (3) DWAN Laptop with internet access;
      (4) Assistance; and
      (5) Supervision;
   b. Denied:
      (1) Nil;
   c. Environmental:
      (1) Any.
3. Standard: The trainee shall lead the bridge team IAW refs by:
   a. Developing the bridge team by:
      (1) Training Bridge Watchkeepers;
      (2) Training the watch on deck; and
      (3) Conducting professional development; and
   b. Practicing BRM by:
      (1) Clarifying roles and responsibilities;
      (2) Resolving ambiguity;
      (3) Advising the OOW;
      (4) Enforcing bridge standards; and
      (5) Reacting to emergencies.
4. Time Required: See Annex K & Annex G
5. Remarks:
   a. Assistance is commensurate with what would be available from the bridge team and fellow navigating officers.

b. Resolving ambiguity (para 3.b.(2)) as it pertains to bridge resource management;

6. Lesson Specifications: Learning objective details and associated references are listed in Annex K Part 2 Activity and Assessment Plan.

7. Assessment Details: Assessment details can be found in Annex K Part 1 (Assessment Overview), Part 2 (AAP) and Part 3 (Assessment Checklists).

PO 004 Provide advice to Command on Navigation Matters

1. Performance: Provide advice to command on navigation matters (FT0003, AT0004).

2. Conditions:

   a. Given:

      (1) References (See Annex K); and

      (2) Supervision;

   b. Denied:

      (1) Assistance;

   c. Environmental:

      (1) Any time of day;

      (2) Any weather conditions; and

      (3) At sea, alongside, or ashore.

3. Standard: The navigator shall provide advice to command on navigation matters (AT0004 & FT0003) IAW references IOT challenge or inform decision making whether informally or formally, by:

   a. Conducting independent research;

   b. Conveying accurate information regarding but not limited:

      (1) International Maritime Laws and Conventions;

      (2) Shiphandling;

      (3) Meteorology;

      (4) Systems and Equipment;

      (5) Bridge Resource Management; and

      (6) Administrative matters.

4. References: See Annex K

5. Task Numbers (TKSA):

   a. **AEEF**: T0003 (FT0003), K0001, K0002, K0004, K0005, K0006, K0007, K0008, K0009, K0010, K0011, K0012, K0013, K0014, K0015, K0016, K0017, K0018, K0019, K0020, K0021, K0023, K0024, K0025, K0026, K0027, K0028, K0029, K0030, K0031, K0033, K0034, K0035, K0036, K0037, K0038, K0040, K0041,

K0042, K0043, K0044, K0045, K0046, K0047, K0048, K0049, K0050, S0001, S0002, S0004, S0005, S0006, S0008; and

b. **AEEA**: T0004 (AT0004), K0001, K0003, K0007, K0008, K0009, K0010, K0012, K0013, K0014, K0015, K0016, K0018, K0019, K0020, K0021, K0022, S0001, S0003, S0005, S0006, S0007, S0010.

6. Training Limitations: Not able to present students with all possible situations or circumstances that would require advice to command.

PO 004 Provide Advice to Command on Navigation Matters

EO 004.01 Conduct Independent Navigation-Related Research

1. Performance: Conduct Independent Navigation-Related Research
2. Conditions:
   a. Given:
      (1) References (See Annex K);
      (2) DWAN Laptop with internet access;
      (3) Phone;
      (4) Chart and Publication Outfit;
      (5) Assistance; and
      (6) Supervision;
   b. Denied:
      (1) Nil;
   c. Environmental:
      (1) Any.
3. Standard: The student shall conduct independent navigation-related research using refs by:
   a. Identifying tools, resources and references available to the Navigator by:
      (1) Relating organizations to the references which they publish and distribute; and
      (2) Acquiring publications and references by:
         (a) Ordering; and
         (b) Purchasing.
   b. Researching information required to advise command by:
      (1) Determining the availability of information required to satisfy the completion of the appraisal and planning checklists; and
      (2) Locating the required information within the references.
4. Time Required: See Annex K & Annex G
5. Remarks:

a. Assistance is commensurate with what would be available from the bridge team and fellow navigating officers.

b. The information gathered during this EO is foundational to the EO - Convey Accurate Information;

c. Purchasing in this context refers to LPO, or by ordering through the supply department.

6. Lesson Specifications: Learning objective details and associated references are listed in Annex K Part 2 Activity and Assessment Plan.

7. Assessment Details: Assessment details can be found in Annex K Part 1 (Assessment Overview), Part 2 (AAP) and Part 3 (Assessment Checklists)

PO 004 Provide Advice to Command on Navigation Matters

EO 004.02 Convey Accurate Information to Command

1. Performance:

2. Conditions:

   a. Given:

      (1) References (See Annex K);

      (2) DWAN Laptop with WECDIS and internet access

      (3) Chart and Publication Outfit;

      (4) Assistance; and

      (5) Supervision;

   b. Denied:

      (1) Nil;

   c. Environmental:

      (1) Any.

3. Standard: The trainee shall convey accurate information from the refs to Command by:

   a. Providing advice formally and/or informally on the following subjects;

      (1) International maritime laws and regulations;

      (2) Shiphandling;

      (3) Meteorology;

      (4) Employment of navigation systems and equipment;

      (5) Bridge Resource Management; and

      (6) Administrative matters pertaining to navigation;

   b. Responding to immediate and/or follow-on questions by:

      (1) Clarifying; and

      (2) Further researching; and

   c. Anticipating the needs and requirements of command.

4. Time Required: See Annex K & Annex G

5. Remarks: Assistance is commensurate with what would be available from the bridge team and fellow navigating officers.

6. Lesson Specifications: Learning objective details and associated references are listed in Annex K Part 2 Activity and Assessment Plan.
7. Assessment Details: Assessment details can be found in Annex K Part 1 (Assessment Overview), Part 2 (AAP) and Part 3 (Assessment Checklists).

PO 005 Manage Navigation Systems and Equipment

1. Performance: Manage Navigation Systems and Equipment (FT0004, AT0005).
2. Conditions:
   a. Given:
      (1) A vessel;
      (2) References (See Annex K);
      (3) Access to DWAN/SHIPLAN/internet;
      (4) Supervision; and
      (5) Assistance;
   b. Denied:
      (1) Nil;
   c. Environmental:
      (1) Onboard a vessel at sea or alongside;
      (2) Any time of day; and
      (3) Any weather conditions.
3. Standard: The navigator shall manage navigation systems and equipment IAW refs (FT0004, AT0005) by:
   a. Mitigate errors and faults;
   b. Troubleshoot errors and faults by:
      (1) Immediately responding to system errors and faults; and
      (2) Conducting basic troubleshooting; and
   c. Liaising with other departments regarding maintenance and repair of systems and equipment.
4. References: See Annex K
5. Task Numbers (TKSA):
   a. **AEFF**: T0002, T0003, T0004, S0003, S0004, K0004, K0005, K0006, K0007, K0008, K0009, K0019, K0026, K0027, K0028, K0029, K0032, K0037, K0038
   b. **AEEA**: T0001, T0003, T0004, T0005, S0002, S0004, K0004, K0009, K0011, K0012, K0018, K0021; and
6. Training Limitations:

a. Errors and faults introduced during NABS training will be limited to the available systems installed for training;

b. Errors or faults introduced during a sea phase will be at the OIC/Commanding Officer's discretion; and

c. The level of assistance provided could mimic personnel available on ship, though may be limited/restricted in order to test the extent of the Navigator's knowledge and abilities.

PO 005 ***Manage Navigation Systems and Equipment***

EO 005.01 **Mitigate Errors and Faults in Navigation Systems and Equipment**

1. Performance: Mitigate Errors and Faults in Navigation Systems and Equipment

2. Conditions:

   a. Given:

      (1) References (See Annex K);

      (2) Navigation Systems and Equipment;

      (3) DWAN Laptop with internet access;

      (4) WECDIS;

      (5) Total Tides software;

      (6) NavPac Software; and

      (7) Supervision; and

   b. Denied:

      (1) Assistance;

   c. Environmental:

      (1) Any.

3. Standard: The student shall mitigate errors and faults in navigational systems and equipment utilizing operator and technical documents, software help functions and setup checklists as performance supports by:

   a. Configuring navigation systems and equipment IAW references and SOPs, including:

      (1) Terrestrial Radio Aids to Navigation;

      (2) Satellite Aids to Navigation;

      (3) Echo Sounders and Speed Logs;

      (4) Attitude and Heading References and Inertial Navigation Systems (INS);

      (5) The Magnetic Compass and Degaussing;

      (6) Digital Chart Systems, Radars and AIS;

      (7) Miscellaneous Navigation Equipment;

      (8) Small Craft Navigation Equipment; and

(9) Asset Tracking; and

b. Describing common errors and faults with navigation systems and equipment by:

(1) Associating the alarm to system or equipment; and

(2) Identifying what the alarm indicates.

4. Time Required: See Annex K & Annex G

5. Remarks:

a. Systems and equipment issued to and fitted in HMC Ships and RCN Vessels.

6. Lesson Specifications: Learning objective details and associated references are listed in Annex K Part 2 Activity and Assessment Plan.

7. Assessment Details: Assessment details can be found in Annex K Part 1 (Assessment Overview), Part 2 (AAP) and Part 3 (Assessment Checklists).

PO 005 ***Manage Navigation Systems and Equipment***

EO 005.02 **Perform Troubleshooting on Navigation Systems and Equipment**

1. Performance: Perform Troubleshooting on Navigation Systems and Equipment

2. Conditions:

   a. Given:

      (1) References (See Annex K);

      (2) Navigation systems and equipment;

      (3) DWAN laptop with internet access;

      (4) WECDIS;

      (5) Total Tides software;

      (6) NavPac software;

      (7) Supervision; and

      (8) Assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) Any.

3. Standard: The trainee shall perform troubleshooting on navigation systems and equipment utilizing operator and technical documents and software help functions as performance supports by:

   a. Reacting immediately to system's alarms and/or indications of equipment errors or faults;

   b. Conducting troubleshooting IAW refs and SOPs to include:

      (1) Advanced troubleshooting WECDIS;

      (2) Basic troubleshooting on other navigation systems and equipment by:

         (a) Performing basic corrective actions including:

            (i) Restarting systems; and

            (ii) Manipulating equipment; and

         (b) Accurately describing the error or fault to technicians and support staff.

(3) Satisfying material Baseline Standards (MBS) by:

(a) Applying knowledge of MBS requirements pertaining to navigation systems and equipment; and

(b) Assisting in risk assessments and mitigation plans.

4. Time Required: See Annex K & Annex G

5. Remarks:

a. Systems and equipment issued to and fitted in HMC Ships and RCN Vessels; and

b. Assistance only from the technician as requested.

6. Lesson Specifications: Learning objective details and associated references are listed in Annex K Part 2 Activity and Assessment Plan.

7. Assessment Details: Assessment details can be found in Annex K Part 1 (Assessment Overview), Part 2 (AAP) and Part 3 (Assessment Checklists).

PO 005 ***Manage Navigation Systems***

EO 005.03 **Liaise with other Departments and Organizations Regarding Navigation Systems and Equipment**

2. Performance: Liaise with other Departments and Organizations Regarding Navigation Systems and Equipment.

3. Conditions:

   a. Given:

      (2) References (See Annex K);

      (3) DWAN laptop with internet access;

      (4) WECDIS;

      (5) Total Tides software;

      (6) NavPac software;

      (7) Supervision; and

      (8) Assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) Any.

4. Standard: The trainee shall liaise with other departments and organizations regarding navigation systems and equipment by:

   a. Following internal reporting procedures for system and equipment errors and faults IAW ship SOPs and references including:

      (1) SSOs; and

      (2) CSOs.

   b. Following up on any installation and/or repairs of navigation systems and equipment by:

      (1) Discussing installations and/or work conducted with technicians; and

      (2) Verifying functionality of systems and equipment.

   c. Engaging with external organizations and resources for support as required by:

(1) Associating the external organizations and resources for the identified problem including:

(a) Fleet Maintenance Facilities (FMF);

(b) In-Service Support Contractors (ISSC);

(c) Hydrographic Service Office (HSO);

(d) Naval Fleet Readiness (NFR);

(e) Life Cycle Material Manager (LCMM);

(f) Fleet Technical Authority (FTA);

(g) Venture Navigation Division (VND); and

(h) Sea Training Group (STG).

(2) Communicating with internal departments regarding any external support received.

5. Time Required: See Annex K & Annex G

6. Remarks:

a. Systems and equipment issued to and fitted in HMC Ships and RCN vessels; and

b. Assistance is commensurate with what would be available from the bridge team and fellow navigating officers.

7. Lesson Specifications: Learning objective details and associated references are listed in Annex K Part 2 Activity and Assessment Plan.

8. Assessment Details: Assessment details can be found in Annex K Part 1 (Assessment Overview), Part 2 (AAP) and Part 3 (Assessment Checklists).

PO 006 Manage navigation aspects of the Maritime Battlespace

1. Performance: Manage navigation aspects of the maritime battlespace (AT0005, FT0006).
2. Conditions:
   a. Given:
      (1) A major warship at sea or alongside;
      (2) An operations team or simulated;
      (3) Access to CSNI;
      (4) References (See Annex K);
      (5) Supervision;
   b. Denied:
      (1) Assistance;
   c. Environmental:
      (1) On board vessel at sea or alongside;
      (2) Any time of day;
      (3) Any weather conditions.
3. Standard: The Fleet Navigating Officer shall manage navigation aspects of the maritime battlespace (T00010) IAW the references, by:
   a. Performing the duties of OOW during Action Stations; and
   b. Providing navigation support for operations.
4. References: See Annex K
5. Task Numbers (TKSA):
   a. **AEFF**: T0005, S0005, K0006, K0010, K0014, K0016, K0026, K0029, K0037, K0049, K0050; and
   b. **AEEA**: T0006, K0004, K0009, K0011, K0013, K0014, K0015, K0018, K0019.
6. Training Limitations: Training will take place in a classroom and simulator setting only.

PO 006 ***Manage the Navigation Aspects of the Maritime Battlespace***

EO 006.01 **Perform the Duties of Action Stations OOW**

1. Performance: Perform the Duties of Action Stations OOW

2. Conditions:

   a. Given:

      (1) References (See Annex K);

      (2) Bridge team;

      (3) Full Halifax Class operations team with CMS in SIM;

      (4) WECDIS;

      (5) Battle orders;

      (6) Assistance; and

      (7) Supervision;

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) Any.

3. Standard: The trainee shall perform as Action Stations OOW IAW references (see Annex K) by:

   b. Taking the watch by:

      (1) Reacting to the action alarm;

      (2) Arriving on the bridge;

      (3) Employing BRM; and

      (4) Setting up bridge communications.

   c. Directing the watch by:

      (1) Responding to the threat including:

         (a) Action conning; and

         (b) Manoeuvring based on the threat.

      (2) Reacting to damage by:

         (a) Supervising bridge rapid survey; and

(b) Liaising with the EOOW in action.

(3) Executing countermeasures including:

(a) Directing return fire/engagement IAW battle orders; and

(b) Directing the launch of material countermeasures.

(4) Enabling fundamental bridge performance by:

(a) Providing manoeuvring and bridge tactical (visual) picture to the CO/ORO;

(b) Liaising with operations on tactical and equipment matters (ORO, SCS, TASSUP); and

(c) Acting as the liaison between Bridge/Ops/MCR.

d. Maintaining situational awareness by:

(1) Operating the combat management system to include:

(a) OOW role selection; and

(b) Threat information display.

(2) Utilising BRM procedures.

4. Time Required: See Annex K & Annex G

5. Remarks:

a. References are SECRET with the exception of SSOs;

b. Assistance is commensurate with what would be available from the bridge team and fellow navigating officers.

c. No tactical trainer is available; and

d. Playbook will detail scenarios to include multi-threat action.

6. Lesson Specifications: Learning objective details and associated references are listed in Annex K Part 2 Activity and Assessment Plan.

7. Assessment Details: Assessment details can be found in Annex K Part 1 (Assessment Overview), Part 2 (AAP) and Part 3 (Assessment Checklists).

PO 006 ***Manage the Navigation Aspects of the Maritime Battlespace***

EO 006.02 **Provide Navigation Support to Operations**

1. Performance: Provide Navigation Support Operations.

2. Conditions:

   a. Given:

      (1) References (See Annex K);

      (2) WECDIS;

      (3) Bridge team;

      (4) Indirect Fire Trainer;

      (5) Supervision; and

      (6) Assistance.

   b. Denied:

      (1) Nil;

   c. Environmental:

      (2) Any.

3. Standard: The trainee shall provide navigation support to operations IAW the refs by:

   a. Advising on Navigation Matters pertaining to:

      (1) NEO/HADR;

      (2) FISHPAT/SARZONE;

      (3) SAR;

      (4) TTW/CZ;

      (5) Manoeuvring Plans;

      (6) MIO;

      (7) Small Boat Ops;

      (8) Force Protection; and

      (9) Information operations;

   b. Contributing to:

      (1) Intelligence Collection including:

(a) Port intelligence reports;

(b) Data recording;

(c) INTREP;

(2) Joint Fires planning;

(3) NGFS by:

(a) Providing spotting support; and

(b) Plotting support;

c. Creating Operational Navigation Products including:

(1) WECDIS Water space management (patrol boxes/moving havens/screens);

(2) Hydrographic survey;

d. Conducting NavWar by:

(1) Utilising WECDIS SPMs;

(2) Utilising W-AIS;

(3) Responding to jamming and/or spoofing;

(4) Employing deceptive Lighting;

(5) Altering NavAids; and

(6) Exploiting PNT opportunities.

4. Time Required: See Annex K & Annex G

5. Remarks:

a. Lessons with SECRET level documents will require an SDA for course material instruction and discussion;

b. Assistance will be provided at the level expected of a normal bridge/ops team at sea;

c. A playbook will detail the aspects of the indirect fire trainer.

6. Lesson Specifications: Learning objective details and associated references are listed in Annex K Part 2 Activity and Assessment Plan.

7. Assessment Details: Assessment details can be found in Annex K Part 1 (Assessment Overview), Part 2 (AAP) and Part 3 (Assessment Checklists).

Annex A – Main References

A = CAF or other official Canadian government publication
B = Official Allied Force or Allied government publication
C = Commercial publication
D = Other

REF CODE	PUBLICATION	QTY
	A References	
A1	CFCD 130 – Canadian Navigation Manual	1
A2	Ship's Standing Orders (SSOs) – accessible via DWAN	1
A3	NPTORDs – accessible via DWAN	1
A4	DAODs – accessible via DWAN	1
A5	QR & Os	1
A6	Captain's Standing Orders	1
A7	NAVORDS – Accessible via DWAN	1
A8	CFCD 106	1
A9	Pronotes – (available on DWAN)	3
A10	HMC Ship's Navigation Data Book	1
A11	Canadian Tide and Current Table Vol 1-7	11
A12	Atlas of Tidal Currents, Vancouver Harbour	11
A13	Public Harbour Regulations	11
A14	Current Atlas (Juan De Fuca Strait)	11
A15	Hydrographic Services Handbook	1
A16	Murray's Tables	11
A18	Canadian Annual Notice to Mariners	1
A19	Chart 1	1
A20	Maritime Command Chart Register	1
A21	MARPIFs – Available on DWAN	1
	B References	
B1	BR45(1) – Admiralty Manual of Navigation, Vol 1 – Principles of Navigation	1
B2	BR45(2) – Admiralty Manual of Navigation, Vol 2 – Astro Navigation	1
B3	BR45(3) – Admiralty Manual of Navigation, Vol 3 – Navigation Systems, Equipment and Instruments	1
B4	BR45(4) – Admiralty Manual of Navigation, Vol 4 – Conduct of Ships at Sea	1
B5	BR45(6) – Admiralty Manual of Navigation, Vol 6 – Shiphandling	1
B6	BR45(8) – Admiralty Manual of Navigation, Vol 8 – WECDIS, Charts and Publications	1

REF CODE	PUBLICATION	QTY
B7	BR45(9) – Admiralty Manual of Navigation, Vol 9 – Operational Navigational Techniques	1
B8	BR 134 – Collisions, Groundings and Other Accidents	6
B9	ATP 1 – Restricted Publication Extracts, Vol 1	6
B10	ATP 4 - Allied Naval Gunfire Support	6
B11	ATP 6 – Mine Warfare Principles	1
B12	ATP 16 – Replenishment at Sea	6
B13	ATP 28 – Allied Anti-Submarine Warfare	6
B14	APP 4 – Allied Maritime Message Formats	6
B15	MTP6 – Mine Warfare Principles	1
B16	MTP24 – Naval Mine Countermeasures	1
B17	ATP4 – Allied Naval Gunfire Support	1
B18	Fleet Guides – Pacific (Item #826)	1
B19	Admiralty Total Tides	11
B20	Admiralty List of Lights	11
B21	Admiralty List of Radio Signals	11
B22	Nautical Almanac	11
B23	NavPac PC	11
C References		
C1	American Practical Navigator (Bowditch), Vol 1/Vol 2	11
C2	"Naval Shiphandling" - Barber	1
C3	NP 100 Mariner's Handbook	2
C4	Hydrographic Services Handbook	2
C5	Nautical Chart Symbols, Abbreviations	11
C6	NP 735 IALA Buoyage System	1
C7	Norries Nautical Tables	11
C8	Sailing directions Complete set of Admiralty, Canadian and American	1 set
C9	Catalogue of Admiralty Charts and Publications	1
C10	Collision Regulations	11
C11	IAMSAR vol 3	11
C12	US Inland Rules	11
C13	A Guide to the Collision Avoidance Rules – Cockcroft and Lameijer (latest edition)	11

REF CODE	PUBLICATION	QTY
C14	"Collisions" – Cahill (latest edition)	11
C15	"Strandings" – Cahill (latest edition)	11
C16	Pilot Charts (Items #476 – 481)	1 ea
C17	NP136 Vol1/Vol2 – Ocean Passages of the World	1
C18	User Manuals for all navigation equipment in use in the fleet	1
D References		

Annex B – Task List

FT#### = AEFF Specific Tasks, AT#### = AEEA Specific Tasks, *Tasks to be added to OSS

TASK DISPOSITION (see note in Foreword (para 2)						
	Task No	**Task Statement**	**Train/No Train Decision**	**Priority**	**Substantiation (Required for NT Decisions & High Priority Items)**	**PO**
AEFF	*FT0001	Navigate an MWS of any class	T	High	Reword - Navigate submarine or MWV of any class	PO 001
	*FT0002	Administer the navigation section of an MWS	T	Med	SB changed to "lead" vice "administer" as this is more CAFJOD (NWO-wide training) - & replace MWS with MWV section as opposed to division due to size of vessel	PO 003
	*FT0003	Provide advice to command on navigation-related matters	T	High	• Critical to success & critical outcome if not trained, instills trust in NavO from Captain • Trained intermediate level (lower level than AEEA) - reword to "Provide advice to command on navigation matters and Shiphandling."	PO 004
	*FT0004	Manage navigational systems and equipment	T	High	New task	PO 005
	*FT0005	Manage navigation aspects of the maritime battlespace	T	High	New task	PO 006
AEEA	AT0001	Navigate an individual ship	T	High	Add "vessel of any class" (OSS update) – an essential part of the job	PO 001
	AT0002	Navigate a formation of ships of any class	T	High	High priority	PO 002
	AT0003	Administer the navigation division of a ship	T	Med	SB changed to "lead" vice "administer" as this is more CAFJOD (NWO-wide training) - division as opposed to section due to size of vessel	PO 003
	AT0004	Provide advice to Commanders on navigation-related staff matters	T	High	• Critical to success & catastrophic outcome if not trained, instills trust in FNO from Captain • Remove "staff" – reword to "Provide advice to command on navigation matters and Shiphandling" – trained at an advanced level (different level than AEFF)	PO 004
	*AT0005	Manage navigation systems and equipment	T	High	New task	PO 005
	*AT0006	Manage navigation aspects of the maritime battlespace	T	High	New task	PO 006

Annex C – Record of Proceedings and Decisions (ROP/ROD)

Dates: 8 Jun 20 – 26 Jun 20

Location: virtually via Zoom.

Board Members:

Position	Rank and name	MOSID/Unit	Dates
TM	LCdr Robert Macquarrie	00207/NTDC(P)	All
Chair	LCdr Robert Macquarrie	00207/NTDC(P)	All
SME	Lt(N) Curtis Dollis Lt(N) Evan Roberts Lt(N) Taylor Workman Capt(N) (ret'd) Clive Butler Capt(N) (ret'd) Kevin Greenwood LCdr (ret'd) Todd Kennedy	00207/NFS(P) 00207/NFS(P) 00207/NTDC(P)	All All All As Req'd As Req'd As Req'd
TDO	Lt Catherine Mason	00211/NTDC(P)	All
Secretary	Lt(N) Roberto Morales	00207/NTDC(P)	All

Board Objective:

This writing board aims to modernize the Fleet Navigating Officer (FNO) QSP accounting for changes to navigation policy and practice, the removal of the Arctic Operations Course and the addition of training material related to operations and equipment acquisition (AOPS and JSS).

Discussions/Decisions

DATE	Discussion/Detail
Day 1: 8 Jun 20	
	Members present: LCdr Macquarrie Lt(N) Dollis Lt(N) Morales Lt(N) Roberts Lt(N) Workman Lt Mason <u>0806 - Introductions</u> Introductions from core board members Not present: Capt(N) (ret'd) Clive Butler Capt(N) (ret'd) Kevin Greenwood LCdr (ret'd) Todd Kennedy <u>0814 - Speech from DCO LCdr Chris Pedrick</u> -4 key points to focus on during the board: • Understanding what performance focus means. Know what performance you want, and how to make that performance happen; • Start building from the assessment. What do you want the end product to look like? Write a clear and focused assessment; • Understand skill and spacing practice; and • Knowing what performance supports you have around you. If there is no need to memorize, why are we training it this way

DATE	Discussion/Detail
	<u>0820 – Speech from C&L Div Mr. Arv Aujla</u> -2 main points to focus on during the board. Trust & Culture -There is a special relationship between a Commanding Officer and their pilot -Many are called to be Navigators, but few are chosen -The element of trust is like glue <u>0828 – TDO Brief from Lt Mason</u> -Try not to leave the meeting, just stop video and mute mic as required -Board chair is the tie breaker. They will keep us on track liaise with OA team as required -TDO will help facilitate the process and draft SITREPS to command -Secretary is responsible to the board chair for ROP, capturing discussions and decisions, actions, suggestions and guidance for development, references and relevant documents, QSP documents and annexes -SMEs will provide content and process. They will participate, keep CFITES in mind as required, supply QSP content and operational requirements. They should come to this from the user's perspective. SMEs should help identify performance standards -Mandate: Modernize the FNO QSP accounting for changes to navigation policy and practice, the removal of the Arctic Operations Course and addition of training material related to operations and equipment acquisition (AOPS and JSS). <u>0852 Expectations</u>: -Lt(N) Dollis: a modernized QSP, and something potentially flexible as we see changes down the road. A QSP that allows future instructors and TDC analysts the ability to be flexible with these changes

DATE	Discussion/Detail
	-Lt(N) Roberts: flexibility in the QSP and alignment to the new NPTORD. A QSP that meets timings, eliminating irrelevant or outdated material. A cleaner QSP that is more closely aligned with what it is like to be a real NavO not just focusing on pilotage. A QSP that also meets transport Canada requirements if we are at all able. -Lt(N) Workman: scaffold model for the QSP, modernizing the QSP, setting up for more realistic training - what the job is day 1. Set up the QSP as a prop for changes down the road. -Lt(N) Morales: learning the process -LCdr Macquarrie: turning things into the best ideas we can. A flexible, aligned and realistic course. Linking course back to warfare and operations. The navigator is part of the warfare team. What the NavO does to help the fight. How does a NavO inside and outside the ship enable the fight. Warrior mentality. -Lt Mason: Guide the process, endeavour to keep to the proposed agenda, build an excellent document for the team to take forward into development -Parking lot in google docs. -Board secretary to manage parking lot <u>0908 – Board Chair Brief</u> -Documentation in Palatino linotype -Board secretary to constantly review parking lot -Safe word for the board: Kiwi -Board rules: • stop video if children/others enter the room • 1000 stand-easy • ignore Cat's dog (Winston) if he comes into the room. -Arcs of fire: we are not rebuilding FNO. We are not allowed to. We cannot submit formal requests for new tasks. We can only feed into the OA. The Navy has said no changes to courses during OA. Info will be passed up to them. 2 weeks is viable for time increase on training. Beyond 2 weeks is doable, but more complicated. Cannot preclude people on threshold exams.

DATE	Discussion/Detail
	-Inputs: FNO PE, RCN Nav Review, letter from CCFP to NFS(P), CAARs, instructor feedback -PE findings: QSP and FNO course schedule are not in alignment. QSP must be the gold standard. There is a lack of sea-time on the course, no MLPs <u>0940 – Break for 20 mins – Board Chair away, will return at 1300. Meeting to continue with no decisions to be made in board chair absence.</u> <u>1000 – meeting resumed without board chair</u> <u>1000 - TDO overview of CFITES</u> -Board SMEs do not have AIT or QSP Mgr -CFITES as guidance -Ensure training systems meets operation goals and requirements -CFITES aims to provide performance-based training -Refining PO scalar -Discuss product, process and combo -Training should always be built towards the job. That's why it can be useful to build the assessment first -Discussion: Lt Mason - Link 2015 and 2018 FNO QSPs in the new document. Lt(N) Roberts noted that the 2015 version of the QSP is what they have been using to teach the course. Lt(N) Workman noted that Annex J has been used to provide some brief ROD. Lt(N) Dollis believes 2018 was merely a format change from MARS to NWO and no actual content changes. <u>1047 – Lessons Learned from COVID-19</u> -Lt(N) Dollis: unresponsiveness from CAF to do work from home initiatives -Lt(N) Roberts: the capabilities have existed. The CAF has just been behind. -Lt(N) Workman: how much someone can get done if you just take the time to sit down and do it. -Lt(N) Morales: CAF finally getting on track with newer technologies

DATE	Discussion/Detail
	-Lt Mason: O365 late delivery and limited capacity, lots of tools that making working virtually, effective and efficient MS Teams, Zoom, Google platform (doesn't work well with DWAN) <u>1053 – Target Population Analysis</u> -Course capacity unchanged from previous QSPs. Discussion on where the maximum number of students came from. Bunks? Math? Can you do what you need to do on the course with more than 10? -Discussion regarding the value in sticking with the course minimum since the course is small anyways. The last FNO course, the Commodore was happy to run the course with such a small amount of people. It was noted that it was better to produce some rather than none. It was discussed that the minimum should be 2 to allow for a better student-instructor ratio. -Discussion on changing the optimal number to 5 instead of 6: Lt(N)s Dollis and Roberts noted the difficulty in working with 5. An even number of students allows them to better divide the teams. 5 students make days in the NABS more difficult. **DECISION**: Keep the course minimum at 4. Waivers may be signed to run a smaller course as required. -Discussion regarding throughput: Lt(N) Dollis noted that 8-10 students a year is what has been going. This throughput seems to be meeting the Fleet's requirements. -Discussion regarding course location: West Coast based on availability of training resources such as NABS, ORCAs etc. In addition, it was noted that the East Coast Geography does not lend itself well to pilotage. -Discussion regarding access to computers – yes, this is critical and a WECDIS enabled computer is required as well. -Discussion regarding average age: it varies greatly especially for reservists. However, the last 4 courses have seen an average age of 24. -Discussion regarding competencies required: education level university level likely, high school minimum. Lt(N) Roberts noted that there is no actual pre-requisite. Reserve Officers do not require a degree prior to coming on the course. In addition, RegF may be on CEOTP and may not hold a university degree when they come to the course. Lt(N) Dollis noted that university-level education is desirable as the course has been set up in the style of a university-level course. Lt(N) Workman questioned the utility of allowing a Reserve Officer to complete the full course even though it is not required for them. Board

DATE	Discussion/Detail
	Chair input is required. A RegF candidate should have completed their NOPQ prior to coming to this course. Lt(N) Roberts suggested that without NOPQ, the cap for candidates should be MWVNO. -Discussion regarding attitude preferences: Lt(N) Workman noted that the student should prefer hands-on learning. He also noted that the student should have an inclination to self-learning and be willing to dig into publications, read a PRONOTE on their own. It was noted that the course does dive into some more advanced theories such as tides and astro. -Discussion regarding reading levels: It was discussed that reading levels should be high/advanced. As it was noted before, students will be expected to read on their own. It was discussed that there was no need for high/advanced writing as most of the writing will be generic military administration. Lt(N) Dollis noted that this is a single language course. Not all references are available in French, and the practical portion of being a NavO occurs in English. -Discussion regarding competencies: dedicated, hard-working, focused, perseverance, resiliency, responsibility, teamwork, honesty, integrity, trust worth and decisive. -Discussion regarding experience with target performance: the FNO PIP was discussed, and it was noted that RegF candidates should ideally be NOPQ qualified and have at least 6 months on their ticket. For ResF candidates, they should have sufficient time on their ticket -Discussion regarding ability to perform any part of a PO: Lt(N) Roberts noted that it should be somewhere between basic and intermediate depending on the students' background. -Discussion regarding knowledge of subject matter: students will have completed the PIP prior to coming to the course. Students will also have completed NWO phases II, IIO and IV. Lt(N) Roberts noted that most students will have seen their NavO perform most of the things that they will be learning on course. -Discussion regarding perception of failure: it was noted that failing the course is not necessarily a career breaker. It does provide some delay as an unsuccessful student will be required to select another D-Level. It was noted that self-impact of failure is likely high as it is a career milestone. -Discussion regarding career stage: D-Level, leading to DP2 -Discussion regarding students' interests: students should be fundamentally interested in matters pertaining to the sea. Students should also be interested in the 4 pillars of Leader, Mariner, Manager, Warrior. Lt(N) Dollis noted that for several students, their interests align with performing the job. It was noted that candidates on the course tend to overwork and put the job before themselves. -Discussion regarding attitudes towards the subject matter: students tend to arrive to the course already interested in the material. Instructors rarely have to try and sell the material to students. Students tend to arrive with respect for the material, knowing how critical it is.

DATE	Discussion/Detail
	-Discussion regarding anxiety levels: it was noted that students tend to arrive with high anxiety levels. Lt(N) Dollis noted that there is a cultural aspect to this as NavO's tend to seek excellence. Lt(N) Workman noted that to an extent, the course helps to quell some of this anxiety as it shows the students what resources they do have to perform the job. -Discussion relating to relationship to peers: it was noted that most students tend to arrive already as some sort of friends. Everyone comes in on equal footing, and very rarely do students come in for the wrong reasons. It was noted that the main differences observed in past courses are between those students that are staying on base, and those that have a family in the city. To a larger extent, single students vs. those married and with children. -Discussion relating to feeling towards authorities: It was discussed that students have a healthy level of respect for authority. Most students understand the role that the NavO plays. -Discussion regarding cooperation or competition: it was noted that during the course there is not much time for competition between students. Students either fail or succeed as a team. As such, students tend to lean towards cooperation -Discussion regarding fitness: it was discussed that the average student fitness is average. -Discussion regarding health status: it was noted that students arrive on the course fit-sea. Relating to mental health, students tend to be able to recognize their limits. -Discussion regarding expected annual throughput: board chair to liaise with Occ. Manager to inform this piece. Will email and have conversation regarding this piece. Suggesting no more than 15 per year. The number of ResF Officers was also discussed. There is a minimum number of seats reserved for ResF Officers. However, if those seats are not filled, they may be filled by RegF personnel. -Discussion regarding submariners on the course: usually submariners will arrive with at least their Dive Dolphins. However, Lt(N) Dollis noted that 2 courses ago, there were submariners that arrived with no Dive Dolphins or Surface Fleet experience. **DECISION:** Board Chair will reach out to Occ. Manager and CANSUBFOR for input -Discussion regarding course pre-requisites: The Board Chair noted that there is a need to maintain the pre-requisite of a 6-month consolidation period for watchkeepers prior to coming to the course. It was recommended that NOPQ be a pre-requisite for RegF Officers prior to coming to the course. It was also discussed that there is no reason to remove MWVNO from this course. -Discussion regarding interests and preference of instructors – it was noted that the interests of instructors greatly reflect that of the students: a fundamental interested in the maritime environment. -Discussion regarding instruction via media and non-traditional methods: it was discussed that less time in the classroom focusing on PowerPoints is better. Instructors look forward to more teaching aids, and creating a more engaging learning environment which is better for both students and instructors.

DATE	Discussion/Detail
	-Discussion regarding instructor experience with technology-based instructions: instructors have extensive experience with the NABS, however, there is not much experience with VR and AR. Instructors also use 3D models, videos and animations as teaching aids. -Discussion regarding existing course/programs which FNO must fit within: it was discussed that there is no real course or program with which FNO must fit within. However, FNO does feed into ORO and CDC. -Discussion regarding hardware commonly available: DWAN computers, NABS, NPTT, smartboards, projectors, ECDIS laptop, sextants etc. All classes of ships are used for. <u>1358 – TPA Complete</u> <u>1423 – TDO brief on task analysis</u> -accurately describe tasks -Avoid duplication -Reduced tasks required by TA -Focus training requirements and resources -Provide guidance for design phase -These tasks must be measurable -"Train/no train decisions" -DIF analysis is the preferred method "difficulty, importance, frequency" -Discussion regarding criteria for Train/No-train: it was discussed that the criteria here should be whether or not a Navigator needs to know this day 1 on the job, and whether or not there is a critical safety aspect to it. <u>1440- Task List Train/No Train</u> -AEEA: Navigate an individual ship Discussion: "of any class" should be added so that it reads "navigate any class of ship." **DECISION:** Train – high priority -AEEA Navigate a formation of ships of any class Discussion: should there be mention of operations in a Task Group at this point. Assessed as high based of DIF analysis, and previously mentioned group criteria. **DECISION:** Train – high priority

DATE	Discussion/Detail
	-AEEA Administer the navigation division of a ship Discussion: there was a discussion regarding the wording of choice. It would be preferable for the task to read "lead" instead of "administer" as there is more to this task than just administering. Assessed as a medium for training; however, this is not the case for on-the-job. It was discussed that the leadership portion of this task is critical to reinforce. **DECISION:** Train – medium priority -AEEA Provide advice to Commanders on navigation-related staff matters Discussion: there was discussion as to whether or not this is a sub-task. An FNO may provide advice while not necessarily in a Navigating role. For example, CCD Analysts and FNI's. It is recommended that the "staff" wording be removed. **DECISION:** Train – high priority -AEFF Navigate MWS of any class Discussion: high priority as this is the core job of MWVNO. **DECISION:** Train – high priority -AEFF Administer the navigation section of an MWS Discussion: There is no navigation division on an MWS. The NavO falls under Ops. **DECISION:** Train – medium priority -AEFF Provide advice to Commanders of navigation-related matters: Discussion: critical to success, instills trust in the NavO. **DECISION:** Train – high priority <u>QSP TASKS</u> -Apply corrections to navigational instruments and electronic fixing aids: Discussion: it was discussed that this is a discrete task. It is not trained anywhere else, only on this course. There is importance in mastering the equipment even if the NavO is not the maintainer of the equipment. There is a distinction in managing and administering navigational equipment. The system management piece is critical to navigation and warfighting. **DECISION:** Train – high priority -Plan coastal, ocean, and pilotage passages Discussion: covered under AEEA Task 1, 2, and 4. It is also covered under AEFF Task 1 and 3. **DECISION:** No train -Prevent collisions at sea in accordance with international maritime law:

DATE	Discussion/Detail
	Discussion: trained under AEEA Task 3 and AEFF Task 2. Recommendation that the NavO as a Bridge Resource Manager be captured in NAVORDS and SSOs. There was also a discussion that when the NavO is not actively navigating the ship, they are not responsible for preventing collisions. The safety piece still resides with the CO. **DECISION:** No train. -Maintain charts, publications and ship's navigation records: Discussion: falls under other tasks. **DECISION:** No train. -Interpret weather charts: Discussion: for HCF, there is usually an embarked MetTech. It was also discussed that this task falls under providing advice to Commanders. The NavO would not interpret a chart for the sake of interpreting it. Rather, they would interpret it with the goal of providing some sort of advice. **DECISION:** No train. <u>1545 – End of day 1</u>
Day 2: 9 Jun 20	
	Members present: LCdr Macquarrie m Lt(N) Dollis Lt(N) Morales Lt(N) Roberts Lt(N) Workman Lt Mason <u>0802 - Agenda for the day - Lt Mason</u> -Discussion regarding scalar and what stage we are at, then going to the jam board in Google docs, followed by performance objectives <u>0803 - Performance task analysis – Lt Mason</u> -Rationale for structure of the job, forms the basis for the PO's -Build standard statement, move into design phase -Start with rough grouping of tasks

DATE	Discussion/Detail
	-Add new tasks if required -Edit tasks as required -Sequence tasks -Avoid the use of subjective qualifiers as much as possible -Discussion: no change to PO's between 2015 and 2018 QSP. -Discussion regarding the OA. Occ. Manager is not overly concerned about new tasks that may arise from JSS. <u>0828 – Review of Day 1 ROP</u> <u>0835 - Discussion on Jam board Google drive</u> -Discussion regarding AT0001 & FT0001: it was discussed that both tasks should be linked to the same PO regardless of the idea that MWV NO may be a pre-requisite for FNO. MWV NO and FNO share the same academic background. It was identified that the main differences between the two are speed and the sea phase. MWV NO is required to navigate up to a slower speed compared to FNO, and they do not have a dedicated sea-phase to the MWV platform. It was discussed that Reserve Officers were taught knowledge and tested on material that pertains only to the Halifax Class on previous courses. It was further discussed that some aspects of the academic phase are common across MWV NO and FNO, such as leadership. **DECISION:** 2 standards may be applied, one specifically for MWV NO and the other for FNO. -Discussion regarding IMO guidance on voyage planning: it was noted that according to IMO guidance, voyage planning should be conducted in 4 distinct phases; appraisal, planning, execution and monitoring. It was discussed that execution and monitoring should be grouped together. A passage is rarely executed while not being monitored. **DECISION:** proceed with the idea of appraisal, planning and execution. -Discussion regarding AT0002: it was discussed that the steps involved are similar to those in AT0001 and FT0001. It was discussed that it was also important to consider this task at the Task Group level. The core components remain appraisal, planning and execution. It was also noted that there was minimal coverage of Q routes and MCM.

DATE	Discussion/Detail
	-Discussion regarding ATT0003 and FT0002: it was discussed the AHR and other documentation was being taught. The AHR does not need to be included as it is not necessarily the responsibility of the Navigating Officer. However, the Ship's Log and OOW Notebook do fall under the responsibility of the Navigating Officer as per the Navigation Manual. **DECISION:** remove the AHR as a lesson from this content. -Discussion regarding AT0004 & FT0003: there was discussion on including content relating to bridge-ops interaction and interaction with other departments. There was also discussion on what could be included in this task, such as delegation of control, and the specific role of the Navigating Officer in certain roles such as SAR. -Discussion regarding T0005: there was a discussion on how to best break down this task. Initial thoughts were electronic and non-electronic instruments. This was followed by discussion of internal and external instruments. There was also discussion relating to the design phase for the task that a section for future navigation technology should be included. -Discussion regarding providing navigation support to operations: there was discussion as to whether this was its own task or not. It was agreed that this should be its own task, as the Navigating Officer is not simply a systems manager. They are also part of the team that provides systems. **DECISION:** rename this task to Act as a Naval Warfighter. 1121 - Jam board version 1 complete Performance Objective Titles PO 001 – Navigate a vessel of any class -No discussion regarding title. -Discussion regarding content of the PO: it was discussed that it would be ideal to follow the IMO guidelines in terms of appraise, plan and execute. The term appraisal was agreed upon as it is in line with the IMO and navigation manuals. It was discussed that the appraisal portion was never explicitly assessed, and that it was an opportunity to allow for a more robust assessment.

DATE	Discussion/Detail
	PO 002 – Navigate a formation of vessels of any class -Discussion regarding title: it was discussed that this PO is similar to PO 001. As such, it would be sufficient to include formation for future EO's. **DECISION:** title to navigate a formation of vessels of any class. PO 003 – Lead the navigation division on any class of vessel Discussion regarding title: there was the discussion over the use of the verb manage or lead. It was discussed that lead has a more positive tone and has more of a tendency to motivate. The 4 pillars of Leader, Manager, Mariner, Warrior were considered. It was discussed that while leading the navigation division, the Navigating Officer may still be conducting divisional and administrative duties. **DECISION:** title to be "Lead the navigation division of any class of vessel. PO004 – Support Command Decision Making on navigation matters -Discussion regarding title: there was discussion regarding the use of the term "Commanders." It was discussed that there was potential to create ambiguity regarding position and rank. The term "Commander" should not just imply the rank but also the position. **DECISION:** the term Commander is to be replaced with Command to avoid ambiguity. -Discussion regarding advising command: there was discussion regarding the term "advise" command. Advising may be done in several manners. Advising may be informal but may also be formal. Depending on the nature of what Command is seeking, there will be different ways to present this information. Some information must first be researched and prepared in a more formal brief. Some advising may also come informally on the bridge during a seamanship evolution. Lt(N) Workman noted that there should be a lesson(s) during the course that teaches the different ways a Navigating Officer may advise Command. **DECISION:** title to be Support command decision making on navigation matters PO005 – Manage navigation systems

DATE	Discussion/Detail
	Discussion regarding title: it was discussed that "manage" was the best action verb for this PO. It was discussed that throughout this PO, the Navigating Officer is involved in the process of setting up, correcting, and monitoring equipment. The Navigating Officer may also mitigate faults relating to navigation systems. They may also perform basic troubleshooting once an error or fault has been detected. **DECISION:** title to be Manage navigation systems <u>PO 006 – Act as a Naval Warfighter</u> Discussion regarding title: the original title for this PO was deemed to be Act as a Naval Warfighter. However, while referencing the Spec-N QSP a more appropriate title was found. **DECISION:** title to be Manage navigation aspects of the maritime battlespace - <u>1244 - Completion of Performance Objective titles</u> <u>1250 - Break for lunch. 30-minute break</u> <u>1320 Meeting resumed</u> <u>1322 - Brief on building performance objectives – Lt Mason</u> -the standard statement is more detailed than the performance statement -performance statement must have an action verb and object. It may have a qualifier. Specific to job level or task -conditions must reflect on the job – not during training -the standard statement should be framed in the "SMART" piece. Should mirror what's done on the job. Must answer what, how, how well <u>Building Performance Objectives</u> <u>PO 001 – Navigate a vessel of any class</u>

DATE	Discussion/Detail
	-Discussion regarding givens: a vessel at sea. It will be possible to clarify what "at sea" means at a later point in the QSP. Further discussion on the navigation team will be provided. As this PO will apply in Open Ocean as well, it is best to use the term "bridge team." There was also discussion as to whether the navigation systems should also be included in the Givens. However, it was determined that navigation systems are included in a vessel at sea. **DECISION:** Givens to include a vessel at sea and a bridge team. -Discussion regarding denied: board members agreed that assistance is denied. There was a debate on whether or not supervision is denied. It was determined that, to an extent, the Navigating Officer is still being supervised. **DECISION:** denied assistance. Supervision should be given. -Discussion regarding the environment: navigation may occur only at sea, at any time of day and in any condition. **DECISION:** environmental to contain at sea, any time of day, any weather conditions. -Discussion regarding the standard statement: having a qualifier in the standard statement is not needed. It is implied that if the Navigating Officer is navigating, that they are doing so safely. **DECISION:** no qualifier in the standard statement. -Discussion regarding the standard: There was discussion on whether to keep a firm speed restriction or whether to use the term "maximum speed" to allow for flexibility. It was found that in the previous QSP there was a distinction between the speed at which an MWV NO and FNO would be expected to navigate the ship. **DECISION:** the difference in speed should be captured in a sub-paragraph of the standard. Navigating an MWV or submarine at speeds up to 15kts, and navigating a vessel at any speed for FNO. -Discussion regarding training limitations: over the course of training, only Orca class and Halifax class vessels are used. FNO or MWV NO may navigate the Kingston Class without having previously navigated on it. **DECISION:** to be captured in training limitations <u>PO 002 – Navigate a formation of vessels</u>

DATE	Discussion/Detail
	-Discussion regarding the standard: it was discussed that it would be unrealistic to expect an MWV NO to navigate a formation of ships at the same level as an FNO. **DECISION:** there will be a separate standard for MWV NO and FNO's similar to PO 001. -Discussion regarding training limitations: it was noted that while training, the assistance of Task Group staff may not be replicated. **DECISION:** this limitation is to be added to training limitations <u>PO 003 – Lead the navigation division of any class of vessel</u> -Discussion regarding the qualifier in the standard statement: there is concern that there is not an appropriate qualifier for the standard statement. **DECISION:** there is no need for a qualifier in the standard statement as it would only serve as a false qualifier. <u>PO 004 – Support Command decision making on navigation matters</u> -Discussion regarding the environment: A Navigating Officer may be required to support command decision-making while not necessarily on board a ship. FNI's, for example. **DECISION:** environmental to include at sea, alongside, or ashore. -Discussion regarding training limitations: it is incredibly difficult, if not impossible, to prepare a Navigating Officer for every scenario that they might encounter. **DECISION:** training limitations to include that candidates may not be prepared for all situations which would require them to advise command. <u>PO 005 – Manage Navigation Systems</u> -Discussion regarding assistance: it was debated whether or not assistance should be a given for this PO. It was agreed on by the board that even in environments where the Navigating Officer may not be able to reach out for external assistance, there are still members on board that may be able to offer some form of assistance.

DATE	Discussion/Detail
	DECISION: assistance is a given for PO 005. <u>PO 006 – Manage navigation aspects of the maritime Battlespace</u> -No further discussion <u>1545 End of Day 2</u>
Day 3: 10 Jun 20	
	Members present: LCdr Macquarrie Lt(N) Dollis Lt(N) Morales Lt(N) Roberts Lt(N) Workman Lt Mason <u>0804 - Discussion regarding ROP from Day 2</u> -There was clarification on naming conventions used in ROP differing between tasks and PO's. It was discussed that there is a benefit to adding "equipment" to PO 005 to align with other POs and task list. **DECISION:** rename PO 005 Manage navigation systems and equipment. <u>0820 - Chapter 1 FNO QSP</u> -Discussion regarding the aim: it was noted that with the QSP template, the aim is acceptable; however it is standard. The 2018 QSP had a more robust aim, and it was noted that the aim from the 2018 QSP is used in the course introduction. **DECISION:** use the aim from the 2018 FNO QSP as a basis and modify as required. -Discussion regarding course duration: initial estimate calculation shows a course duration of roughly 135 days. This number will be initially included however it will be refined later.

DATE	Discussion/Detail
	DECISION: include a 135-day provisional course duration with the knowledge that this will be later refined. -Discussion regarding resources: it was discussed that a LCdr is still required, though not necessarily in the role of Senior Instructor. The title of Senior Evaluator was discussed. It was also discussed that the LCdr position is usually the VENTURE Div. Commander. It was discussed that the N-Qual AEEA is required, and AEEG is preferred. **DECISION:** Sr Evaluator – Venture Div Cdr. -Discussion regarding resources: it was discussed that it is preferred that the FNI's are AEEA and past EQ. It was also discussed that it is preferred that 1 FNI is from the West Coast and the other from the East Coast. <u>0845 – Capt(N) (ret'd) Greenwood joined – joined breakout room with Board Chair.</u> <u>0849 – Capt(N) (ret'd) Butler joined – joined breakout room with Board Chair.</u> -Discussion regarding guest speakers as a resource: it was discussed that it is beneficial to have the guest lecturers listed as a resource. It was discussed that it was worth including guest speakers as a resource to supplement the knowledge of the FNI's. It will also allow for substantiation when sending formal requests. **DECISION:** include guest speakers as a critical resource. -Discussion regarding MetTech as guest speaker: it was discussed that it is preferred to have a MetTech that has sea time to come in as a guest speaker. This will allow them to better understand what the needs are of the students on this course as they will have interacted with FNO's prior to coming in as a guest. Discussion regarding JSS Instructor: As the RCN prepares for the JSS, it is beneficial that the FNO course have NavO with JSS experience come in to discuss the platform. It was discussed that, initially, it is likely that this Officer will come from the German Navy with Berlin Class experience. As the RCN develops its own experience, this will likely transition to a Canadian Officer. -Discussion regarding Senior mentor as a guest speaker: it was discussed that the senior mentor should be a rank minimum of Cdr (or Cdr ret'd) and be a former Commanding Officer. It is also preferred to have a senior mentor from a navigation background.

DATE	Discussion/Detail
	-Discussion regarding class-specific instructor: it was discussed that the class-specific instructor would supplement the knowledge of the FNI's. It was discussed that this class-specific instructor is different from the JSS instructor in that this speaker will come from the RCN. It would be possible to have an MWV NO come in and discuss shiphandling from an MWV perspective. <u>0915 – LCdr (ret'd) Kennedy joined – joined breakout room with Board Chair.</u> -Discussion regarding critical resources: it was discussed that the Orca class vessel should be listed as a critical resource. However, it was discussed that it would still be possible to conduct training on another platform such as the MCDV's. **DECISION:** naval training vessel (Orca Class preferred). -Discussion regarding breakout room discussion with Board Chair: it was discussed that, overall, senior leadership was happy with the breakdown of tasks and POs. It was discussed that under PO 004, shiphandling should be brought back. It was discussed that shiphandling is a subset of seamanship. In the eyes of the CO, the Navigating Officer will also advise them on matters pertaining to shiphandling in addition to other evolutions. There was also discussion regarding the incorporation of BRM into different POs or the possibility of having BRM as its own subtask. <u>1048 – Follow up: reach out to DNav P&T regarding the PLAR process</u> -Discussion regarding PLAR process: in addition to reaching out to DNav P&T, it was discussed that the VEN Div. Cdr should review PLARs with consultation from FNI's. **ACTION:** engage DNav P&T for PLAR process. -Discussion regarding pre-requisites: it was discussed that the pre-requisites in this QSP should not be more containing than those in the Occ. Spec. It was discussed that for FNO, one of the pre-requisites should be a minimum of 6 months of a major warship ticket. This is different from the Occ. Spec which specifies 6 months of consolidation post NOPQ. **DECISION:** recommendation to the Occ Mgr to amend the pre-requisite. -Discussion regarding top student: it was discussed that there should still be an award for the top student of the course, as it provides an additional level of excellence for students to aspire to. **DECISION:** create an annex outlining the guidelines for nominating the top student of the course.

DATE	Discussion/Detail
	<u>1309 – Review of PO's prior to sending to HQ for provisional approval</u> -Discussion: no changes to PO 001 and PO 002. -Discussion regarding PO 003: included in the standard: leading the bridge team. It is more than just leading the navigation section. This should include the bridge team. -Discussion regarding PO 004: addition of BRM, adding systems and equipment to the standard. -Discussion regarding PO 005: no changes to PO 005. -Discussion regarding PO 006: no changes to PO 006 <u>1330 – TDO Brief on Design Phase</u> **DECISION:** proceed with creating an assessment plan first <u>Draft PC Assessment for PO 001</u> -Discussion regarding draft PC for PO 001: there was significant discussion regarding the PC for PO 001. Discussion started with the current assessment (PC 006 in the 2015 QSP). Board members were generally content with the current structure on the PC. The discussion then tended towards rebuilding the PC from the beginning. It was noted that there is no assessment of other seamanship evolutions such as departures or alongsides. There was significant discussion regarding whether the pilotage aspect of the PO is what should be assessed. There was further discussion on which aspect to assess and based on what metric. There was discussion regarding which should take priority between open ocean navigation, coastal navigation and pilotage navigation. While there was disagreement between which should take priority, all members agreed that the assessment should include all stages (appraisal, planning and execution). **DECISION:** the board will come back to PC 001. Proceed with the PC for PO 006.

DATE	Discussion/Detail
	<u>1420 – Draft PC Assessment for PO 006</u> -Discussion regarding draft assessment for PO 006: there was significant discussion regarding the priority for the PC. There was discussion of the fact that the Navigating Officer played a critical part in supporting operations other than war. Conversely, there was discussion that the NavO , while acting as OOW while at action stations, should be the priority-based partly on the fact that this is outlined in RCN Doctrine. It was ultimately agreed that the course should provide FNO's a sufficient level of knowledge that when they arrive to their Ship, they will know where to look for references and know the personnel that should be consulted when dealing with operations other than war. However, given the level of immediacy with action stations, a NavO is unlikely to have the time to look for references and consult with other personnel. **DECISION:** act as OOW at action stations to be the priority. -Discussion regarding draft assessment for PO 006: initial discussion for the PC tended to lean towards a practical assessment of the student while at sea. The ship at sea would be brought to action stations, and the assessment of the student would begin when they arrived on the bridge. It was discussed that there was no need to bring the full ship to the action station but to just close up the bridge and ops (unless the CO felt otherwise). It was discussed that assessing the student at sea allowed the SECRET content of the course to be taught and examined. The discussion then shifted to the training time impact of this assessment method. It was determined that this method would not be practical given the time constraint of the Halifax Class sea phase. **DECISION:** PC for PO 006 to be a written exam. -Discussion regarding draft assessment for PO 006: it was discussed that the structure of the PC should be written and comprise both providing support to warfare and OOTW and acting as OOW while at action stations. It was discussed that supporting operations should remain at the UNCLASS level. There was a follow-up discussion regarding the value of just teaching the UNCLASS material. It was discussed that the SECRET material could still be taught in the appropriate environment and tested in an appropriate environment to ensure that lesson content was hoisted in. **DECISION:** non-critical SECRET EC to be tested in an appropriate environment. PC to remain UNCLASS and incorporate both supporting warfare and OOTW and act as OOW while at action stations.

DATE	Discussion/Detail
	-Discussion regarding the impact of failure: it was discussed that in the previous QSP, 3 consecutive failures would result in the student being unsuccessful on the course. There was a discussion regarding the NPTORD, which sought to provide guidance on training failures. It was discussed that the impact of training failure might be noted in the QSP under special instructions. **DECISION:** a training failure will lead to review and re-write of the exam. 2 consecutive failures to result in a TRB. <u>1545 – End of day 3</u>
Day 4: 11 Jun 20	
	Members present: LCdr Macquarrie Lt(N) Dollis Lt(N) Morales Lt(N) Roberts Lt(N) Workman Lt Mason <u>Opening the discussion with a recap from communication last night with Occ Mgr and HQ</u> -Discussion regarding the PLAR process: It was noted from an email with the Occ manager that the current process for PLARs is that they go to the Occ Manager and are then forwarded to NPTGHQ. -Discussion regarding the number of students on the course: it was discussed from an email with the Occ Manager that there are no reserved seats on the course for Reserve members of CANSUBFOR members. Discussion regarding the naming of Supporting command decisions: In preliminary talks with HQ, the term advise command is preferred to support command. There was discussion that the board prefers the term support, as the role of the NavO is unique in that they do more than just advise. It was discussed that several members might advise Command, but there are few that support it. It was also noted that the NavO role also provides a challenge function to Command, which is more than just advising Command. **DECISION:** revisit the naming at a later point. <u>0809 - Review of ROP – No further discussion</u>

DATE	Discussion/Detail
	<u>0812 – Draft assessment of PC 004</u> -Discussion regarding PC 004: there was discussion as to how to best break down this PC. It was discussed that given the amount of content in this PO, the PC would be very large and runs the risk of not being able to assess all the key points. There was then discussion of potentially breaking down the PC into different parts (parts 1, 2, 3 etc.). There was then discussion of the possibility of having this PC assessed as part of PC 001. There was then discussion of giving the students a problem to take home and analyse, or providing them with a problem and then giving them sufficient time to prepare a brief for Command. The possibility of conducting this PC while at sea was discussed. It was discussed that this would be suitable as it could be conducted on a training vessel so as to include MWV NO and FNO. It was then discussed that EC could be written and used to build to build-up to the PC, which will take place at sea. **DECISION:** PC to be practical and written at sea. Students will be required to provide a brief to Command on a shiphandling problem. Students will then be required to advise Command during the execution of the plan. <u>0848 – Draft assessment for PC 003</u> -Discussion regarding the format of the exam: it was initially discussed that the PC should be a written exam. It was then discussed that the written portion could be in the ECs, but the PC should be a performance-based assessment. There was discussion that performing administrative duties is difficult to assess in any other form that through written exams. There was also discussion on how to assess the other portion of the PO, which is leading a bridge team. It was discussed that it would be possible to have 2 separate PCs for this PO. There was also discussion as to whether references should be allowed or not. It was discussed that there are some things that the NavO must know off-hand, while other questions will need the support of references. **DECISION:** 2 PCs, part A relating to administrative duties of the NavO, part B relating to leading a bridge team. 3-hour exam, the first 2 hours students will not be allowed to access references. In the last hour, students will be allowed to access references in order to confirm previous answers and to finish the remainder of the exam. -Discussion relating to part B (leading a bridge team): the appropriate time to conduct this assessment was discussed. It was discussed that it would be possible to assess this while also assessing PC 001. There was also discussion of conducting this assessment while in the NABS. It was discussed that while it would be possible to conduct this assessment in the NABS, there was a risk over having this artificiality to the assessment. The conversation was then shifted to assessing this PO while at sea.

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	There was then discussion that part of the difficulty in assessing leading a bridge team was that a strong bridge team could theoretically help keep the NavO afloat and that if the bridge team was spot on, it would be difficult to assess the student. There was then discussion that part of the challenge on being a NavO is managing the transition from coastal passages to pilotage and closing up SSD. It was then discussed that a scenario could be developed so that the NavO was in their cabin or classroom working on other correspondence. They would then have to answer questions from the OOW, head to the bridge and have to deal with BRM issues and consequently navigate the ship until SSD was secured. **DECISION:** part B of the PC will consist of a NavO away from the bridge and then having to come to the bridge to deal with BRM and lead the bridge team. Not to be conducted concurrently with PC 001. <u>1009 - PC 005</u> -Discussion regarding the format of the exam: it was discussed that the exam should be similar to the 2015/2018 FNO QSP. There was discussion on whether there should be 2 separate exams, one for MWV NO and another for FNO. There was follow on discussion of the possibility of having the PCs build on each other. A concern was raised that the FNO students would be potentially be tested twice in this construct. There was follow on discussion regarding eliminating the use of major/minor warship and replacing that with basic and complex navigation systems. It was discussed that under this construct, the students would then have the knowledge to ask the right questions when dealing with a complex navigation system regardless of the class of ship. Following this, there was discussion regarding the format of the exam for MWVNO and FNO. It was discussed that there could potentially be 4 exams in total. 1 knowledge exam for MWVNO followed by a practical exam. In addition to these exams, for FNO students, another knowledge exam and another practical assessment. The time of the assessment was also discussed. It was discussed whether this should be assessed in conjunction with another PC, or whether this should be assessed on its own. There was follow on a discussion that this could be assessed while a student is acting in the capacity of duty navigator so as to encompass dealing with the system and advising command on the loss of the system. **DECISION:** one PC across MWVNO and FNO. The standard must be maintained throughout for FNO students. The knowledge exams will be critical ECs. -Discussion regarding class packages: it was discussed that given the breakdown between basic and complex navigation systems, it would, in theory, be possible to scale back this PO as required in the future. It would be possible to create class-specific packages relating to the navigation systems on each platform.

DATE	Discussion/Detail
	ACTION: board secretary to determine which TDC is responsible for the Nav Yeoman's course. <u>1114 – Draft assessment for PC 001</u> -Discussion regarding the format of PC: it was discussed that this PC should be similar to PC 006 from the 2015/2018 FNO QSP. It was discussed that in addition to the previous assessment, there should be an assessment for a seamanship evolution at the end of the pilotage run. It was discussed that the PC should follow the model of navigation appraisal, planning and execution for open ocean, coastal and pilotage navigation. It was discussed that open ocean was not suitable for assessment on execution as the transits are generally OOW tracks. It was discussed that open ocean and coastal navigation were better suited towards planning. It was discussed that the duty NavO could be assessed during coastal navigation. There was follow on discussion that FP should be an inject for pilotage runs. It was initially discussed that assessing this in the NABS would prove too artificial. However, it was then discussed that the main concern of the NavO would be keeping the ship navigationally safe. Therefore, it would be possible to assess the student in the NABS as the other FP measures are less critical in this assessment. It was then discussed whether a high-speed and high-risk pilotage run were required for FNOs. It was discussed that prior to navigating a major warship in extremely confined waters, students should first be assessed on their ability to determine safety. **DECISION:** 1 PC for MWVNO, 1 PC for FNO (at high speed, confined waters), 1 critical EC for FNO (high speed, high risk run to serve as safety check prior to going to sea), an EC for coastal navigation (MWVNO and FNO), EC for open ocean passage (MWVNO and FNO). <u>1315 – Board Chair and DCO in a breakout room</u> -DCO desire to use an activity and assessment plan (AAP) <u>Draft assessment for PC 002</u> -Discussion regarding assessment format: it was discussed that the assessment should focus on operational formations and not just ceremonial formations. It was discussed that the assessment would take place in the NABS are there are often not sufficient resources for the course to have an entire TG during the sea phase. It was discussed that the assessment should focus on the student navigating the formation and not the individual navigation of the ship they are on. It was discussed that 4 NABS cubicles could be running and 1of the 4 cubicles would be designated as the formation flagship. The student in the flagship cubicle would be responsible for the formation while the other cubicles conducted their own pilotage. There was then

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	discussion on whether there would be a different standard for MWVNO and FNO. It was discussed that academically, there was little difference in navigating a formation. The main difference would be in the speed at which the students would be required to navigate the formation. It was discussed that there could be 1 PC done at 12-15kts. FNO students would be required to maintain this standard. Moreover, during the FNO portion of the course, they would be able to apply the lessons from navigating a single ship at 30kts to formation navigation. **DECISION:** 1 PC assessed during the MWVNO course. -Discussion regarding PC 006: after considering the content in the PO, it was agreed that there are some aspects that MWVNO can be involved in. Specifically support to OOTW (SAR, MIO etc.) **DECISION:** MWVNO EC for PO 006 <u>1451 – grouping and sequencing brief from TDO</u> <u>1503 - EO's from PO 001</u> -Board worked individually in Jam boards to create EOs for PO 001. <u>1545 – End Meeting</u> -Discussion regarding Nav Yeoman course: through the RCN MITE code list, it was found that the Nav Yeoman course is owned by NTDC(A). Given that the course is taught by FNI's and is a navigation-centric course, it was discussed that NTDC(P) should be responsible for the course. **DECISION:** board chair will request that the Nav Yeoman course be transferred to NTDC(P).
Day 5: 12 Jun 20	
	Members present: LCdr Macquarrie Lt(N) Dollis Lt(N) Morales Lt(N) Roberts Lt(N) Workman Lt Mason

DATE	Discussion/Detail
	Review of Day 4 ROP -No further discussion on Day 4 ROP. -Lt Mason noted that the board had received an email containing provisional approval from HQ TM Lt(N) Robinson, to continue on to the design phase. 0822 – Review of jam board -Discussion regarding EO 001.01: content for EO 001.01 was discussed, and how it should be broken down. It was discussed that it is important for students to know who to ask and how to ask for information that the NavO does not have. There was discussion regarding whether liaising with Command should be its own task prior to commencing the planning stage. It was discussed that it is important for Command to be aware of the NavO's plan prior to the NavO commencing the full planning stage. There was discussion as to what to name the task. After consulting BR45, it was determined that it would be best to call this the Command Appraisal. **DECISION:** EO 001.01 Perform a navigation appraisal 0845 – Chair away, board to carry on with the remainder of the process -Discussion regarding EO 001.02: There was discussion regarding the term "notebook." It was noted that even when the NavO is not working on the pilotage notebook, a notebook of sorts is still produced for open ocean and coastal navigation. After consulting with BR45, it was determined that the term "NavPlan" should be utilized. The NavPlan includes pilotage, open ocean and coastal navigation. It was also noted that a job-aid (planning checklist) might be included. **DECISION:** EO 001.02 Conduct navigation planning 0933 – Board Chair returned -Discussion regarding EO 001.03: there was discussion that mental math was worth including as its own task. It was discussed that mental math is used in a variety of ways, and like ColRegs needed emphasis on the course. It was discussed that mental

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	math is important but merely serves as a tool to assess the ship's position. It was discussed that mental math could be front-loaded on the course and taught throughout the content to emphasize its importance. **DECISION:** EO 001.03 Execute the navigation <u>0959 - DGPR happy with an amendment to the task list (provisionally approved)</u> <u>1026 – EOs for PO 003</u> -Discussion regarding EO 003.01: it was discussed that learning points like RITCAF, CFPAS, CAFJODs etc., fall under the realm of general Officer duties and would not be trained on this course. It was discussed that some content would be better suited under the administrative duties EO rather than divisional duties. **DECISION:** CFPAS, CAFJODs etc. not trained on this course. EO 003.01 Conduct divisional duties, including -Discussion regarding EO 003.02: it was discussed that some content under this EO would also need to be covered under the "Advise Command" EO. Content such as BRM, watch rotation etc. It was discussed that there could be 1 lesson that applies to both EOs. **DECISION:** EO 003.02 Conduct administrative duties including: -Discussion regarding EO 003.03: there was discussion regarding the role of the NavO in certain BRM situations. It was discussed that during open ocean and coastal navigation, the role of the NavO consists of conflict resolution and advising the OOW. **DECISION:** EO 003.03 Manage the bridge team by: -Discussion regarding EO 004.01: there was discussion regarding the naming of the PO and EO. There was a discussion that the phrasing in the standard statement was convoluted and could lead to redundancies. It was discussed that there could be 2 EOs, 1 relating to research and the other relating to conveying the advice. **DECISION:** PO named Support command decision making on navigation matters. Split into 2 EOs, "Conduct independent navigation-related research" and "convey accurate information" on the following. EO 004.02 Convey accurate information on the following: -Discussion regarding EO 005.01: there was discussion that the way the EO is setup is currently not how things are necessarily conducted on the ships.

DATE	Discussion/Detail
	DECISION: EO005.01 ~~detect or identify errors and faults~~. -Discussion regarding EO 005.02: it was discussed that, unlike most other equipment, WECDIS errors are distinctly a NavO issue as the NavO will do the troubleshooting for it. While techs will normally conduct advanced troubleshooting on other equipment like the echo sounder. **DECISION:** EO 005.02 ~~Mitigate errors and faults~~. -Discussion regarding EO 005.03: there was discussion that after internally reporting faults and having the techs conduct troubleshooting, it is important for the NavO to re-verify the functionality of the equipment before allowing the techs to sign off on the correction of the fault. **DECISION:** EO 005.03 Liaise with other departments/organizations regarding ~~equipment and sensors~~. -Discussion regarding EO 006.01: there was discussion about including a content piece about taking the watch from the OOW. It was discussed that the main role of the NavO is to direct the watch and maintain SA. **DECISION:** EO 006.01 Perform the duties of OOW during action stations, including: -Discussion regarding EO 006.02: there was discussion over including a piece regarding advising other warfare directors. It was discussed that the NavO is not necessarily advising on the warfare piece but the navigation-related to it. **DECISION:** EO 006.02 Provide navigation support to operations by: <u>1500 – End of Day 5</u>
Day 6: 15 Jun 20	
	Members present: LCdr Macquarrie Lt(N) Dollis Lt(N) Morales Lt(N) Roberts Lt(N) Workman Lt Mason

DATE	Discussion/Detail
	ACTION: Resend task list with new tasks to DGPR for approval <u>0807 - Review of Day 5 ROP</u> -No further discussion on day 5 ROP <u>0811 – EO Template Review –Lt Mason</u> **ACTION**: regarding pre-requisites for the course: CANSUBFOR has requested that there be a minimum time on a BWK for a Submariner prior to coming to the course. This pre-requisite will be similar to the pre-requisite for Surface Fleet 6-month post NOPQ. <u>0825 – EO 001.01</u> -Discussion regarding givens: there was discussion on whether paper charts should be included in the givens. It was discussed that the NavO will often use paper charts prior to moving to the planning phase. <u>0827 – Board chair away – board to continue progressing EO development</u> -Discussion regarding the run sheet: it was discussed that the FNIs have created run sheets for Commanders intent. It was discussed that the run sheet may be described in an annex. It was also discussed that the specifics of the chart and publication outfit may be described in a separate remark or annex. -Discussion regarding assistance as a given: there was discussion as to whether assistance and supervision should be given at the EO/EC level. There was concern over the use of the word assistance. In the 2015/2018 version of the QSO, assistance was listed as denied for most EOs. It was discussed that on the job, there is no guarantee of assistance. It was discussed that the students would sometimes informally bounce an idea off of the FNIs during training. It was also discussed that the amount of assistance might be limited, and this may be specified as a separate remark. **DECISION:** assistance will be listed under givens for this EO. -Discussion regarding denied: nil denied.

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	-Discussion regarding environmental: any environment. -Discussion regarding standard: there was discussion regarding which checklist to reference. It was noted that there was no checklist in the approved CFCD 130. It was noted that BR45 contained 3 checklists which could be used during the appraisal and planning phase. **DECISION:** reference the BR45 Vol I checklists. -Discussion regarding the time required: it was discussed that the time required for the EO might be filled in at a later point once the LOs for the EO have been decided. -Discussion regarding remarks: chart and publication outfit as issued to HMC Ships and run sheet will be described in a separate annex. -Discussion regarding conducting navigation research: it was discussed that despite the items that should be researched being in a checklist, it is worth having them listed here in the QSP. It was noted that having the items specifically listed out would provide easy access to a student or instructor that needed information. The checklists will still be listed in the QSP. It was also discussed that considerations for IALA buoyage should be moved to the planning phase. The board will work on the EOs separately but will still go over the EOs as a group once they are complete. <u>1102 – Board Chair returned. Board to continue working individually on EOs. Board will reconvene at 1330</u> -Discussion regarding EO 001.02: it was discussed that navigation tools would be provided. Navigation tools will be specified at a later point in Annex F (or separate annex) -Discussion regarding standard: there was discussion that the checklist may be created at a later point taking reference from BR45. **ACTION:** once the checklist is complete, checklist items are to be removed from the standard.

DATE	Discussion/Detail
	ACTION: Board Chair has decreed that Lt(N) Roberts is not to conduct an assessment for the FNO course after completing the quarterly PDW. -Discussion regarding EO 001.03: there was discussion that in the remarks section, it must be noted that navigation systems and equipment may be taken away during an assessment. This statement will also apply to the bridge team. It was discussed that the level of supervision provided will be limited and will vary depending on the navigation regime. -Discussion regarding EO 002.01: no additional discussion. -Discussion on EO 002.02: there was discussion regarding the removal of advanced RELVEL. It was discussed that the advanced RELVEL piece would be an exposure piece given that the navigation systems and equipment are capable of performing this task. **DECISION:** advanced RELVEL removed from the QSP. The theory will still be taught; however, it will not be in-depth. -Discussion regarding EO 002.03: discussion regarding the standard: there was discussion that there was no need for the qualifier "safely." It was discussed that it is an implied factor that the NavO is navigating the ship in a safe manner. There was discussion that there should be content relating to the management of NavComms. It was discussed that OOW manoeuvres should be discussed but that there was no need to cover the topic in depth. -Discussion regarding EO 003.01: it was discussed that supervision and assistance should be givens. It was discussed that The XO onboard should be able to assist the NavO with questions relating to the conduct of divisional duties. -Discussion regarding EO 003.02: it was discussed that for some navigation administration, such as sailing orders and occasions to call are important and should be discussed during the course. It was, however noted that it is possible for an FNO or MWVNO student to be successful in the course without having prepared some of the administration. It was discussed that it would be important to expose the candidates to this administration but would not be covered in depth. <u>1550 - End of Day 6</u>
Day 7: 16 Jun 20	
	Members present: LCdr Macquarrie

DATE	Discussion/Detail
	Lt(N) Dollis Lt(N) Morales Lt(N) Roberts Lt(N) Workman Lt Mason <u>0818 - Review of Day 6 ROP</u> -No additional discussion of ROP -**Clarification:** CANSUBFOR pre-requisites to be aligned with MWVNO. **ACTION:** Lt(N) Roberts to create checklists adapted from BR45. -Discussion regarding EO 003.03: it was discussed that it would be possible for FNO/MWVNO students to select a PD piece and then present it to NWO students while they are at VENTURE. It was discussed that there should be a qualifier for conflict resolution as conflict resolution in this case mainly refers to interactions between watchkeepers or conflict between Ops/OOW. **DECISION:** conflict resolution renamed to clarifying roles and responsibilities on the bridge. -Discussion on EO 004.01: it was discussed that on the job, the NavO often makes phone calls to other agencies as part of appraisal and planning. It was discussed that under resources, a phone should be included. There was discussion that there was no need to teach the students how to conduct the research. The key point is in showing the students where they can find resources to help them in their research. It was discussed that the information gathered during the EC would be used at a later point, and the students would not be conducting research for the sake of conducting research. **ACTION:** phone to be listed in resources (or communications device). -Discussion regarding EO 004.02: it was discussed that the information conveyed will be from the references, but there is no reference for how to convey it. It was discussed that the student should be able to anticipate the needs and requirements of command.

DATE	Discussion/Detail
	-Discussion regarding EO 005.01: there was discussion that BR45 groups navigation equipment and systems into chapters. Those specific chapters will be referenced in the EO. **DECISION:** group navigation equipment and systems as per BR45. -Discussion regarding EO 005.02: it was discussed that there is a difference between re-flash and re-start. The definition may be included in remarks or annex F. -Discussion regarding EO 005.03: it was discussed that the main external organizations that the NavO will often liaise with should be listed in the EO. -Discussion regarding EO 006.01: it was discussed that to allow for flexibility, acting as OOW should be done IAW battle orders since battle orders will not only vary from class to class but also ship to ship. **ACTION:** Liaise with N92 regarding the use of SDA for the classified portions of the course. -Discussion regarding EO 006.02: it was discussed that in supporting warfare, the main role of the NavO was to produce navigation products that would support warfare. It was also discussed that SAR and SARZONE should be included given that they are 2 separate components. <u>1028 – Completion of EO building</u> <u>1057 – Discussion regarding the creation of ECs</u> -Discussion regarding EC for 002: it was discussed that one of the ECs under PO 002 could be a group EC. -Discussion regarding ECs for PO 001: it was discussed that given the importance and safety aspect of ColRegs, it should be a CEC. **DECISION:** CEC for ColRegs -Discussion regarding ECs for PO 001: there was discussion regarding the EC taking place while a student is acting as Duty Nav. It was discussed that while acting as Duty Nav, other EOs could be assessed under the EC. There was discussion as to whether there would be one EC for both MWVNO and FNO or if there would be one assessment for each. It was discussed that

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	there should be one EC that would be assessed during MWVNO, but that standard would need to be maintained moving forward for FNO students. -Discussion regarding whether the EC should be a CEC: there was discussion as to whether the EC should be a CEC. It was noted that the Duty Nav portion is usually what occurs in the day-to-day aspects of the job while acting as the actual pilot for pilotage accounts for a smaller portion. It was discussed that there was difficulty in conducting the EC in the NABS as there would be difficulty in replicating some of the at-sea portions of being Duty Nav. It was discussed that it would theoretically be possible to conduct a CEC in the NABS, but the ideal environment would be at sea. It was discussed that there would potentially be an insufficient amount of sea time during the MWVNO portion of the course to allow for a CEC if the course was at max load. It was discussed that IOT support a CEC. For a course of over 10 students, there would be a requirement for an additional Orca. It was also discussed that a failure in the CEC for an FNO student could be re-tested during the 2nd or 3rd sea phase. **DECISION & ACTION:** it should be amended in Chapter 1 of the QSP that for a course load of more than 8 students, a 2nd Orca will be required. It should also be noted in the training management details that it would be possible to consider outcomes outside of the normal bounds. <u>1300 – Meeting Resumed – SLt Seo (OJT) observing</u> -Discussion regarding ECs for PO 002: it was discussed that there could be a group EC for PO 002. It was discussed that the FNIs could teach a ceremonial day and an operational day regarding formation navigation. It was discussed that the EC would be non-critical to cover the non-pilotage aspects of the PO. -Discussion for ECs for PO 003: it was discussed that PC 003B is going to run concurrently though over other POs. There was discussion that for the EC, the students could pick a topic from a list and deliver a PD session to NWO students that are in training. -Discussion for ECs for PC 004: it was discussed that the met exam should be a CEC as the NavO will be the only Officer that will receive additional training on storm avoidance which contains a safety aspect. It was discussed that the EC may contain a combination of theory and met products that the students will be tested on. **DECISION:** Met exam will be a CEC

DATE	Discussion/Detail
	-Discussion regarding ECs for PO 004: it was discussed that the BRM PD briefs could be tied into providing Command with advice. There was also discussion that is important for the course is that students learn where to find CRRs that they will eventually become responsible for and that there was no need to test their knowledge on the CRRs as this could be learned on the job. -Discussion regarding ECs for PO 005: it was discussed that all students would complete an EC for this PO. It was discussed that FNO students would have to maintain the standard that they had met once they move into the FNO phase of the course. It was discussed that the PC would remain a written exam and that the PC would be evaluated in conjunction with another PO. **ACTION:** The creation of playbooks for the scenarios that students will encounter in the NABS. -Discussion regarding ECs for PO 006: there was discussion that a hydrographic survey should be an EC at the MWVNO level, as it is more likely to conduct a hydrographic survey onboard MWVs. **ACTION:** board to continue working individually through draft EC assessment forms. The board will go over the draft assessment forms as a group the following morning. <u>1411 – End of Day 7</u>
Day 8: 17 Jun 20	
	Members present: LCdr Macquarrie Lt(N) Dollis Lt(N) Morales Lt(N) Roberts Lt(N) Workman Lt Mason SLt Seo (TDO OJT) <u>0802 - Review of Day 7 ROP</u>

DATE	Discussion/Detail
	-Discussion that N92A should be listed as a critical resource. It was noted that staff and students will also need VCRs prior to accessing the building. **DECISION:** N92A – Critical resource for SDA -No further discussion regarding ROP from day 7. <u>0807 – Review of Draft EC Assessments</u> -<u>Discussion regarding CEC 001.02 (execute an open ocean passage by celestial fixing):</u> **DECISION:** CEC based on safety aspect Discussion regarding time: it was discussed that for scheduling purposes, the exam should be scheduled for 4 hours. This includes a 3-hour exam with an additional hour for questions as required. **DECISION:** exam time, 4 hours -Discussion regarding TEL: it was discussed that a star chamber such as one from Kongsberg Marine should be a TEL option (https://www.kongsberg.com/digital/solutions/maritime-simulation/celestial-navigation) -<u>Discussion regarding CEC – 001.03 (ColRegs):</u> **DECISION:** CEC based on safety aspect -Discussion regarding standard: it was discussed that after passing the CEC, students are expected to maintain the standard throughout the course. -Discussion regarding scenarios: it was discussed that an NSSC should be created so as to keep track of the scenarios that each student has been presented with to ensure uniformity. **ACTION:** Creation of NSSC <u>-Discussion regarding CEC 001.04 (Act as Duty Navigator):</u> **DECISION:** CEC based on safety aspect

DATE	Discussion/Detail
	-It was discussed that the minimum time for a student to act as Duty Nav should be 16 hours. This accounts for the departure and alongside on a Monday or Friday. It was discussed that a checklist with the types of scenarios that the Duty Nav will be required to assist the OOW should be created. It was discussed that on a previous sea phase, transiting locks proved to be one of the most valuable evolutions. It was discussed that transiting locks should be captured in the QSP. **DECISION:** specify as a resource that there should be an evolution of transiting locks <u>-Discussion regarding PC 001A: Conduct a basic SSD pilotage run (MWVNO):</u> -There was discussion regarding the maximum speed that the students would be allowed to proceed at. There was discussion that the maximum speed should be 15kts, given that it is the top speed of MCDVs. However, it was discussed that for certain legs of a passage, it would be possible to navigate at high speed so long as it was planned accordingly and not used due to poor planning. There was significant discussion regarding the number of runs a student must pass IOT be considered an MWVNO. It was discussed that simply passing 1-day run and 1-night run would be insufficient as it would not demonstrate consistency. It was discussed that the MWVNO sea phase is 2 weeks long, and it would be possible to reduce the developmental runs at sea and start assessments earlier. This would provide students the opportunity to conduct more runs at sea. It was discussed that students should be assigned 7 pilotage runs. All 7 runs will be assessed. Of the 7 runs, 5 must be passed, including a minimum of 2 day and 2 night runs. **DECISION:** students will be assigned 7 nav passages. They must pass at least 5, with a minimum of 2 successful day and 2 successful night runs. <u>-Discussion regarding CEC 001.05 (Conduct high-speed pilotage run):</u> **DECISION:** CEC based on safety aspect -There was discussion that a CEC is required to ensure safety prior to allowing students to progress to the sea phase. It was discussed that the students should be able to assess safe water and keep the ship in safe water. It was discussed that the students would be assigned 4 passages, and all passages would be assessed. Students would need to pass at least 2 runs (1 day and 1 night). **DECISION:** students must pass at least 2 runs (1 day and 1 night).

DATE	Discussion/Detail
	-There was discussion that the focus on the assessments at sea should be on BRM matters and bridge team management. It was noted that at least 1 day was required for students to adjust to being on a major warship. There was concern that for courses being run over the summer, there would potentially not be enough hours of darkness during the sea phase to accommodate a full course load. There was discussion regarding the number of runs students should be given and how many the students would be required to pass. It was discussed that students would be assigned 4 passages and that they would be assessed on all passages. It was discussed that students should be required to pass 3 of 4 runs consisting of 2 day runs and 1-night run. **DECISION:** students must pass 3 of 4 runs (minimum 2 day and 1 night). **ACTION:** max course load should be reduced to 8 FNO students. The max course load for MWVNO may remain 10. It is not feasible to conduct high speed pilotage runs with the 5-day major warship sea phase. Additional students would require extra sea time. <u>-Discussion regarding EC 002.01 (Conduct Task Group Operations in Coastal Waters):</u> -There was discussion that this could be incorporated into the planning for an open ocean transit. **DECISION:** EC to be included in EC 001.01 <u>-Discussion regarding EC 001.01 (Plan and Brief an Open Ocean Passage):</u> -There was discussion that as part of this EC, students would have to plan a leg of the transit while operating in a TG. Students will submit a draft plan for feedback from the FNIs. After receiving feedback from the FNIs, the students would move to brief their plan. It was discussed that students should be given dedicated time in class to work on the EC so as to prevent them from having an excessive amount of take-home assignments. It was discussed that while this is not a group EC, students would still be allowed to discuss with each other. **DECISION:** scope of the EC will include EO 002.01 and EO 004.01 <u>-Discussion regarding PC 002 (Navigate a Formation of Ships in an Operational Environment):</u> -There was discussion that the PC should be assessed at the MWVNO level. It was discussed that for FNO students, they would be able to organically absorb the considerations for navigating a TG at higher speeds as the course progresses. It was discussed that the objective of the PC is not to assess a student's ability to navigate their individual ship, but rather their ability to oversee the navigation of the formation. It was discussed that the developmental runs for the PC should be repeats of the runs that they

DATE	Discussion/Detail
	conducted while conducting development for PC 001A. It was discussed that there should be consideration to conducting this PC after PC 001. **DECISION:** students will not be expected to navigate their own vessel during the PC. <u>-Discussion regarding EC 002.01 (Navigate a Formation of Ships in a Ceremonial Environment)</u>: -There was discussion that the focus of the EC should be on the appraisal, planning and oversight of the execution of the formation navigation. It was discussed that under time constraints, it would also be possible to conduct this as a group EC. **DECISION:** students will be expected to individually navigate the ship that they are on. <u>-Discussion regarding EC 003.01 (Respond to a Complex Navigation Policy or Procedural Issue:</u> -There was discussion that all correspondence that the students would be expected to create for the EC would be equal in their complexity so that creating a MARPIF, for example, would not be more taxing to create than a briefing note. **ACTION:** DS solution to be created. <u>-Discussion regarding EC 003.02 (Lead Mariner Professional Development):</u> -There was discussion that for PD, there should be a focus on BRM as it may be briefed to any level (NWO trainees, peers, CDC students etc.). It was discussed that it would be preferable to have students lead PD for NWO students as NavO are expected to teach and mentor their subordinates and peers on board. **ACTION:** Liaise with Occ Mgr to potentially have NavOs take the instructional techniques course on DLN. <u>-Discussion regarding PC 003B (Lead the Bridge Team):</u> -There was discussion that the PC will be assessed under CEC 001.04 as a separate checklist item. **DECISION:** assessed under CEC 001.04. <u>-Discussion regarding PC 004 (Provide Advice to Command on Navigation Matters):</u>

DATE	Discussion/Detail
	-There was discussion that the PC should be assessed separately from CEC 001.04 (act as duty nav.). It was discussed that there would be a reasonable amount of opportunity to assess students on briefs as seamanship evolutions could be incorporated into the Flex. It was discussed that some material for CDC assessments could be used when developing this assessment. <u>-Discussion regarding CEC 004.01 (Shiphandling Theory):</u> **DECISION:** CEC based on safety aspect. -No additional discussion. <u>-Discussion regarding CEC 004.02 (Met Theory):</u> **DECISION:** CEC based on safety aspect. -No additional discussion. <u>-Discussion regarding PC 005A Demonstrate Knowledge of Basic Navigation Systems (MWVNO)</u>: -There was discussion that the PC for MWVNO should still be a written portion as there is a knowledge piece that should be assessed. The practical portion will be tested under CEC 001.04. **DECISION:** separate exam for MWVNO. <u>-Discussion regarding PC 005B Demonstrate Knowledge of Complex Navigation Systems (FNO):</u> -There was discussion that the setup would be the same as that of MWVNO, but the content would be different. There remains a knowledge piece that should be assessed. The practical portion will be assessed under CEC 001.04. **DECISION:** separate exam for FNO. <u>-Discussion regarding EC 005.01 (Navigation System Presentation:</u>

DATE	Discussion/Detail
	-There was discussion that the EC would be mainly activity-based. Students will research and present the information that they have found. It was discussed that there should be a DS solution to ensure that students had covered all the pertinent information of the system that they are presenting. **ACTION:** creation of DS solution for the activity. <u>-Discussion regarding EC 005.02:</u> -There was discussion that EC 005.02 would be covered under acting as Duty Nav. **DECISION:** EC covered under CEC 001.04. <u>-Discussion regarding PC 006 (Manage Navigation Aspects of the Maritime Battlespace):</u> -There was discussion that this PC would take place during the MWVNO phase and would be applicable to both MWVNO and FNO students. It was discussed that there would be no SECRET material on the PC. **DECISION:** PC will be tested during the MWVNO phase. <u>-Discussion regarding EC 006.01 (Action Stations OOW):</u> -There was discussion that the 1-hour block for the EC would include time for feedback and reset. It was discussed the EC should be done IAW a ship's bridge pack or IAW battle orders that would be created specifically for the FNO course. It was discussed that given the level of urgency required for the action stations OOW, references would be denied. **DECISION:** references during this EC will be denied. <u>-Discussion regarding EC 006.02 (Nav Support to Operations):</u> -There was discussion that this EC would take place at the MWVNO level as it was applicable to both MWVNO and FNO. It was also discussed that the EC could be an activity instead of an assessment. **DECISION:** EC will be conducted as an activity. <u>-Discussion regarding EC 006.03 (Tactics and Doctrine):</u>

DATE	Discussion/Detail
	-There was discussion that this exam will be for FNO students only. It was discussed that the exam still needs to be developed at that it would be SECRET. <u>-Discussion regarding EC 006.04 (Hydrographic Survey):</u> -There was discussion that the EC would be a group EC. It was discussed that the students could work together to create a small chart correction of WECDIS. **DECISION:** EC 006.04 will be a group EC. <u>1530 – Completion of PC and EC Building</u> <u>1545 – End of Day 8</u>
Day 9: 18 Jun 20	
	Members present: LCdr Macquarrie Lt(N) Dollis Lt(N) Morales Lt(N) Roberts Lt(N) Workman Lt Mason <u>Review of Day 8 ROP</u> -PC 001B was missing from teams - will be uploaded -Discussion regarding CEC 001.05: there was discussion that the standard was lower on the CEC and then was higher on the PC. It was discussed that the standard was lower on the CEC as, in theory, it would be harder since there are more problems than the students can be faced with in the NABS. It was discussed that given that this is a CEC, it would prevent an unsafe student from proceeding to sea. <u>0816 – End of Review of ROP</u>

DATE	Discussion/Detail
	-Discussion on reducing the maximum number of students on FNO: there was discussion that having 9 or 10 FNO students on the course is, in theory, possible. However, it pushes the boundaries for sustainability while at sea on a major warship. It was discussed that it was better to have the limit at 8, and if there is a requirement for more students, there would be a requirement for more sea time. -Discussion on CEC 004.01: there was discussion that the safety aspect in keeping this a CEC was the fact that the CEC and practical PC will only cover a limited number of shiphandling activities. It would not be possible to assess students on all shiphandling evolutions. In addition, NavOs are required to be competent in shiphandling across all platforms. -Discussion on tides: It was noted that in the previous QSP, there was a separate assessment for tides. It was discussed that the exam is the one that is most often failed on the course. It was discussed that students do not often struggle with medium or advanced tidal theory but rather with safety clearances over/under, which is taught during NWO training. It was discussed that given the safety aspect of safety clearances, it might be worth having tides as its own EC again. It was discussed that it would be possible to include tidal questions in the threshold knowledge exam to determine the knowledge of tides the students were coming to the course with. It was noted that the objective of the threshold exam is not to fail a student off the course, but rather for instructors and students to gauge their knowledge prior to commencing the course. There was discussion that tides should be covered in the PIP. It was discussed that the students could be given a worksheet to consistently assess their knowledge of tidal theory. The first few developmental runs in the NABS could also be pre-planned so as to have a DS solution regarding tides. **DECISION:** there will not be an EC for tides. Students will be given worksheets and activities during the early stages of the course. The first few developmental runs in the NABS will be pre-planned and will have a DS solution with tides. Students will be required to submit their tidal calculations, and FNIs will be able to assess their work and provide them with feedback. -Discussion regarding the PIP: there was discussion that the difficulty with the PIP is that not all students will have access to a NavO. Some COs may be too busy to sign off on portions of the package. There was discussion that the PIP should be modernized and made available online through DLN. It was also discussed that tidal theory would be included in the PIP. It was discussed that a MITE code should be attached to the PIP, and students would not be loaded on the course without having the MITE code. It was also discussed that access to the PIP would be controlled so that students would not have early access to the PIP prior to coming to the course. **DECISION & ACTION:** PIP modernization should be included as a TEL option so that it may be made available on the DLN.

DATE	Discussion/Detail
	-Discussion regarding CEC 004.02: there was discussion that tidal theory could be assessed with met theory. It was also discussed that other sciences relating to navigation could be grouped with the CEC as they incorporate safety aspects. **DECISION:** rename CEC 004.02 to Navigation (and marine?) Sciences Written Exam <u>0918 - Board Chair Away - Board to continue with grouping LOs</u> <u>1048 -</u> **DECISION:** Degaussing and magnetism removed from the QSP; it is more of an engineering and UWWO consideration. <u>1300 – Board Chair returned</u> <u>1400 – Board Chair & secretary away-board will continue working on AAP template</u> <u>1500 – End of Day 9</u>
Day 10: 19 Jun 20	
	Members present: LCdr Macquarrie Lt(N) Dollis Lt(N) Morales Lt(N) Roberts Lt(N) Workman Lt Mason SLt Seo <u>0803 – Review of Day 9 ROP</u> -No additional discussion <u>0812 – Presentation from SLt Seo on AAP template for EO 006.01.01</u> -Board to break out and work on AAP sheets individually

DATE	Discussion/Detail
	-TDO provided tips and tricks on how to develop activity plans -Breakout room audience received a brief from Mr. Jason Knowledge. -Discussion between TDO and Lt(N) Dollis regarding activity development for assigned PO <u>1430 – Reconvene to check on AAP progress</u> -Discussion regarding daily quiz or worksheets: there was discussion that it should be captured in the QSP that there will be daily quizzes or worksheets throughout the course. It was discussed that it would be possible to include this at the beginning of an activity or LO or to have it as a note in the AAP. -Discussion regarding industry tours: there was discussion that industry tours may be included as an activity in the AAP. It was discussed that it would also be important to highlight in the AAP what the students would be getting out of the tour. <u>1500 – End of Day 10</u>
Day 11: 22 Jun 20	
	Members present: LCdr Macquarrie Lt(N) Dollis Lt(N) Morales Lt(N) Roberts Lt(N) Workman Lt Mason SLt Seo <u>0807 - Review of Day 10 ROP</u> -No further discussion <u>0811 – Review of AAPs</u>

DATE	Discussion/Detail
	-Discussion regarding AAP 003.01.02: -There was discussion that a floating flip chart activity could be used to describe the NOPQ process. -Discussion regarding AAP 003.02.01: -No additional discussion. -Discussion regarding AAP 003.02.02: -Discussion regarding the NIR activity: it was discussed that it would be beneficial for every student to fill out an NIR as a point of consistency. Even if the student did not have a near-miss, they would be able to take examples from their fellow students. -Discussion regarding AAP 003.03.01: -There was discussion that activity 2 could be a rough floating flip chart to help generate discussion for the students and FNIs. It was also discussed that activity 1 could be a homework assignment to save time in the course schedule. -Discussion regarding AAP 003.02.04: -There was discussion that activities will provide exposure to MARPIFS, PRONOTES and other documentation. This will lead up to EC 003.01. <u>0900 – Capt(N)s (ret'd) Butler and Greenwood and LCdr (ret'd) Kennedy joined – breakout room discussion with the board chair</u> -Discussion regarding AAP 003.02.05: -There was discussion that the activity could be included with tour day. The students would bring the checklist with them and follow along. Students will spend some time covering the NAI and then carry on with the rest of the frigate tour. -Discussion regarding AAP 003.02.06:

DATE	Discussion/Detail
	-There was discussion that students could brainstorm what secondary duties they think they are responsible for and then have a DS solution. Follow up with an actual lecture to cover any points that were missed during discussion. -Discussion regarding AAP 003.03.01: -There was discussion that it would be hard to have this as a dedicated activity in the classroom. It was discussed that it is something that is best done in the NABS. Students will have to come into the NABS to see a poorly run watch, and then they will have to come in and solve it. **DECISION:** Conducted during PO 001 developmental runs. -Discussion regarding activity 3: a coastal nav run leading into SSD. This will lead into PC 003B, which leads to act as duty NavO (CEC 001.04) <u>1000 – Lt Buffa (TDO OJT) joined</u> -Discussion regarding AAP 003.03.02: -There was discussion that activity 1 will be the PD portion and will be covered under the EC for delivering PD. It was discussed that for activity 2, students would be given a lecture on administering bridgemanship and ColRegs exams. <u>1025 – Recap from breakout room discussion:</u> -There was discussion that now would be the time to request a 2nd week for the FNO Halifax Class sea phase. It was discussed that adding the second week would allow for greater training flexibility and allow the course to provide more context to topics while at sea. -There was discussion that there should be more nav runs in and out of the harbour. It was discussed that these entrances and exits could be done in the NABS at a minimum. There was also discussion that some entrances or exits could be assessed runs.

DATE	Discussion/Detail
	-There was discussion that the minimum SOA should also be stated within the PO. It was discussed that the performance statement explicitly states a maximum SOA, but a minimum SOA should also be added for consistency. -There was discussion that it should be stated in the QSP that the NavO would have control of the ship's movement for some nav runs. This would be to allow for more opportunity to show the students what it would be like to have control of the ship's movement during a nav run. -There was discussion that an FP escort or FP integration scenario should be added into a run (NSSC). **ACTION:** include FP run in the NSSC -There was discussion that degaussing and sound ranging should be discussed. It was discussed that there is value in discussion shiphandling while in sounding or degaussing ranges. -There was discussion that the CDC team would be happy to assist with the BRM piece. It was discussed that BR134 should be added as a reference. It was discussed that BRM content should include error tracking and trapping, the psychology of teams relating to BRM and how to identify when a ship is standing into danger. It was discussed that this could be achieved through readings, discussions and case studies. There was also discussion that the course should explore international mariner regulations to understand what standard other professional mariners are expected to maintain. -There was discussion that a current or former CO should come in to discuss the night orders and CSO piece to the students. **ACTION:** add current or former CO to the guest speaker list (specifically for CSOs) -Discussion regarding the NavO and shiphandling: there was discussion that the main role of the NavO in alongsides is to get the ship onto the approach track, and the CO or designated Officer will drive the alongside. The role of the Navigator is keeping the ship safe, knowing what water is being passed over and precise shiphandling issues. It was discussed that students should not be assessed on how they drive an alongside. -Discussion regarding the role of the NavO in BRM: there was discussion as to what role the NavO plays in BRM. It was identified that being the BRM SME is not currently a task assigned to the NavO. It was agreed that the NavO certainly has a role to play in BRM, but the extent of that role should be explicitly defined.

DATE	Discussion/Detail
	ACTION: add BRM to the task list for DGPR approval -Discussion regarding AAP 005.01.01: -There was discussion that a DS solution must be created for each system. If there aren't enough students, they can still get all the info that they need for the other systems that weren't covered. -Discussion regarding AAP 005.01.02: -As above. -Discussion regarding AAP 005.02.01: -There was discussion that activity 1 could take place in the NABS. It was discussed that while some students were in the NABS conducting setup with the FNIs, other students could be doing a concurrent activity such as working on in-class or take-home assignments. **DECISION:** revisit this activity once sequencing and timing are complete. -Discussion regarding AAP 005.02.02: -There was discussion that a problem that is be solvable by using the WECDIS Help Functions should be added. -Discussion regarding AAP 005.02.03: -There was discussion that a risk mitigation assessment could be assigned as homework. A guided discussion or slide sort could also be done. Students would pick from a pile of cards and be required to present the situation, impact options and recommendations on the loss of that piece of equipment. -Discussion regarding AAP 005.03.01: -There was discussion that this would be done in conjunction with runs during the NABS and at sea. There was also discussion that there should be a guided discussion on how to interact with MSE and CSE.

DATE	Discussion/Detail
	-Discussion regarding AAP 005.03.02: -There was discussion that this was a no-train activity. Students will be continually assessed on this -Discussion regarding AAP 005.03.03: -There was discussion that the activity could be a group discussion to brainstorm what external agencies are available. A DS solution would be required. -Discussion regarding AAP 006.01.01: -There was discussion that the main point of focus should be on what you are thinking about during an action alarm as a NavO if you're somewhere else on the ship. With regard to routing to the bridge based on the location of attack, damage etc. -Discussion regarding AAP 006.01.02: -There was discussion regarding whether a guided discussion in the classroom would be of value or if the students would just absorb the material throughout the course. It was discussed that IOT add value the ST(P) OpsO or WpnsO should be brought in as a guest lecturer to discuss the bridge-ops liaison piece. There was also discussion that OROCO or -2 should also be involved. **DECISION:** include STG as a guest lecture in resources OROCO as well as WpnsO or OpsO -Discussion regarding AAP 006.01.03: -There was discussion that the key for this LO is for the students to understand who is on the bridge and what function they serve. It was discussed that the BRM at action stations piece falls under one of the earlier LO as it involved communication. -Discussion regarding AAP 006.02.01: -There was discussion that the FNIs will talk about SAR since it is an important piece. Students will talk about the others, and there will be a DS solution to ensure all pertinent points are covered. Students will then have a team quiz at the end.

DATE	Discussion/Detail
	-Discussion regarding AAP 006.02.02A: -A lot of secret stuff -Discussion regarding AAP 006.02.02B: -There was discussion that this is when the students should be brought to the indirect fire trainer at the Bay Street Armoury. -Discussion regarding AAP 006.02.03: -There was discussion that the resources should be captured under annex h. -Discussion regarding AAP 006.02.04: -There was discussion that the students could create a deceptive Nav Aid plan for a harbour or passage. It was also discussed that the students could also conduct a run through the modified passage and have the FNIs conduct NAVWAR against the students in the NABS. There was also discussion that there should be a classified and unclassified portion for the lectures. -Discussion regarding AAP001.01.01A: -No additional discussion -Discussion regarding AAP 001.01.01B: -There was discussion that this is when the students should go on a field trip to CHS to see how charts are made. -Discussion regarding AAP001.01.01C: -There was discussion that this is when the students would go on a field trip to HSO.

DATE	Discussion/Detail
	-Discussion regarding AAP 001.01.02: -There was discussion that the students would be given a run sheet and would be going through the exercise with the FNIs on an ongoing basis until its completion. -Discussion regarding AAP 001.01.03: -There was discussion that this could be a fishbowl activity. A student would present to an FNI, and other students will observe. It was also discussed that a student could brief a Command qualified Officer, and the other students could observe. 1545 – End of Day 11
Day 12: 23 Jun 20	
	Members present: LCdr Macquarrie Lt(N) Dollis Lt(N) Morales Lt(N) Roberts Lt(N) Workman Lt Mason SLt Seo Review of Day 11 ROP -No further discussion Discussion regarding AAP 001.02.01: -There was discussion that for the activity, the students will continue to follow along with the run sheet. Discussion regarding AAP 001.02.02:

DATE	Discussion/Detail
	-There was discussion regarding whether risk mitigation is a NavO function? It was discussed that risk mitigation is conducted throughout the appraisal and planning process. There was also discussion that risk management is still an important piece that needs to be discussed as the NavO should be able to defend their work from a risk management perspective. Discussion regarding AAP 001.02.03: -There was discussion that the students could brainstorm pre-sail checks. FNIs would then provide a DS solution and tips and tricks afterwards to ensure all points are covered. Discussion regarding AAP 001.02.04: -There was discussion that the students could brainstorm what elements should be included in the nav brief. There would be a follow on PPT with policy, minimum requirement, good and bad nav briefs and a DS solution. Discussion regarding Annex K: -There was discussion that Annex K should be uploaded in teams. It would then be possible to go back and revisit as required. Discussion regarding AAP 001.03.01: -There was discussion that the daily COLREG quizzes (already specified) had already been specified but it could be mentioned again here. Discussion regarding AAP 001.03.02: -There was discussion that that tug element could be removed from this LO and included later in the shiphandling LOs. There was also discussion regarding the BRM piece. It was discussed that as far as BRM is concerned during pilotage, the main concern would be interacting with key personnel such as the OOW, embarked pilot etc. **DECISION:** LO name change to "interact with key personnel during pilotage." Discussion regarding AAP 001.03.03: -There was discussion that the material covered in this LO would not necessarily be tested in a PC or EC, but would help prepare the students for the on-the-job requirements. It was discussed that content here would help to add context to operating in EMCON and satellite denied environments. Discussion regarding AAP 001.03.04: -There was discussion that the focus on the LO would be on coastal and open ocean navigation. It was discussed that information flow during pilotage would be better covered in determining the ship's position. Discussion regarding AAP 001.03.05: -There was discussion that the LO should be split into parts A (relative navigation) and B (celestial navigation). There was discussion that some of the lectures could be conducted at the planetarium.

DATE	Discussion/Detail
	Discussion regarding AAP 001.03.06 -There was discussion that the LO would introduce advanced concepts and reinforce previous knowledge from NWO IV. Discussion regarding AAP 002.01.01 -There was discussion that some of the run sheets from other activities could be used during the LO with the addition of the formation navigation piece. Discussion regarding AAP 002.01.02 -There was discussion that the content under this LO could be covered elsewhere. Discussion regarding AAP 002.02.01 -There was discussion that both LOs to be combined into one. There was discussion that the scenario run sheet could be used here but that it would be best to create a new scenario. Discussion regarding AAP 002.03.01 -There was discussion that both LOs combined under one. It was also discussed that the NABS practical time would be included here. Discussion regarding AAP 004.01.01 -There was discussion that the students would follow along during an online tour, bookmarking resources as they go through. **ACTION: What is the navigator's role?** Creation of a new LO under PO 001 (separate from this LO). Discussion regarding AAP 004.01.02: -There was discussion that the content would help highlight how useful just picking up the phone and calling people can be. There was discussion that the students would be given a homework assignment and be required to give a 5-minute brief on a resource that was not discussed in class. Discussion regarding AAP 004.02.01 -There was discussion that the LO would include guest lectures from AJAG and UEnvO. Discussion regarding AAP 004.02.02 -There was discussion that there would not be a need to re-train because the students should have learned the material on NWO IV. There was also discussion that given the potential 2 year or more gap from having learned the material, there may be a need to re-teach. There was also discussion that the students could be given worksheets separate from the intended daily worksheets. **ACTION:** add an academic piece on conducting a RAS from the supply ship perspective.

DATE	Discussion/Detail
	Discussion regarding AAP 004.02.03: -There was discussion that the LO should be split between A/B/C Discussion regarding 004.02.03B: -There was discussion that an informative piece on WECDIS integration should be included. Discussion regarding 004.02.03C: -There was discussion that the time zone piece would include the theory of time zones and less on the application of time zones. Discussion regarding 004.02.04: -There was discussion that this is when the students would have the opportunity to explore and discuss new tech in the nav world. Discussion regarding 004.02.05: -There was discussion that most topics under this can be covered under other LOs at various points. There was discussion that this LO could be used at the end of the course as a "first day on the job" discussion. 1700 – End of Day 12
Day 13: 24 Jun 20	
	Members present: LCdr Macquarrie Lt(N) Dollis Lt(N) Morales Lt(N) Roberts Lt(N) Workman Lt Mason SLt Seo Review of Day 12 ROP -No additional discussion -Discussion regarding minimum SOA:

DATE	Discussion/Detail
	There was discussion that a minimum SOA should be included in the performance statement. There was discussion that an SOA "window" could be created. This would state the maximum and minimum SOA. There was also discussion that a target SOA could be established. There was discussion that in legacy run sheets, the SOA window was between 10-12kts for MWVNO and 14-16kts for FNO. There was discussion that it would be best to proceed with an SOA window. FNIs would be able to control the SOA by injecting interfering contacts in nav runs. **DECISION:** SOA between 10 – 12kts for MWVNO assessed runs. SOA between 16-20kts for FNO assessed runs. **ACTION:** NTDC(P) to follow up with FNIs for data on high-speed runs by the end November 2020. <u>0835 – Board chair away for OAG</u> **DECISION:** refer to the students as "students" throughout the QSP. Will refer to them as NavO in the PO portion. -Discussion regarding PC001B: there was discussion that in the QSP, it was written that a student would receive a remedial run after 3 failures. It was discussed that this would not be appropriate for the course and could pose a safety risk. There was clarification that the failure of 2 runs would lead to a TRB. -Lt(N) Dollis away **DECISION:** PIP will be included in the QSP as a new annex. **DECISION:** removal of high, medium, and low, priorities for development. AAP will inform development. **ACTION:** Liaise with NTPGHQ to receive a MITE code for the FNO PIP. -Discussion regarding draft course scheduling: -There was discussion that a dedicated mental math period should be included. This will be in addition to the daily quiz. **DECISION:** addition of a metal math period. -There was discussion that a period should be dedicated to discussion Command, Charge and Control and how it applies to the NavO. This will be added into the LO interacting with key personnel.

DATE	Discussion/Detail
	DECISION: addition of an activity covering Command, Charge, and Control as it applies to the NavO. -There was discussion that the role of the NavO lecture should be extended to include 2 periods. **DECISION:** increase the time allotted to the role of the NavO lecture. -There was discussion that the heads-up navigation exercise in the NABS had been removed. This activity is valuable and should be added as an activity. **DECISION:** ensure heads-up nav activity is included in the AAP. -There was discussion that the allotted time at the ocean sciences institute was inadequate. It should be scheduled as a forenoon to account for travel time. **DECISION:** increase the time allotted to the ocean sciences institute tour. -There was discussion that some industry tour activities had not been included in the AAP. Those tours should be added to the AAP. **DECISION:** industry tours to be added to the AAP. -There was discussion that students should be given dedicated review periods where possible. This would allow students last-minute access to FNIs to answer any questions. **DECISION:** include a dedicated review period before assessments where possible. -There was discussion that there is no longer a paper chart assessment. There was discussion that this would be acceptable as navigation is now done almost entirely electronically. There was discussion that there should still be an exposure piece in the content as allied navies may still be using paper charts for navigation. **DECISION:** include a lecture or activity in the AAP for exposure to paper charts and specifically how allied navies may utilize them. -There was discussion that time must be allocated for the Command appraisal activity for open ocean passage planning. **DECISION:** Command appraisal activity to be added to the AAP.

DATE	Discussion/Detail
	-There was discussion that passage planning time should be allocated for students. There was discussion that during the developmental phase, every Wednesday could be made a shiphandling day. The students would divide into 2 groups. One group would be in the NABS in the forenoon while the other group would use the time to catch up on planning. The 2 groups would then switch in the afternoon. **DECISION:** shiphandling Wednesdays during the developmental phase with time built in for planning. -There was discussion that there should be a graduation ceremony for MWVNO. **DECISION:** MWVNO students are to receive their own grad ceremony vice being invited back for the FNO course graduation. -Discussion regarding the time allotted to shiphandling theory. There was discussion that 1 period of shiphandling theory may be insufficient. It was discussed that in addition to the dedicated period, students would also be given worksheets throughout the course. Students would also have time in the NABS to work on shiphandling principles. -Discussion regarding foreign Officers on the course: -There was discussion that foreign Officers participate in the course. There was concern over SECRET material that is covered during the FNO phase. It was discussed that the foreign student could be given the week during which SECRET material is discussed as a leave period. The student would still receive the FNO qualification with the exception of specific POs. **DECISION and ACTION:** foreign officers who successfully complete the course will still receive the qualification of FNO with the exception of the SECRET POs. A note is to be added to remarks to highlight this. <u>1620 – End of Day 13</u>
Day 14: 25 Jun 20	
	<u>25 Jun 20 – Day 14</u> Members Present: LCdr Macquarrie Lt(N) Dollis Lt(N) Morales Lt(N) Roberts

DATE	Discussion/Detail
	Lt(N) Workman Lt Mason <u>Review of Day 13 ROP</u> **ACTION:** NTDC(P) to investigate actual costing during and after the pilot course. <u>0823 – LSC Div Cdr talk</u> <u>0900 – NTDC(P) CO talk</u> <u>0928 – End CO and LSC Div Cdr talk</u> -There was discussion regarding the number of homework the students are assigned throughout the course. There was discussion that there is no tracking for the amount of take-home work the students are given. **ACTION:** during the pilot course, FNIs are to monitor the amount of time students are spent working on assignments at home. Discussion from breakout room: -There was discussion that 15 minutes in the morning may not be enough time to take up the worksheets that were assigned the night before. There was discussion that on the CDC course, it usually takes an hour to get through the worksheet. -There was discussion that wind shear and measuring your ship's performance should be added to the shiphandling serials. -There was discussion that the NOAA visit should be added to the AAP. -There was discussion that the time at sea in a frigate should be focused on pilotage. There was discussion that by trying to focus on too many other points, the students may lose sight of the focus of being at sea, which is pilotage. -There was discussion for the open ocean nav passage that the initiating individual should also be the individual that receives the brief for a point of consistency. -There was discussion that the time in the NABS spent on anchorages could be spent elsewhere. That time could be turned into maneuvering in confined waters. Anchoring stuff will still be covered but will be done on the worksheets and at sea. -There was discussion that a transit of the Fraser River needs to be included in the AAP as an activity. -There was discussion that a handout of best practices for duty nav could be created. **ACTION:** Update stores requirement loan card to reflect current QSP

DATE	Discussion/Detail
	DECISION: pass criteria for exams is 90% for the collision regulations exam 80% for all other written exams.
Day 15: 26 Jun 20	Members present: LCdr Macquarrie Lt(N) Dollis Lt(N) Morales Lt(N) Roberts Lt(N) Workman Lt Mason <u>Review of Day 14 ROP</u> -No additional discussion -Discussion that crewing should be listed in course resources -Added to critical resources <u>1110 – Review of QSP</u> -There was discussion that checklists would be included in Annex K. – Part 3 **DECISION:** Nav planning, nav execution and NSSC to be added to Annex K when complete. <u>1205 – End of QSP Review</u> <u>1210 – Parting words</u> **Board members**: Newfound respect for TDOs, best board I've ever been on, the pilot course will tell if we have done what we intended, we have a much better product, great experience, Nav Elder Brethren will be proud, 10/10 would definitely do a QSP board again, great to finish on a high note, dream product, great first board,

DATE	Discussion/Detail
	TDO: excellent working with all of you, set a standard for excellence in QSPs, leaving a good example for TDOs at NTDC (P) helps that we had a great membership to do the heavy lifting, very respectful board, challenged each other, worked hard and we have an excellent product that will serve to improve the course **Board Chair**: This was a first in terms of conducting a remote QSP WB conducted entirely online, and there is no other way to describe it but as a success. It was challenging at times, but all of you came prepared each day with passion, good ideas, and dedication to the task. This shows clearly in the final product and will be reflected in the quality of future RCN NavOs who complete this course, so I thank you for this. 1215 – QSP Board Adjourned

Lessons Learned

TDO Lessons Learned

ITEM	WWW (What Went Well)	WCBB (What could be better)	ACTION (Going forward)
Preparation	• Board Chair Brief via Zoom • Comprehensive Agenda created and agreed upon by Board Chair • Channel Created in MS Teams, loaded available files, templates, ppts, job aids, and began to organize folders for future documents • Welcome email draft sent to board chair for dissemination	• More notice on use of AAP • AAP template upload to MS teams • Templates were not preformatted to what was going to be used in the final document, which led to much work on the TDO for final formatting	Ensure all templates are formatted in advance to the final format you would like; that way, they are easier to upload to the QSP final version. This includes Navigation Pane headings so that they will match the final version of the QSP. (might require an advanced MS Word course, but would be ideal, as the template is not formatted well)
Process	• At each stage of the process had a quick presentation about Analysis, Design, Scalar creation, POs, EOs, Assessments and AAP – showed 1	• AAP was introduced after the board had started; while the board chair and TDO agreed to implement it, it was an arduous	• Have all your presentations prepped in advance and ready to go • Take members through each stage and show examples and work

ITEM	WWW (What Went Well)	WCBB (What could be better)	ACTION (Going forward)
	example using one of the POs and walked them through it so that the board members could continue on with minimal guidance (it also allowed them to see the format that I was looking for, which meant minimal format effort in AAP/PO/EO templates etc.) • Agenda was followed almost perfectly (excellent board chair, keeping us on track) • AAP added a considerable amount of time, however, will inform the development process and was successfully used (at a high level) and will be able to be used as a base for MLPs (possibly replacing them if this becomes a robust document during development) • Board chair and TDO worked well together to keep the process moving according to the agenda • Secretary kept an excellent ROP/ROD and kept up with Annex A, C, F • Annex F – 2 parts to show terms as well • Annex G – 2 parts (part 1 with template, part 2 ideal schedule)	process, which taxed all board members and meant late nights IOT stay on track for completion within the timeline of the board • The template could use some reworking • Annex G part one was not completed by the end of the board (it was almost an afterthought)	through one real item (PO, EO, Draft Assessment, LO) together so that they have the tools and format you want them to use, so they can work on their own effectively with minimal supervision • AAP template should be formatted in advance to avoid having to go back and change it. • Knowledge and familiarity with this template are necessary to effectively and efficiently employs its use within the QSP. • Understanding that it can take the place of writing EOs and POs at the top level would have saved time in the Design phase. • Annex H should be done concurrently with Critical Resources in Chapter 1 • Annex G should be done concurrently with the AAP (have secretary or other board members continually update this so that it is done at the end) as well it will inform the table in Chap 1 for BAT, TAT and ATT etc. • Recommend 3-part Annex K • Keep Annex L handy during AAP, and add to it as you go along

ITEM	WWW (What Went Well)	WCBB (What could be better)	ACTION (Going forward)
	• Annex K – 3 parts (Assessment Overview, AAP, Assessment Checklists)		
QSP	• TDO recommended starting from scratch to build a new QSP • Allowed board to build the training to the performance requirements • Better product • The rebuilding of the QSP, not using the template but creating a new document made the formatting a tidier process	• A bit more work considering we had to review the old QSP to ensure we captured all teaching points, but also because of the AAP addition • Used the format of the old template, but with a lot of copy-paste	• Take them through the process (reviewing old ways is not always going to work). • Have the AAP locked down in advance of the board and enter information as you do it: ○ POs entered during Analysis • PCs/CECs/ECs and EOs & LOs entered during design, • Create a better template with proper heading functionality; this will save you a headache later.
Zoom Virtual Meeting Platform	• Advanced notice to protocols and some ground rules made the experience painless • TDO procured Zoom Pro • Provided ability to have no limit on meeting time • Provided breakout room and advanced features for co-hosting etc. • Zoom platform overall is very user-friendly and enabled board members to collaborate by sharing screens. • Breakout rooms provided a space for additional SMEs to converse with Board Chair, for TDO to meet with OJTs to explain items, and for	• Probably could have used just Zoom free (would have forced breaks) • Some guests that came in did not use headphones, causing feedback (the beauty is the ability to mute people)	• Continue to use Zoom as the virtual meeting space for virtual boards • Recommend Pro for additional capabilities • Possibly look into MS teams training (while the video and audio are not as high quality it, has much the same functionality as zoom) • Other platforms are useful too, I am sure, Google Hangouts, Saba Meeting • Build in health breaks • Build in extra time for the AAP and for document review at the end

ITEM	WWW (What Went Well)	WCBB (What could be better)	ACTION (Going forward)
	members to work on separate tasks in groups • Audio and video quality were consistently good • Very few issues with the technology		
MS Teams	• Created a team for the division and a private channel the team which only members of the board had access to • Invited members to org, team and channel • Included a quick overview on how to use teams in the welcome email and used share screen on the first day to show navigation through MS Teams • TDO used MS Office 2019 • One repository for all documents • All board members had access rights so they could upload, download and edit. • Enabled collaboration within documents while on share screen, and multiple team members can be in a document collaborating • Good to have version control in renaming files	• MS teams on O365 did not have the capabilities that we needed to create new teams/channels • Not all features for formatting are available, and it made it miserable for formatting. Mitigated by downloading documents, modifying them and then re-uploading to the team. There was some worry about version control, but we mitigated that as well • It would have been good to figure out a standard naming convention, but this group was quick to adapt to my many changes (renaming was easy in MS Teams)	• More features on O365 are necessary to use MS Teams there. • Version control is very important, so ensure that your naming convention allows for this (and decide on that naming convention at the outset) • Folders are your friend; organize your documents so they are easy to find • Ensure your members have the rights to do what you need them to do • Delete old files once they are no longer required (ensure they have been captured in a new version) IOT to keep it from becoming too overwhelming
Google Jam Board	• This was useful as a collaboration tool for Scalaring and brainstorming throughout the Analysis and Design Phases	• More familiarity with how Jam Board works would have been helpful, as we only discovered after 2 days that you could copy-paste stickies	• Continue to use excellent tool • Play with it beforehand to ensure functionality for your needs

ITEM	WWW (What Went Well)	WCBB (What could be better)	ACTION (Going forward)
	• Jam Board was preloaded with the POs from the task list • Worked really well as we could all work in the same Jam Board, and one person could share their screen as well. We were able to do concurrent activities and work in real-time. • Stickies didn't move unless we wanted them too • Resizing stickies as well as colour coding them • Easy to refer back to when we were questioning where we had placed items	• Restriction on size of board you can have so we had several frames • Would have been good to use for the scalar diagrams had it been more flexible with sticky size (only squares available) but would have saved some time on scalar diagram creation	• Research other options that have collaboration but more editing functions?
Mind Manager	• Used to create the scalars • Has a lot of cool functions • Was visually pleasing for the scalars • Easy to use for the scalars but so many more capabilities that require more experience to use.	• Mind manager trial version expired before the end of the board, so the TDO did a screen capture and used that to put into the QSP – not ideal as those images will not be easily manipulated	• Mind manager functionality that could be used well during a board. The ability to assign work, create the scalar and track progress on a Gant chart as well as it produces word documents and a whole bunch of other stuff that requires some familiarity with the program and a license for the product.
Virtual Environment	• TDO used same personal meeting space daily, so that members always had the same meeting ID and PW or link • Settings in Zoom were pre-done so that there was no waiting room,	• Possible distractions of private chats, but this did not seem to be an issue for our board • People sometimes forgot to unmute themselves (but this made for some laughs as well)	• Ensure you prep your board members with some group ground rules • Headphones are a must for this environment

ITEM	WWW (What Went Well)	WCBB (What could be better)	ACTION (Going forward)
	members could join before the host if necessary • Board members used headphones throughout the 3 weeks which reduced feedback • The virtual platform, by its nature, requires that only one person is speaking at a time and reduces side conversation • The chat function provided a place to put ideas that were immediate, links to documents, links to the jam board, notifications when AFK was necessary, or to have private conversations (TDO to OJT to explain) etc. • This was a far less stressful environment as the board was only interrupted virtually (scheduled/expected guests), and with breakout rooms, members could talk virtually in another area, so that remaining board members were able to continue our flow of work free from side conversations and distractions • The board members were all hardworking, respectful and keen to create an excellent product that will serve to improve the training going forward	• All members luckily had reliable internet connectivity; however, it was all done on personal computers with personal internet (so competing family members cause some lagging when using the same connection) • We worked too much. It is important to leave the office at the end of the day and try to build in some PT. While we took breaks as needed, we often had the propensity to work after our meeting had concluded IOT to implement the AAP working into the evenings because we are not leaving our place of work	• Ensure your settings are pre-calibrated for sharing, waiting room options, chat and polls if necessary • Set up a personal meeting space with the same meeting ID for each day of the board so that members will never have to search for it • Make sure that all visitors to your meeting (SMEs, CoC, OJTs, others) are scheduled into your agenda • Disable the waiting room and put on the chime function when someone joins (as this will alert you if you are sharing your screen, and it lets everyone know that someone is joining)

Board Chair Lessons Learned:

1. The ability to conduct an entirely remote QSP WB proved effective. The ability to bring in SMEs for consult throughout the process was hugely beneficial and easy to do. The schedule followed (below) by the WB was highly effective and could be used as a starting point for future QSP WBs.

2020	June (Milestones in Green) Timings are 0800 – 1500 daily w/ breaks & 1 hour for lunch **Analysis Phase Purple, Design Phase Orange, Staff Work Blue**					
MONDAY	**TUESDAY**	**WEDNESDAY**	**THURSDAY**	**FRIDAY**	**WEEKEND**	
08 • Board start • **TDO Brief w/ Introductions Agenda & Milestones** • Set Expectations • FLOOR – Ground Rules • Board Chair brief • CoC Brief? Introductions? • **TDO brief - CFITES process overview** • Review Relevant Docs **TDO Brief on Analysis Phase** • TPA (Annex M), Task Disposition T/NT decisions (Annex B)	09 • ROP review/refine (Annex C) • Task Analysis continued (scalar) • ID PO as Product/Process/ or Combo • PO Scalars (Annex D) • Write POs	10 • ROP review/refine (Annex C) • Clive and Kevin - PO refinement • Write POs continued • Review Chapter 1 of existing QSP • Update QSP Chap 1 & 2 • Liaise with Occ Advisor to ensure alignment to Occ Spec • SEND Chap 1 & 2 (with PO scalars) to HQ for provisional approval • (end of Analysis) • SITREP to CoC	11 • ROP review/refine (Annex C) • **TDO Brief on Design Phase** • Review TPA and further define Learner Characteristics • Refine POs if necessary • Draft Assessment Plan (PC) - summative assessments (Annex K) ○ Capture assessment criteria ○ Draft assessment details	12 • ROP review/refine (Annex C) • Continue with Assessment Plan (Annex K) • Assessment Criteria and details • Identify EO's ○ Group and sequence (EO scalars) ○ ID EO's as product/process/combo ○ Refine • SITREP to CoC	13 WEEKEND	14 WEEKEND
15	16	17	18	19	20	21

• ROP review/refine (Annex C)
• Continue with EOs
  ○ ID LO's (learning objectives) – teaching points
• Write EOs for QSP
	• ROP review/refine (Annex C)
	• Continue with EOs
	• Write EOs for QSP
	  ○ ID LO's (learning objectives) - teaching points
		• ROP review/refine (Annex C)
		• Draft assessment plan (EC) - formative assessments (Annex K)
		  ○ Capture assessment criteria
		  ○ Draft assessment details
		• Update EOs
			• ROP review/refine (Annex C)
			• Finalize Assessment Plan (Annex K)
			• Draft activity plan
			  ○ Separate EOs/LOs into activities
			  ○ MASTER M
			  ○ Identify supporting KSAs
			• SITREP to CoC
				• ROP review/refine (Annex C)
				• Update EOs
				• Sequencing and Timing (Annex G)
				• Detailed Resources (Annex H) - in concert with Chapter 1 of QSP *critical resources
					WEEKEND	WEEKEND

22
• ROP review/refine (Annex C)
• Chapters 1 & 3
• Cross Check References (Annex A)
	23
	• ROP review/refine (Annex C)
	• TEL (Annex L)
	• Abbreviations & Acronyms (Annex F)
		24
		• ROP review/refine (Annex C)
		• Course History (Annex J) - suggest link to old QSP
		• Costing (Annex I)
			25
			• ROP review/refine (Annex C)
			• OJPR (Annex E)
			• Verify:
			  ○ QSP for completeness
			  ○ All Annexes (A thru M)
			  ○ QSP for formatting
				26
				• ROP review/refine (Annex C)
				• Verify complete QSP
				• Final SITREP to CoC
				• Submit for Approval to NTDC (P) CO
					27
					WEEKEND
						28
						WEEKEND

SME Lessons Learned:

1. Involving a spread of SMEs across different timelines (i.e. who completed the job at different times over the recent past) was effective to ensure we were not missing anything from lessons learned in the past or holding onto/dropping specific learning objectives based on narrow experience. Engaged and passionate SMEs made the process much easier and resulted in a high-quality product.

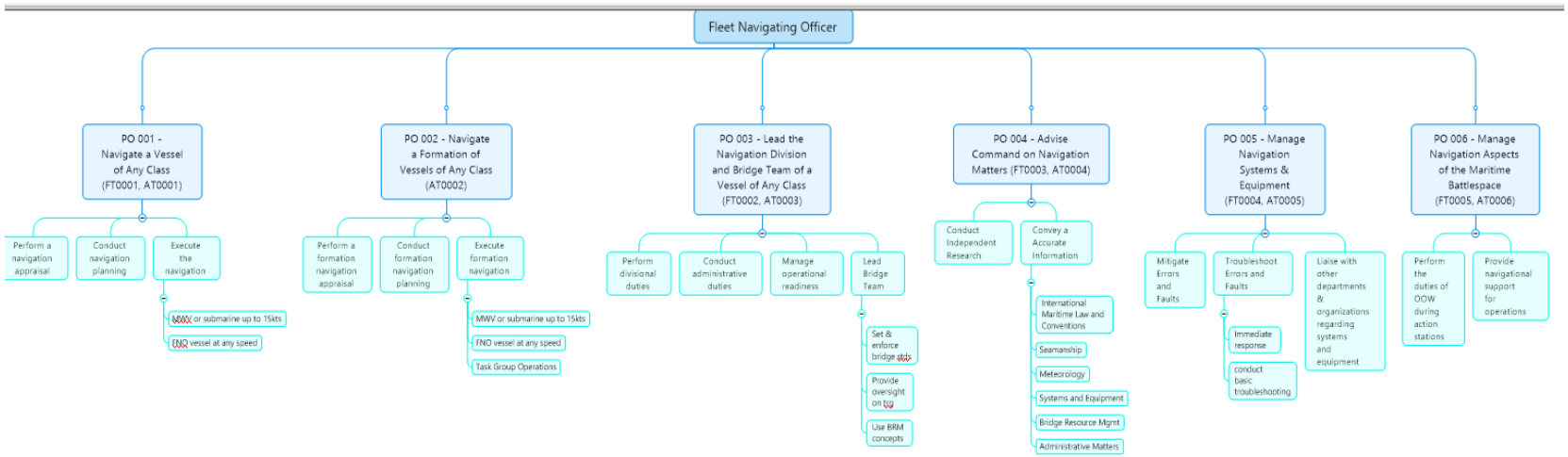
Guidance to Instructional Staff

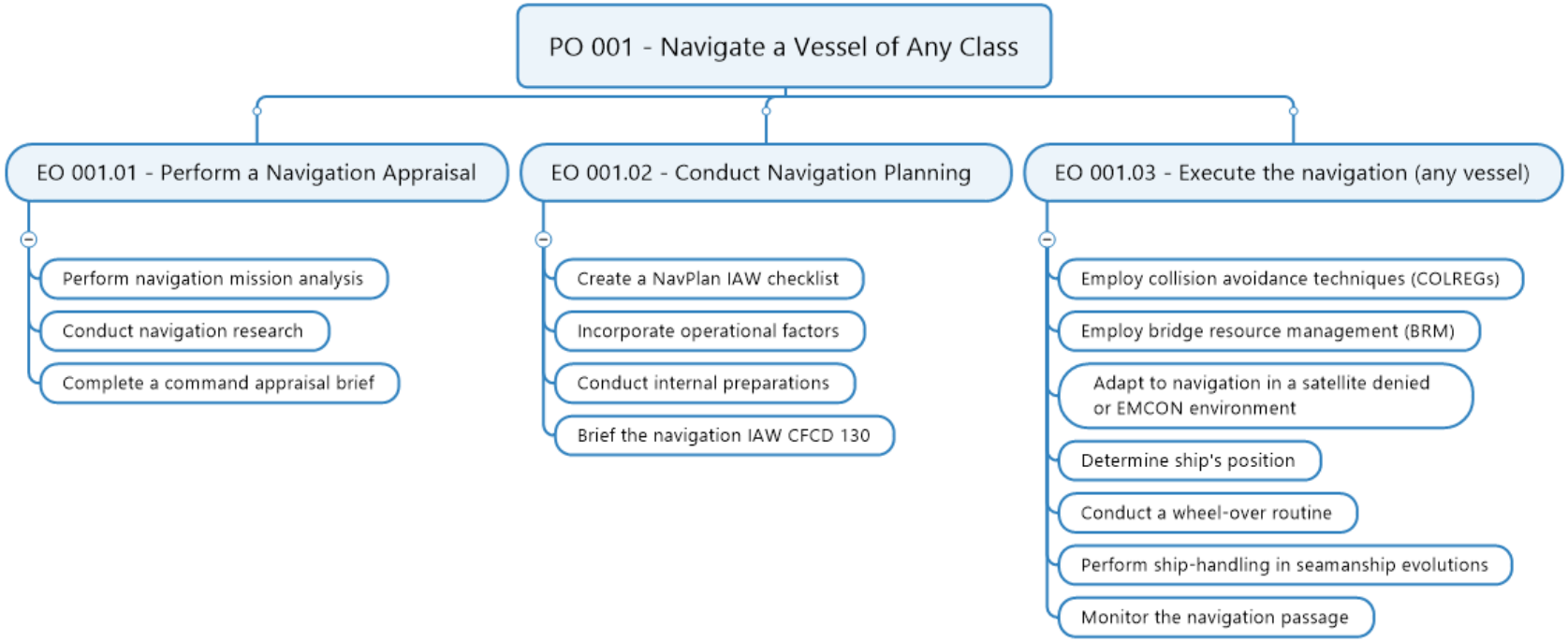
1. The AAP at Annex K Part 2 will inform Development Phase.

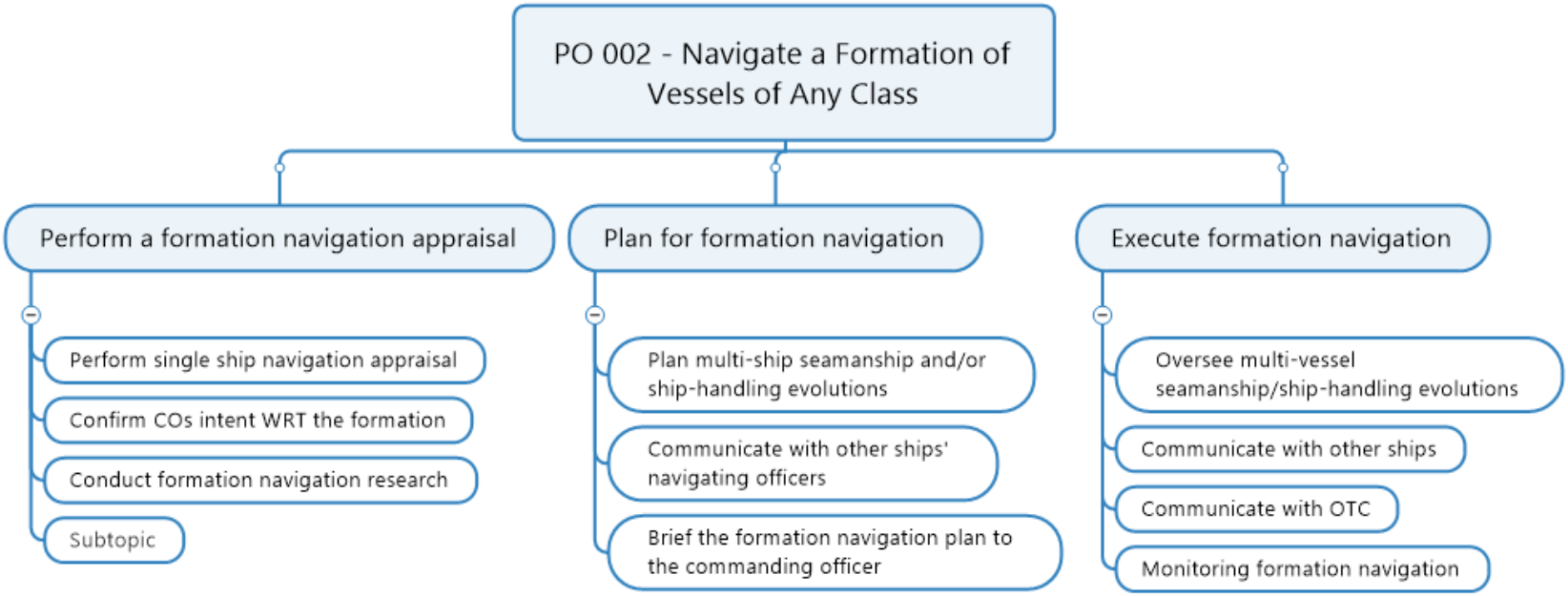
Follow Up Action

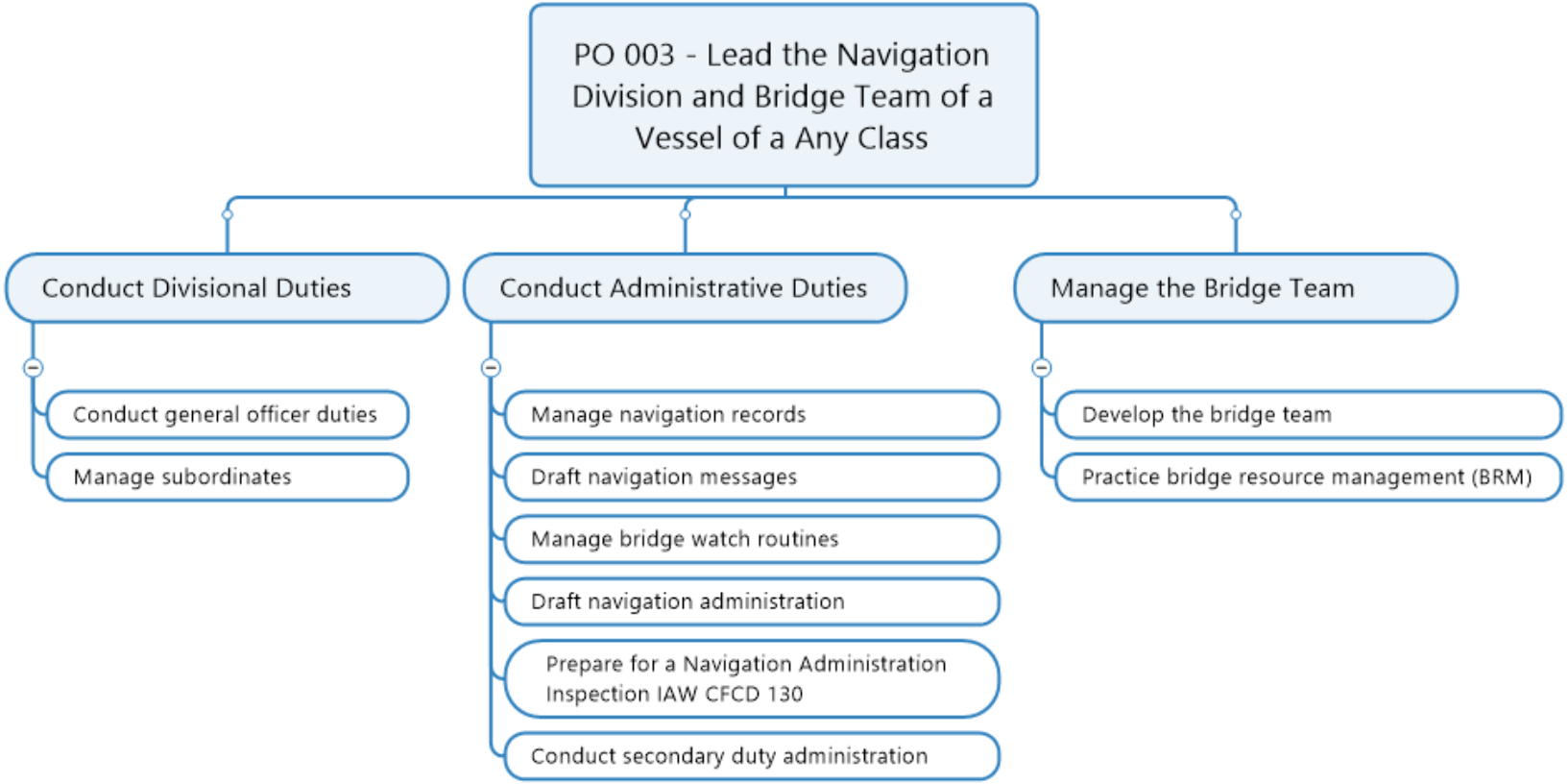
Item	Action	OPI	Expected Date of Completion
1	Chair to contact Occ Mgr and CANSUBFOR for input	Board Chair	12 Jun 20
2	TDO completing formatting of document and ensuring alignment throughout	TDO (Lt Mason)	27 Jun 20
3	Development according to items captured in the AAP	Development Team	31 Dec 20
4	Post Pilot Course Update to QSP	CCD Analyst (Lt(N) Workman)	January 2022
5			
6			

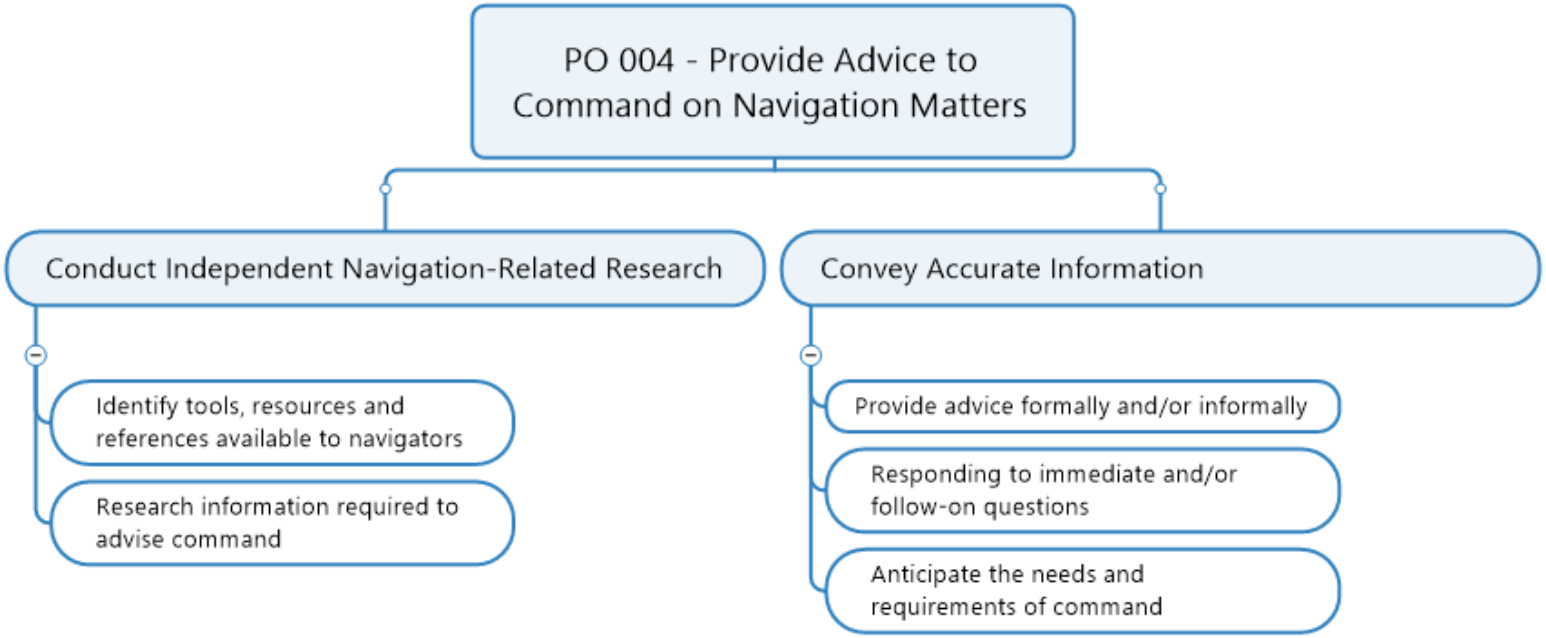
Annex D – Scalar (POs & EOs)

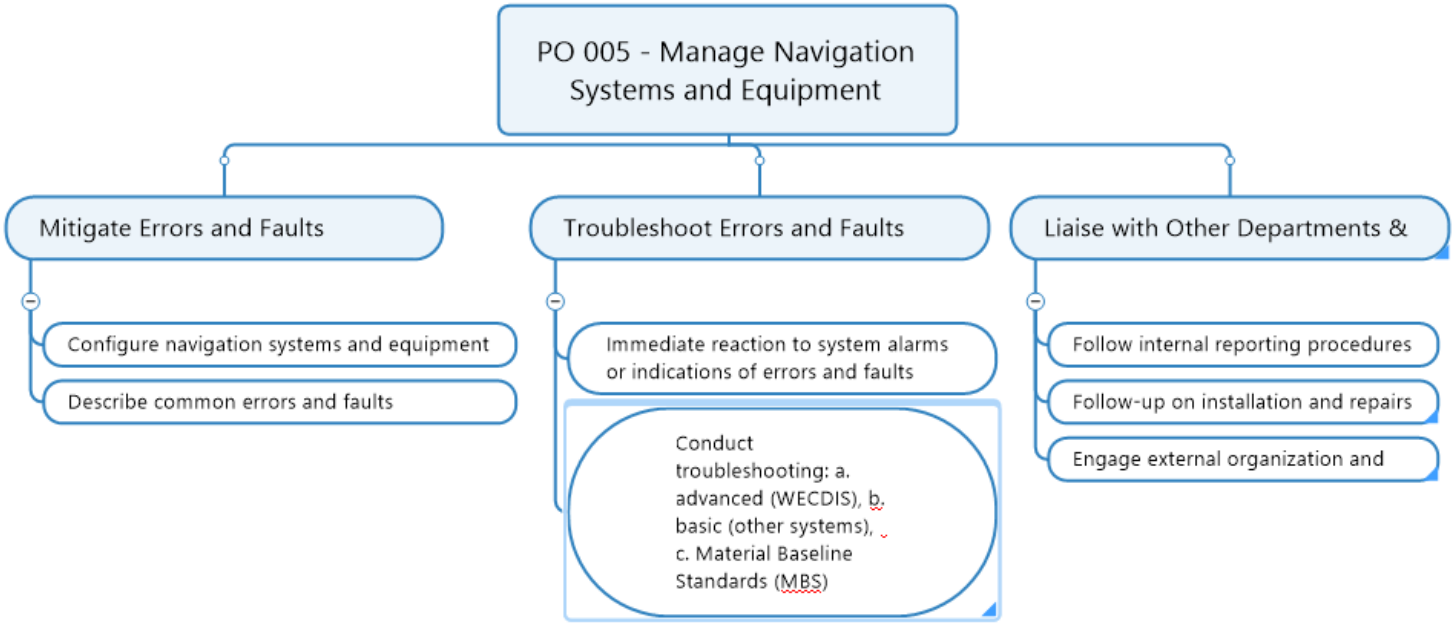

Fleet Navigating Officer
PO 001 - Navigate a Vessel of Any Class (FT0001, AT0001)
Perform a navigation appraisal
Conduct navigation planning
Execute the navigation
MWV or submarine up to 15kts
FNO vessel at any speed
PO 002 - Navigate a Formation of Vessels of Any Class (AT0002)
Perform a formation navigation appraisal
Conduct formation navigation planning
Execute formation navigation
MWV or submarine up to 15kts
FNO vessel at any speed
Task Group Operations
PO 003 - Lead the Navigation Division and Bridge Team of a Vessel of Any Class (FT0002, AT0003)
Perform divisional duties
Conduct administrative duties
Manage operational readiness
Lead Bridge Team
Set & enforce bridge stds
Provide oversight on trg
Use BRM concepts
PO 004 - Advise Command on Navigation Matters (FT0003, AT0004)
Conduct Independent Research
Convey a Accurate Information
International Maritime Law and Conventions
Seamanship
Meteorology
Systems and Equipment
Bridge Resource Mgmt
Administrative Matters
PO 005 - Manage Navigation Systems & Equipment (FT0004, AT0005)
Mitigate Errors and Faults
Troubleshoot Errors and Faults
Immediate response
conduct basic troubleshooting
Liaise with other departments & organizations regarding systems and equipment
PO 006 - Manage Navigation Aspects of the Maritime Battlespace (FT0005, AT0006)
Perform the duties of OOW during action stations
Provide navigational support for operations

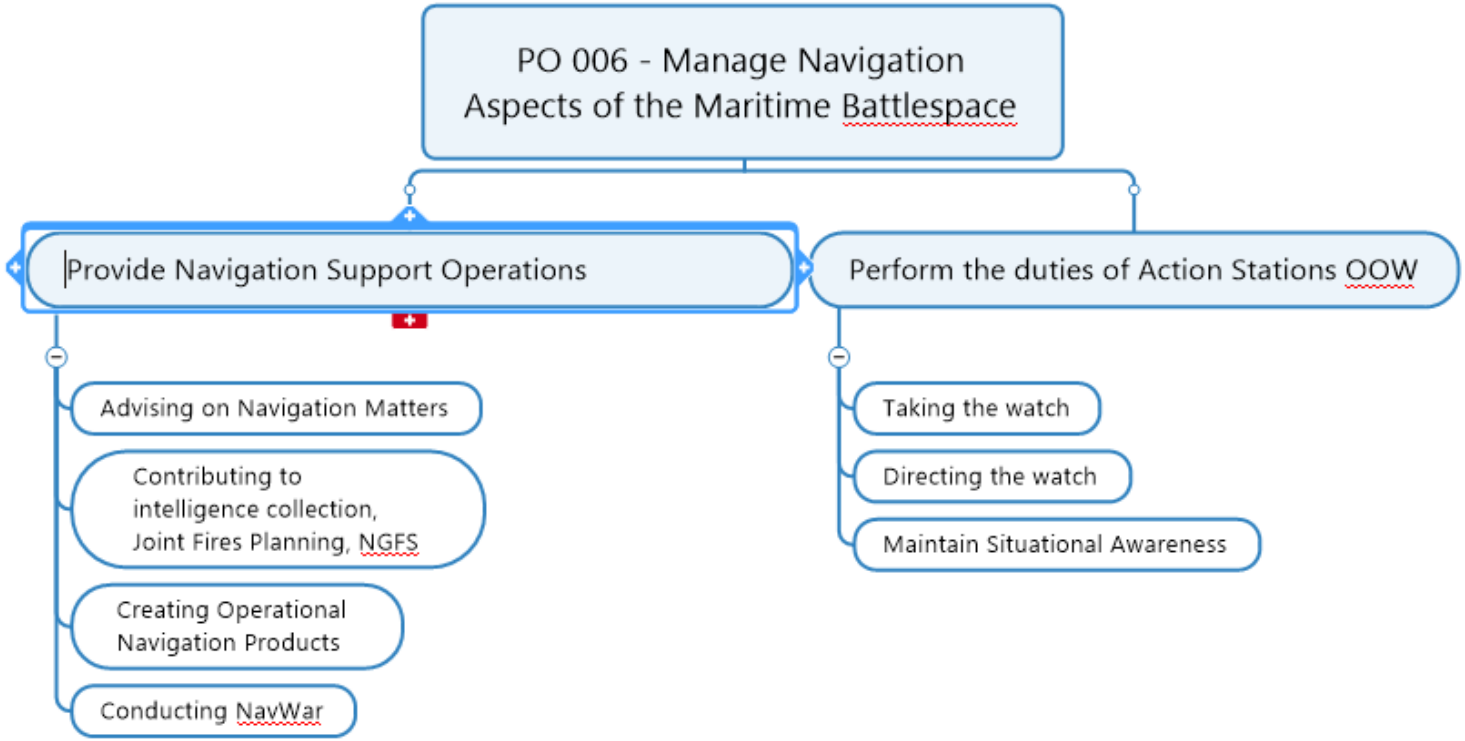

PO 001 - Navigate a Vessel of Any Class
EO 001.01 - Perform a Navigation Appraisal
Perform navigation mission analysis
Conduct navigation research
Complete a command appraisal brief
EO 001.02 - Conduct Navigation Planning
Create a NavPlan IAW checklist
Incorporate operational factors
Conduct internal preparations
Brief the navigation IAW CFCD 130
EO 001.03 - Execute the navigation (any vessel)
Employ collision avoidance techniques (COLREGs)
Employ bridge resource management (BRM)
Adapt to navigation in a satellite denied or EMCON environment
Determine ship's position
Conduct a wheel-over routine
Perform ship-handling in seamanship evolutions
Monitor the navigation passage


PO 002 - Navigate a Formation of Vessels of Any Class
Perform a formation navigation appraisal
Perform single ship navigation appraisal
Confirm COs intent WRT the formation
Conduct formation navigation research
Subtopic
Plan for formation navigation
Plan multi-ship seamanship and/or ship-handling evolutions
Communicate with other ships' navigating officers
Brief the formation navigation plan to the commanding officer
Execute formation navigation
Oversee multi-vessel seamanship/ship-handling evolutions
Communicate with other ships
Communicate with OTC
Monitoring formation navigation


PO 003 - Lead the Navigation Division and Bridge Team of a Vessel of a Any Class
Conduct Divisional Duties
Conduct general officer duties
Manage subordinates
Conduct Administrative Duties
Manage navigation records
Draft navigation messages
Manage bridge watch routines
Draft navigation administration
Prepare for a Navigation Administration Inspection IAW CFCD 130
Conduct secondary duty administration
Manage the Bridge Team
Develop the bridge team
Practice bridge resource management (BRM)


PO 004 - Provide Advice to Command on Navigation Matters
Conduct Independent Navigation-Related Research
Identify tools, resources and references available to navigators
Research information required to advise command
Convey Accurate Information
Provide advice formally and/or informally
Responding to immediate and/or follow-on questions
Anticipate the needs and requirements of command


PO 005 - Manage Navigation Systems and Equipment
Mitigate Errors and Faults
Configure navigation systems and equipment
Describe common errors and faults
Troubleshoot Errors and Faults
Immediate reaction to system alarms or indications of errors and faults
Conduct troubleshooting: a. advanced (WECDIS), b. basic (other systems), c. Material Baseline Standards (MBS)
Liaise with Other Departments &
Follow internal reporting procedures
Follow-up on installation and repairs
Engage external organization and


PO 006 - Manage Navigation Aspects of the Maritime Battlespace
Provide Navigation Support Operations
Perform the duties of Action Stations OOW
Advising on Navigation Matters
Contributing to intelligence collection, Joint Fires Planning, NGFS
Creating Operational Navigation Products
Conducting NavWar
Taking the watch
Directing the watch
Maintain Situational Awareness

Annex E – OJPR

Not applicable for this iteration of the QSP for FNO
See Annex N for the Pre-Instruction Package (PIP)

Annex F – Acronyms, Abbreviations & Terms

This annex is composed of two (2) tables. Table 1 will detail acronyms and abbreviations. Table 2 will detail terms.

List 1 - Acronyms and Abbreviations

Acronym/Abbreviation	Term or Definition
ACP	Allied Communications Publication
AIS	Automatic Identification System
AJAG	Assistant Judge Advocate
ALP	Ambulance Loading Point
ALRS	Admiralty List of Radio Signals
AML	Additional Military Layer
AMN	Admiralty Manual of Navigation
AOR	Auxiliary Oiler Replenishment vessel
ARPA	Automatic Radar Plotting Aid
Art	Article
ASW	Anti-submarine Warfare
ASuW	Anti-surface Warfare
ATP	Allied Tactical Publications
ATR	Any Trade/Rank
BRM	Bridge Resource Management
BWK	Bridge Watch Keeper
C2D2E	Command and Control in a Communications Denied or Degraded Environment
CANHYDROPAC/LANT	Canadian Hydrographic Notices Pacific/Atlantic
CAPN	Computerized American Practical Navigator
CBT	Computer Based Training
CCA	Curriculum Control Authority
CCIS	Command, Control and Information System
CCS	Command and Control System
CCG	Canadian Coast Guard
CDP	Command Decision Point
CF 377	Canadian Forces Course Report
CFAO	Canadian Forces Administrative Orders
CFCD	Canadian Forces Controlled Document
CFOO	Canadian Forces Organization Order
CHP	Canadian Hydrographic Publication
CHS	Canadian Hydrographic Services
CIP	Calling in Point
CMS	Combat Management System
CNPTG	Commander Naval Personnel and Training Group
CO	Commanding Officer
COALT	Co Altitude
CODEC	Co Declination

Acronym/Abbreviation	Term or Definition
COLAT	Co Latitude
COLREGS	Collision Regulations
Comm Pkg	Communications Package
CRR	Combat Readiness Requirements
CSO	Captain's Standing Orders
CSNI	Consolidated Secure Network Infrastructure
CTG	Commander Task Group
CTO	Course Training Officer
CZ	Contiguous Zone
D Geo Ops	Director of Geographic Operations
DAOD	Defence Administrative Orders and Directives
Dec	Declination
DGPS	Differential Global Positioning System
DNC	Distance to New Course/Digital Nautical Chart
DND	Department of National Defence
DNav P&T	Director Naval Personnel and Training
DNR	Directorate of Naval Requirements
DMil C	Director Military Careers
DR	Dead Reckon
DST	Daylight Savings Time
DWAN	Defence Wide Area Network
EC	Enabling Check
ECDIS	Electronic Charting Display & Information System
ECL	Enabling Checklist
ECS	Environmental Chief of Staff
EEZ	Exclusive Economic Zone
EFA	Electronic Fixing Aids
EFZ	Exclusive Fishing Zone
EMCON	Emissions Control
ENC	Electronic Navigation Chart
EO	Enabling Objective
EP	Estimated Position
EPIRB	Emergency Position Indicating Radio Beacon
ETA	Estimated Time of Arrival
ETD	Estimated Time of Departure
FFH	Fast Frigate Helicopter
FLS	Forward Logistics Support
FMF	Fleet Maintenance Facility
FNI	Fleet Navigating Instructor
FNO	Fleet Navigating Officer
FP	Force Protection
FPO	Force Protection Officer
FTA	Fleet Technical Authority
GHA	Greenwich Hour Angle
GIS	Global Information Systems

Acronym/Abbreviation	Term or Definition
GMDSS	Global Maritime Distress and Safety System
GMT	Greenwich Mean Time
GPS	Global Positioning System
HCI	Harbour Control Instructions
HMCS	Her Majesty's Canadian Ship
HS	Sextant Altitude
HSA	Horizontal Sextant Angle
HSO	Hydrographic Services Office
HW	High Water
HYDROPAC/LANT	Hydrographic Note Pacific/Atlantic
IALA	International Association of Lighthouse Authorities
IAW	In Accordance With
IHO	International Hydrographic Organization
IMO	International Maritime Organization
INMARSAT	International Maritime Satellite System
INS	Inertial Navigation System
IOS	International Oceanographic Survey
ISE	Independent Ship Exercise
ISSC	In Service Support Contractors
JMCIS	Joint Maritime Command Information System
JRCC	Joint Rescue Coordination Centre
LCMM	Life Cycle Material Manager
LDL	Limiting Danger Line
LG	Landing Ground
LHA	Local Hour Angle
LMT	Local Mean Time
LOGREQ	Logistics Report
LORAN C	Long Range Radio Aid to Navigation System "C"
LOP	Line of Position
LW	Low Water
Mag	Magnitude
MANICE	Manual of Ice Observing
MANMAR	Manual of Maritime Observations
MARPAC	Maritime Forces Pacific
MARPIF	Maritime Port Information File
MBS	Material Baseline Standard
MCDV	Maritime Coastal Defence Vessel
MCE	Mapping and Charting Establishment
MCOIN	Maritime Command Operational Information Network
MCR	Machinery Control Room
MCTS	Marine Communications and Traffic Services
MERPASS	Meridian Passage
Met	Meteorological
Met Tech	Meteorological Technician
METOC	Meteorological Operations Centre

Acronym/Abbreviation	Term or Definition
MEU	Marine Expeditionary Unit
MIO	Maritime Interdiction Operations
MLP	Master Lesson Plan
MOVREP	Movement Report
MWVNO	Minor War Vessel Navigating Officer
NABS	Navigation Bridge Simulator
NAI	Navigation Administration Inspection
NATO	North Atlantic Treaty Organization
Nav HOD	Navigation Head of Department
NAV NOSTP	Navigation Naval Operations School Training Publication
NAV PIM	Navigational Projected Intended Movements
NAVAREAS	Navigational Areas
NavPac	The software package **NavPac** runs on an IBM PC or compatible and enables navigators to compute their position at sea.
NAVRESHQ	Naval Reserve Headquarters
NAVTEX	Navigational Transition Exchange
NAVWAR	Navigation Warfare
NB	Notebook
NDHQ	National Defence Headquarters
NFR	Naval Force Readiness
NFS(P)	Naval Fleet School (Pacific)
NGFS	Naval Gun Fire Support
NI	Needs Improvement
NIMA	National Imaging and Mapping Agency
NLT	No Less Than
NMEA	National Marine Electronics Association
NMT	No More Than
NOPQ	Naval Officer Professional Qualification
NPTORD	Naval Personnel and Training Orders
NPTT	Navigational Part Task Trainer
NSSC	Navigaton Student Standard Card
NTDC(P)	Naval Training Development Centre (Pacific)
NWO	Naval Warfare Officer
OCE	Officer Conducting Exercise
OCS	Officer Conducting Serial
OGD	Other Governmental Departments
OOW	Officer of the Watch
OPDEF	Operational Deficits
OPSKED	Operational Schedule
ORO	Operations Room Officer
OSQ	Operational Standard Qualification
OTC	Officer in Tactical Command
OTSR	Optimum Track Ship Routing
PC	Performance Check
PCL	Performance Checklist

Acronym/Abbreviation	Term or Definition
PI	Parallel Index
PIM	Position and Intended Movement
PIP	Pre-Instructional Package
PNR	Point of no Return
PO	Performance Objective
PONZAB	Time Zone Acronym
PPS	Precise Positioning System
PVM	Permanent Vertical Magnetism
PZX	Vertices of a Spherical Triangle
QM	Quartermaster
QSP	Qualification Standard and Plan
R/V	Rendezvous
RA	Right Ascension
RACON	Radar Beacon
RAMARK	Radar Marker
RAS	Replenishment At Sea
RCMP	Royal Canadian Mounted Police
ROR	Rules of the Road
SACLANT	Supreme Allied Commander Atlantic
SAR	Search and Rescue
SART	Search and Rescue Transponder
SAT	Satisfactory
SDA	Secret Discussion Area
SDM	SHINNADS Dual-Military
SENC	System Electronic Nautical Chart
SHA	Sidereal Hour Angle
SHINNADS	Shipboard Integrated Navigation and Display System
SHM	Ship's Head Marker
SI	Safety Index
SINS	Shipboard Inertial Navigation System
SIR	Significant Incident Report
SMG	Speed Made Good
SNI	Senior Navigation Instructor
SOA	Speed of Advance
SOLAS	Safety of Life at Sea
SOP	Standard Operating Procedure
SPM	Self-protective Measures
SPS	Special Purpose Segment
SR	Speed Required
SSD	Special Sea Duty Personnel
SSO	Ship's Standing Orders
SST	Sea Surface Temperature
STANAG	NATO Standardization Agreement
STCW	Standards of Training, Certification and Watchkeeping
STD	Speed Time Distance

Acronym/Abbreviation	Term or Definition
STG	Sea Training Group
SWBAT	At the end of this learning objective, "Student Will Be Able To"
SWE	Shallow Water Effect
TAS	Towed Array Sonar
TBD	To Be Determined
TRB	Training Review Board
TS	Training Standard
TTP	Tactics Techniques Procedures
TTW	Territorial Waters
TZD	True Zenith Distance
UCR	Unsatisfactory Condition Report
UHF	Ultra-High Frequency
UNSAT	Unsatisfactory
USN	United States Navy
UT	Universal Time
VCR	Visit Clearance Request
VDR	Voice Data Recorder
VHF	Very High Frequency
VSA	Vertical Sextant Angle
VSAT	Very Satisfactory
VTMS	Vessel Traffic Management System
W/O	Wheel over
WAIS	Warship Automatic Identification System
WDO	Weapons Directing Officer
WECDIS	Warship Electronic Chart Display & Information System

List 2 - Terms

Term	Definition
Chart and Publication Outfit	The publication and chart holdings carried on board HMC ships and vessels.
Multi-threat	A scenario or occurrence where there exists the presence of enemy capability in more than one area of warfare (anti-air, anti-surface, anti-submarine).
Navigation Tools	Tools used to assist the navigator in determining the ship's position. I.e. parallel ruler, rolling ruler, dividers, etc.
Navigator's Notebook	A notebook used in the execution of pilotage that has critical information as per CFCD 130.
Navigator's Workbook	A document (electronic or physical) that captures key appraisal and planning information used by the navigator in all regimes of navigation.
NavPac	The software package NavPac runs on an IBM PC or compatible and enables navigators to compute their position at sea.
NSSC	A card detailing the experience requirements that a student must complete IOT to meet the training objectives.
Run Sheet	A sheet detailing a scenario will be given to students. The sheet will include:

Term	Definition
Total Tides	ADMIRALTY TOTAL TIDE Software provides fast, accurate tidal height and tidal stream predictions for over 7,000 ports and more than 3,000 tidal stream stations worldwide. It can be integrated with NavMaster ECDIS.

Annex G – Suggested Sequencing & Timings

Part 1 – Scheduling Resources

Activity or Assessment	Enabling or Performance Objective	Personnel Resources	Required Resources (for scheduling)	Trg Time/resource (mins)	EC Time (mins)	CEC/PC Time (mins)
PO 001: Navigate a vessel				Total PO time:		34699.75
PO ATT:	29965	PO Admin Time:	2037.9	PO PT time:	2696.85	
EO 001.01	Perform a Nav Appraisal			1185		
EO 001.02	Conduct Nav Planning			675		
EO 001.03	Execute the Navigation			15925		
PC 001A	MWVNO Assessment					870
PC 001B	FNO Assessment					870
EC 001.01	Plan and Brief Open Ocean Passage				660	
CEC 001.02	Celestial Nav Written					180
CEC 001.03	ColRegs					180
CEC 001.04	Duty Nav					8640
CEC 001.05	High Speed NABS					780
PO 002: Formation Navigation				Total PO time (mins):		4138.5
PO ATT:	3390	PO Admin Time:	443.4	PO PT time:	305.1	
EO 002.01	Perform a Formation Nav Appraisal			45		
EO 002.02	Conduct Formation Nav Planning			180		
EO 002.03	Execute the Formation Nav			1125		
PC 002	Formation Nav - Operational					1020
EC 002.01	Formation Nav - Ceremonial				1020	
PO 003: Lead Navigation Div				Total PO time (mins):		1815.5
PO ATT:	1370	PO Admin Time:	322.2	PO PT time:	123.3	
EO 003.01	Conduct Divisional Duties			90		
EO 003.02	Conduct Administrative Duties			495		
EO 003.03	Manage Bridge Team			395		
PC 003A	Nav Policy Written					120

Activity or Assessment	Enabling or Performance Objective	Personnel Resources	Required Resources (for scheduling)	Trg Time/resource (mins)	EC Time (mins)	CEC/PC Time (mins)
PC 003B	Assessed under Duty Nav					0
EC 003.01	Group EC				90	
EC 003.02	Lead PD				180	
PO 004: Support Command Decision Making				Total PO time (mins):		5633.5
PO ATT:	4690	PO Admin Time:	521.4	PO PT time:	422.1	
EO 004.01	Determine Avail Resources			450		
EO 004.02	Convey Accurate Info			3190		
PC 004	Provide Advice					810
CEC 004.01	Shiphandling Theory					120
CEC 004.02	Nav & Marine Sciences					120
PO 005: Manage Navigation Systems				Total PO time (mins):		2609
PO ATT:	2060	PO Admin Time:	363.6	PO PT time:	185.4	
EO 005.01	Mitigate Errors and Faults			165		
EO 005.02	Troubleshoot Errors and Faults			855		
EO 005.03	Liaise			650		
PC 005A	Basic Nav Systems					120
PC 005B	Complex Nav Systems					90
EC 005.01	Presentation				180	
PO 006: Maritime Battlespace				Total PO time (mins):		3034.5
PO ATT:	2430	PO Admin Time:	385.8	PO PT time:	218.7	
EO 006.01	OOW Action Stations			405		
EO 006.02	Nav Support to Ops			945		
PC 006	Maritime Battle Space					120
EC 006.01	OOW Action Stations				360	
EC 006.02	Tactics and Doctrine				120	
EC 006.03	Survey				480	

Actual Training Time	ATT	43905
Base Administrative Time	BAT	
Training Administrative Time	TAT	4074.3
Training Travel Time	TTT	
Physical Fitness Training	PT	3951.45
	Total	51930.75

Activity or Assessment	Enabling or Performance Objective	Personnel Resources	Required Resources (for scheduling)	Trg Time/resource (mins)	EC Time (mins)	CEC/PC Time (mins)
			Total Days	129.826875		
			Rounded	130		

Part 2 - Ideal Course Schedule

The ideal course schedule should be read in conjunction with the Activity and Assessment Plan at Annex K. This ideal course schedule is provided to give a sense of the pace of the course and to aid FNIs in creating a schedule. FNIs should also seek to utilize previous course schedules (archived) to learn from lessons learned regarding sequencing, etc. The course schedule provided below will be updated after the course is piloted in its entirety and reflect any feedback on sequencing provided from students and instructors.

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Monday	Instructor Prep and Arrival Time												
Tuesday													
Wednesday	Course Intro & Introductions		In Routine & Stores (BAT)	Stand Easy	Threshold Knowledge Exam/PIP Review		Dinner	In Routine & Stores (BAT)	Div Interviews (TAT)	Stand Easy	Div Interviews (TAT)	CO Welcome Address (BAT)	Review Role of NavO references (SSO, CFC 130, BR45(4)
Thursday	Quiz/ Worksheet	LO 001.03.01 The Role of the Navigating Officer		Stand Easy	LO 001.01.01B Geodetic Systems and Charts		Dinner	LO 003.01.01 Collision Avoidance		Stand Easy	LO 003.01.01 Collision Avoidance	HW/Debrief (TAT)	BR45(4), Assign Lessons Learned as NavO, Best Practices and things to talk to CO/ BWKs)
Friday	Quiz/ Worksheet	LO 003.01.01 Collision Avoidance		Stand Easy	LO 001.01.01B Geodetic Systems and Charts	LO 001.01.01B Geodetic Systems and Charts	Dinner	LO 001.01.01C Ordering Charts and Publications	LO 001.01.01C Ordering Charts and Publications	Sports/PT			Review Tide and Met Theory

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Saturday	Weekend												
Sunday													
Monday	Quiz/ Worksheet	LO 003.01.01 Collision Avoidance		Stand Easy	Ordering Charts and Publications (HSO Visit)	LO 001.01.01C Ordering Charts and Publications	Dinner	LO 004.02.03B Navigation and Marine Sciences (Tides)		Stand Easy	LO 001.01.01C Ordering Charts and Publications	HW/Debrief (TAT)	BR45(1)
Tuesday	Quiz/ Worksheet	LO 004.02.03B Navigation and Marine Sciences (Tides)	LO 004.02.03B Navigation and Marine Sciences (Tides)	Stand Easy	LO 004.02.03B Navigation and Marine Sciences (Tides)		Dinner	LO 004.02.03B Navigation and Marine Sciences (Tides)		Sports/PT			Readings on GNSS denial
Wednesday	Geodetic Systems and Charts (Ocean Sciences Institute Visit)						Dinner	LO 004.02.03A Navigation and Marine Sciences (Met)		Stand Easy	LO 004.02.03A Navigation and Marine Sciences (Met)	HW/Debrief (TAT)	Review Astro, Read Celestial Navigation Handouts, BR45(2)

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Thursday	Quiz/ Worksheet	LO 004.02.03A Navigation and Marine Sciences (Met)	LO 004.02.03A Navigation and Marine Sciences (Met)	Stand Easy	LO 004.02.03A Navigation and Marine Sciences (Met)	LO 003.01.03 Navigating in GNSS Denied/EMCON Environment	Dinner	LO 003.01.03 Navigating in GNSS Denied/EMCON Environment	LO 004.02.03C Navigation and Marine Sciences (Astro)	Stand Easy	LO 004.02.03C Navigation and Marine Sciences (Astro)	HW/Debrief (TAT)	Read Celestial Navigation Handouts, BR45(2)
Friday	Travel to Royal Dominion Observatory		LO 001.03.05B Determine the Ship's Position (CELNAV) @RDO	Stand Easy	LO 001.03.05B Determine the Ship's Position (CELNAV) @RDO		Dinner	LO 001.03.05B Determine the Ship's Position (CELNAV) @RDO	LO 001.03.05B Determine the Ship's Position (CELNAV) - NavPac	Stand Easy	LO 001.03.05B Determine the Ship's Position (CELNAV) - NavPac	004.02 Review (TAT)/Sextant Practical	Review UNCLOS
Saturday	Weekend												Review BR45(6) Shiphandling, Complete Worksheets
Sunday													Review BR45(6) Shiphandling, Complete Worksheets
Monday	CEC 004.02 Navigation and Marine Sciences			Stand Easy	LO 001.03.05B Determine the Ship's Position (CELNAV) - NavPac		Dinner	LO 004.02.01 International Maritime Law and Conventions (AJAG)		Stand Easy	LO 004.02.01 International Maritime Law and Conventions (AJAG)	HW/Debrief (TAT)	Review BR45(6) Shiphandling, Complete Worksheets
Tuesday	Quiz/ Worksheet	LO 004.02.01 International Maritime Laws (FNI)	LO 004.02.01 International Maritime Law and Conventions (UEnvO)	Stand Easy	LO 001.03.05B Determine the Ship's Position (CELNAV) - NavPac		Dinner	LO 001.03.05B Determine the Ship's Position (CELNAV) - NavPac		Sports/PT			Review BR45(6) Shiphandling, Complete Worksheets
Wednesday	LO 001.03.05B Determine the Ship's Position (CELNAV) - NavPac		CEC 001.02 Ocean Passage by Celestial Fixing				Dinner	LO 004.04.02 Shiphandling and Seamanship		Stand Easy	LO 004.04.02 Shiphandling and Seamanship	HW/Debrief (TAT)	Review BR45(6) Shiphandling, Complete Worksheets

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Thursday	Sports		LO 004.04.02 Towing	Stand Easy	LO 004.04.02 Shiphandling and Seamanship (Heavy Weather)		Dinner	LO 004.04.02 Shiphandling and Seamanship (Heavy Weather)	Shiphandling and Seamanship (Tow Approaches)			HW/Debrief (TAT)	Review Tug Employment
Friday	Quiz/ Worksheet	Shiphandling and Seamanship (RAS)		Stand Easy	Shiphandling and Seamanship (Employment of Tugs)		Dinner	Shiphandling and Seamanship (Employment of Tugs)		Stand Easy	004.01 Review/Rewrite (TAT)		Review Ship Design Principles
Saturday	Weekend												Review ECPINS Manual
Sunday													Review ECPINS Manual
Monday	CEC 004.01 Shiphandling Theory		Shiphandling and Seamanship (Field Trip - QHM Tugs)				Dinner	Shiphandling and Seamanship (Ship Design - NavArc)	LO 005.01.01 FNI EFA presentation/Assignments	Stand Easy	LO 004.04.02 Shiphandling and Seamanship (Worksheet)	HW/Debrief (TAT)	BR45(6) Berthing and Slipping, Create 1-page cheat sheets
Tuesday	Quiz/ Worksheet	Shiphandling and Seamanship (Berthing and Slipping)		Stand Easy	Shiphandling and Seamanship (Berthing and Slipping)		Dinner	Shiphandling and Seamanship (Berthing and Slipping)		Sports/PT			BR45(x) Nav Systems
Wednesday	Quiz/ Worksheet	LO 005.01.02 Describe common errors and faults	LO 005.02.02 Conduct Troubleshooting	Stand Easy	LO 005.02.02 Conduct Troubleshooting	LO 003.02.05 Prepare for a NAI	Dinner	LO 005.02.01 Minor War Vessel navigation systems and equipment tour					Prepare EFAs

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Thursday	Sports			Stand Easy	LO 005.02.03 Describe MBS	LO 005.03.01 Internal Reporting of system Errors and Faults	Dinner	LO 005.02.02 Conduct Troubleshooting	LO 005.01.01 Prepare for EFA presentation	Stand Easy	Prepare EFA/Exam Rewrite (TAT)		Prepare EFAs
Friday	Quiz/ Worksheet	EC 005 EFA Presentation		Stand Easy	EC 005 EFA Presentation		Dinner	EC 005 EFA Presentation		Stand Easy	Review 005	HW/Debrief (TAT)	Review Navigation Systems
Saturday	Weekend												
Sunday													
Monday	Quiz/ Worksheet	LO 005.02.01 React to alarms		Stand Easy	LO 005.02.01 React to alarms		Dinner	LO 005.02.01 React to alarms		Stand Easy	LO 005.02.01 React to alarms		Assign EC 003.01 and EC 003.02
Tuesday	Quiz/ Worksheet	LO 004.02.04 Employment of Navigation Systems and Equipment		Stand Easy	LO 004.02.04 Employment of Navigation Systems and Equipment	LO 003.02.03 Manage Watch Routines	Dinner	PC 005A Demonstrate Knowledge of Basic Navigation Systems		Sports			
Wednesday	Quiz/ Worksheet	LO 003.02.04 Navigation Administration		Stand Easy	LO 003.02.01 Maintain Navigation Records	LO 003.01.02 Manage Subordinates	Dinner	LO 003.03.01 BRM Concepts (CDC)	LO 003.03.01 BRM Error Analysis (CDC)	Stand Easy	LO 003.03.01 Prepare BRM Case Studies		BRM Case Studies

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Thursday	Quiz/ Worksheet	LO 003.03.02 Develop the bridge team	LO 003.03.01 Deliver BRM Case Studies	Stand Easy	LO 003.03.01 Deliver BRM Case Studies		Dinner	LO 003.03.01 Deliver BRM Case Studies	LO 003.03.01 Guided Best Practices & List Building	Stand Easy	LO 003.03.01 Guided Best Practices & List Building	HW/Debrief (TAT)	Read BR45(1) Voyage Planning/RAS as Supply)
Friday	Sports		Shiphandling Worksheet (RAS) Review	Stand Easy	Shiphandling and Seamanship (RAS as Supply Ship)		Dinner	LO 001.01.01A Navigation Mission Analysis		Stand Easy	LO 001.01.01A Navigation Mission Analysis	HW/Debrief (TAT)	Read BR45(1) Voyage Planning
Saturday	Weekend												
Sunday													
Monday	Quiz/ Worksheet	LO 004.01.01 Determine Available Resources		Stand Easy	LO 001.01.03 Complete a Command Appraisal Quarters Brief (Informal)		Dinner	LO 004.01.02 Locate information when deployed		Stand Easy	LO 005.03.03 Engage external agencies/resources for support	HW/Debrief (TAT)	Read BR45(1) Voyage Planning
Tuesday	Group EC 003.01 Respond to complex Navigation Policy or Procedural Issues			Stand Easy	LO 001.02.01 Create a Nav Plan	LO 001.02.02 Include Operational Factors in the Nav Plan	Dinner	LO 003.01.05A Determine the Ship's Position (Methods to Keep the Ship Safe)		Sports			Read BR45(1) Fixing
Wednesday	Sports			Stand Easy	LO 003.01.05A Determine the Ship's Position (Methods to Keep the Ship Safe)		Dinner	LO 003.01.05A Determine the Ship's Position (Methods to Keep the Ship Safe)	LO 003.01.05A Determine the Ship's Position (Fixing EP+DR)	Stand Easy	LO 003.01.05A Determine the Ship's Position (Fixing EP+DR)	HW/Debrief (TAT)	Read BR45(1) Fixing

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Thursday	Quiz/ Worksheet	LO 003.01.05A Determine the Ship's Position (Fixing EP+DR)		Stand Easy	LO 003.01.05A Determine the Ship's Position (Fixing/EP+DR)		Dinner	LO 003.01.05A Determine the Ship's Position (Fixing EP+DR)	LO 001.03.07 Conduct Wheelover Routine	Stand Easy	LO 001.03.07 Conduct Wheelover Routine	HW/Debrief (TAT)	Read BR45(1) Fixing, Plan Wheelover Routine
Friday	EC 003.02 Lead Mariner Professional Development			Stand Easy	LO 001.03.07 Conduct Wheelover Routine (Visual)		Dinner	LO 001.03.07 Conduct Wheelover Routine (Visual)		Stand Easy	LO 003.02.02 Draft Navigation Messages	HW/Debrief (TAT)	Read BR45(1) Fixing, Assign Rest Vis Nav Planning
Saturday	Weekend												
Sunday													
Monday	NABS Prep	LO 001.03.07 Conduct Wheelover Routine (Restricted Vis)		Stand Easy	LO 003.01.02 Interaction with Key Pers during Pilotage		Dinner	LO 001.03.01 Command, Charge and Control		Stand Easy	LO 003.02.06 Conduct NavO Secondary Duties	HW/Debrief (TAT)	Read BR45(1) Pilotage
Tuesday	NABS Prep	LO 001.03.06A (Heads Up Navigation)		Stand Easy	LO 001.03.06A (Heads Up Navigation)		Dinner	LO 001.02.03 Conduct Internal Preparations		Stand Easy	Review PC 003		Read BR45(1) Pilotage
Wednesday	PC 003 Respond to various administrative and policy questions and problems			Stand Easy	LO 001.01.03 Complete a Command Appraisal	LO 002.01.04 The Navigation Brief	Dinner	LO 002.01.04 The Navigation Brief	Supervised Pilotage Planning for Run 1	Stand Easy	Supervised Pilotage Planning for Run 1		Prepare Nav Runs

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Thursday	Nav Brief	Developmental Pilotage (Frigate, Basic Nav Routine)		Stand Easy	Developmental Pilotage (Frigate, Basic Nav Routine)		Dinner	Developmental Pilotage (Frigate, Basic Nav Routine)		Stand Easy	Developmental Pilotage (Frigate, Basic Nav Routine)	HW/Debrief (TAT)	Review and Revise Nav Run
Friday	Nav Brief	Developmental Pilotage (Frigate, Basic Nav Routine) Repeat Run		Stand Easy	Developmental Pilotage (Frigate, Basic Nav Routine) Repeat Run		Dinner	Developmental Pilotage (Frigate, Basic Nav Routine) Repeat Run		Stand Easy	Developmental Pilotage (Frigate, Basic Nav Routine)	HW/Debrief (TAT)	Prepare Nav Runs
Saturday	Weekend												
Sunday													
Monday	Nav Brief	Developmental Pilotage (Frigate, Restricted Vis)		Stand Easy	Developmental Pilotage (Frigate, Restricted Vis)		Dinner	Developmental Pilotage (Frigate, Restricted Vis)		Stand Easy	Developmental Pilotage (Frigate, Restricted Vis))	HW/Debrief (TAT)	Prepare Nav Runs
Tuesday	Nav Brief	Developmental Pilotage (MWV, Traffic)		Stand Easy	Developmental Pilotage (MWV, Traffic)		Dinner	Developmental Pilotage (MWV, Traffic)		Stand Easy	Developmental Pilotage (MWV, Traffic)	HW/Debrief (TAT)	Prepare Shiphandling (Med Moor, Turning Short, Dredging etc.)
Wednesday	Quiz/ Worksheet	LO 004.04.02 Shiphandling and Seamanship (Worksheet)	Shiphandling and Seamanship (Berthing and Slipping)	Stand Easy	Shiphandling and Seamanship (Free play)		Dinner	EC 001.01 Plan an Ocean Passage		Stand Easy	Pilotage Review	HW/Debrief (TAT)	Prepare Shiphandling (Med Moor, Turning Short, Dredging etc.)

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Thursday	Sports			Stand Easy	Shiphandling and Seamanship (Confined Waters)		Dinner	EC 001.01 Plan an Ocean Passage		Stand Easy	LO 003.01.04 Monitor the Navigation Passage	HW/Debrief (TAT)	Prepare Shiphandling (Med Moor, Turning Short, Dredging etc.)
Friday	NABS Prep	Shiphandling and Seamanship (Free play)		Stand Easy	EC 001.01 Plan an Ocean Passage		Dinner	EC 001.01 Plan an Ocean Passage		Stand Easy	Nav Planning Time	HW/Debrief (TAT)	Review COLREGS
Saturday	Weekend												
Sunday													
Monday	CEC 001.03 Collision Regulations			Stand Easy	LO 004.04.02 Shiphandling and Seamanship (Worksheet)	Shiphandling and Seamanship (Multi-Class)	Dinner	Shiphandling and Seamanship (Multi-Class)		Stand Easy	Shiphandling and Seamanship (Multi-Class)		Prepare Nav Runs
Tuesday	Nav Brief	Developmental Pilotage (Frigate, Equipment Loss)		Stand Easy	Developmental Pilotage (Frigate, Equipment Loss)		Dinner	Developmental Pilotage (Frigate, Equipment Loss)		Stand Easy	Developmental Pilotage (Frigate, Equipment Loss)	HW/Debrief (TAT)	Prepare Nav Runs
Wednesday	Nav Brief	Developmental Pilotage (Frigate, Equipment Loss)		Stand Easy	Developmental Pilotage (Frigate, Equipment Loss)		Dinner	Developmental Pilotage (Frigate, Equipment Loss)		Stand Easy	Developmental Pilotage (Frigate, Equipment Loss)	HW/Debrief (TAT)	Review Paper Navigation in CFCD 130

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Thursday	Sports			Stand Easy	LO 002.02.01 Paper Nav & Chartwork		Dinner	Nav Planning Time (Entry/Exit)		Stand Easy	Nav Planning Time (Entry/Exit)	HW/Debrief (TAT)	Prepare Nav Runs
Friday	Nav Brief	Developmental Pilotage (Harbour Entries and Exits)		Stand Easy	Developmental Pilotage (Harbour Entries and Exits)		Dinner	Developmental Pilotage (Harbour Entries and Exits)		Stand Easy	Developmental Pilotage (Harbour Entries and Exits)	HW/Debrief (TAT)	Prepare Nav Runs
Saturday	Weekend												Assigned PC 004 Runs
Sunday													Review PC 004 Runs
Monday	Nav Brief	Developmental Pilotage (Frigate, Equipment Loss)		Stand Easy	Developmental Pilotage (Frigate, Equipment Loss)		Dinner	Developmental Pilotage (Frigate, Equipment Loss)		Stand Easy	Developmental Pilotage (Frigate, Equipment Loss)	HW/Debrief (TAT)	Assigned PC 004 Runs
Tuesday	NABS Prep	PC 004 Provide Advice to Command on Navigation Matters		Stand Easy	PC 004 Provide Advice to Command on Navigation Matters		Dinner	PC 004 Provide Advice to Command on Navigation Matters		Stand Easy	PC 004 Provide Advice to Command on Navigation Matters		Prepare Shiphandling (Med Moor, Turning Short, Dredging etc.)
Wednesday	NABS Prep	LO 004.04.02 Shiphandling and Seamanship (Worksheet)	Shiphandling and Seamanship (Maneuvering in Confined Waters)	Stand Easy	Shiphandling and Seamanship (Free play)		Dinner	Shiphandling and Seamanship (Free play)		Stand Easy	LO 001.01.03 Complete a Command Appraisal (Coastal)		Prepare Nav Runs

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Thursday	Nav Brief	Developmental Pilotage (Orca)		Stand Easy	Developmental Pilotage (Orca)		Dinner	Developmental Pilotage (Orca)		Stand Easy	Developmental Pilotage (Orca)	HW/Debrief (TAT)	Prepare Nav Runs
Friday	Nav Brief	Developmental Pilotage (Orca)		Stand Easy	Developmental Pilotage (Orca)		Dinner	Developmental Pilotage (Orca)		Stand Easy	(TAT)		Prepare Nav Runs
Saturday	Weekend												Prepare Nav Runs
Sunday													Prepare Nav Runs
Monday	EMBARK	MWVNO AT SEA DEVELOPMENTAL											
Tuesday	MWVNO AT SEA DEVELOPMENTAL												
Wednesday	MWVNO AT SEA DEVELOPMENTAL												

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Thursday	MWVNO AT SEA DEVELOPMENTAL							LOCK TRANSIT					
Friday	MWVNO AT SEA DEVELOPMENTAL							NOAA VISIT					
Saturday	Foreign Port Visit												
Sunday													
Monday	MWVNO AT SEA ASSESSMENTS												
Tuesday	MWVNO AT SEA ASSESSMENTS												
Wednesday	MWVNO AT SEA ASSESSMENTS												

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Thursday	MWVNO AT SEA ASSESSMENTS												
Friday	MWVNO ASSESSMENTS									Disembark (TAT)			
Saturday	Weekend												Prepare Nav Runs
Sunday													Prepare Nav Runs
Monday	Return from Sea (TAT)			Stand Easy	LO 002.01.01 Formation Navigation Appraisal	LO 002.02.01 Formation Navigation Planning	Dinner	LO 002.02.01 Formation Navigation Planning	LO 002.02.01 Formation Navigation Planning	Stand Easy	LO 002.03.01 Formation Navigation Execution	HW/Debrief (TAT)	Prepare Nav Runs
Tuesday	Nav Brief	Execute Formation Navigation		Stand Easy	Execute Formation Navigation		Dinner	Execute Formation Navigation		Stand Easy	Execute Formation Navigation		Prepare Nav Runs
Wednesday	Nav Brief	EC 002.01 Navigate a formation of ships in a ceremonial environment		Stand Easy	EC 002.01 Navigate a formation of ships in a ceremonial environment		Dinner	EC 002.01 Navigate a formation of ships in a ceremonial environment		Stand Easy	EC 002.01 Navigate a formation of ships in a ceremonial environment		Prepare Nav Runs

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Thursday	Nav Brief	Execute Formation Navigation		Stand Easy	Execute Formation Navigation		Dinner	Execute Formation Navigation		Stand Easy	Execute Formation Navigation		Prepare Nav Runs
Friday	Nav Brief	PC 002 Navigate a formation of ships in an operational environment		Stand Easy	PC 002 Navigate a formation of ships in an operational environment		Dinner	PC 002 Navigate a formation of ships in an operational environment		Stand Easy	PC 002 Navigate a formation of ships in an operational environment		Review Br45(x) Surveying for Non-Surveyors
Saturday	Weekend												Review Br45(x) Surveying for Non-Surveyors
Sunday													Review Br45(x) Surveying for Non-Surveyors
Monday	PC 002 Reruns			Stand Easy	PC 002 Reruns		Dinner	LO 006.02.03 Survey Theory	LO 006.02.03 Survey Lead-through	Stand Easy	LO 006.02.03 Survey Lead-through	HW/Debrief (TAT)	
Tuesday	Quiz/ Worksheet	LO 006.02.01 Advise on Operational Navigation Matters		Stand Easy	LO 006.02.03 Create Operational Navigation Products		Dinner	LO 006.02.04 Conduct NAVWAR (Plan NAVWAR)		Stand Easy	Review 006 Survey (TAT)	HW/Debrief (TAT)	
Wednesday	Sports			Stand Easy	LO 006.02.04 Conduct NAVWAR		Dinner	LO 006.02.04 Conduct NAVWAR		Stand Easy	LO 006.02.04 Conduct NAVWAR	Review 006 Op Products(TAT)	

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Thursday	PC 006 Manage the Navigation Aspects of the Maritime Battlespace			Stand Easy	Sports		Dinner	EC 006.03 Conduct a Hydrographic Survey		Stand Easy	EC 006.03 Conduct a Hydrographic Survey	HW/Debrief (TAT)	
Friday	Travel	EC 006.03 Conduct a Hydrographic Survey		Stand Easy	EC 006.03 Conduct a Hydrographic Survey		Dinner	EC 006.03 Conduct a Hydrographic Survey		Stand Easy	EC 006.03 Conduct a Hydrographic Survey	HW/Debrief (TAT)	
Saturday	Weekend												
Sunday													
Monday	PC 006 Rewrite/Sports			Stand Easy	EC 001.01 Ocean Passage Review and Submission		Dinner	Travel to Vancouver (Tour of BC Ferry Bridge)					Overnight in Vancouver
Tuesday	Travel	BCIT Marine Campus		Stand Easy	Planetarium		Dinner	Pilotage Authority		Stand Easy	Vancouver Port Authority		Overnight in Vancouver
Wednesday	Travel	CCG Tour		Stand Easy	CCG Hovercraft Practical		Dinner	OSI Tour		Stand Easy	OSI Tour		Travel to Victoria, Finalize Ocean Passages

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Thursday	Quiz/ Worksheet	EC 001.01 Brief an Ocean Passage (MWVNO priority)		Stand Easy	EC 001.01 Brief an Ocean Passage (MWVNO priority)		Dinner	LO 005.02.02 Conduct Troubleshooting FMF Lecture		Stand Easy	Review Complex Nav Systems	HW/Debrief (TAT)	Review Complex Nav Systems
								Out Routine (MWVNO)					
Friday	Quiz/ Worksheet	LO 005.02.01 Major Warship navigation systems and equipment tour		Stand Easy	LO 005.02.01 Major Warship navigation systems and equipment tour	LO 004.02.05 Expectations and Role as NavO	Dinner	MWVNO GRADUATION CEREMONY					
	MWVNO	Out Routine (MWVNO)			Out Routine (MWVNO)								
Saturday	Weekend MWVNOs Depart												
Sunday													
Monday	PC 005B Demonstrate Knowledge of Complex Navigation Systems			Travel to N92	LO 006.01.01 Take the Watch		Dinner	LO 006.01.02 Direct action stations watch (N92A)		Stand Easy	LO 006.01.02 Direct action stations watch (N92A)		
Tuesday	Travel to N92	Review 006 Direct Action Stations Watch	LO 006.01.02 Direct action stations watch (N92A)	Stand Easy	LO 006.02.02A Contribute to Intelligence Collection and Joint Fires (N92A)		Dinner	LO 006.01.03 Maintain Situational Awareness (N92 SEACOT)		Stand Easy	Review 006 Action Stations (TAT)	Travel from N92	

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Wednesday	Travel to N92	EC 006 Review		Stand Easy	EC 006.02 Tactics and Doctrine (N92)/Travel from N92		Dinner	Sports	Sports/005B Rewrite	Stand Easy	Sports/005B Rewrite		
Thursday	Travel to Dockyard	EC 006.01 Action Stations OOW PC		Stand Easy	EC 006.01 Action Stations OOW		Dinner	EC 006.01 Action Stations OOW		Stand Easy	EC 006.01 Action Stations OOW		
Friday	Travel to Bay Street	LO 006.02.02B Contribute to NGFS (Bay Street Armoury)		Stand Easy	LO 006.02.02B Contribute to NGFS (Bay Street Armoury)		Dinner	LO 006.02.02B Contribute to NGFS (N92A)		Stand Easy	LO 006.02.02B Contribute to NGFS (N92A)		
Saturday	Weekend												
Sunday													
Monday	Nav Brief	FNO DEVELOPMENTAL (Frigate, High Speed)					Dinner	FNO DEVELOPMENTAL (Frigate, High Speed)					
Tuesday	Nav Brief	FNO DEVELOPMENTAL (Frigate, High Speed)					Dinner	FNO DEVELOPMENTAL (Frigate, High Speed)					

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Wednesday	Nav Brief	FNO DEVELOPMENTAL (Frigate, High Speed)					Dinner	FNO DEVELOPMENTAL (Frigate, Narrows)					
Thursday	Nav Brief	FNO DEVELOPMENTAL (Frigate, High Speed)					Dinner	FNO DEVELOPMENTAL (Frigate, High Speed, Restricted Vis)					
Friday	Nav Brief	FNO DEVELOPMENTAL (Frigate, Harbour Entry)					Dinner	FNO DEVELOPMENTAL (Frigate, Harbour Entry)					
Saturday	Weekend												
Sunday													
Monday	Nav Brief	FNO DEVELOPMENTAL (MULTI-CLASS)					Dinner	FNO DEVELOPMENTAL (MULTI-CLASS)					
Tuesday	Nav Brief	FNO DEVELOPMENTAL (MULTI-CLASS)					Dinner	FNO DEVELOPMENTAL (MULTI-CLASS)					

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Wednesday	Nav Brief	FNO DEVELOPMENTAL (MULTI-CLASS)					Dinner	FNO DEVELOPMENTAL (MULTI-CLASS)					
Thursday	Nav Brief	FNO DEVELOPMENTAL (MULTI-CLASS)					Dinner	FNO DEVELOPMENTAL (MULTI-CLASS)					
Friday	Nav Brief	FNO DEVELOPMENTAL (Multi-Class, Harbour Entry)					Dinner	FNO DEVELOPMENTAL (Multi-Class, Harbour Entry)					
Saturday	Weekend												
Sunday													
Monday	Nav Brief	FNO DEVELOPMENTAL (Frigate, hell week)					Dinner	FNO DEVELOPMENTAL (Frigate, hell week)					
Tuesday	Nav Brief	FNO DEVELOPMENTAL (Frigate, hell week)					Dinner	FNO DEVELOPMENTAL (Frigate, hell week)					

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Wednesday	Nav Brief	FNO DEVELOPMENTAL (Frigate, hell week)					Dinner	FNO DEVELOPMENTAL (Frigate, hell week)					
Thursday	Nav Brief	FNO DEVELOPMENTAL (Frigate, hell week)					Dinner	FNO DEVELOPMENTAL (Frigate, hell week)					
Friday	Nav Brief	FNO DEVELOPMENTAL (Frigate, hell week, harbour entry)					Dinner	FNO DEVELOPMENTAL (Frigate, hell week, harbour entry)					
Saturday	Weekend												
Sunday													
Monday	Nav Brief	FNO NABS ASSESSED					Dinner	FNO ASSESSED					
Tuesday	Nav Brief	FNO NABS ASSESSED					Dinner	FNO ASSESSED					

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Wednesday	Nav Brief	FNO NABS ASSESSED					Dinner	FNO ASSESSED					
Thursday	Nav Brief	FNO NABS ASSESSED					Dinner	FNO ASSESSED					
Friday	Nav Brief	FNO NABS ASSESSED					Dinner	FNO ASSESSED					
Saturday	Weekend												
Sunday													
Monday	EMBARK	FNO AT SEA DEVELOPMENTAL											
Tuesday	FNO AT SEA DEVELOPMENTAL												

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Wednesday	FNO AT SEA SHIPHANDLING												
Thursday	FNO AT SEA DEVELOPMENTAL												
Friday	FNO AT SEA DEVELOPMENTAL										DISEMBARK (TAT)		
Saturday	Weekend												
Sunday													
Monday	EMBARK	FNO AT SEA FFH DEVELOPMENTAL											
Tuesday	FNO AT SEA ASSESSED												

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Wednesday	FNO AT SEA ASSESSED												
Thursday	FNO AT SEA ASSESSED												
Friday	FNO AT SEA ASSESSED									DISEMBARK (TAT)			
Saturday	Weekend												
Sunday													
Monday	Disembark FFH/Return from Sea (TAT)						Dinner	LO 004.02.05 Expectations and Role as NavO		Stand Easy	Return Stores (TAT)		
Tuesday	Sports			Stand Easy	EC 001.01 Brief an Ocean Passage (FNO - if req'd)		Dinner	EC 001.01 Brief an Ocean Passage (FNO - if req'd)		Stand Easy	Course Critique (TAT)		

	0800-0815	0815-0900	0900-1000	1000-1015	1015-1100	1100-1200	1200-1300	1300-1345	1345-1430	1430-1445	1445-1545	1545-1600	**Homework**
Wednesday	Out Routine (TAT)			Stand Easy	Course Reports (TAT)		Dinner	FNO GRADUATION CEREMONY					
Thursday	Instructor Prep and AAR/Students Travel												
Friday													
Saturday	Weekend												
Sunday													

Annex H – Additional Training Resources

(**Annex H not to be deleted**)

This annex is a compliment to the staffing and critical resources listed in chapter 1 para 3, 4 & 5 of this QSP.

FC = Full Course, A/R = As Required

STORES REQUIREMENTS		
Item Description	Quantity	Notes
Handheld GPS	1 per course	
Stuart's Distance Meter	2 per course	
Weymouth-Ross Distance Meter	2 per course	
Plastic Parallel Ruler	1 per student	*loan card to be created for each student with all resources given for the duration of the course.
Compass Set	1 per student	
Brass Dividers	1 per student	
Portland Speed Time Distance (STD) Ruler	1 per student	
Sextant (metal)	1 per student	
Sextant (plastic)	3 per course	
3 Arm Protractor (Station Pointer)	1 per course	
Publications	1 per student	Publications in Annex A.
DND USB memory stick	1 per student	*see loan card note above
Portable DVD drive/USB	1 per course	
Plain Printer Paper (box)	10 per course	
Binders 3 ring – 8 ½ x 11"	1 per student	*consumable items not on loan card
OOW Notebook	1 per student	
Pens	3 per student	
Pencils (2B leads)	1 box per student	
Erasers	3 per trainee	

Annex I – Costing Information

Annex I – Costing Information

All numbers in this table are in CAD currency.

Personal Equipment	Quantity	Cost Per Student	Initial TE Set-Up	Each Successive Course	Cost per FY	5 Year Estimate
Laptop	10	1500	27000	0	N/A	27000
Laptop Case and Power Supply	10	90	900	0	N/A	900
Admiralty Total Tides Licence	10	118	1180	1180	2360	11800
Publications and Instruments	10	528	5280	0	0	5280
PPNS	10	50	900	500	1000	5000
Sub-total		2286	35260	1680	3360	49980

Note 1. Costing based on 2016 set-up costs for NWO IV based on lack of historical data for FNO.
Note 2. Quantity based on maximum course loading.
Note 3. Sea-phase costs are not captured in this annex and must be sourced from F3 Ops to reflect up-to-date pricing.
Note 4. Industry Tour costing has not exceeded $10,000 for engagement with OGD/industry in support of training objectives. See NFS(P) Comptroller for historical data.

Annex J – Course History

This annex shall NOT be deleted.

\It must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details on course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications shall be captured in the table below. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Reason for amendment (with date)	Details on course history
QSPWB 08 June 20 – 26 June 20	See Record of Proceedings/Decisions (ROP/ROD) at Annex C for a summary of proceedings and pertinent decisions that were made. A detailed summary of the minutes and discussions can be found in the ROP/RDP within this QSP.
FNO QSP Rev 2018 V 2.3	See RDIMS 392011
FNO QSP 18 Aug 2015 V 2.2	See RDIMS 392011

Annex K – Assessment Plan

Part 1 - Assessment Overview

The scope of the assessment represents portions of the EOs.
This assessment plan shall be read in conjunction with the Annex K Part 2 AAP and Annex G Part 2 Ideal Course Schedule

ASSESSMENT OVERVIEW							
Scope	**Purpose of Check**	**Method Used**	**When and How Conducted**	**Resources**	**Time**	**Limitations**	**Impact & Remedial Instruction**
PC001A **MWVNO (all students)** PO001 PO003 PO004 PO005	To assess the student's ability to Conduct MWV Pilotage	Practical at Sea	Performed at the end of MWVNO Sea Phase **See Annex K Part 2 and Annex G Part 2** Execution of Pilotage in a Minor War Vessel or Naval Training Vessel **Remarks**: During execution assessment, the minimum equipment available to the student will be a WECDIS laptop, navigator's notebook, and timepiece.	Scenario: Minor War Vessel or Naval Training Vessel, Chart and Publication Outfit, Navigation Tools, Supervision of a Commanding Officer. WECDIS laptop, DWAN and Internet access	240min planning time 60min execution + 30min feedback/student (540min for ideal class size of 6) Total time 780min	SOA Minimum: 10kts Maximum: 12kts Max speed as per class of vessel being navigated	Candidates must complete 4 assessed passages (Min 2 day & 2 night). Candidate must pass 3, (min 1 day & 1 night). *See note in Para 25 regarding "maintaining standards" Failure of 2 runs results in a TRB.IAW NPTORD 4500-54
PC001B **FNO only** PO001 PO003 PO004 PO005	To assess the student's ability to Conduct a High-Speed Pilotage Run at Sea	Practical at Sea	Performed at the end of FNO Sea Phase **See Annex K Part 2 and Annex G Part 2** **Remarks:** During the execution assessment, the minimum equipment available to the student will be a WECDIS laptop, navigator's notebook, and timepiece. The focus is BRM and team management, with a lower emphasis placed on constricted waters and speeds.	Major Warship, Chart and Publication Outfit, Navigation Tools, Supervision of a Commanding Officer. WECDIS laptop, DWAN and Internet access. Full Bridge Team	240min planning time 60min execution + 30min feedback/student (540min for ideal class size of 6) Total time 780min	SOA minimum: 16kts Maximum: 20kts, Max speed as per class of vessel being navigated	Candidate must complete 4 pilotage passages (min 2 day & 2 night). Candidate must pass 3 pilot passages (min 1 day & 1 night). *See note in Para 25 regarding "maintaining standards" Failure of any 2 results in a TRB IAW NPTORD 4500-54
EC 001.01 Scope: EO 001.01-02 EO 002.01-02 EO 003.02 EO 004.01-02	To check the student's ability to: Plan and Brief an Open Ocean Passage	Written planning report assessment IAW checklist Brief to FNI acting as command	**See Annex K Part 2 and Annex G Part 2** **Remarks:** Assigned as homework after completion of lessons. Students have NLT than 4 weeks to complete it as homework. 8 hours are assigned as classroom planning time. See Annex K – Part 2 AAP EC 001.01 for assessment details	WECIDS and DWAN laptop with internet access, Chart and Publications Outfit, Navigation Tools. Access to external agencies, instructors, and fellow officers.	Allotted 8 hours of classroom planning time 480min See AAP for additional info	Students must manage their time IOT complete this assignment Not all outside agencies should be contacted as the passage is simulated. Instructors will simulate some outside agencies.	Completed IAW checklist Draft report to FNI for feedback and resubmit corrections as necessary Formative Feedback from FNI after completion of brief
CEC 001.02 Scope: EO 001.03	To confirm the student's ability to Employ Celestial Navigation Software to Execute a	Written Exam	**See Annex K Part 2 and Annex G Part 2** NavPac PC software used IOT calculate the times and azimuth of rising and setting of heavenly bodies, and fixing the	Classroom, computer with NavPac PC and WECDIS, pens, exam papers	180min exam 60min fudge factor (questions, extra time?) should be a finite amount of time	No actual star or sun sights are to be taken by sextant; measurements are provided to students.	Must answer 80% of questions correctly. Review, Re-write (supplemental CEC)

ASSESSMENT OVERVIEW							
Scope	**Purpose of Check**	**Method Used**	**When and How Conducted**	**Resources**	**Time**	**Limitations**	**Impact & Remedial Instruction**
EO 004.02 Critical due to safety.	Simulated Open Ocean Passage		ship's position using MERPASS, Sun – Run – Sun, and Star sights.				2 consecutive failures results in a TRB IAW NPTORD4500-54 Candidate expected to maintain standard throughout the course.
CEC 001.03 Scope: EO 001.03 EO 004.02 Critical due to safety	To confirm the student's ability to apply collision regulations (COLREGs)	Written Exam Multiple choice and short answer	**See Annex K Part 2 and Annex G Part 2** Written CEC consisting of multiple choice and short answer questions including a multimedia component that demonstrates shapes, lights, and sound signals a Navigator would be expected to identify at sea.	Classroom, computer, projector, speakers, pens, exam papers, writing instrument	180min exam	Nil	90% pass mark Review, rewrite (supplemental CEC) Unsatisfactory result on supplemental CEC will result in TRB IAW NPTORD4500-54. A second supplemental PC will not be authorized Candidate expected to maintain standard throughout the course.
CEC 001.04 Scope: EO 001.02 EO 001.04 EO 003.02-03 EO 004.01-02 EO 005.01-03 Critical due to safety	To confirm the student's ability to Act as a Duty Navigator	Practical Experiential – acting as Duty Nav Sea Phase including day and night	**See Annex K Part 2 and Annex G Part 2** Duty Nav rotation, reaction to a least one system or equipment failure & advice to command See Annex K – Part 2 AAP EC 001.04 for assessment details	Vessel at sea, DWAN Laptop with WECDIS, references, Chart and publications outfit.	24-hour rotation per student (no less than 16 hours) must include an overnight portion	No less than 16 hours to account for days where the vessel departs or arrives alongside	Formative feedback Re-attempt 2nd unsatisfactory attempt results in a TRB IAW NPTORD4500-54. *opportunities for re-assessment may be given during subsequent training sea-phases at the discretion of the TRB (available MWVNO and FNO).
CEC 001.05 Scope: EO 001.01-03 Critical due to safety	To confirm the student's ability to conduct a high-speed pilotage run.	Practical NABS Trainer Simulated Major Warship	**See Annex K Part 2 and Annex G Part 2** FNI input for mission and at least 1 evolution	Simulated Major Warship, Chart and Publication Outfit, Navigation Tools, Supervision of a Commanding Officer. WECDIS laptop, DWAN and Internet access.	240min appraisal and planning (per student concurrent) 60min execution of navigation scenario per student 30min feedback per student	SOA: Minimum: 16kts Maximum: 20kts Max speed as per class of vessel being navigated FNO level assessed runs will have a minimum of: • 1 multi-contact situation;	Candidate must complete 5 simulated pilotage passages (min 1 day & 1 night, at least 1 in foreign waters). The fifth day of assessments can be tailored to the students day/night requirements. Candidate must pass 3/5 simulated pilotage passages (min 1 day & 1 night) but will complete all 5 runs.

ASSESSMENT OVERVIEW							
Scope	**Purpose of Check**	**Method Used**	**When and How Conducted**	**Resources**	**Time**	**Limitations**	**Impact & Remedial Instruction**
						• 1 narrows/choke point situation; • 3 other interfering contacts; • Non-interfering contacts are not regulated as to how many or location of them; • A GPS loss; • a minimum of 2 other equipment failures (IAW NSSC) • Restricted visibility for a minimum of 2nm • Where possible, every effort should be made to ensure an FNO run has at least one area where safe water (as defined by LDL to LDL) is less than 300x • End in an anchorage run where the FNO student must assess the ship's position accurately within 50x OR end with a harbour entrance and alongside	Failure to pass 3/5 runs (as above) or any observed single "Automatic Safety Failure Criteria" IAW "FNO NAVIGATION ASSESSMENT FORM PC 001A/B & CEC 001.05" will result in TRB IAW NPTORD4500-54.
PC 002 **Scope :** **PO 001** **PO 002** **PO 003** **PO 004** **PO 005**	**To confirm the student's ability to** **Navigate a Formation of Ships in an Operational Environment**	Practical NABS Trainer Simulated Scenario with at least 2 other simulated vessels in formation	**See Annex K Part 2 and Annex G Part 2** Assignment is given to student NLT 24hrs prior to execution. FNI input, appraisal, planning, oversee formation during execution & monitoring Not expected to navigate the individual ship.	Any simulated vessel, Chart and Publication Outfit, Navigation Tools, Supervision of a Commanding Officer. WECDIS laptop, DWAN and Internet access. Full Bridge Team, access to FNIs acting as all outside agencies.	**Total Time 450min** 300min prep time 30min brief 90 min execution 30 min debrief	Task group staff and other external agencies will be emulated by FNIs. SOA Minimum: 10kts Maximum: 12kts Max speed as per class of vessel being navigated	Formative feedback Re-attempt 2nd unsatisfactory attempt results in a TRB IAW NPTORD4500-54.
EC 002.01 Scope:	To check the student's ability to Navigate a Formation of Ships in a	Practical NABS Trainer	**See Annex K Part 2 and Annex G Part 2** Conducted during the PC 002 Development Week	Any simulated vessel, Chart and Publication Outfit, Navigation Tools,	**Total Time 450min** 300min prep time	Task group staff and other external agencies will be emulated by FNIs.	Formative Feedback

ASSESSMENT OVERVIEW							
Scope	**Purpose of Check**	**Method Used**	**When and How Conducted**	**Resources**	**Time**	**Limitations**	**Impact & Remedial Instruction**
EO 001.01-03 EO 002.01-03	Ceremonial Environment	Simulated Scenario with at least 1 other simulated vessel in formation	FNI input Scenario: Candidate controls a formation of ships into Halifax harbour for a ceremonial anchorage.	Supervision of a Commanding Officer. WECDIS laptop, DWAN and Internet access. Full Bridge Team, access to FNIs acting as all outside agencies.	30min brief 90 min execution 30 min debrief	SOA Minimum: 10kts Maximum: 12kts Max speed as per class of vessel being navigated	
PC 003A **Scope:** **PO 003**	To confirm the student's ability to Respond to Various Administrative and Policy Questions and Problems	Written Exam Policy guided questions	**See Annex K Part 2 and Annex G Part 2** Scenario: as a navigator alongside, student is presented with various administrative and policy questions and problems, quick response to matters they should already be familiar with, access to references for verification/clarification for their answers only during the last hour.	Classroom, WECDIS laptop, exam paper, pen, pencil, eraser, references available for the last hour.	120 min exam 90 min no refs/30 min access to references	No subordinates, division or full bridge team is available.	Review, Re-write (supplemental PC) 2 consecutive failures result in a TRB IAW NPTORD4500-54
PC 003B **Scope :** **PO 003**	To confirm the student's ability to Lead the Bridge Team	Practical Experiential	**Run concurrently with CEC 001.04 (Duty Navigator)** **See Annex K Part 2 and Annex G Part 2** Evaluated as part of CEC 001.04 Duty Navigator Checklist	Full bridge team	Run concurrently with CEC Duty Navigator	Same as CEC 001.04	Formative feedback Re-attempt 2nd unsatisfactory attempt results in a TRB IAW NPTORD4500-54. *opportunities for re-assessment may be given during subsequent training sea-phases at the discretion of the TRB (available to both MWVNO and FNO students).
EC 003.01 Scope : EO 003.01-03	To check the student's ability to Respond to a Complex Navigation Policy or Procedural Issue	Homework Assignment Presentation to peers	**See Annex K Part 2 and Annex G Part 2** Delivery of content by students during EO 003.01 Students assigned scenario Present product for peer review, back brief FNI feedback, debrief	FNI input (scenario), Classroom, WECDIS laptop, DWAN and internet access. Charts and Publication Outfit, Navigation Tools	1 week to do research 15min/student presentation 15min/student feedback	Students must work on their own time to complete this assignment.	Peer and Instructor Formative Feedback Rewrite only if the information is not IAW current Nav Policy.
EC 003.02 Scope :	To check the student's ability to	Independent Work with Presentation	**See Annex K Part 2 and Annex G Part 2**	Classroom, WECDIS laptop, DWAN and Internet access.	1 week to do research 30min/student presentation	Students must work on their own time to complete this assignment	Formative Feedback

ASSESSMENT OVERVIEW							
Scope	**Purpose of Check**	**Method Used**	**When and How Conducted**	**Resources**	**Time**	**Limitations**	**Impact & Remedial Instruction**
EO 003.01-03 EO 004.01	Lead Mariner Professional Development		Students assigned BRM case study to be delivered to personnel outside the FNO course Present product for peer review, back brief FNI feedback, debrief	A group of officers to present to below the FNO level.	10min/student discussion/feedback/questions	Ideally presented to junior officers (NWO II or IV students)	
PC 004 Scope PO 004	To confirm the student's ability to Provide Advice to Command on Navigation Matters, Including Seamanship and Shiphandling Evolutions	Practical NABS trainer	**See Annex K Part 2 and Annex G Part 2** Scenario: Navigator in NABS (or at sea). The ship will perform a seamanship evolution, and the Navigator will be required to brief command on the conduct of the evolution and advise command in real-time during the conduct of the evolution.	NABS (or patrol vessel at sea), WECDIS laptop, pen, pencil, eraser, references (TBD)	120min planning 30min brief 45min evolution	Only evolutions capable of being completed by the vessel in use will be assigned. **Remarks**: Separate from Duty Nav evaluation and PC 001 evaluation (pilotage run)	Summative Feedback Re-attempt 2nd unsuccessful attempt results in TRB IAW NPTORD 4500-54
CEC 004.01 EO 004.02 → specifically LO 004.02.02 Critical Due to Safety	To check the student's Knowledge of Shiphandling Theory	Written Exam Short/Long Answer MC Questions Diagrams to draw	**See Annex K Part 2 and Annex G Part 2** Various questions on Shiphandling theory	Exam Paper Writing Instrument	120 min	No references Closed book exam No support to NavO	80% pass mark Review, Rewrite (supplemental CEC) 2nd unsuccessful attempt results in a TRB IAW NPTORD 4500-54
CEC 004.02 EO 001.03 → specifically LO 001.03.06B EO 004.02 Critical Due to Safety	To check the student's Theoretical Knowledge of Navigation Sciences	Written Exam Short/Long Answer MC Questions Diagrams to draw	**See Annex K Part 2 and Annex G Part 2** Upon completion of all EOs, which include the following: Meteorology, Tides, Geodesy, Astro, Celestial	Exam Paper Writing Instrument	120 min exam	No references Closed book exam No support to NavO	80% pass mark Review, Rewrite (supplemental CEC) 2nd unsuccessful attempt results in a TRB IAW NPTORD 4500-54
PC 005A **MWVNO** Scope : PO 005	To confirm the student's Theoretical Knowledge of Basic Navigation Systems	Written Exam Short/Long Answer Diagrams to draw	**See Annex K Part 2 and Annex G Part 2** Assessed during MWVNO phase after completion of EO 005.01 & EO 005.02 (Basic Navigation Systems - Only)	Exam Paper Writing Instrument References (only last 30min of exam)	120 min (60min no refs, 60min access to refs)	Basic navigation systems only (MWVNO). No references for first 60min	80% pass mark Review, Rewrite (supplemental PC) 2nd unsuccessful attempt results in TRB IAW NPTORD 4500-54

ASSESSMENT OVERVIEW							
Scope	**Purpose of Check**	**Method Used**	**When and How Conducted**	**Resources**	**Time**	**Limitations**	**Impact & Remedial Instruction**
PC 005B **FNO** Scope : PO 005	To confirm the student's Theoretical Knowledge of Complex Navigation Systems	Written Exam Short/Long Answer Diagrams to draw	**See Annex K Part 2 and Annex G Part 2** Assessed during FNO phase after completion of EO 005.01 & EO 005.02 (Complex Navigation Systems) as prep for CEC 001.04	Exam Paper Writing Instrument References (only last 30min of exam)	90 min (60min no refs, 30min access to refs)	Complex navigation systems only (FNO only). No references for the first 60min	80% pass mark Review, Rewrite (supplemental PC) 2nd unsuccessful attempt results in TRB IAW NPTORD 4500-54
EC 005.01 EO 004.01 EO 005.03	To check the student's Theoretical Knowledge of a Navigation System	Presentations	**See Annex K Part 2 and Annex G Part 2** As content delivery in conjunction with EO 005.01 & EO 005.02 as prep for CEC 001.04 Assigned as homework to research Nav System and present the following week	Refs, guidance from FNIs regarding which info to include/cover, laptop with internet access, smartboard projectors and other required presentation materials	1 week to prepare 20-30min/student presentation (180 min total for 6 students)	Students must work on their own time to complete this assignment	Formative Feedback Missing critical information filled in by student (further research)
PC 006A **MWVNO** Scope : PO 006 material taught during MWVNO	To confirm the student's Management of Navigation Aspects of the Maritime Battlespace	Written PC with WECDIS component	**See Annex K Part 2 and Annex G Part 2** Assessed during MWVNO phase. Various questions with scenario of NavO on deployment, simulated provision of support to operations.	Classroom, WECDIS laptop, exam paper, writing instrument, paper, references, support FNI mimicking ORO.	120 min exam	No SECRET material on written PC.	80% pass mark Review, Rewrite (supplemental PC) 2nd unsuccessful attempt results in TRB IAW NPTORD 4500-54
PC 006B **FNO** Scope: PO 006	To check the student's Theoretical Knowledge of Tactics and Doctrine	Written Exam MC and Short Answer Diagrams to draw	**See Annex K Part 2 and Annex G Part 2** Assessed during FNO phase	VCR required, SDA classroom, Writing instrument, Exam, Paper	90 min	No references for the first 60mins, reference for the final 30mins. SDA classroom required	80% pass mark Detailed debrief and review, rewrite (supplemental PC)
EC 006.01 Scope : EO 006.01-02	To check the student's ability to Act as the OOW during Action Stations	Practical on bridge of Halifax Class vessel at sea or alongside (preferred during FFH tours)	**See Annex K Part 2 and Annex G Part 2** Completed during FNO phase	Halifax class ship, bridge team (or simulated), Halifax Class Ops Room with CMS in SIM, playbook.	Preferred is 60 min/student includes feedback and SIM reset during Fleet IPT for full Action Stations simulation and effect with multiple players, noise and comms nets. Minimum is 60min/class with simulated out stations and a walk-through of NavO responsibilities when the ship is brought to Action Stations, including interaction with comm nets and CMS.	References will not be available due to the urgent responses required in a tactical situation.	Detailed debrief/formative feedback and review upon completion.

ASSESSMENT OVERVIEW							
Scope	**Purpose of Check**	**Method Used**	**When and How Conducted**	**Resources**	**Time**	**Limitations**	**Impact & Remedial Instruction**
EC 006.02 Scope : EO 006.02	To check the student's ability to Complete a Minor Hydrographic Survey	Group EC Produce a survey	**See Annex K Part 2 and Annex G Part 2** Groups of 2 or more Completed during MWVNO Phase	GPS datum/mark, Sextant, blank paper, stopwatch, toilet paper roll, pencil, ruler, lead line/hand-held depth sounder, WECDIS	480min total time 240min survey 240min create product	Only provided initial datum	Formative feedback on product Correct errors in the survey (if time allows)

Part 2 - Activity and Assessment Plan (AAP)

PO 001 – Navigate a Vessel of Any Class

Details the performance standard is as follows:

PO 001:
The navigator shall navigate a vessel (AT0001 & FT0001), IAW refs CFCD 130 & BR45 by:

a. Performing a navigation appraisal;
b. Conducting navigation planning; and
c. Executing the navigation by:
    i. Navigating a minor war vessel (MWV) or submarine at speeds up to 15kts (MWVNO);
    ii. Navigating any vessel at any speed (FNO).

Notes for planning next delivery of PO 001

PC 001A – Conduct a Basic Pilotage Run (MWVNO)

Overview	Objective(s)	Assessment Details	Lessons Identified
Media: WECDIS Laptop Ship at sea **Method**: Practical **Time**: 870min 330mins total for planning, execution and feedback. Each pilotage run will take 45-60mins to execute. X 6 students = 540min **Environment**: At Sea	• The goal of this PC is to assess a student's ability to navigate at a MWVNO level.	**<u>Before the assessment:</u>** 1. FNIs will give the student the intended start and endpoints for an assigned navigation run as well as a basic mission scenario and list of constraints. For each run, the student will have to factor in at least one evolution on top of the required pilotage (i.e. seamanship, force protection, shiphandling, etc.). Student will follow the appraisal, plan and execute/monitor process IAW BR45. Student will have 4hrs to conduct their appraisal and planning. Student's appraisal and planning will be assessed by the FNI as per the evaluation sheet. **<u>During the assessment:</u>** 1. Student will brief CO as req'd prior to taking over the navigation from another student or from the OOW. Student will execute their planned pilotage run and respond to navigation casualties and any changing meteorological conditions throughout the run. Student will be assessed throughout with a focus on keeping the ship safe. **<u>Assessment feedback:</u>** 1. After the student completes their assigned pilotage run, they will turnover the navigation to the next student or back to the OOW. The FNIs and CO of the ship will complete the assessment form and determine if the student met the standard or not before providing feedback to the student.	-
			To be developed
			- Run sheet - Assessment Checklist
	Impact and Limitations	**Resources**	
	Impact: 7 runs assigned, all of which student must complete. Student must achieve the MWVNO standard on 5 runs - 2 of which must be day and 2 of which must be night. If student does not meet the above criteria a TRB will be convened. **Limitations/Remarks**: - SOA between 10-12kts - Max speed as per the class of vessel the student is navigating - Student to conduct no more than 2 runs per day (1 day, 1 night) - Any meteorological conditions - Minimum equipment during the execution will be a WECDIS laptop, navigator's notebook and a timepiece.	**For student:** - WECDIS laptop - Printer - Navigation notebook - Chart and publication outfit - Guidance from CO (as req'd) **For instructor:** - Run sheet - Assessment Checklist **References:** A1, A2, A5, A12, A15, B1, C15	

PC 001B – Conduct an Advanced Pilotage Run (FNO)

Overview	Objective(s)	Assessment Details		Lessons Identified
Media: WECDIS Laptop Major Warship at sea **Method**: Practical **Time**: 870min 330mins total for planning, execution and feedback. Each pilotage run will take 45-60mins to execute. X 6 students **Environment**: At Sea	1. The goal of this PC is to assess the student's ability to navigate at an FNO level.	**<u>Before the assessment:</u>** 1. FNIs will give student the intended start and endpoints for an assigned navigation run as well as a basic mission scenario and list of constraints. Student will follow the appraisal, plan and execute/monitor process IAW BR45. Student will have 4hrs to conduct their appraisal and planning. Student's appraisal and planning will be assessed by the FNI as per the evaluation sheet. **<u>During the assessment:</u>** 1. Student will brief CO as req'd prior to taking over the navigation from another student or from the OOW. Student will execute their planned pilotage run and respond to navigation casualties and any changing meteorological conditions throughout the run. Student will be assessed throughout with a focus on BRM and keeping the ship safe. **<u>Assessment feedback:</u>** 1. After the student completes their assigned pilotage run, they will turnover the navigation to the next student or back to the OOW. The FNIs and CO of the ship will complete the assessment form and make a satisfactory/unsatisfactory determination before providing feedback to the student.		-
				To be developed
				- Run sheet
	Impact and Limitations	**Resources**		
	Impact: 4 runs assigned (2 day and 2 night), all of which student must complete. Student must achieve the FNO standard on 3 runs - 1 of which must be day and 1 of which must be night. If student does not achieve the FNO standard on any 2 runs, a TRB will be convened. **Limitations/Remarks**: - SOA between 16-20kts - Max speed as per the class of vessel the student is navigating - Student to conduct no more than 2 runs per day (1 day, 1 night) - Any meteorological conditions - Minimum equipment during the execution will be a WECDIS laptop, Navigator's notebook and a timepiece.	**For student:** - WECDIS laptop - Printer - Navigation notebook - Chart and publication outfit - Guidance from CO (as req'd)	**For instructor:** - Run sheet - Evaluation sheet **References:** A1, A2, A5, A12, A15, B1, C15	

EO 001.01 – Perform a Navigation Appraisal

LO 001.01.01A – Navigation Mission Analysis

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Instructor Led, Student following **Media**: Computer with projector **Method**: EDIP **Time**: 2 x 45 min (90min) **Environment**: Classroom	1. Overall Voyage Planning Checklist (4 step process) (20min) 2. 4 step process – Appraisal/Planning/Execution and Monitoring (20min) 3. Part 1 of the Voyage Planning Checklist - Navigation Appraisal (5min) 4. The navigation mission analysis process (first section of the Navigation Appraisal) with considerations (45min) <u>Teaching points:</u> Liaise with outside agencies Consider threat areas Port visits Determine jurisdiction/territorial waters (further lesson with AJAG)	<u>Activity #1 – Sample Navigation Mission Analysis</u> Example OP order or Run Sheet – 1 example that follows through the course Sample navigation mission analysis (coastal, pilotage), FNIs takes students through the process going over how to: a. Determine commander's intent b. Consider the operational situation c. Determine what information is available Students are following along on their own laptops, looking up information and references/resources as FNI instructs	-
	Why	**Resources**	**To be developed**
	- Part of the Voyage Planning Process (part of the appraisal process) - Reduces time and errors in the planning process - Will inform final ECs - And PC	**For student:** - Checklist (To Be Created) - WECDIS laptop - DWAN laptop with internet access - Draft Example Run Sheet - Chart and Publication Outfit **For instructor:** Checklist (To Be Created) Computer w/ Projector (or smartboard) Draft Example Run Sheet Chart and Publication Outfit **References:** A1, A6, A7, A10, A14, B1, B4, B6, B19, C3, C4, C5, C9, C10, C14, C15, C21	- Part 1 Voyage Planning Checklist - Part 1 - Navigation Appraisal - Master Lesson Plan - Example Scenario (that will flow through the course – includes Op order, run sheet, different locations etc.)

LO 001.01.01B – Geodetic Systems & Charts

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Guided Discussion/Lectures Field Trip **Media**: Computer w/ projector Example Charts Lectures **Method**: IL **Time**: 8 X 45min (360min) **Environment**: Classroom and CHS office	• Geodetic Systems (135min – 3x45min periods) • Chart Activity (45mins) • Chart Creation (180min – 4 x 45min periods) Teaching Points: ***Geodetic Systems*** a. Terrestrial coordinates b. The shape of the Earth c. Common reference points d. Chart Datums e. Nautical measurements ***Chart Creation*** a. Survey Methods b. Reliability c. Projections d. Chart organizations e. Chart types	<u>Activity 1: Guided Discussion and Lectures</u> (3 x 45min) Part 1: Geodesy and Geodetic Systems (45mins) Part 2: Terrestrial Coordinate Systems and Chart Projections (45mins) Part 3: Charts and Chart Reliability (45mins) <u>Activity 2: Compare and Contrasts Charts Activity</u> (45min) - Students look at example charts and find errors (compare and contrast the good with the bad). - Examples of good and bad charts are held within the FNO classroom. Instructors to guide students through this activity and allow students to take the lead on discussion identifying good and bad aspects of the example charts. - Activity may not take the full 45mins. Ideally, this is scheduled before a Sports Period or Admin Period to allow students more time for those activities if Activity 2 does not require the full 45min. <u>Activity 3: Navigation Warnings and Chart Corrections</u> (45min) <u>Activity 4: Field trip to CHS</u> (forenoon – 4 x 45min) - See chart creation, survey work, and how survey information is converted to useable charts. - Review current technologies in use for conducting surveys and discuss the process of surveying an area. - Review different types of ENCs in use and under development	-
	Why	**Resources**	**To be developed**
	- Understanding reliability and safety for chart creation and how that impacts navigation - Understanding datums is critical to safety in navigation - Will inform Marine Sciences EC and PC001	**For student:** Chart and Publication Outfit **For instructor:** Examples of bad charts Examples of good charts Computer w/ projector Chart and Publication Outfit **References:** A1, A15, B1, B3, B6, C1, C3	- Ppt already created - Charts already exist (collected bad examples)

LO 001.01.01C – Ordering Charts & Publications

Overview	Objective(s)	Activity Details		Lessons Identified
Strategy: Lecture EDIP Leadthrough **Media**: Computer w/ projector CD outfit ADC software **Method**: IL **Time**: 2x45min + 2x30min (150min) **Environment**: Classroom Dockyard Field Trip	1. Describe the HMCS Chart & Publication Outfit 2. How to Order charts and publications 3. HSO Experience	<u>Activity 1: Charts and Publications</u> (30min) - MDFs - Core Outfit - SCA/SLoC - NavO Binder/Chart discs/Pub discs - Leads into Activity 2 <u>Activity 2: Explain Demonstrate Imitate Practice (EDIP) Activity for Ordering Products (45min)</u> - EDIP for creating a basket file leading into Activity 3 <u>Activity 3: HSO Lead through</u> (30min) Practical lecture using the HSO website (no PPT lecture) Topics to be covered include: - HSO Handbook - Available products - Best practices as to where to find items on the HSO website - Contact info - Students to have a brief opportunity to explore the HSO Website to familiarize and if they have any questions, they can ask the instructor <u>Activity 4: HSO Tour</u> (1 x 45min) Can be combined with other dockyard tours or conducted as a standalone activity - Observe the back end of chart and publication printing and ordering - Interact with HSO staff		-
	Why	**Resources**		**To be developed**
	- Understanding reliability and safety for chart creation and how that impacts navigation - Understanding datums is critical to safety in navigation - Will inform Marine Sciences EC and PC001	**For student:** Laptop with DWAN access/per student ADC software-enabled laptop/per student	**For instructor:** Computer w/ projector Chart and Publication Outfit ADC software-enabled laptop DWAN laptop PPT Guest Lecturer: list of points to cover **References:** A1, A5, A6, A7, A15, A20, B19, B20, B21, C3, C4, C9	- Guest lecturer points to cover - Ppt already exists

LO 001.01.02 – Conduct Navigation Appraisal

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Guided Activity **Media**: Laptop Projector **Method**: Lead through **Time**: 5x45min (225min) **Environment**: Classroom	SWBAT • Find nav related resources to assist with planning • Complete a navigation appraisal by incorporating nav related research	<u>Activity #1 – Identifying Resources Instructor-Led (1x45min)</u> Examples of resources to include: a. HSO/HSO after-hours availability b. Admiralty website/dealers c. CHS website and pubs d. USN pubs (fleet guides) e. Forward Logistic Support Teams f. RJOC/MCC g. MARPIFS h. FNIs i. Pilotage Authorities j. Harbour Authorities <u>Activity #2 – Identifying Resources Student Led (1x45min)</u> Students to spend time conducting independent research to add to the list of resources introduced by the instructor. <u>Activity #3 – Lead through of Navigation Appraisal</u> (3x45min) Example OP Order or Run Sheet – 1 example that follows through the course Sample navigation research to advise the appraisal process (coastal, pilotage) take them through the process: a. Consult Nav related publications and other references; b. Create the initial routing with consideration of traffic patterns; and c. Calculate Fuel consumption and SOA	-
			To be developed
			- Run sheet - DS Solution
	Why	**Resources**	
	- Informs EC - Informs PC - step in the process (part 1)	**For student:** WECDIS laptop DWAN laptop with internet access Chart and publication outfit Run sheet Checklist **For instructor:** Run sheet DS solution **References:** A1, A6, A9, A11, A13, A18, A21, B1, B5, B7, B19, B20, B21, C1, C2, C3, C6, C8, C9, C16, C17	

LO 001.01.03 – Complete a Command Appraisal (informal)

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Interactive **Media**: Laptop Projector **Method**: Demonstration/Walkthrough **Time**: 4 x 45min (180mins) **Environment**: Classroom	SWBAT 1. Present an informal brief to command regarding the navigation appraisal 2. *Still informing* final step in part 1 of the Voyage Planning Checklist – Navigation Appraisal (5min) Teaching points: Routing information Fuel calculations Traffic patterns SOA Command brief (informal)	<u>Activity#1 – Lead Through Exercise (45mins)</u> Example OP order or Run Sheet – 1 example that follows through the course Sample navigation research to advise the appraisal process (coastal, pilotage) take them through the process: <u>Activity 2 - Fishbowl</u> (15min) 2 FNI do the informal brief perfect scenario – while other students observe Back brief, debrief, discussion points <u>Activity 3: Brief preparation</u> (30min) Prepare brief (informal) – notes/charts/laptops Have all items to brief command Checklist as reference Work in pairs with FNI facilitating Answer questions and note changes from command Request approval to continue with the planning process Back brief/debrief Open ocean, coastal, pilotage comparison <u>Activity 4: Walkthrough of Open Ocean Command Appraisal</u> (2x45mins) FNI led walkthrough with each student highlighting the requirements and process for an open ocean command appraisal	-
			To be developed
			- Run sheet - FNI perfect demo

Why	Resources	
- Informs EC - Informs PC - step in the process (part 1)	**For student:** WECDIS laptop DWAN laptop with internet access Chart and publication outfit Run sheet Checklist	**For instructor:** Run sheet DS solution **References:** A1, A6, A9, A11, A13, A18, A21, B1, B5, B7, B19, B20, B21, C1, C2, C3, C6, C8, C9, C16, C17

EO 001.02 – Conduct Navigation Planning

LO 001.02.01 – Theory and Application of Navigation Planning

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Interactive **Media**: Projector WECDIS laptop **Method**: EDI **Time**: 9x45 min (390min) **Environment**: Classroom	SWBAT 1. Create a Nav Plan 2. Step 2 of the Process, 2nd part of the Voyage Planning Checklist 3. Incorporate Appraisal Research into the Nav Plan 4. Incorporate COs recommendations	Example OP order or Run Sheet – 1 example that follows through the course Sample navigation research to advise the appraisal process (coastal, pilotage) take them through the process: <u>Activity 1: Basics of the Nav Plan</u> (1x45min) Compare and contrast available planning checklists Review BR 45 checklist <u>Activity 2: Theory of Nav Planning (4x45min)</u> Powerpoint lead through going through each step of a Nav Plan. Introduce a variety of notebook types (hard copy), and decide individually which works best for them. Discussion on benefits and drawbacks of each type <u>Activity 3: Supervised Pilotage Planning</u> (3x45min) FNIs will supervise and mentor students through their first development pilotage passage plan. Ideally, first passages are assigned Friday at noon for execution Monday and Friday afternoon will be in class supervised planning with the FNIs present. Last hour of period FNIs to individually discuss planning process with students and identify command concerns <u>Activity 4: Paper Navigation</u> Part 1 – FNI lead demonstration (1x45min) Part 2 – NABs developmental runs	-
	Why	**Resources**	**To be developed**
	- Informs EC 001.01 - Informs PC 001A/B - Important step in the Voyage Planning process	**For student:** WECDIS Laptop DWAN laptop with internet access Chart and publication outfit **For instructor:** Projector WECDIS laptop **References:** A1, A9, A11, A14, A16, A18, B1, B4, B5, B6, B9, B19, B20, B21, C2, C3, C8, C16	- The checklist already exists; however, it needs to be updated to reflect IMO 4 step process

LO 001.02.02 – Include Operational Factors

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Direct **Media**: Laptop Projector **Method**: IL **Time**: 2x45 min 1x10min 100min **Environment**: Classroom	SWBAT 1. Incorporate Operational Factors into the Planning Checklist 2. Describe SAT considerations 3. Interpret PIM messages Satellite Communication Mgmt. - Pre-plan alternate methods to keep the ship safe in a SAT denied or EMCON Env Risk Management - Force protection - Traffic & Timing Waterspace Management	<u>Activity 1: PIM Management</u> (1x45min) WGS PPT Lecture PIM Waterspace Management <u>Activity 2: Using WECDIS</u> (1x45min) Pull-up navigation data for input into SATCOM plans Bandwidth, spot beams, locating and connections <u>Activity 3: Risk Mgmt. – Jig Saw</u> (10min) Jig Saw with different risk management models Compare to the checklist that has inherent risk mgmt. (back brief/debrief)	-
	Why	**Resources**	**To be developed**
	- Informs EC 001.02 - Informs PC 001A/B - step in the process (part 1) - informs EO 002.02	**For student:** WECDIS laptop DWAN laptop with internet access Paper Writing utensil **For instructor:** WECDIS laptop Projector **References:** A1, A6, B1, B4, B6	- The checklist for planning needs to be updated - Risk mgmt. models (2 or 3) - DS solution for comparison - PPT slide deck exists

LO 001.02.03 – Internal Preparations

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="3">Strategy:
Direct
Indirect

Media:
PPT
Notebooks

Method:
IL

Time:
90mins

Environment:
Classroom</td><td>SWBAT
1. Conduct internal preparations
2. Create a Navigator’s Notebook
3. Liaise with OpsO regarding ship’s plan
4. Reliability and condition of systems and equipment

Navigator’s Notebook
Timing considerations (breakfast, logistical planning for families)</td><td colspan="2"><u>Activity 1: Scenario – Flex Injects/SOM</u> (2x45mins)
Detailed breakdown of 3-4 examples of scenarios where the NavO would need to make flex injects. Guided discussion on walking back timings and how to complete flex injects accounting for things such as: Special Parties, Boat Launch, etc., in a variety of Nav regimes (i.e. Coastal, Open Ocean, Pilotage).</td><td>-</td></tr>
<tr><td></td><td></td><td></td><td>To be developed</td></tr>
<tr><td>Why</td><td colspan="2">Resources</td><td rowspan="2">- PPT already exists for notebooks
- Notebooks already exist
- Scenario slides – need to be created for A2
- DS solution for A3</td></tr>
<tr><td>- Informs EC 001.02
- Informs PC 001A/B
- step in the process (part 2)
- informs EO 005.01/02</td><td>For student:
WECDIS laptop
DWAN laptop with internet access
Whiteboard
Dry Erase Markers</td><td>For instructor:
WECDIS laptop
Projector
DS solutions
PPT slide decks

References:
A1, A6, B1, B6</td></tr>
</table>

LO 001.02.04 – The Navigation Brief

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: interactive **Media**: PPT Nav Briefs **Method**: IL **Time**: 45 min **Environment**: Classroom	SWBAT 1. Prepare a Navigation Brief 2. Components of the Navigation Brief 3. Identify good and bad briefs	<u>Activity 1: Brainstorm a list of items that should be in a Nav Brief</u> (1x45mins) Brainstorm requirements at the beginning to get students thinking PPT brief with minimum requirements of a Navigation Brief and associated policy Provide examples of PPT briefs Tips and tricks for preparing PPT briefs Guided discussion about the good the bad <u>Activity 2: Quarters Brief</u> (1x45min)	-
			To be developed
			- PPT already exists briefs - DS solution

Why	Resources	
- Informs EC 001.03 - Informs PC 001A/B - step in the process (part 2)	**For student:** WECDIS laptop DWAN laptop with internet access Whiteboard Markers	**For instructor:** WECDIS laptop Projector DS solutions PPT slide decks **References:** A1, A6, B1

EO 001.03 – Execute the Navigation

LO 001.03.01 – Role of the Navigating Officer

Overview	Objective(s)	Activity Details	Lessons Identified

Strategy:
Direct
 	
Media:
Laptop
Projector
 	
Method:
Guided Discussion
 	
Time:
3x45 min
(135min)
Homework/Ongoing
 	
Environment:
Classroom
	• SWBAT
	  • Identify the role of the NavO
	  • Identify areas of responsibility of the NavO
	  • Understand the role of the RCN NavO in relation to other Navies and Civilian NavOs
	  • NavO and Command, Charge and Control
		<u>Activity 1 – Guided Discussion and Lectures</u> (2x45min)
		Part 1 – Responsibilities of the RCN NavO (45min)
		• FNI led lecture to set the stage for the course covering what the "Role of the RCN Navigating Officer" is.
		
		Part 2 – Relationship of the NavO with external agencies (45min)
		• The history of Navigation Governance
		• Contemporary Navigation issues/discussions
		• Interaction with NFR, Nav Working Group, Naval Fleet School and Naval Training and Development Centre (Pacific)
		
		<u>Activity 2 – Command, Charge and Control Lecture (2x45min)</u>
		Guided discussion on Command, Charge and Control and how the NavO falls into this construct (specifically control) (Post Command guest lecturer preferred)
		
		<u>Activity 3- Ongoing record of lessons learned</u>
		Students to create and maintain their own list of items to discuss on day 1 of the job with their CO/good ideas learned throughout the course (homework/ongoing)
		
		<u>Activity 4 – Ongoing PD reading assignment</u>
		Students to be assigned to read through SSO, CFCD 130, BR45 Roles of NavO
			-
			
			To be developed
			
			-PPT on "Role of the Navigating Officer"

	Why	**Resources**	
	-Students need to know who/what the NavO is and why their role exists	**For student:** Pen/Pencil, Paper (notetaking) **For instructor:** PPT **References:** A1, A2, A5, A6, A7, B1, B2, B3, B4, B5, B6, B7, B8	

LO 001.03.02 – Collision Avoidance

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Direct Interactive **Media**: NABS Laptop Projector **Method**: Practical/Lecture **Time**: 6x45min (270min) **Environment**: NABS Classroom	SWBAT Execute Collision Avoidance	<u>Activity 1: Simulator COLREGs fundamentals</u> (4x45min) - Simulated runs involving various COLREGs scenarios with increasing difficulty and complexity - Scenarios are pre-developed and contained within the instructor's "practical playbook." - Students to act as OOW and come up with solutions to COLREGs situation. After the scenario is finished, the run will be re-set, and a different solution can be trialled. Follow on students are not to use the same solution as the previous student in an attempt to force the exploration of various actions/solutions to the COLREGs scenario that is taken. - Tabletop/debriefs to follow each run. - Ideally, these are split up over two days (2x45min/day) so as to maintain student interest and readiness to learn <u>Activity 2: Advanced COLGREGs Theory and Application - Guest Lecture</u> (2x45) Advanced theoretical discussion on COLREGs and application from a Command Qualified Officer Preferably guest lecturer has at sea command experience	-
	Why	**Resources**	**To be developed**
	- Informs CEC 001.03 – COLREGs - Worksheets daily - Informs PC 001A/B - Step in the process (step 3) - Use these rules in every run	**For student:** Pen/Pencil Paper (notetaking) **For instructor:** WECDIS laptop Projector DS solutions PPT slide decks NABS Worksheets **References:** A1, B1, C10, C13	- PPT already exists briefs - DS solution - worksheets

LO 001.03.03 – Interactions with key personnel during pilotage

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Direct Interactive **Media**: PPT NABS **Method**: IL **Time**: 4x45min (135min) Ongoing during development runs 6x45 (based on a preferred course load of 6 students) (270min) **Environment**: Classroom NABS	SWBAT 1. Interact with key personnel during pilotage, including: a. Internal Pers: i. Bridge Team ii. EOOW iii. MSEO iv. TOW v. CSEO vi. LogO vii. XO viii. CO ix. Ops x. ORO xi. FPO xii. Image Tech b. External i. MCTS ii. Pilot iii. QHM/Harbour Authorities	<u>Activity 1: Lecture (Internal Interactions (1x45mins)</u> Part 1: Have students brainstorm a list of internal pers who the NavO would interact with during pilotage Part 2: Divide the list evenly amongst all the students and assign them (15mins) to develop a quick overview of how each interaction would look Part 3: DS solution as required to fill in any gaps <u>Activity 2 – Lecture (External Interactions) (1x60mins)</u> Students to be given a homework assignment of creating their own template/checklist for interacting with a pilot Part 1: Have students brainstorm a list of external pers who the NavO would interact with during pilotage Part 2: Guided discussion giving a brief overview of how each interaction would look, focusing on the pilot interaction Part 3: Take up/review students' homework assignment (template/checklist for pilot interaction) Part 4: FNIs to demonstrate (very basic) pilot interaction where one is acting as NavO and the other as Pilot. Have students follow through with the demonstration and check off aspects of their template/checklist and discuss anything that was missed. <u>Activity 3: Developmental Run with Pilot Interaction (45min/student)</u> As part of one of their development runs (NABS) – Enter/Exit Scenario – incorporate pilot into the plan, interact with a pilot (using NSSC) – concurrent with foreign port developmental	-
	Why	**Resources**	**To be developed**
	- Informs CEC 001.05 (pilotage) - Worksheets daily - Informs PC 001A/B - Informs PO - Step in the process (step 3) - Use these rules in every run	**For student:** WECDIS laptop DWAN laptop with internet access Pen/pencil Paper (notetaking) NSSC **For instructor:** WECDIS laptop Projector DS solutions PPT slide decks NABS **References:** A1, A2, A5, A6, B1, B4, B5, B7, B8, C1, C3, C13	

LO 001.03.04 – Navigate in a Satellite Denied and/or EMCON Environment

Overview	Objective(s)	Activity Details		Lessons Identified
Strategy: Direct Interactive **Media**: PPT WECDIS laptop **Method**: IL **Time**: 2x45min (90min) **Environment**: Classroom	SWBAT 1. Navigate in a Satellite Denied and/or EMCON Environment	<u>Activity 1: Navigating in a GNSS Denied/EMCON Environment Lecture</u> (2x45min) PPT available covering topics such as: - What a Satellite denied environment is/means to the NavO. - What an EMCON environment is/means to the NavO. - What do you do with no access to GPS? - What is Pool of errors? - How to conduct navigation in a GNSS denied/EMCON environment. - Deceptive attacks, jamming and spoofing Lecture includes a brief Discovery Activity (5min to search, 15min to discuss) where students will search for recent occurrences of EW attacks and GNSS issues. - This lecture serves as a transition from Tides and Met to Cell Nav and informs why Cell Nav is important today		-
	Why	**Resources**		**To be developed**
	- On the job performance requires that you are able to effectively navigate in SAT denied/EMCON env - Informs PC 001A/B	**For student:** WECDIS laptop Pen/pencil Paper (notetaking)	**For instructor:** WECDIS laptop Projector DS solutions PPT slide decks NABS **References:** A1, A6, B1, B2, B3, B6, B7, C1	- PPT needs to be created (borrow/adapt from Dived Nav course CANSUBFOR) - Playbook for the NABS

LO 001.03.05 – Monitor the Navigation Passage

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Strategy:
Interactive

Media:
Laptop
Projector

Method:
IL

Time:
1 x 45min

Ongoing during Duty Navigator CEC

Environment:
Classroom</td><td>SWBAT
1. Monitor the navigation passage during execution
2. Build a routine that works</td><td colspan="2"><u>Activity 1: Navigator's Battle Rythmn</u> (60min)
Guided Discussion building on student experience and insert FNI experience covering day to day routine outside of pilotage

Monitoring and adapting to:
- SAR and PIM management
- Moving haven (water space) management – still in the box?
- Communications
- Weather updates
- Time Zone changes
- Speed required – how fast you need to go to arrive on time, accounting for delays
- Fuel consumption (fuel report)
- Flow of information</td><td></td></tr>
<tr><td></td><td></td><td></td><td>To be developed</td></tr>
<tr><td></td><td></td><td></td><td rowspan="2">- Playbook for the NABS
- DS solution (points to cover)</td></tr>
<tr><td></td><td>Why</td><td colspan="2">Resources</td></tr>
<tr><td></td><td>- This is the navigator's routine
- Informs all navigation runs
- Informing CEC 001.04 Considerations as Duty Nav
- Informs PC 001A/B</td><td>For student:
DWAN laptop with internet access
Pen/pencil
Paper (notetaking)</td><td>For instructor:
DS solutions

References:
A1, A2, A6, B1, B4, B6, B7</td><td></td></tr>
</table>

LO 001.03.06A – Determine the Ship's Position (Relative Nav)

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Direct Experiential **Media**: Laptop Projector NABS **Method**: IL Practical **Time**: 19 x 45min (855min) **Environment**: Classroom NABS	SWBAT 1. Determine Ship's Position 2. Understand position lines 3. Understand and Take a fix 4. Understand and apply DR and EP 5. Use safety methods to clear dangerous situations. 6. Use paper charts	<u>Activity 1: Lecture</u> (1 x 45min) Theory of fixing and position lines <u>Activity 2: Guided discussion</u> (2 x 45 min) Fixing methods & advanced fixing methods Part 1 – Basic Part 2 – Advanced <u>Activity 3: DR/EP Lecture</u> (1 x 45 min) DR and EP <u>Activity 4: NABS practical</u> (3 x 45 min) Consolidation activity in the simulator covering fixing - including DR and EP <u>Activity 5: Lecture</u> (4 x 45 min) Methods to keep the ship safe - including clearing bearings, safety ranges, RADAR methods including PI/SI, VRMs, Offset Dead Ranges, etc Part 1 – LDL, CBs, Part 2 – Clearing Depths, CBs, EAP, Part 3 – Radar Methods : PI/SI, VRM <u>Activity 6: NABS practical</u> (3 x 45 min) Consolidation activity on methods to keep the ship safe (methods as covered in A5) Part 1 – Visual Part 2 – Restricted Vis <u>Activity 7: Lecture</u> (2 x 45 min) Part 1 – Basic MM & Shiphandling (to be completed before shiphandling practical NABs) Part 2 – Advanced MM Advanced Mental Math Concepts and employment during pilotage <u>Activity 8: NABS practical</u> (4 x 45 min) Heads up nav exercise in Barkley Sound using visual, soundings, POE <u>Activity 9: Paper Navigation and Chart Work</u> (45min) Lecture on fundamentals of paper chart work, planning, and minimum criteria for executing a paper navigation passage.	**To be developed** - PPTs already exist (can use some modification – DR & EP update) - Simulator Playbooks (update)
	Why	**Resources**	

Overview	Objective(s)	Activity Details		Lessons Identified
	- Informs all navigation runs - Informs CEC 001.05 - Informs PC 001A/B	**For student:** WECDIS laptop DWAN laptop with internet access Chart and publication Outfit (including paper charts) Navigation tools Navigator's Notebook	**For instructor:** WECDIS laptop **References:** A1, A6, B1, B3, B4, B6, B7, C1, C3	

LO 001.03.06B – Determine the Ship's Position (Celestial Nav)

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Interactive Experiential **Media**: PPTs/Videos Worksheets **Method**: Field Trip IL EDIP **Time**: 14 x 45min (630min) **Environment**: Classroom Planetarium Outdoors at night	SWBAT 1. Fix the ship's position using Celestial Methods 2. Describe the Celestial Sphere 3. Use NAVPAC to solve problem sets 4. Night celestial sextant use	<u>Activity 1: NAVPAC Leadthrough</u> (5 x 45 min) Part 1 – NavPAC install (45mins) - A period with instructors to ensure NavPac PC is installed correctly on all student laptops Part 2 – NavPac Problems Leadthrough (4x45min) Instructor to lead through NavPac problem sets (4x – each one allotted 45mins) and students to follow along using their NavPac PC Software Problem sets to include plotting determined position on WECDIS Part 3 – Intro to Stellarium FNI Roberts Video to be shown (Video File: Intro to Stellarium) (3min) <u>Activity 2: Cel Nav Theory Lectures (2x45min; 2x30min, 1x15min = 165min)</u> Part 1 – Introduction to Cel Nav Lecture (15min) - To cover why we conduct Cel Nav, how, what is a sextant, etc. - To follow LO 001.03.04 Part 2 – The Solar System and Navigational Stars Lecture (45min) - To cover the solar system, motion, inferior and superior planets, conjunction and opposition, naming conventions, etc. Part 3 – Terrestrial Time Lecture (60min) - To cover the importance of time, systems of time, types of time (solar, standard, lunar, etc.) - Time and Coordinates video to be viewed (Video File: Celestial Nav 2) (3min) Part 4 – Celestial Sphere (60min) -To cover basic concepts, celestial poles and equator, declination, meridians, horizon, etc - Celestial Sphere video to be viewed (Video File: Celestial Nav 1A) (3.5mins) - Parts of the Celestial Sphere video to be viewed (Video File: Celestial Nav 1B) (4mins) Part 5 – PZX Triangle (60min) - PPT lecture - PZX Triangle video to be viewed (Video File: Celestial Nav 3A) (3min)	**To be developed** - Update PPT for seamless look and feel with video products - Worksheet/quiz (TB Created) A1 - Worksheets for Activity 2 already created

Overview	Objective(s)	Activity Details		Lessons Identified
		- PZX Triangle video to be viewed (Video File: Celestial Nav 3B) (4min) <u>Activity 3: Cel Nav Practical</u> (4x45min) Part 1 – Use of the Sextant Lecture (45min) Part 2 – Sextant Practical (45min) - Outdoors (Gunroom Lawn) taking horizontal and vertical sextant angles as a class Part 3 - Sight Reduction Leadthrough (2x45min) - Instructor to lead students through (using PPT) how to complete a sight reduction - Plotting Line of Position video to be shown (Video File: Celestial Nav 4) (3min) Part 4 – Bearing Amplitude Leadthrough (45min) - Instructor to lead students through (using PPT) how to calculate the bearing amplitude <u>Activity 4: Field trip to the Planetarium</u> (5 x 45 min) Instructor to establish contact with Planetarium in advance to determine the types of activities and focus of learning during the field trip. Contact number for Dominion Astrophysical Observatory 250-886-5977 - Instructors to ensure students use the sextant under nighttime conditions to take simulated sight		
	Why	**Resources**		
	- Informs all navigation runs - Informing CEC 001.05 Considerations as Duty Nav - Informs PC 001A/B When GPS fails, this may be the only way to fix a ship's position (super exciting!)	**For student:** WECDIS laptop DWAN laptop with internet access Celestial Navigation Software (NAVPAC) Celestial terminology handout – worksheet/quiz Worksheets for Activity 2 Sextant	**For instructor:** WECDIS laptop DWAN laptop with internet access Celestial Navigation Software (NAVPAC) Worksheets (DS solutions) Sextant **References:** A1, A2, B1, B2, B6, B22, B23, C1, C3	

LO 001.03.07 – Conduct Wheelover Routine

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Interactive Experiential **Media**: PPTs/Videos NABS **Method**: IL Practical **Time**: 10x45min (450min) **Environment**: Classroom NABS	SWBAT 1. Conduct the Wheelover routine 2. Monitor the ship throughout a turn 3. Applying considerations for reduced visibility	<u>Activity 1: FNO cable countdown</u> (10min) Review 5 cable countdown and show template in CFCD 130 Discuss key concepts: 1) Prioritization – 5cbl over other info flow OR the idea of "zip lip" nav (when you may want to be quiet) 2) Checking ships head/rudder 3) Assessing the turn (Quartering bearings/Check bearings) 4) Speed – how fast things happen/how much time you have to make your reports to introduce the idea of pre-briefing when necessary <u>Activity 2: Review of Advanced Concepts</u> (1x45min) Finer points about advanced concepts using diagrams covering: - Quartering bearings - Turn safety bearings - Time gates - Wheelover bearings - Wheelover PIs - EP and DR <u>Activity 3: NABS practical</u> (9x45min) Concurrent with 001.03.06A Activity 4 Consolidation activity on conducting the wheel over routine Part 1 - Visually Part 2 - Non-visually (reduced visibility)	**To be developed** - Slide Sort activity (use existing PPT slides) - DS solution for SS activity - Playbook already exists for NABS A2 - Rights for HRO vid clip
	Why	**Resources**	
	- Informs all navigation runs - Informs PC 001A/B As the NavO you need to have this Wheelover routine nailed down!	**For student:** WECDIS laptop **For instructor:** Slides for Activity 1 DS solution Playbooks for NABS Hunt for Red October video clip **References:** A1, B1	

LO 001.03.08 – Develop Minor Warship Pilotage Practical Skills

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr>
<td rowspan="4">Strategy:
Interactive
Experiential

Media:
NABS

Method:
IL
Practical

Time:
80x45min
(3,600min)
+5 Sea Days

(4,800min) On own time navigation planning

Environment:
Classroom
NABS</td>
<td>SWBAT
1. Conduct Pilotage in NABs and at Sea
2. Transit a lock at Sea
3. Conduct a river transit
4. Conduct a port visit</td>
<td colspan="2">Activity 1: Pilotage Developmental in NABS (Total for optimum load 80x45min)
Students will be assigned run sheets for various pilotage runs throughout the world and have a minimum of 4 hours to prepare them and NLT an hour to execute. Briefing and executing one Nav passage a day in any conditions to prepare them for PC 001. One student per day will act as Duty Navigator. A minimum of 2 paper passages will be assigned to give the students a chance to become familiar with paper nav planning, execution and FixO duties when using paper.

Activity 2: Pilotage Developmental at Sea (5 sea days)
Students will be assigned run sheets for various pilotage runs in an MWV and have a minimum of 4 hours to prepare them and NLT than an hour to execute. Executing 1 or more Nav passages a day in any conditions to prepare them for PC001.

Activity 3: Transit a Lock (During MWVNO Sea Phase)
Students will as a group, plan and execute a lock transit at sea. Recommended Ballard Lock.

Activity 4: Conduct a river transit (During MWVNO Sea Phase)
Students will as a group, plan and execute a river transit. Recommend Fraser River.

Activity 5: Conduct a port visit (During MWVNO Sea Phase)
Students will as a group, plan and execute a port visit. The intention is to closely replicate the planning and administration that has to be completed to conduct a foreign port visit, and every effort should be made to make this an American, or unfamiliar port. Recommend Seattle or Vancouver.</td>
<td rowspan="4">To be developed
- Run sheets
- NSSC</td>
</tr>
<tr><th>Why</th><th colspan="2">Resources</th></tr>
<tr>
<td>- Prepare for PC001</td>
<td>For student:
Run Sheet
WECDIS laptop
Nav Tools
Chart and Publications Outfit
NSSC</td>
<td>For instructor:
Run Sheets

References:
A1, B1</td>
</tr>
</table>

LO 001.03.09 – Develop Major Warship High-Speed Pilotage Practical Skills

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Interactive Experiential **Media**: NABS **Method**: IL Practical **Time**: 80x45min (3,600min) +5 Sea Days (4,800min) On own time navigation planning **Environment**: Classroom NABS	SWBAT 1. Conduct Major Warship, High-Speed Pilotage in NABs and at Sea	<u>Activity 1: Pilotage Developmental in NABS</u> (Total for optimum load 120x45min) Students will be assigned run sheets for various pilotage runs throughout the world and have a minimum of 4 hours to prepare them and an NLT an hour to execute. Briefing and executing one Nav passage a day in any conditions to prepare them for PC 001. Each passage should include some sort of seamanship, shiphandling, or operational component. With at least 3 harbour entries or departures. One student per day will act as Duty Navigator. <u>Activity 2: Pilotage Developmental at Sea</u> (5 sea days) Students will be assigned run sheets for various pilotage runs and have a minimum of 4 hours to prepare them and NLT than an hour to execute. Executing 1 or more Nav passages a day in any conditions to prepare them for PC001. Preferably this is conducted in a Major Warship, or at the least, a minor warship, with the OIC being the CO of the major warship for the assessed phase. Every effort should be made to provide time for ship handling and advice to command.	
			To be developed - Run sheets - NSSC
	Why	**Resources**	
	- Prepare for PC001	**For student:** Run Sheet WECDIS laptop Nav Tools Chart and Publications Outfit NSSC **For instructor:** Run Sheets **References:** A1, B1	

EC 001.01 – Plan and Brief an Open Ocean Passage

Overview	Objective(s)	Assessment Details		Lessons Identified
Media: Computer Projector **Method**: Practical **Time**: Total time = 660min Student to have no less than 8 weeks to complete this assignment as homework with 480mins of in-class time assigned. Presentation to be 20-30 minutes (180 min – 6 students @30min each) **Environment**: Classroom/Homework	1. The goal of this EC is to expose the student to the process of planning and briefing an open ocean passage, including the amount of time required, resources available, and factors to consider.	**<u>Before the assessment:</u>** 1. FNIs will provide student with the details of an Open Ocean Passage, including start and end location, constraints and any specific requirements to include in their plan. Student will conduct the bulk of research and planning independently using the FNIs as simulated external resources. **<u>During the assessment:</u>** 1. Student will prepare an Open Ocean Passage plan written package to include details and discussions of their planning factors, recommendations/choices. This written package will be submitted to the FNIs for review and feedback prior to the student preparing and delivering a brief of their Open Ocean Passage plan. Student should be prepared to answer any questions regarding their proposed plan. **<u>Assessment feedback:</u>** 1. Written feedback on their Open Ocean Passage plan written package, which is to inform the student's brief. Formative feedback in the form of open discussion with the student upon completion of their brief. This is not a critical EC therefore, there is no standard to meet however the FNIs are expected to guide the student to fill in any critical information missing from their written package or brief.		-
				To be developed
				- DS solution on critical information which much be covered
	Impact and Limitations	**Resources**		
	Impact: Formative feedback. If there is any critical information missing, the student will resubmit work after being given additional time. **Limitations/Remarks**: Students to treat this as a real Open Ocean Passage plan; however should not engage external agencies/resources without FNI approval.	**For student:** WECDIS Laptop DWAN Laptop with internet access Chart and publication outfit Navigation tools Access to external agencies/simulated access via FNIs	**For instructor:** Run sheets DS solution with critical information which students must cover in their presentation and package **References:** A1, A2, A6, A7, A13, A21, B1, B2, B4, B5, B6, B7, B9, B18, B19, B20, B21, C3, C4, C6, C8, C9, C16, C17	

CEC 001.02 – Execute a Simulated Open Ocean Passage by Employing Celestial Navigation Software

Overview	Objective(s)	Assessment Details	Lessons Identified
Media: Paper WECDIS laptop Computer with Celestial Navigation Software **Method**: Written Exam (80%) **Time**: 240min **Environment**: Classroom	1. The goal of this CEC is to assess the student's ability to employ Celestial Navigation Software (NAVPAC) to execute a simulated Open Ocean Passage. Inputting sun and star sights, calculating rising and setting times, maintaining an accurate DR and transferring positions to WECDIS will be critical to this assessment.	**<u>Before the assessment:</u>** 1. Students will ensure their WECDIS laptop and Celestial Navigation Software (NAVPAC) are functioning and available for use during the exam. FNIs will provide a copy of the Exam to each student. **<u>During the assessment:</u>** 1. Students to work independently through their exam. Use of WECDIS and Celestial Navigation Software (NAVPAC) is authorized. Measurements will be provided to students simulating actual sights conducted in the open ocean. 2. Students can ask questions to FNIs 3. FNIs will clarify questions without providing the answer to the student. **<u>Assessment feedback:</u>** 1. Exam papers and Celestial Navigation Software (NAVPAC)/WECDIS products will be evaluated. Exam papers with grading and comments will be returned to students. Review (take-up) of the exam will be conducted.	-
			To be developed
			- Exam - Answer Key
	Impact and Limitations	**Resources**	
	Impact: If a student is unsuccessful on the first attempt - review of graded exam/comments and course material as required. Automatic re-write granted. If a student is unsuccessful in re-write, then a TRB will be convened. **Limitations/Remarks**: No actual star or sun sights will be taken using a sextant. The exam assumes the sky is visible, and measurements simulating actual sights will be provided—students to work independently.	**For student:** - WECDIS laptop - Celestial Navigation Software (NAVPAC) - Exam paper (provided by FNI) - Pen/pencil - Blank paper **For instructor:** Exam paper (IOT to clarify Questions) Answer Key **References:** A1, B2, B23	

CEC 001.03 – Collision Regulations

Overview	Objective(s)	Assessment Details	Lessons Identified
Media: Paper Computer Projector Speakers **Method**: Written Exam (90%) **Time**: 180min **Environment**: Classroom	2. The goal of this CEC is to assess the student's knowledge and understanding of the "Rules of the Road" COLREGs.	**<u>Before the assessment:</u>** 1. Students will ensure they have all required materials to write the exam, including verifying the setup and functionality of the Computer, Projector and Speakers required for the multi-media aspect of the exam. FNIs will provide a copy of the Exam to each student. **<u>During the assessment:</u>** 1. Students to work independently through their exam. FNIs will oversee the multi-media portion of the exam. 2. Students can ask questions to FNIs 3. FNIs will clarify questions without providing the answer to the student. **<u>Assessment feedback:</u>** 1. Exam papers will be evaluated. Exam papers with grading and comments will be returned to students. Review (take-up) of the exam will be conducted.	-
			To be developed
			-

Impact and Limitations	Resources	
Impact: If a student is unsuccessful on the first attempt - review of graded exam/comments and course material as required. Automatic re-write granted. If a student is unsuccessful in re-write, then a TRB will be convened. **Limitations/Remarks**: This EC is deemed critical due to the safety implications regarding a student's understanding and application of the "Rules of the Road." - Students are expected to maintain the standard demonstrated during this CEC throughout PC001A and PC001B. - Students to work independently.	**For student:** - Exam paper (provided by FNI) - Pen/pencil - Blank paper	**For instructor:** - Multi-media portion of the exam (PPT) **References:** A1, A18, B1, B4, B9, C13

CEC 001.04 – Act as Duty Navigator

Overview	Objective(s)	Assessment Details	Lessons Identified
Media: Nil **Method**: Practical **Time**: 8640min total time Minimum 16hrs Maximum 24hrs x 6 students **Environment**: At Sea	1. The goal of this CEC is to assess a student's ability to act as Duty Navigator onboard a ship at sea. Managing the overall transit plan for the day including: speed management, timings, overseeing the bridge team personnel, responding to emergencies, responding to equipment errors and faults, as well as briefing, are all critical to this assessment.	**<u>Before the assessment:</u>** 1. Student will know in advance which time period their Duty Navigator assessment will take place over. Prior to this, they will have the opportunity to plan and prepare any deliverables and set up for the period of time in which they are Acting as Duty Navigator. **<u>During the assessment:</u>** 1. FNIs will assess the student through the period of time in which they are Acting as Duty Navigator. Management of the overall transit and execution of the mission for the day will be the responsibility of the student Duty Navigator. The student's BRM, specifically bridge team management, responding to emergencies and responding to equipment errors and faults in the role as the Duty Navigator will all be assessed. Any briefs which the student provides to Command or to the ship will also be assessed. **<u>Assessment feedback:</u>** 1. After the student completes their time as Duty Navigator, they will turnover to the next student. The FNIs and CO of the ship will complete the assessment form and make a satisfactory/unsatisfactory determination before providing feedback to the student.	-
			To be developed
			-

Impact and Limitations	Resources	
Impact: Review key concepts, roles and responsibilities and Re-attempt. If a student does not meet the standard during 2 consecutive attempts, a TRB will be convened. **Limitations/Remarks**: This EC is deemed critical due to the safety aspect of being a Navigator at sea during times other than pilotage. - Time as Duty Navigator must include an overnight period.	**For student:** - WECDIS Laptop - Chart and publication outfit - Navigation tools	**For instructor:** - Duty Nav evaluation sheet **References:** A1, A2, A4, A5, A6, A7, A9, A11, A12, A13, A14, A16, A18, A21, B1, B3, B4, B5, B6, B7, B9, B19, B20, B21, C1, C2, C3, C8, C10, C11, C16, C18

CEC 001.05 – Conduct a High-Speed Pilotage Run (FNO)

Overview	Objective(s)	Assessment Details	Lessons Identified
Media: Nil **Method**: Practical **Time**: 780min 240min planning (concurrent) 60min ex + 30min fb/Student x 6 students = 540min **Environment**: NABS	1. The goal of this CEC is to assess the student's potential to navigate a major warship at high speed. Maintaining situational awareness and keeping the ship safe will be critical to this assessment.	**<u>Before the assessment:</u>** 1. FNIs will give students the intended start and end points for an assigned high-speed pilotage run as well as a basic mission scenario and list of constraints. Student will follow the appraisal, plan and execute/monitor process IAW BR45. Students will have at a minimum 4hrs to conduct their appraisal and planning. Student's appraisal and planning will be assessed by the FNI as per the evaluation sheet. **<u>During the assessment:</u>** 1. Students will brief CO as req'd prior to starting their pilotage run. Students will execute their planned pilotage run and respond to navigation casualties and any changing meteorological conditions throughout the run. Students will be assessed throughout with a focus on their ability to maintain situational awareness, execute the pilotage according **<u>Assessment feedback:</u>** 1. After the student completes their assigned pilotage run, they will turnover the navigation to the next student or back to the OOW. The FNIs and CO of the ship will complete the assessment form and make a satisfactory/unsatisfactory determination before providing feedback to the student.	-
			To be developed
			-
	Impact and Limitations	**Resources**	
	Impact: 4 runs assigned (at least 1 day and 1 night), all of which students must complete. Students must achieve the FNO standard on 2 runs - 1 of which must be day and 1 of which must be night. If a student does not achieve the FNO standard on any 3 runs, a TRB will be convened. **Limitations/Remarks**: - SOA between 16-20kts - Max speed as per the class of vessel the student is navigating - Any meteorological conditions - Minimum equipment during the execution will be a WECDIS	**For student:** - WECDIS laptop - Printer - Navigation notebook - Chart and publication outfit - Navigation tools - Guidance from CO (as req'd) **For instructor:** - Run sheet - Evaluation sheet **References:** A1, A6, A11, A13, A14, A16, A18, A21, B1, B3, B4, B5, B6, B7, B9, B19, B20, B21, C8, C10, C18	

PO 002 – Navigate a Formation of Vessels (of any class)

Details the performance standard is as follows:

PO 002

The navigator shall lead the navigation of a formation of vessels of any class at sea (AT0002), IAW CFCD130 & BR45 by:

a. Performing a formation navigation appraisal;
b. Conducting formation navigation planning; and
c. Executing the formation navigation including:
    i. Navigating a minor war vessel (MWV) or submarine at speeds up to 15kts (MWV NO);
    ii. Navigating any vessel at any speed (FNO);
    iii. Pilotage; and
    iv. Task Group Operations.

Notes for planning next delivery of PO 002

PC 002 – Navigate a Formation of Vessels (of any class) in an Operational Environment

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Assessment Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Media:
Nil

Method:
Practical

Time: 1020min
300min in-class prep
30min brief
90min execution
(Per student)

Environment:
Classroom
NABS</td><td rowspan="2">The goal of this PC is to assess a student's ability to navigate a formation of ships in an operational environment. Maintaining formation safety (navigationally) will be critical for this assessment.</td><td colspan="2"><u>Before the assessment:</u>
No less than 24hrs before execution, the FNIs will give students the intended start and end points for an assigned formation navigation run as well as a basic mission scenario and list of constraints. Students will follow the formation appraisal, plan and execute/monitor process IAW BR45. Student will have 5hrs in-class time to conduct their appraisal and planning, and 30 mins to brief. Students' appraisal and planning will be assessed by the FNI as per the evaluation sheet.

<u>During the assessment:</u>
The student will execute their planned formation pilotage run and will be assessed throughout with a focus on keeping the formation safe. Each run will be 90mins.

<u>Assessment feedback:</u>
After the student completes their assigned formation pilotage run, the FNIs will complete the assessment form and determine if the student met the standard or not before providing feedback to the student.</td><td>-</td></tr>
<tr><td colspan="2"></td><td>To be developed</td></tr>
<tr><td></td><td colspan="2"></td><td rowspan="3">-</td></tr>
<tr><td>Impact and Limitations</td><td colspan="2">Resources</td></tr>
<tr><td>Impact: Review key concepts and principles, automatic re-attempt. If student is unsuccessful on re-attempt, a TRB will be convened.

Limitations/Remarks: SOA minimum 10kts, maximum 12kts. Maximum speed as per the class of vessel being navigated. Task group staff and external support will be emulated by FNIs.</td><td>For student:
WECDIS laptop
Chart and publication outfit
Navigation tools</td><td>For instructor:
- Run sheet
- Evaluation sheet

References:
A1, A2, A5, A6, A7, A9, A10, A11, A13, A14, A15, A16, A18, A21, B1, B3, B4, B5, B6, B7, B9, B18, B19, B20, B21, C2, C6, C8, C10, C16, C18</td></tr>
</table>

EO 002.01 – Perform a Formation Navigation Appraisal

LO 002.01.01 – Formation Navigation Appraisal

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Strategy:
Interactive

Media:
White Board

Method:
Brainstorm

Time:
1x45 min

Environment:
Classroom</td><td rowspan="2">SWBAT
• Account for differences in size and manoeuvring characteristics of vessels
• Understand navigation policies of other countries (TTPs)
• Apply considerations for civilian vessels
• Task Group Operations/Sailing with Consorts</td><td colspan="2"><u>Activity 1: Builds on Info from EO 001.01</u> (45min)
10 min remind them of the original scenario
Students brainstorm on the board other considerations when given 4 crazy class vessels, to add to the scenario – can they go here?– group activity, Using the Navigation Appraisal Planning part of checklist Voyage Planning Checklist – back brief
FNI debrief – DS solution

Different classes of vessel – characteristics
Navigation policies of other countries
Considerations for civilian vessels</td><td>-</td></tr>
<tr><td colspan="2"></td><td>To be developed</td></tr>
<tr><td colspan="2"></td><td colspan="2"></td><td rowspan="2">- Scenario (from EO 001.01 that threads through and is expanded upon here)
- DS solution</td></tr>
<tr><td>Why</td><td colspan="2">Resources</td></tr>
<tr><td></td><td>- Avoid embarrassment for Canada by keeping allied nations safe.
- Informs PC 002</td><td>For student:
WECDIS laptop
DWAN laptop with internet access
Chart and publication outfit
Navigation tools</td><td>For instructor:
Scenario
DS solution

References:
A1, A2, A5, A6, A7, A9, A10, A11, A13, A14, A15, A16, A18, A21, B1, B3, B4, B5, B6, B7, B9, B18, B19, B20, B21, C2, C6, C8, C10, C16, C18</td></tr>
</table>

EO 002.02 – Perform a Formation Navigation Appraisal

LO 002.02.01 – Formation Navigation Planning

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Strategy:
Interactive

Media:
Scenario

Method:
PPT – scenario + work products and briefing products

Time:
4 x 45min
(180min)

Environment:
Classroom</td><td>SWBAT
• Draft formation messages
• Plan operation formation navigation
• Plan ceremonial formation navigation</td><td colspan="2"><u>Activity 1: FNI Lead Through of Task Group Planning</u> (2x45min)
Scenario-based, FNI to talk through the planning process to include:
- Timing and considerations
- Lead time
- Battle rhythm
- Example of messages (PIM and OCS intentions)
- Data on ships
- Patrol of an area
- 3 ships
- Waterspace mgmt.
- Ceremonial anchorage
- Photo ex considerations
- Q-routes
- Lead throughs
- Screens
- Force LDL</td><td>-</td></tr>
<tr><td></td><td></td><td></td><td>To be developed</td></tr>
<tr><td></td><td></td><td></td><td rowspan="2">- DS solution
- Scenario
- PPT slide deck</td></tr>
<tr><td></td><td>Why</td><td colspan="2">Resources</td></tr>
<tr><td></td><td>Avoid embarrassment for Canada by keeping allied nations safe.

Informs PC 002</td><td>For student:
WECDIS laptop
DWAN laptop with internet access
Chart and publication outfit (including paper charts)
Navigation tools</td><td>For instructor:
PPT slide deck
Scenario
DS solution
3d model ships (foreign, civilian, RCN)

References:
A1, A2, A5, A6, A7, A9, A10, A11, A13, A14, A15, A16, A18, A21, B1, B3, B4, B5, B6, B7, B9,B10, B11, B12, B14, B18, B19, B20, B21, C2, C6, C8, C10, C16, C18</td><td></td></tr>
</table>

EO 002.03 – Execute Formation Navigation

LO 002.03.01 – Formation Navigation Execution

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Discuss then Demonstrate **Media**: Laptop Projector **Method**: Lecture Practical **Time**: 25 x 45min (1,125min) **Environment**: Classroom NABS	SWBAT • Monitor the safety of a formation of vessels • Manage signals • Use WECDIS to facilitate station keeping	<u>Activity 1: Lecture Formation Nav Execution</u>(1 x 45min) Tips and tricks on formation navigation in the real world – how to pass signals, control units, brief command, direct the team, direct other units <u>Activity 2: NABS practical</u> (1 x 90min/run x 6 students) NABS practical, formation navigation in an operational environment - building on developmental runs from PO 001 (redo with operational aspects). Formation mine transit led by MCDV to include 3 additional ships. Required: OPTASK MINE WARFARE, Transit plan <u>Activity 3: NABS practical</u> (1 x 90min/run x 6 students) NABS practical, formation navigation in a ceremonial environment - building on developmental runs from PO 001 (redo with ceremonial aspects). SGI transit ending in RV with unit for follow-on serial (PHOTOEX) to incorporate transition from one serial to the next. Formation led by CPF with 2 additional ships for pilotage portion and RV with Asterix and 2 additional ships for a total of 6 ship formation. Student is OCS for both serials. Required: OCS Intentions <u>Activity 4: Formation Entrance in Foreign Ports (1x60min/Student)</u> NABs practical, formation navigation in a foreign environment – building on developmental runs from PO 001. Foreign ports to include Boston, Pearl Harbour, San Diego. CPF lead ship with 3 additional ships (at least 1x CPF for second bridge and 1x foreign ship, 1x asterisk) <u>Remarks</u>: Schedule after individual runs	-
	Why	**Resources**	**To be developed** - WECDIS video (already exist)

Overview	Objective(s)	Activity Details		Lessons Identified
	Avoid embarrassment for Canada by keeping allied nations safe. Informs PC 002	**For student:** WECDIS laptop Chart and publication outfit Navigation tools NSSC	**For instructor:** WECDIS laptop & Video **References:** A1, B1, B3, B4, B5, B6, B7	

EC 002.01 – Navigate a Formation of Vessels (of any class) in a Ceremonial Environment

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Assessment Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Media:
Nil

Method:
Practical

Time: 1020min
300min in-class prep
30min brief
90min execution
(per student)

Environment:
Classroom
NABS</td><td rowspan="2">The goal of this PC is to assess a student's ability to navigate a formation of ships in a ceremonial environment.</td><td colspan="2"><u>Before the assessment:</u>
No less than 24hrs before execution, the FNIs will give the student the intended start and end points for an assigned formation navigation run as well as a basic ceremonial scenario and list of constraints. Student will follow the formation appraisal, plan and execute/monitor process IAW BR45. Student will have 5hrs in-class time to conduct their appraisal and planning, and 30 mins to brief. Student's appraisal and planning will be assessed by the FNI as per the evaluation sheet.

<u>During the assessment:</u>
Student will execute their planned formation pilotage run and will be assessed throughout with a focus on keeping the formation safe and executing the ceremonial aspect of the run. Each run will be 90mins.

<u>Assessment feedback:</u>
After the student completes their assigned formation pilotage run, the FNIs will complete the assessment form and provide formative feedback.</td><td>-</td></tr>
<tr><td colspan="2"></td><td>To be developed</td></tr>
<tr><td></td><td colspan="2">Impact and Limitations</td><td colspan="2">Resources</td><td rowspan="2">- Assessment checklist
- Run sheet
- Playbook</td></tr>
<tr><td></td><td>Impact: Formative feedback. This EC is not critical; therefore, there will not be any re-attempts.

Limitations/Remarks: SOA minimum 10kts, maximum 12kts. Maximum speed as per the class of vessel being navigated. Task group staff and external support will be emulated by FNIs.</td><td>For student:
- WECDIS laptop
- Chart and publication outfit
- Navigation tools</td><td>For instructor:
- Run sheet
- Assessment Checklist

References:
A1, A2, A5, A6, A7, A9, A10, A11, A13, A14, A15, A16, A18, A21, B1, B3, B4, B5, B6, B7, B9, B10, B11, B12, B14, B18, B19, B20, B21, C2, C6, C8, C10, C16, C18</td></tr>
</table>

PO 003 – Lead the Navigation Division and Bridge Team of Any Class of Vessel

Details the performance standard is as follows:

PO 003
The navigator shall lead the navigation division and bridge team on any class of vessel (FT0002, AT0003), IAW refs by:

a. Performing divisional duties;
b. Performing administrative duties;
c. Managing the operational readiness of the division; and
d. Leading the bridge team, to include:
    i. Setting and enforcing bridge standards;
    ii. Providing oversight on training; and
    iii. Leading the team using Bridge Resource Management concepts.

Notes for planning next delivery of PO 003

PC 003A – Respond to Administrative and Policy Problems and Questions

Overview	Objective(s)	Assessment Details		Lessons Identified
Media: Nil **Method**: Written Exam (80%) **Time**: 120mins **Environment**: Classroom	The goal of this PC is to assess a student's ability to recall administrative and policy details as well as identify references and locate information within references.	**<u>Before the assessment:</u>** Students will ensure they have all required materials to write the exam, including verifying the setup and functionality of their computer, including access to references required for the exam. FNIs will provide a copy of the Exam to each student. **<u>During the assessment:</u>** Students to work independently through their exam. During the first 90 minutes of the exam, students will not have access to references. During the last 30 minutes of the exam, students will have access to references. The FNIs will oversee the portion of the exam when access to references is granted. Students can ask questions to FNIs FNIs will clarify questions without providing the answer to the student. **<u>Assessment feedback:</u>** Exam papers will be evaluated. Exam papers with grading and comments will be returned to students. Review (take-up) of the exam will be conducted.		-
				To be developed
	Impacts and Limitations	**Resources**		
	Impact: If a student is unsuccessful on first attempt - review of graded exam/comments and course material as required. Automatic re-write granted. If a student is unsuccessful on re-write, then a TRB will be convened. **Limitations/Remarks**: - Students to work independently - There will not be a simulated division or subordinates available for student use - Policy questions will be navigation policy questions from CFCD130 and BR45	**For student:** Exam paper (provided by FNI) Pen/pencil Blank paper References for final 30min of exam	**For instructor:** **References:** A1, A2, A3, A4, A5, A7, A18, B1, B2, B3, B4, B5, B6, B7, C3	

PC 003B – Lead the Bridge Team

Overview	Objective(s)	Assessment Details		Lessons Identified
Media: Nil **Method**: Practical **Time**: Concurrent with CEC 001.04 **Environment**: At Sea		*This PC will be evaluated during CEC 001.04 – Act as Duty Navigator* **<u>Before the assessment:</u>** **<u>During the assessment:</u>** **<u>Assessment feedback:</u>**		-
				To be developed
	Impacts and Limitations	**Resources**		
	Impact: **Limitations/Remarks**:	**For student:**	**For instructor:**	

EO 003.01 – Conduct Divisional Duties

LO 003.01.01 – Conduct general officer duties

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Strategy:
Experiential

Media:
nil

Method:
Observed throughout the course

Time:
Concurrent with other activities throughout the course

Environment:
Any</td><td>• SWBAT</td><td colspan="2">• No-train observed through the course as the officers are expected to be able to do this task.</td><td>-</td></tr>
<tr><td></td><td></td><td></td><td>To be developed</td></tr>
<tr><td></td><td></td><td></td><td rowspan="3">-</td></tr>
<tr><td>Why</td><td colspan="2">Resources</td></tr>
<tr><td>-</td><td>For student:</td><td>For instructor:</td></tr>
</table>

LO 003.01.02 – Manage Subordinates

Overview	Objective(s)	Activity Details		Lessons Identified
Strategy: Direct/Experiential Interactive **Media**: Laptop Projector **Method**: Lecture/Lead-through/ Flip Chart **Time**: 90 min 2x45min **Environment**: Classroom	SWBAT: - Liaise with the BTL Manager - Manage Met Techs - Control records - Manage NOPQ Training	<u>Activity 1 – Lecture (1x45min)</u> Instructor lead lecture reviewing NWO career progression references, common SLt problems onboard ship, and how to liaise with the BTL manager, including recent BTL manager feedback. This is followed by example questions regarding managing subordinates which the students have to work through to answer and present to the class. <u>Activity 2 - Met Tech Guest Lecture (1x45min)</u> Met Tech Guest lecturer briefly explains Met Tech career progression and common Met Tech-related problems onboard. <u>Activity 3 - Monitor Mass Lead Through (1x30min)</u> A "follow along" guide through of monitor mass functions as they apply to NOPQ and watchkeeping <u>Activity 4 – Managing NWOPQ Discussion (1x30min)</u> A floating flip chart activity where students talk and explain their experience through NWOPQ and how best to approach and lead officers through it, as the NavO. <u>Activity 5 – Homework Develop In-Routine for Subordinates</u> Students to develop an in-routine checklist to include a list of important documents, interview questions and the student's leadership philosophy.		-
	Why	**Resources**		**To be developed**
	- You are the Divisional Officer for a large number of officers and have a responsibility to effectively lead and train them - Doing so properly will make your job easier	**For student:** Laptop with Monitor Mass Problem Scenarios Whiteboards	**For instructor:** Guest Instructor PPT Lecture Laptop with MM and Projector Problem scenarios Debrief questions DS solution **References:** A1, A2, A4, A5, A6, A7	- Problem Scenarios - Lecture needs to be adjusted from the old curriculum - Common divisional problems need to be brainstormed

EO 003.02 – Conduct Administrative Duties

LO 003.02.01 – Maintain Navigational Records

Overview	Objective(s)	Activity Details		Lessons Identified
Strategy: Experiential Independent **Media**: Laptop, References **Method**: Interactive Activity – Collaborative Problem Solving Learner back briefs Instructor debriefs **Time**: 90min 2x45 min **Environment**: Classroom	SWBAT • Maintain Paper Records(30) • Maintain Electronic Records (15) Teaching Points	<u>Activity 1 – Group Research (2x45 min)</u> Students are divided into groups and then each assigned one of the following topics to research the policies in CFCD 130 and present to their fellow classmates: a. Ship's Log, b. OOW Notebook c. VDR d. MCCR e. Navigation Data Book f. Nav DVD Binder (charts + pubs) g. Captain's Ship Book h. Marine Mammal Log i. NAVORD 1000-6 The instructor clarifies each teams' learning by prompting them with further questions on the policies by referring to the DS solution. <u>Activity 2 – VDR (EDIP) (30 min)</u> The instructor leads the class through the process to save and backup a VDR using WECDIS on the projector in front of the class, as well as showing them how to configure automatic recording, screen capture functionality, and how to transfer the files. The activity is complete when the students are able to email a VDR file to each other for review.		-
				To be developed
				-
	Why	**Resources**		
	- Essential to recreate incidents and to protect yourself and the institution against false claims. - Great tool for debriefing an alongside or another manoeuvre - Assists in the production of PRNOTES and makes you look good. - Helps to practice researching and to get into CFCD 130	**For student:** Laptop with WECDIS CFCD 130	**For instructor:** Laptop with WECDIS and Projector DS solution to policy questions on PPT Confirmation Questions **References:** A1, A2, A4, A5, A6, A7, A10, A15, A20, B1, B3, B6, C4	

LO 003.02.02 – Draft Navigation Messages

Overview	Objective(s)	Activity Details		Lessons Identified
Strategy: **Media**: **Method**: **Time**: 1x45min **Environment**:	SWBAT - Draft common navigation messages - Input navigation information into the LOGREQ - Understand the Navigation Incident Report Format(5min)	<u>Activity 1 - Message Creation (30min)</u> Students are given access to the message archive; the reference, and they are each assigned a message type and scenario to create and pass to the left for review. a. OOW Designator List b. BWK Qualification c. NOPQ Nomination d. PIM <u>Activity 2 - Review LOGREQ (15min)</u> Students are shown a blank example LOGREQ and brainstorm what sort of navigational information might be important to include. Instructor provides any items they may be missing when complete. <u>Activity 3: Navigation Incident Report (10min)</u> Students are shown the template and then instructed that they will be required to create a navigation incident report following their first navigation incident in the simulator.		-
	Why	**Resources**		**To be developed**
	- Correspondence is highly valued in our organization - Proper LOGREQs will ensure you get the resources and information you need to pilot the ship - You may be part of an incident investigation on one side or the other!	**For student:** DWAN laptop with internet access MCOIN (Message drafter software) Message Archive (DWAN) Chart and publication outfit Printer Pen/pencil	**For instructor:** DS solution for messages DS solution for LOGREQ examples and “tips and tricks.” **References:** A1, A2, A4, A5, A6, A7, A10, A15, A20, B1, B3, B6, C4	- Messages exist - Message Scenarios - DS LOGREQ solution

LO 003.02.03 – Manage Watch Routines

Overview	Objective(s)	Activity Details		Lessons Identified
Strategy: **Media**: **Method**: **Time**: 1 x 45min **Environment**: Outside/ Spinnaker's Pub	SWBAT • Understand the principles of good watch keeping schedules • Recommend watch routines and regimes to Command Teaching Points	<u>Activity 1 - Assigned reading as homework.</u> The reading should cover the effects of poor sleep on watchkeeping or case studies that involve poorly organized watch routines. DRDC articles on watchkeeping and sleep are suggested, with students discussing the readings at the start of this class. <u>Activity 2: (FFC) Discuss as a group: principals for creating a good watch routine</u> (45min) As watchkeepers themselves, students should be encouraged to discuss the best strategies for creating a watch routine. If required, they should be prompted by the instructor to discuss the impact of the following factors: • Command Evening Briefs and other scheduling conflicts • Early morning and late-night evolutions • The manpower implications of closing up an SSD watch • Sleep and safety implications • SSO policies for watchkeeping		-
				To be developed
				- DS solution - Example Watch Routine
	Why	**Resources**		
	- Watchkeeping is core to the safety of the ship - It affects the morale of your officers	**For student:** Assigned readings DWAN laptop with internet access Chart and publication outfit	**For instructor:** DS solution for topics Whiteboard **References:** A1, A2, A5, A6, A9, B1, B3, B4, C3	

LO 003.02.04 – Navigation Administration

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Strategy:
Indirect
Independent

Media:
Publications

Method:
Homework
Brainstorm
Scenario-Based

Time:
2x45min
(90min)

Environment:
Classroom
Homework</td><td rowspan="2"><ul><li>Be exposed to MARPIFs and PRONOTES</li><li>Discuss CSOs, Occasions to Call NavO/CO</li><li>Draft Night Orders</li></ul></td><td colspan="2" rowspan="2"><u>Activity 1 - Exposure to MARPIFS and PRONOTES (homework)</u>
Students will be given access to the MARPIF library and both the digital and print versions of PRONOTES and asked to peruse them as homework. Each student will present a short (5min/1slide) highlight of something they found interesting or most relevant from the MARPIF or PRONOTE they read. (i.e. the lesson learned in the PRONOTE or a piece of significant Nav info that they wouldn’t have gotten without reading the MARPIF).

<u>Activity 2: Occasions to call the NavO/CO (45min)</u>
Brainstorm occasions to call the NavO/CO and discuss why they exist. Discuss CSOs and the pros and cons of certain policies.

<u>Activity 3 – Night orders worksheet (2x45min)</u>
Scenario for the student to write the ship’s Night Orders. Completed in class with the FNIs available to provide feedback and support. FNIs to provide examples and students to discuss examples.</td><td>-</td></tr>
<tr><td></td><td></td><td>To be developed</td></tr>
<tr><td rowspan="2">Why</td><td colspan="2">Resources</td><td rowspan="3">- Example (bad and good) CSOs
- Worksheets for Night Orders
- DS solution to brainstorming of occasions to call the NavO</td></tr>
<tr><td>Your occasions to call will be critical and possibly save you from disaster, or at the very least, embarrassment.

Your Captain’s need to make sense; you will be their keeper.</td><td>For student:
Assigned readings
DWAN with internet access
Chart and publication outfit</td><td>For instructor:
DS solution for topics
Whiteboard

References: A1, A2, A6, A7, A9, A21, B1, B8</td></tr>
</table>

LO 003.02.05 – Prepare for a Navigation Administration Inspection

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Strategy:
Independent
Experiential

Media:
Tour
Checklist

Method:
Field Trip

Time:
4x45min
(180min)

Environment:
Ship in Dockyard</td><td>SWBAT
• Conduct an NAI
• Advise Command on NAI
• Understand

Teaching Points
IAW CFCD 130 NAI inspection checklist
Review Ships Navigating Outfit</td><td colspan="2"><u>Activity 1 – NAI Walkthrough – (4x45min – forenoon/afternoon)</u>
Part 1 – Prepare for an NAI (Lecture 1x45min)
The class will review the NAI inspection checklist
Part 2 – Ship tour and mock inspection (2x45)
Proceed to an HMC ship alongside to conduct a mock NAI, led by their FNI. (Conducted on “tour day” in conjunction with LO 005.02 Troubleshooting Faults – Nav Equipment Tour Activity)</td><td>-</td></tr>
<tr><td></td><td></td><td></td><td>To be developed</td></tr>
<tr><td></td><td>Why</td><td colspan="2">Resources</td><td></td></tr>
<tr><td></td><td>Your NAI will likely be one of the first things you do, it contributes to your success as a navigator and ensures your equipment is working properly.</td><td>For student:
HMC Ship alongside
Assigned readings
DWAN laptop with internet access
Chart and publication outfit
CFCD 130 NAI Checklist</td><td>For instructor:
Transport
NAI Inspection checklist and clipboard

References: A1</td><td></td></tr>
</table>

LO 003.02.06 – Conduct NavO Secondary Duties

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Strategy:
Interactive

Media:
PPT
Whiteboard

Method:
Brainstorm
Guided Discussion
Interactive Lecture

Time:
1x45min

Environment:
Classroom</td><td>SWBAT
• Understand what a Secondary Duty is
• Have a sense of the variety of secondary duties that their subordinates could have
• Understand the NavOs secondary duties</td><td colspan="2"><u>Activity 1 –Secondary Duties (Subordinates and NavO)(45min)</u>
Part 1: Guided Discussion on Secondary Duties (general)(10mins)
- What are Secondary Duties
- Who assigns Secondary Duties
- Is being a Divisional Officer considered a Secondary Duty?
Part 2: Brainstorm Secondary Duties (Subordinates)(5mins)
- Students to create a list of secondary duties that they think their subordinates could have
Part 3: Guided Discussion on Secondary Duties for your subordinates(10mins)
- DS solution provided (list of secondary duties your subordinates could have) → ask the class if they came up with anything not on the list
Part 4: Brainstorm Secondary Duties (NavO)(5mins)
- Students to create a list of secondary duties that they think their subordinates could have
Part 5: Guided Discussion on NavO Secondary Duties (15mins)
- To include: Unit Historical Records Officers, Mail Officer, Food Services Officer and Administering a SLoc</td><td>-</td></tr>
<tr><td></td><td></td><td></td><td>To be developed</td></tr>
<tr><td></td><td></td><td></td><td rowspan="2">-</td></tr>
<tr><td>Why</td><td colspan="2">Resources</td></tr>
<tr><td>Secondary duties are a necessary part of life on ship. It is important to understand what your subordinates may be tasked with as you are responsible for supervising them and managing their effort between their primary duty (Training) and any secondary duties—understanding what Secondary Duties will likely be assigned to you as NavO is important to prepare for your future role.</td><td>For student:
DWAN laptop with internet access
Pen/pencil
Paper(notetaking)</td><td>For instructor:
Laptop
Projector

References: A1, A2, A4, A5, A6, A7</td></tr>
</table>

EO 003.02 – Manage the Bridge Team

LO 003.03.01 – Conduct BRM

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Independent Experiential Interactive **Media**: PPT **Method**: Guided Discussion Fishbowl **Time**: 305 min total 1x45min 1x45min 1x135min prep + 30min presentation per student (180 minutes) 2x45min ICW PO 001 **Environment**: Classroom NABS	SWBAT • Respond to BRM situations Teaching Points Advise OOW Conflict Resolution Occasions to call the NavO Error Trapping & Tracking Breaking the Error Chain	<u>Activity 1 - Lecture on BRM Concepts and The Role of the NavO</u> (45min) Guided lecture reviewing BRM concepts with a focus on the role of the NavO in BRM. <u>Activity 2 - Lecture on Error Analysis</u> (45min) Error Chain, Root Cause and Event Mapping to be covered. FNIs to provide real-world examples of poor BRM examples they have experienced/witnessed. <u>Activity 3 – BRM Case Studies</u> (135min prep time (in class or homework) + 30 min presentation) Part 1 – Assign and prepare a case study. Students will prepare and present, in syndicate (2 pers teams), BRM case studies Part 2 - Syndicates selected by DS to both highlight key issues and review relevant recent accidents Each case study to provide: • Overview of scenario • Findings of Relevant Investigation • Event Map created by students <u>Activity 4 - Guided Discussion on Best Practices and List Building</u> (2x45min) Students will review suggested best practices and, based on their experience and the case studies, discuss their implementation/suitability for these, and any new ones they can identify, to HMC ships <u>Activity 5 - Conducted during PO 001 Nav Runs: (ICW PO 001)</u> The students are exposed to a poorly run bridge scenario in NABS and at sea, and they are prompted to react. The students are faced with a variety of scenarios as the NavO. “Scenario cards” are given to the bridge team who play roles and practice poor BRM. This completed the 003.03.01 BRM portion of the *NSSC*. • Contact misidentification • Bad rules assessment • Poor fixing • Distracted Bridge • Poorly timed maintenance • Failure to call IAW CSOs • Improper bridge setup, etc. <u>Activity 6 - Coastal Navigation Run into SSD (leads into 003B PC and CEC 001.04 Duty Nav)</u> (45mins)	- **To be developed** - *NSSC* - BRM injects - Poor BRM examples - Coastal nav/pilotage scenario

Overview	Objective(s)	Activity Details		Lessons Identified
		One of the assigned runs takes students from coastal waters into pilotage waters. Student simulates starting in their cabin and must manage the bridge and ship as it approaches the harbour without SSD closed up. The bridge team encounters problems and practices poor BRM. The student must refocus the Bridge team and bring them together. Conduct as a fishbowl (example FNI run) if time does not allow. The emphasis here is to focus on BRM rather than pilotage.		
	Why	**Resources**		
	Your ability as a NavO and SWK will steer the ship clear of danger, and the bridge team's practices and habits are yours to mould. Will prepare you for your BTT.	**For student:** WECDIS laptop DWAN laptop with internet access Chart and publication outfit Navigation tools Coastal Nav/Scenario NSSC	**For instructor:** BRM Injects Coastal Nav/Scenario **References:** A1, A2, A4, A5, A6, A7, A9, A12, A13, A16, A17, A21, B1, B3, B4, B5, B6, B7, B19, B20, B21, C1, C2, C3, C13	

LO 003.03.02 – Develop the Bridge *Team*

Overview	Objective(s)	Activity Details		Lessons Identified
Strategy: Independent Experiential Interactive **Media**: **Method**: **Time**: (180min) 4x45min **Environment**:	SWBAT Deliver PD Create a training plan Administer Bridgemanship and Rules Exam Training of Bridge Pers Manage and deliver CRRs Teaching Points	<u>Activity 1 – PD Presentation</u> The PD portion of this is covered under EC 003.02, with the presentation of BRM being considered the topic of professional development. Emphasis should be placed on ensuring that the officer can impart information, teaching, guidance and mentoring; if possible, the PD brief should be given to personnel more junior than the officer. (To NWO 2,3 or 4 students if possible). <u>Activity 2 – Brainstorming/Lecture and lead-through of CRRs (45 min)</u> Students to look up CRRs the NavO is responsible for in CFCD 102, and then the FNI delivers a lecture and lead through of best practices. <u>Activity 3 – Guided discussion – Bridge Team Employment (45min)</u> Discussion on watch regimes/manning requirements, employment of lookouts, POOW, Bosn Mate, helm and throttles, and NavComm in order to establish an effective bridge team <u>Activity 4– Guided Discussion Ticketing Policy and Training (1x45 mins)</u> Discuss BWK ticketing policy and how to advise command on the granting of qualifications, and training of NCMs on the bridge team		-
	Why	**Resources**		**To be developed**
	Training and developing the bridge team is your responsibility. A well-trained bridge team enables you to perform at a higher standard and reduce the chance of mistakes being made.	**For student:** Assigned case studies DWAN laptop with internet access Chart and publication outfit	**For instructor:** Lecture DWAN laptop with projector **References:** A1, A2, A4, A5, A6, A7, A9, A12, A13, A16, A17, A21, B1, B3, B4, B5, B6, B7, B19, B20, B21, C1, C2, C3, C13	- Lecture (can be adapted from existing lectures) - PD topics (can be adapted from existing BRM case stud

EC 003.01 – Respond to a Complex Navigation Policy or Procedural Issue (Group EC)

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Assessment Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Media:
WECDIS Laptop
DWAN laptop with internet access
Projector

Method:
Research and Present

Time:
7days to research and create a 15min presentation
(90 min)

Environment:
Classroom and at home</td><td rowspan="2">The goal of this EC is to assess the student’s ability to conduct research and provide a brief on a complex navigation policy or procedural issue.</td><td colspan="2"><u>Before the assessment:</u>
FNI will provide a prompt to student regarding a complex navigation policy or procedural issue which the student must conduct research about and provide a brief on.

<u>During the assessment:</u>
FNIs will be available as a resource, and students can ask questions throughout the research phase. The final product submitted will depend on the type of issue the student is tasked to research but could include a PRONOTE, MARPIF, Briefing Note, CSOs Entry or Navigation Incident Report. After submitting the written response, students will give a 15min presentation to the class and be open to questions, comments and discussion.

<u>Assessment feedback:</u>
FNIs will participate in the question and discussion period following the student’s presentation. FNIs will review and provide feedback on the written submission.</td><td>-</td></tr>
<tr><td colspan="2"></td><td>To be developed</td></tr>
<tr><td>Impacts and Limitations</td><td colspan="2">Resources</td><td rowspan="2"></td></tr>
<tr><td>Impact:
Formative feedback. Possibility for the student to have to re-submit work if the information is not IAW current nav policy.
Limitations/Remarks: Nil</td><td>For student:
Prompt from FNI
WECDIS laptop
DWAN laptop with internet access
Chart and publication outfit
Navigation tools</td><td>For instructor:

References: A1, A2, A4, A5, A6, A7, A9, A12, A13, A16, A17, A21, B1, B3, B4, B5, B6, B7, B19, B20, B21, C1, C2, C3, C13</td></tr>
</table>

EC 003.02 – Lead Mariner Professional Development

Overview	Objective(s)	Assessment Details		Lessons Identified
Media: WECDIS Laptop DWAN laptop with internet access Projector **Method**: Research and Present **Time**: 7days to research and create a 30min presentation (180 min) **Environment**: Classroom and at home	The goal of this EC is to assess the student's ability to lead mariner professional development.	**<u>Before the assessment:</u>** FNI will provide a prompt to student regarding a Professional Development topic on which student will conduct research and prepare a brief for. The ideal setting for this EC is for the student to instruct an NWO class on a Nav Related Topic. Student can use prepared NWO briefs but should review material and add/expand as they see fit. **<u>During the assessment:</u>** FNIs will be available as a resource, and students can ask questions throughout the research phase. The final product will be some sort of Professional Development Activity (preferred that the student teaches an NWO class) **<u>Assessment feedback:</u>** FNIs will participate in the question and discussion period following the student's professional development lesson. FNIs will provide feedback and comments at the end of the question and discussion period.		-
				To be developed
	Impacts and Limitations	**Resources**		
	Impact: Formative feedback. **Limitations/Remarks**: Nil	**For student:** WECDIS laptop DWAN laptop with internet access Chart and publication outfit Navigation tools	**For instructor:** **References:** A1, A2, A4, A5, A6, A7, A9, A12, A13, A16, A17, A21, B1, B3, B4, B5, B6, B7, B19, B20, B21, C1, C2, C3, C13	

PO 004 – Provide advice to command on navigation matters

Details the performance standard is as follows:

PO 004

The navigator shall provide advice to command on navigation matters (AT0004 & FT0003) IAW References IOT challenge or inform decision making whether informally or formally by:

a. Conducting independent research;
b. Conveying accurate information

Assessment

Practical assessments to confirm performance when conveying information in the simulator and at sea
Theoretical assessments to confirm supporting knowledge

Other learning activities

Practical development by performing research throughout the course
Learning activities that support performance

Admin points

Guest Lecturers will augment training in this PO

Other considerations

Students will be required to work outside of course hours IOT complete some assignments. This accurately reflects their role as navigator, as they will be required to conduct research and convey information outside of normal working hours.

Situate Learning:
It is important for Navigators to perform independent research and convey accurate information to command in their role on a daily basis, to maintain the safety of all RCN, allied and civilian vessels.

Notes for planning next delivery of PO 004:

Main learning events

Conducting independent research;
Conveying accurate information

Assessment

Practical assessments to confirm performance when conveying information in the simulator and at sea
Theoretical assessments to confirm supporting knowledge

Other learning activities

Practical development by performing research throughout the course
Learning activities that support performance

Admin points

Guest Lecturers will augment training in this PO

Other considerations

Students will be required to work outside of course hours IOT complete some assignments. This accurately reflects their role as navigator, as they will be required to conduct research and convey information outside of normal working hours.

PC 004– Provide Advice **to Command on Navigation Matters**

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Assessment Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Media:
NABS/at sea

Method:
Practical

Time: 570min total
180-240min
Prompt given 2 hours before planned evolution

Environment:
Day/Night, in any conditions of visibility, in NABS (or at sea during MWVNO) Remarks: Assessment time will vary depending on the assigned evolution</td><td>SWBAT
Completeness? A brief for command that includes all pertinent aspects of the shiphandling evolution
Decisions? Selection of an appropriate course of action and ability to assess and make effective recommendations to command during the execution.
Accuracy? Brief must contain no errors of fact or policy. During the execution, assessment of the direction of the ship’s movement and position relative to dangers must be correct.
Speed? Any assessment of danger must be given in sufficient time that command is able to correct the position of the ship.</td><td colspan="2"><u>Before the assessment (setup/preparation):</u>
1. NABS set up/or at sea bridge preparation
2. Prepare assessment checklist for applicable student

<u>During the assessment:</u>
1. Prompt given by FNI to student
2. Student given 120min for planning
3. Activity: Practical PC in NABS (or at sea). Students will be required to provide a brief to command on a shiphandling problem while in NABS (or at sea on Training Vessel) Draft Assessment Plan (EC 005.01) Demonstrate Knowledge of Simple Navigation Systems (MWVNO)
4. Scenario: Navigator in NABS (or at sea). The ship will perform a seamanship evolution, and the Navigator will be required to brief command on the conduct of the evolution and advise command in real-time during the conduct of the evolution. Student will close meeting with follow-up information
5. Student will complete the meeting in XX minutes

<u>Assessment feedback:</u>
1. Feedback is given individually after the student’s performance
2. Student will take notes of the areas of improvement
3. Student will sign the assessment checklist</td><td rowspan="2">To be developed</td></tr>
<tr><th>Impact and Limitations</th><th colspan="2">Resources</th></tr>
<tr><td>Impact: Review, Re-attempt. 2 consecutive failures results in a TRB IAW NPTORD 4500-54
Limitations/Remarks:
Only evolutions capable of being completed by the vessel in use will be assigned.
* Separate from Duty Nav evaluation and PC 001 evaluation (pilotage run)</td><td>For student:
NABS (or patrol vessel at sea)
WECDIS laptop
Chart and publication outfit
Pen/pencil</td><td>For instructor:
Run sheet
Assessment Checklist
References: A1, A2, A4, A5, A6, A7, A9, A11, A13, A16, A18, A21, B1, B2, B3, B4, B5, B6, B7, B18, B19, B20, B21, C1, C2, C3, C8, C10, C18</td></tr>
</table>

EO 004.01 – Conduct Independent Navigation Related Research

LO 004.01.01 – Liaise with External Agencies

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Strategy:
Direct

Media:
Tour

Method:
In-person tour

Time:
270mins

Environment:
Facility tour</td><td>SWBAT
Recall from their experience what resources are available at external agencies</td><td colspan="2"><u>Activity 1 – Tour of Port Authority</u> (2 x 45 min)
Students to visit a working Port Authority Office to liaise with civilian experts on matters important to visiting ships navigators. Vancouver/Fraser River Port Authority preferred

<u>Activity 2 – Tour of NOAA and NWS</u> (4 x 45 min)
Students to visit NOAA facility and NWS facility to understand the basics of US chart and weather product production, to highlight the differences and similarities of different countries hydrographic and meteorological standards and products.</td><td></td></tr>
<tr><td></td><td></td><td></td><td>To be developed</td></tr>
<tr><td>Why</td><td colspan="2">Resources</td><td rowspan="2"></td></tr>
<tr><td>- Important for students to know what support exists beyond internal resources and agencies</td><td>For student:</td><td>For instructor:

References: A1, A2, A4, A5, A6, A7, A9, A11, A13, A16, A18, A21, B1, B2, B3, B4, B5, B6, B7, B18, B19, B20, B21, C1, C2, C3, C8, C10, C16, C17, C18</td></tr>
</table>

EO 004.02 – Convey Accurate Information

LO 004.02.01 – International Maritime Law and Conventions

Overview	Objective(s)	Activity Details		Lessons Identified
Strategy: Direct Experiential **Media**: PPT Guest Lecture WECDIS **Method**: IL Practical **Time**: 4.5 x 45min (200min) **Environment**: Classroom	SWBAT 1. Provide advice to command regarding: a. UNCLOS b. Environmental Regulations c. COLREGs d. International Naval Conventions e. Certificates f. Port Authorities g. Territorial Waters 2. Apply territorial water information to a chart	<u>Activity 1 – AJAG Lecture</u> (135min) International Maritime Law, TTW, CZ <u>Activity 2 – UEnvO/MSEO Lecture</u> (45min) SEMS, MARPOL areas <u>Activity 3 - Practical application of maritime law using WECDIS (1x30min)</u> Review Port authorities, CUES, treaties. EDIP – FNI lead students through plotting information on a chart in WECDIS, including AMLs		**To be developed**
	Why	**Resources**		
	- Advise command PC 004 and throughout the course. - Don't want to break any laws or violate any treaties. - NavO is responsible for this (especially with no JAG on board).	**For student:** WECDIS laptop DWAN laptop with internet access Pen/pencil Paper (notetaking)	**For instructor:** WECDIS laptop TTW AML CD **References:** A1, A2, A4, A5, A6, A7, A9, A11, A13, A16, A18, A21, B1, B2, B3, B4, B5, B6, B7, B18, B19, B20, B21, C1, C2, C3, C8, C10, C16, C17, C18	

LO 004.02.02 – Shiphandling and Seamanship

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Interactive Experiential **Media**: Guest Lectures Worksheets NABS **Method**: IL Practical **Time**: (2000min) A1: 15min/day x 5days = 125min A2: 540min A3-10: 840min A11: 45min A12: 270min A13: 180min A14 : 45min A15 : 90min A16: 180min A17: 45min **Environment**: Classroom NABS Field Trip	SWBAT 1. Advise command on the following matters: a. Ship design concepts b. Berthing and slipping i. Employment of tugs ii. Use of lines/anchors c. Shiphandling in heavy weather d. Anchoring e. RAS f. Tow approaches g. JSS Specific h. Buoy work i. Manoeuvring in confined waters j. Shallow water effects 2. Understand the principles of shiphandling & ship characteristics	<u>Activity 1 – Introduction to Shiphandling and Seamanship</u> Part 1 – Threshold Knowledge quiz on shiphandling and seamanship concepts (45min) - The intent of this quiz is to ground the students with the recognition that our shiphanding and seamanship training to date has been limited and that we collectively have the knowledge to fill - Brief 15-20min quiz with equal time for take up Part 2 – Explanation of Shiphandling and Seamanship Instruction Concepts (45min) - Instructors to introduce the various methods and approaches to learning shiphandling and seamanship to be employed in this course - Intro worksheets, shiphandling problems, shiphandling practical exercises <u>Activity 2 - Worksheets (15min/day x # number of students)</u> Each day students are assigned a 1x worksheet and associated during the last period of the day with time allotted for work. On a rotating basis, students will present the worksheet/topic the following morning, and the class will take it up Worksheet topics to be a prompt for the following day's NABS activities. Topics are sequenced by increasing difficulty/unfamiliarity (generally speaking) and are to include: 1. Alongsides/Departures 2. Shiphandling with Tugs 3. Anchoring 4. Mooring 5. RAS 6. Tow Approach 7. Heavy Weather (No associated NABS practical) 8. Confined Waters (No associated NABS practical – students to gain experience during Pilotage developmental) 9. Transiting Locks (No associated NABS practical – done at sea) <u>Activity 3 – Shiphandling Problems (15min/student)</u> Part 1 - Each student is assigned a unique scenario for berthing or slipping, and they are to research the situation and present a shiphandling plan that includes details on major considerations, risks/concerns, and employment of tugs if applicable. Part 2 - Students to present their shiphandling problem and solutions to the class and facilitate a short discussion/answer any questions from instructors or peers	**To be developed**

Overview	Objective(s)	Activity Details	Lessons Identified
		Activity 4 – NABS Practical 1. Alongsides/Departures- 2. Shiphandling with Tugs- 3. RAS 4. Anchoring (concurrent with 001.03 developmental runs) 5. Tow Approach 6. Buoy Approach (CPF Demo for blind area considerations, MCDV model) Activity 5: Alongsides and Departures (2 x 45min) Lecture to include choosing alongside position, accounting for enviromentals, identifying approach to be used, planning and execution Activity 6: Shiphandling with Tugs (2 x 45min) + (4x45min field trip) Part 1 – Lecture (2x45min) Part 2 – QHM Tour and Tugs Activity 7: Multiclass (5x45min) Assign readings prior to Free-play NABs Activity 8 – Shallow Water Effect (1 x 45min - *to be given before activity #14 RAS considerations) Discussion on the limitations of the new squat tables, and walkthrough of how to interpret and use them. EDIP Shallow water effects, concepts and exercise in comparing the Cb's and squat calcs of various classes of ships Activity 9: Guest Lecture (Heavy Weather) – Kevin Greenwood (3x45min) Activity 10: RAS Considerations (1x45min) Worksheet activity with PPT lecture Shallow water effect, force onset depth, load, trim, stability as a Supply Ship (JSS) Activity 11: Tow Approach and Towing – Guest Lecture Kevin Greenwood (1x45min) Lecture to include planning considerations, approach planning, execution, and follow on towing requirements Activity 12: Guest Lecture – NavArch (1 x60min)	

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="3"></td><td></td><td colspan="2">Ship design concepts

<u>Activity 13: Shiphandling at sea (4 x45min)</u>
Conduct shiphandling evolutions at sea, including each student executing an alongside and departure with control of the vessel. Recommend this occur in Dockyard or at Nanoose.

<u>Activity 14: Anchoring and the Anchor as a Ship Handling tool (45mins)</u>
Lecture on anchoring and use of anchor
- Types of anchorage, anchorage position selection, execution, formation anchorage,</td><td rowspan="3"></td></tr>
<tr><th>Why</th><th colspan="2">Resources</th></tr>
<tr><td>Need to convey accurate information IOT to keep the ship safe
High visibility activity</td><td>For student:
WECDIS Laptop with internet connection
Chart and publication outfit
Navigation tools
NSSC
Worksheets (provided by FNI)</td><td>For instructor:
WECDIS laptop
Projector
Worksheet DS solutions
Use of Tugs video

References: A1, A2, A4, A5, A6, A7, A9, A11, A13, A16, A18, A21, B1, B2, B3, B4, B5, B6, B7, B18, B19, B20, B21, C1, C2, C3, C8, C10, C16, C17, C18</td></tr>
</table>

LO 004.02.03A – Navigation and Marine Sciences (Meteorology)

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Direct **Media**: Laptop Projector PPTs OERs **Method**: SSP **Time**: 1x60min 3x45min 1x30min 1x15min (240min) **Environment**: Classroom	SWBAT 1. Advise command on Meteorology to include: a. Climate & met theory b. Storm avoidance c. Ocean currents d. Meteorological theory e. Impact of weather on operations 2. Where to find weather charts 3. Read and interpret weather charts 4. Deliver a Met Brief	<u>Activity 1 - Climate & Meteorological Theory Lecture (2x45mins)</u> PPT lecture covering various topics including: definitions, atmosphere, circulation, air masses, pressure, winds, fronts, clouds and fog. <u>Activity 2 – Severe Weather Lecture and Storm Avoidance Case Studies (guided discussion) (45min)</u> 11. PPT including various storm avoidance case studies. Note not all case studies need to be covered in class with the instructor, but PPT can be reviewed as homework to review the ones which are not discussed in class. <u>Activity 3 - Impact of Weather on Operations (30min)</u> 12. Students to spend 10minutes brainstorming a list of all the various impacts of weather on their job (operations, watchkeeping, transit plans, etc.) 13. Class to spend remaining time reviewing the brainstorming 14. The intent is to get students thinking about all the possible impacts of weather on their role as NavO (i.e. storm avoidance, seakeeping, fuel consumption, watchkeepers being unable to stand watch due to seasickness, etc.) <u>Activity 4: Demonstration of Met Brief (15mins)</u> 15. FNIs to deliver a sample Met Brief as the "Demonstrate" portion of EDIP learning. To be scheduled in the course anytime before developmental NABS runs start <u>Activity 5: Guest Lecture: Met Tech (with Sea-going experience) – Concurrent with LO 003.02.06 (60mins)</u> 17. Briefly review Met Tech role and responsibilities onboard ship (what they do, what they can provide, battle rhythm on ship, best briefing practices, available external supports (METOC, etc.) 18. Opportunity to ask a Met Tech any unresolved questions regarding met theory The first half of this activity is meant to be a quick review of Met Techs roles relating to Met onboard ship, but the primary focus is on being the DivO for Met Techs reviewing topics such as: 19. Career progression 20. Coursing/Training Performance reporting	**To be developed**
	Why	**Resources**	

Overview	Objective(s)	Activity Details		Lessons Identified
	Informs CEC 001.04 - Act as Duty Nav (assess the brief part) Informs CEC 004.02	**For student:** Pen/pencil Paper (notetaking)	**For instructor:** **References:** A1, A2, A4, A5, A6, A7, B1, C1, C3, C16	

LO 004.02.03B – Navigation and Marine Sciences (Tides)

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Interactive Experiential **Media**: PPT Total Tides software **Method**: IL Practical **Time**: 6 x 45min (270min) **Environment**:	SWBAT Advise command on tides & currents Use Total Tides software	<u>Activity 1 – Lecture with videos (4 x 45 min)</u> PPT lecture covering various topics including: 22. Tide theory 23. Equilibrium theory 24. Tidal rising forces 25. Interaction with Sun and Moon <u>Activity 2 - Total tides set-up and WECDIS integration (1 x 45 min)</u> 26. PPT lecture - "Tides and Currents, Creating a Current Graph using Excel" available as a guide however, this is a lead through (EDIP) activity 27. Instructor to lead students through setting up Total tides, including clearances, and integration into ECPINs <u>Activity 3 - Worksheet - Calculate tides and currents using Total Tides software (45min)</u> 28. Assigned as homework at the end of A1 and A2. 29. 45min period is dedicated to taking up the worksheet and answering any questions	
	Why	**Resources**	**To be developed**

Overview	Objective(s)	Activity Details		Lessons Identified
Classroom	Important to be accurate when advising command Ensure safety of the ship Informs CEC 004.02	**For student:** WECDIS laptop Total Tides Software DWAN Laptop with internet access	**For instructor:** **References:** A1, A12, A16, B1, B19, C1, C3	

LO 004.02.03C – Navigation and Marine Sciences (Astronomics)

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Direct Interactive **Media**: Laptop PPT **Method**: EDIP IL **Time**: 2 x 45min (90min) **Environment**:	SWBAT Advise command on Astronomics	<u>Activity 1 – Time Zones: Theory and Application (1 x 45min)</u> 30. PPT lecture covering topics such as: "the day," daylight savings time, time zones, dateline, converting time zones and some sample problems for Speed Required calculations. <u>Activity 2 – Lead through of Navigation Software (focusing on astronomy) (1 x 45min)</u> Instructor to lead students through Total Tides, NAVPAC and ECPINS 31. For Total Tides, topics to cover include: changing time zones, steps to getting astronomic data (sunrise, sunset, BA, etc.) 32. For NAVPAC, topics to cover include: setting up time zones, steps to getting astronomic data, etc 33. For ECPINS, topics to cover include: how to change time zones, how to use route calculator with respect to time zones	
	Why	**Resources**	

Overview	Objective(s)	Activity Details		Lessons Identified
Classroom	Important to be able to advise command on astronomic phenomena. Safety of vessel.	**For student:** WECDIS laptop DWAN laptop with internet access Celestial Navigation Software (NAVPAC) Total Tides Software	**For instructor:** WECDIS laptop DWAN laptop with internet access Celestial Navigation Software (NAVPAC) Total Tides Software **References:** A1, A12, A16, B1, B19, C1, C3	

LO 004.02.04 – Employment of Navigation Systems and Equipment

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Indirect **Media**: Independent study Student presenters **Method**: Independent study Presentations **Time**: 4 x 45min (180min) (Based on an ideal course load of 6	SWBAT Advise command on the employment of new navigation systems and equipment	<u>Activity 1 - Research Presentation (10min/student)</u> Students to choose a new piece of navigation equipment or navigation equipment used effectively by other organizations (that is not currently used by our org) presentation focus on employment of that equipment in the RCN, benefits, risks, cost, etc. student topic to be approved by FNI 30min presentation/student (5min present, 5min Q&A + debrief) <u>Activity 2 – Tour of WECDIS production facility</u> (3 x 45 min) Students to tour a WECDIS production facility in order to understand the newest features and capabilities being produced prior to their introduction to the fleet and have an opportunity to discuss any questions with WECDIS design and production experts. The preferred facility is OSI Ltd of Burnaby, BC.	
			To be developed
	Why	**Resources**	

Overview	Objective(s)	Activity Details		Lessons Identified
students @30min each) **Environment**: Classroom	Important to advise command with accuracy and potential for innovation	**For student:** WECDIS laptop DWAN laptop with internet access Chart and publication outfit	**For instructor:** Criteria sheet (what needs to be covered by the student) **References:** A1, A2, A3, A5, A6, A7, A9, B1, B2, B3, B4, B5, B6, B7, B8, C1, C3, C8, C13	
Overview	**Objective(s)**	**Assessment Details**		**Lessons Identified**

LO004.02.05 – Expectations and Role as NavO

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Interactive **Media**: Whiteboard **Method**: Brainstorm GD points **Time**: 2 x 45 min (90min) **Environment**: classroom	SWBAT • First day on the job, developing the command relationship • Clarifying expectations and role as NavO • Understanding the status quo	<u>Activity 1 – Road-mapping the course (2 x 45min)</u> Brainstorm– pull out your list of lessons learned and key points that was started at the beginning of the course "Roadmap" of the course (ocean) The instructor will draw the road (just a line or something more intricate or have the students draw their road (or make it an open ocean that they have to draw the buoys and nav aids along the way students write the highlights and memories of the course along the timeline All the considerations from speaking with command on the first day What can you fix that you hated as a SLt? Your navigation philosophy The big take-aways from the course Guided discussion and debrief (PPT to guide)	
			To be developed
	Why	**Resources**	

Overview	Objective(s)	Activity Details		Lessons Identified
	Important to develop a shared mental model with command asap Informs Advice Command PC	**For student:** Whiteboard Markers Notes from course	**For instructor:** Whiteboard Markers GD points **References:** nil	

CEC 004.01 – Shiphandling Theory Exam

Overview	Objective(s) Goal of the assessment	Assessment Details		Lessons Identified
Media: Paper **Method**: Written Exam (80%) **Time**: 120min **Environment**: Classroom	SWBAT • Demonstrate understanding of the fundamentals of shiphandling theory	**<u>Before the assessment (setup/preparation):</u>** 1. Review session on the day before the assessment 2. FNI will read invigilator notes to students and clarify any questions that arise **<u>During the assessment:</u>** 1. Students will follow invigilator notes as directed by FNI 2. No references are provided during the exam 3. Student will hand in exam paper to FNI and proceed to break area 4. Student will have 120 minutes to complete the exam **<u>Assessment feedback:</u>** 1. Feedback is given individually in the form of marks on the exam 2. FNI will take up questions as required		
	Impact and Limitations	**Resources**		**To be developed**
	Impact: 80% pass mark Failure = review and rewrite. Two consecutive failures = TRB **Limitations/Remarks**: Students will have no access to references (closed book)	**For student:** Exam Paper (provided by FNI) Pen/pencil Squat Tables	**For instructor:** Exam papers Answer key	

Overview	Objective(s) Goal of the assessment	Assessment Details		Lessons Identified
			References: A1, A2, A4, A5, A6, A7, A9, A11, A13, A16, A18, A21, B1, B2, B3, B4, B5, B6, B7, B18, B19, B20, B21, C1, C2, C3, C8, C10, C18	

CEC 004.02 – Navigation Sciences Exam

Overview	Objective(s)	Assessment Details		Lessons Identified
Media: Paper **Method**: Written Exam (80%) **Time**: 120min **Environment**: Classroom	SWBAT • Demonstrate understanding of navigation sciences, including: a. Meteorology b. Tides c. Geodesy d. Astronomics e. Celestial	**<u>Before the assessment (setup/preparation):</u>** 1. Review session on the day before the assessment 2. FNI will read invigilator notes to students and clarify any questions that arise **<u>During the assessment:</u>** 1. Students will follow invigilator notes as directed by FNI 2. No references are provided during the exam 3. Student will hand in exam paper to FNI and proceed to break area 4. Student will have 120 minutes to complete the exam **<u>Assessment feedback:</u>** 1. Feedback is given individually in the form of marks on the exam 2. FNI will take up questions as required		
	Impact and Limitations	**Resources**		**To be developed**
	Impact: 80% pass mark Failure = review and rewrite. Two consecutive failures = TRB **Limitations/Remarks**: Students will have no access to references (closed book)	**For student:** Exam Paper (provided by FNI) Pen, pencil, eraser Squat tables	**For instructor:** Exam papers Answer key **References:** A1, A2, A4, A5, A6, A7, A9, A11, A13, A16, A18, A21, B1, B2, B3, B4, B5, B6, B7, B18, B19, B20, B21, C1, C2, C3, C8, C10, C18	

PO 005 – Manage Navigation Systems

Details the performance standard is as follows:

PO 005

The navigator shall Manage Navigation Systems (FT005) IAW refs by;

a. Mitigating errors and faults;
b. Troubleshooting errors and faults; and
c. Liaise with other departments and organizations regarding navigation systems and equipment

Notes for planning next delivery of PO 005: nil

PC 005A – Demonstrate a Knowledge of Basic Navigation Systems

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Assessment Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Media:
Paper

Method:
Written Exam (80%)

Time:
120mins

Environment:
Classroom</td><td rowspan="2">The goal of this PC is to assess a student's ability to recall troubleshooting procedures and research details of basic navigation equipment

Written Exam to include:
Short/Long Answer
Diagrams to draw</td><td colspan="2"><u>Before the assessment:</u>
1. Students will ensure they have all required materials to write the exam, including verifying the setup and functionality of their computer, including access to references required for the exam. FNIs will provide a copy of the Exam to each student.

<u>During the assessment:</u>
1. Students to work independently through their exam. During the first 60 minutes of the exam, students will not have access to references.
2. During the last 60 minutes of the exam, students will have access to references. The FNIs will oversee the portion of the exam when access to references is granted.
3. Students can ask questions to FNIs.
4. FNI shall clarify question intents without providing the answer to the exam question.</td><td></td></tr>
<tr><td colspan="2"><u>Assessment feedback:</u>
Exam papers will be checked against the answer key. Exam papers with grading and comments will be returned to students. Review (take-up) of the exam will be conducted.</td><td>To be developed</td></tr>
<tr><th>Impacts and Limitations</th><th colspan="2">Resources</th><td rowspan="2"></td></tr>
<tr><td>Impact: If a student is unsuccessful on first attempt - review of graded exam/comments and course material as required. Automatic re-write granted. If a student is unsuccessful on re-write, then a TRB will be convened.

Limitations/Remarks:
- Students to work independently
- Troubleshooting scenarios will use technical manual of navigation equipment used in Minor War Vessels</td><td>For student:
Exam paper (provided by FNI)
Pen/pencil
Blank paper</td><td>For instructor:
Exam paper (IOT clarify questions)
Answer key for exam

References: A1, A2, A4, A5, A6, A7, B1, B2, B3, B4, B5, B6, B7, C1, C3, C18</td></tr>
</table>

PC **005B – Demonstrate a Knowledge of Complex Navigation Systems**

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Assessment Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Media:
Paper

Method:
Written Exam (80%)

Time:
90mins

Environment:
Classroom</td><td rowspan="2">The goal of this PC is to assess a student's ability to recall troubleshooting procedures and research details of complex navigation equipment

Written Exam to include:
Short/Long Answer
Diagrams to draw</td><td colspan="2"><u>Before the assessment:</u>
1. Students will ensure they have all required materials to write the exam, including verifying the setup and functionality of their computer, including access to references required for the exam. FNIs will provide a copy of the Exam to each student.

<u>During the assessment:</u>
1. Students to work independently through their exam. During the first 60 minutes of the exam, students will not have access to references.
2. During the last 30 minutes of the exam, students will have access to references. The FNIs will oversee the portion of the exam when access to references is granted.
3. Students can ask questions to FNIs
4. FNI shall clarify question intents without providing the answer to the exam question.

<u>Assessment feedback:</u>
1. Exam papers will be evaluated. Exam papers with grading and comments will be returned to students. Review (take-up) of the exam will be conducted.</td><td></td></tr>
<tr><td></td><td></td><td>To be developed</td></tr>
<tr><td>Impacts and Limitations</td><td colspan="2">Resources</td><td></td></tr>
<tr><td>Impact: If a student is unsuccessful on first attempt - review of graded exam/comments and course material as required. Automatic re-write granted. If a student is unsuccessful on re-write, then a TRB will be convened.

Limitations/Remarks:
- Students to work independently
- Troubleshooting scenarios will use technical manual of navigation equipment used in Major Warships</td><td>For student:
Exam paper (provided by FNI)
Pen/pencil
Blank paper</td><td>For instructor:
Exam Paper (IOT clarify questions)
Answer Key

References: A1, A2, A4, A5, A6, A7, B1, B2, B3, B4, B5, B6, B7, C1, C3, C18</td><td></td></tr>
</table>

EO 005.01 – Mitigate Errors and Faults

LO 005.01.01 – Preparation for Electronic Fixing Aid (EFA) Presentation

Overview	Objective(s)	Activity Details		Lessons Identified
Strategy: Independent **Media**: Computer with internet access **Method**: Explain, Demonstrate, Imitate (EDI) **Time**: 20min instructor presentation 4 days to conduct research and prepare a presentation (20min presentation) 120min for presentation based on the ideal course **Environment**: Classroom	**SWBAT:** • Describe the assigned navigation system in detail, to include: A) Basic Function B) Setup C) Employment D) Tuning E) Determining Fixed and Variable Errors **Teaching Points:** Requirements for EFA Lectures Where to find info for EFA lectures	<u>Activity 1– Instructor Example of EFA Presentation (20min)</u> Instructor to present an example EFA presentation (covering one of the systems/pieces of equipment not assigned to a current student) in order to provide students with a template of how they will do their independent EFA presentations 34. Instructor to assign EFA presentation topics upon conclusion of example presentation <u>Activity 2 – EFA Research</u> Students will perform independent research on the assigned EFA system from a list of suggested systems, including: A) Gyro Compass B) Inertial Navigation System C) Speed Logs D) Echo Sounder E) Distance Meters/Range Finding Equipment (including Sextant) F) WECDIS G) Radars and ARPA H) W-AIS I) GMDSS J) RF-DF/DSC Students will prepare EFA presentations during assigned windows and for homework Presentations will be scheduled later in the course Group feedback/Debrief from the instructor after each presentation		**To be developed**
	Why	**Resources**		
	Informing EC 005.01	**For student:** DWAN laptop with internet access Chart and publication outfit	**For instructor:** DS solutions **References:** A1, A2, A4, A5, A6, A7, B1, B2, B3, B4, B5, B6, B7, C1, C3, C18	

LO 005.01.02 – Describe Common Errors and Faults

Overview	Objective(s)	Activity Details		Lessons Identified
Strategy: Interactive **Media**: Paper **Method**: Brainstorm/Jigsaw Puzzle **Time**: 45min **Environment**: Classroom	**SWBAT:** Identify common errors and faults Associate common errors and faults with various systems and equipment **Teaching Points:** Common errors for Nav Systems and Equipment to include: Gyro Compass Inertial Navigation System Speed Logs Echo Sounder Distance Meters/Range Finding Equipment (including Sextant) WECDIS Radars and ARPA W-AIS GMDSS RF-DF/DSC	<u>Activity 1 – Brainstorming and Group Discussion (45min)</u> Students to be broken up into groups; each group will brainstorm as many common errors and faults as they can for various navigation systems and equipment, writing each down on a separate piece of paper (sticky note) Groups will then swap the collection of errors and faults which they brainstormed, and the other group will work to identify which system or equipment each error and fault is associated with After listing errors and faults and grouping them with associated systems and equipment, all students will work together and review each error and fault and discuss and describe: a) How to identify the error or fault; b) Possible causes; and c) Share real-world experiences of the error or fault		
	Why	**Resources**		**To be developed**
	Informing LO 005.01.01 Informing EC 005.01 Informing P0 001 and P0 004	**For student:** Pen/pencil/markers Sticky Notes Whiteboard Dry erase markers	**For instructor:** DS solutions of errors and faults **References:** A1, A2, A4, A5, A6, A7, B1, B2, B3, B4, B5, B6, B7, C1, C3, C18	

EO 005.02 – Troubleshoot Errors and Faults

LO 005.02.01 – React to Alarms

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Interactive **Media**: NABS **Method**: Role Play **Time**: 12 x 45min (540min) (Not including Activity 3 time) (Activity 1 time based on an ideal course load of 6 students) **Environment**: NABS Ship Alongside	**SWBAT:** • Set up bridge systems enabling alarms • Establish alarm parameters • Recognize or identify alarms • Proper initial reaction to an alarm **Teaching Points:** Basic set up for bridge equipment How to enable alarms on various systems How to change the alarm parameters Understanding policy and best practices on alarms and alarm parameters Familiarization with MWV and Major Warship bridge/Nav systems and equipment	<u>Activity 1 – Bridge Set-Up (4x60min)</u> Part 1 – Bridge Set-Up (1x45min) FNIs to demonstrate comprehensive bridge setup for NABs highlighting common mistakes Part 2 – Bridge Mess Up (1x60min) In groups, students complete a setup with mistakes making notes of any setup issues the team has introduced. After 30 mins, teams will switch and attempt to identify the errors from the other teams. Part 3 – React to Alarms (2x60min) Coastal transit with errors induced by control – students to react and rectify issues, brief command, identify solutions and mitigation strategies. Debrief on completion <u>Activity 2 – Minor War Vessel Tour of Bridge/Navigation Systems (2 x 45min)</u> LO 003.02.05 Conduct an NAI will be grouped in with this LO <u>Activity 3 – Pilotage Runs in NABS/At Sea (in conjunction with PO 001)</u> During NABS and At Sea Pilotage Runs, the instructors will inject system errors and faults, causing the students to have to react to the alarms in the midst of a pilotage run <u>Activity 4 – Major Warship Tour of Bridge/Navigation Systems (4 x 45min)</u> LO 003.02.05 Conduct an NAI will be grouped in with this LO Topics to ensure students are shown: • GPS Crypto loading and taught how to verify if crypto is loaded • How to verify which INS is selected to feed NDDS • Shown where the INS (gyros) are located, discuss kalmin filter,	**To be developed**
	Why	**Resources**	
	Informing PO 001 and PO 004 Informing EC 001.04 Informing LO 005.02.02 Students need to know how to set up the bridge properly and enable alarms/change alarm parameters.	**For student:** Bridge set up checklist/equipment set up checklist NAI Checklist **For instructor:** NAI Checklist Tour guidebook of MWV and Major Warship bridge/nav systems and equipment **References:** A1, A2, A4, A5, A6, A7, B1, B2, B3, B4, B5, B6, B7, C1, C3, C18	

LO 005.02.02 – Conduct Troubleshooting

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Direct **Media**: Computer, Projector/Smartboard **Method**: Guided Discussion **Time**: 4 x 45min (180min) **Environment**: Classroom	**SWBAT:** • Describe advanced troubleshooting techniques for WECDIC errors or faults • Describe basic troubleshooting techniques for navigation systems and equipment, including: - Gyro Compass - Inertial Navigation System - Speed Logs - Echo Sounder - Distance Meters/Range Finding Equipment (including Sextant) - WECDIS - Radars and ARPA - W-AIS - GMDSS - RF-DF/DSC **Teaching Points:** -WECDIS troubleshooting tips and tricks - How to handle a sextant and determine errors (correctable and non-correctable) - Applying corrections to sextant - Complex Nav Systems overview (exposure piece)	<u>Activity 1 – WECDIS Advanced Troubleshooting Lecture (2 x 45min)</u> Instructor to provide presentation and lead guided discussion on WECDIS advanced troubleshooting Students are given a WECDIS problem which is solvable using the WECDIS Help Function, and each will conduct troubleshooting to address the error <u>Activity 2 – Sextant Errors (45min)</u> Students to determine sextant errors and correct any correctable errors conducted alongside LO regarding "Use Marine Sextant." <u>Activity 3 – Guest Lecture: Senior Sonar FMF Presentation (60min)</u> FNIs to provide parameters to Senior Sonar about points to cover	**To be developed**

Overview	Objective(s)	Activity Details		Lessons Identified
	Why	**Resources**		
	Informing PO 001 and PO 004 Informing EC 001.04 Students will be required to conduct troubleshooting onboard ship during day-to-day and pilotage activities	**For student:** Chart and publication outfit Sextant	**For instructor:** WECDIS advanced troubleshooting presentation Sextant **References:** A1, A2, A4, A5, A6, A7, B1, B2, B3, B4, B5, B6, B7, C1, C3, C18	

LO 005.02.03 – Describe Material Baseline Standard (MBS) Requirements

Overview	Objective(s)	Activity Details		Lessons Identified
Strategy: Interactive/Independent **Media**: Computer with internet access, references **Method**: Interactive Learning – Collaborative Problem Solving/Written submission **Time**: 45min Activity 2 - 1 day (homework) Activity 3 – Ongoing PD reading **Environment**: Classroom/Home	**SWBAT:** • Be able to identify navigation system and equipment specific MBS requirements • Conduct a risk assessment and provide mitigation strategies **Teaching Points:** What is MBS Where to find details regarding Nav specific MBS requirements What to do when MBS not met	<u>Activity 1 – Research and Guided Discussion (45min)</u> Class will break into groups and work together to research MBS requirements (nav systems and equipment specific) using NAVORD ref for various ship classes. Students will then have a guided discussion with FNIs about risk mitigation when MBS requirements are not satisfied. (Not about how to risk mitigate each specific item but what risk mitigation is, how to assess risk, discuss risk and mitigation strategies) Students will draw a card indicating a system or piece of equipment and conduct research and a short presentation about how they would discuss the issue with the CO. <u>Activity 2 – Risk Assessment and Mitigation (Homework)</u> 2. Students will be assigned a nav system or equipment deficiency leading to MBS requirements not being satisfied and have to produce a risk assessment and provide mitigation strategies <u>Activity 3 – Ongoing Homework (PD Readings)</u> BR45(8) - Chapter 1, Chapter 2 (sections 1-2), Chapter 7 Ecpins Operator Manual "Appendix F "vessel shape" CFCD 130 "ecpins setup checklist CFTO – Halifax Class Offests CFTO – MCDV Offsets ECPINS Checklist		**To be developed**
	Why	**Resources**		
	Informing PO 001 and PO 004 Informing EC 001.04 Students need to know about MBS requirements regarding Nav systems and equipment Risk Assessments drafted by NavO on MWV and serves to inform for ORO role writing risk assessments	**For student:** Chart and publication outfit Risk Assessment template	**For instructor:** Computer Projector **References:** A1, A2, A4, A5, A6, A7, B1, B2, B3, B4, B5, B6, B7, C1, C3, C18	

EO 005.03 – Liaise with Other Departments and Organizations Regarding Navigation Systems and Equipment

LO 005.03.01 – Internal Reporting of System Errors and Faults

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Strategy:
Interactive

Media:
NABS

Method:
Role Play

Time:
20mins

#2 and #3: In conjunction with PO 001, developmental and assessment runs

Environment:
Classroom
NABS
At Sea</td><td rowspan="2">SWBAT:
Report System Errors and Faults Internally

Teaching Points:
Fights you'll have on ship with various personnel</td><td colspan="2"><u>Activity 1 – Guided Discussion on Working with CSE/MSE (30min)</u>
1. FNI guided discussion on how to interact and work with various ships personnel regarding Nav Systems and Equipment. UCRs, SOCD, EC, and OPDEFs to be covered (introduced but students not expected to be able to draft these)

<u>Activity 2 – NABS Runs</u>
1. As students are conducting pilotage runs in the NABS and instructors inject system and equipment errors and faults, the student is to simulate immediate reporting of the problem to techs while maintaining SA on the bridge and keeping the ship safe. (Not just FIRE AND FORGET, but manage the real-time reporting of the issue, documentation and briefing to the CO)

<u>Activity 3 – At Sea Runs (Pilotage/Duty Nav)</u>
2. Similar to Activity #2 but At Sea and involving the full ship's team (techs, department heads, etc.)

Needs to be noted on NSSC</td><td rowspan="2">To be developed</td></tr>
<tr><th>Why</th><th colspan="2">Resources</th></tr>
<tr><td>Informing PO 001 and PO 004
Informing EC 001.04
Important to understand internal reporting procedures</td><td>For student:
WECDIS laptop
DWAN laptop with internet access
Chart and publication outfit
Navigation tools</td><td>For instructor:
PPT
Playbook

References: A1, A2, A4, A5, A6, A7, B1, B2, B3, B4, B5, B6, B7, C1, C3, C18</td><td></td></tr>
</table>

LO 005.03.02 – Re-Verify Functionality of Systems and Equipment

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Strategy:
Experiential

Media:
Systems/Equipment Technicians

Method:
Concurrent learning throughout the course

Time:
Concurrent with other activities

Environment:
Any</td><td>Follow up with technicians

Monitor repair timelines and fight for equipment to be returned to service</td><td colspan="2">Note: Students will be continually assessed on their ability to follow up on repairs to systems and equipment (during NABS and At Sea), notably during Duty NavO assessments.</td><td></td></tr>
<tr><td></td><td></td><td></td><td>To be developed</td></tr>
<tr><th>Why</th><th colspan="2">Resources</th><td></td></tr>
<tr><td>Important to follow up with technicians to ensure that what you asked for has been done.

Overseeing is part of the NavOs responsibilities</td><td>For student:
Nil</td><td>For instructor:
Nil
References: A1, A2, A4, A5, A6, A7, B1, B2, B3, B4, B5, B6, B7, C1, C3, C18</td><td></td></tr>
</table>

LO 005.03.03 – Engage External Agencies/Resources for Support

Overview	Objective(s)	Activity Details		Lessons Identified
Strategy: Interactive **Media**: Laptop and Projector **Method**: Guided Discussion **Time**: 14 x 45 min (630min) **Environment**: Classroom	**SWBAT:** Identify which external agencies can assist with which type of problems Know the general procedures for interacting with/reporting with external agencies/resources **Teaching Points:** What external agencies and resources exist, including: HSO CHS FMF LCMM MCTS ISSC How to contact and work with an external agency	<u>Activity 1 - Instructor-led guided discussion (45min)</u> Students will brainstorm available external agencies/resources. The instructor will fill in any that are missed As a class, discussion to follow on what type of assistance each external agency/resource can provide. Instructor to inform students on how to engage/proper reporting procedure for each identified external agency/resource <u>Activity 2 – Visits to External Agencies</u> (13 x 45 min) Students will visit several External agencies in order to form relationships with other maritime stakeholders and gain an appreciation for how their organizations work and what resources they offer the RCN/navigators. Preferred visits include: a) A voyage on a large commercial vessel to include a period of "SSD" equivalent a transit of Active Pass, in order to examine the differences in civilian and RCN BRM practices. Recommended tours include a transit of Active pass on the bridge of a BC ferry or a voyage to/from the pilot stations at Victoria and Vancouver or Nanaimo. b) A visit to a civilian marine academy, in order to understand the differences in civilian marine education and the RCN training system. Recommended tours include BCIT Marine Campus in North Vancouver. c) A visit to a pilotage authority or pilot association in order to understand better what it is that commercial pilots offer to the navigator and the differences in coastal navigation practices in the civilian industry and the RCN. Recommended tours include the Pacific Pilotage Authority in Vancouver. d) A visit to a Canadian Coast Guard station, in order to learn about SAR response capabilities and procedures as well as what other resources the CCG can offer the Government and RCN. Ideally, a tour at sea in a coast guard vessel will be conducted to better examine the differences in CCG and RCN practices. Recommended tours include CCG station Sea Island.		**To be developed**
	Why	**Resources**		
	Important to know who to turn to for assistance outside of the ship	**For student:** Pen/pencil Paper (notetaking)	**For instructor:** Laptop Whiteboard Dry erase markers **References:** A1, A2, A4, A5, A6, A7, B1, B2, B3, B4, B5, B6, B7, C1, C3, C18	

EC 005.01 – Navigation Systems Presentation

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Assessment Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Media:
WECDIS Laptop
DWAN laptop
with internet access
Projector

Method:
Research and Present

Time: 180min
4days to research and create a 20min presentation
X 6 students

Environment:
Classroom and at home</td><td rowspan="2">The goal of this EC is to assess the student's ability to conduct research and provide a brief on a basic navigation system used in HMC ships and vessels</td><td colspan="2"><u>Before the assessment:</u>
1. FNI will provide a prompt to the student regarding a navigation system that the student must conduct research about and provide a brief on.

<u>During the assessment:</u>
1. FNIs will be available as a resource, and students can ask questions throughout the research phase. The final product submitted will depend on the type of system the student is tasked to research but could include ECPINS, RADAR, AIS, Echo Sounder, Speed Logs, VHF/RF-DF/DSC. Students will give a 20min presentation to the class and be open to questions, comments and discussion.

<u>Assessment feedback:</u>
1. FNIs will participate in the question and discussion period following the student's presentation. FNIs will review and provide feedback on written submissions.</td><td></td></tr>
<tr><td colspan="2"></td><td>To be developed</td></tr>
<tr><td></td><td colspan="2"></td><td rowspan="2">DS solutions systems/equipment
Criteria for presentation</td></tr>
<tr><td>Impacts and Limitations</td><td colspan="2">Resources</td></tr>
<tr><td>Impact:
Formative feedback. Possibility for the student to have to re-submit work if the information is not IAW references.

Limitations/Remarks: Nil</td><td>For student:
Criteria for the presentation
Prompt from FNI
WECDIS laptop
DWAN laptop with internet access
Chart and publication outfit
Navigation tools</td><td>For instructor:
DS Solutions for each system or piece of equipment

References: A1, A2, A4, A5, A6, A7, B1, B2, B3, B4, B5, B6, B7, C1, C3, C18</td><td></td></tr>
</table>

PO 006 – Manage the navigation aspects of the Maritime Battlespace

Details the performance standard is as follows:

PO 006

The Fleet Navigating Officer shall manage navigation aspects of the maritime battlespace (T00010) IAW the references, by:

a. Performing the duties of OOW during Action Stations; and
b. Providing navigation support for operations.

Notes for planning next delivery of PO 006

PC 006A – Manage navigation aspects of the Maritime Battle Space (MWVNO)

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Assessment Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Media:
Paper
WECDIS Laptop

Method:
Written Exam with WECDIS component (80%)

Time:
120min

Environment:
Classroom</td><td>• The goal of this PC is to determine the student’s ability to create operational navigation products and support basic operational planning through the proper use of WECDIS.

Written Exam will include:
Various questions with a scenario of NavO on deployment.
Simulated provision of support to operations.</td><td colspan="2"><u>Before the assessment:</u>
Students will ensure that their WECDIS laptop is functioning and available during the exam. FNIs will provide a copy of the exam to each student.

<u>During the assessment:</u>
Students will work independently throughout the exam. Use of WECDIS and references are authorized. The exam will have a scenario built-in, with answers and WECDIS products required to assist the student’s notional unit through an operational area. Questions may be posed of an FNI to mimic ORO-level support.

<u>Assessment feedback:</u>
Exam papers and any WECDIS files will be submitted to the FNIs. Exams will be graded, with a section of comments on the WECDIS support provided. A review of the exam will be conducted.</td><td></td></tr>
<tr><td></td><td></td><td></td><td>To be developed</td></tr>
<tr><td></td><td>Impact and Limitations</td><td colspan="2">Resources</td><td></td></tr>
<tr><td></td><td>Impact: Detailed review and re-write. An automatic re-write is provided. A second failure results in a TRB IAW NPTORD 4500-54.

Limitations/Remarks: This exam allows students access to references that they would reasonably have with them at sea. Assistance can be asked of an FNI, but peer assistance is not allowed.</td><td>For student:
Exam paper (given by FNI)
Pen/pencil
Blank Paper
WECDIS laptop
Chart and publication outfit
Support FNI mimicking ORO</td><td>For instructor:
Exam marking guide
WECDIS Laptop

References:
A1, A2, A4, A5, A6, A7, A8, B1, B4, B7, B9, B10, B11, B12, B13, B14, B15, B16, B17, C2,</td><td></td></tr>
</table>

PC 006B – Theoretical Knowledge of Tactics and Doctrine (FNO)

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Assessment Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Media:
Paper

Method:
Written Exam
(80%)

Time:
90min

Environment:
Classroom
(VCR required for access to secret space)</td><td rowspan="2">• The goal of this PC is to assess the knowledge of Tactics and Doctrine</td><td colspan="2"><u>Before the assessment:</u>
Students will ensure they have all the required materials to write the exam. FNIs will provide a copy of the Exam to each student.

<u>During the assessment:</u>
Students are to work independently through their exams. During the first 60 minutes of the exam, students will not have access to references.

During the last 30 minutes of the exam, students will have access to references. The FNIs will oversee the portion of the exam when access to references is granted.

Students can ask questions to FNIs. FNI shall clarify question intents without providing the answer to the exam question.

<u>Assessment feedback:</u>
Exam papers will be checked against the answer key. Exam papers with grading and comments will be returned to students. Review (take-up) of the exam will be conducted, and exams will be re-collected and destroyed/stored to maintain security.</td><td rowspan="2">To be developed</td></tr>
<tr><th>Impact and Limitations</th><th colspan="2">Resources</th></tr>
<tr><td>Impact: Detailed review and re-write. An automatic re-write is provided. A second failure results in a TRB IAW NPTORD 4500-54.

Limitations/Remarks: This exam allows students access to references that they would reasonably have with them at sea. Assistance can be asked of an FNI, but peer assistance is not allowed.</td><td>For student:
Exam paper (given by FNI)
Pen/pencil
Blank Paper</td><td>For instructor:
Exam marking guide

References:
Classified references</td><td></td></tr>
</table>

EO 006.01 – Perform the Duties of Action Stations OOW (FNO)

LO 006.01.01 – Take the Watch (FNO)

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Experiential **Media**: DWAN/CSNI Computer Projector Printed case studies Printed problem scenarios **Method**: Interactive PPT Lecture Interactive Activity - Case Study Interactive Activity – Collaborative Problem Solving Learner back briefs Instructor debriefs **Time**: 2 x 45 mins (90min) **Environment**: Classroom	SWBAT • Develop a Bridge comms plan; and • React to an action alarm; **Activity Objectives** • By completing Activity 1 -Case Study, learners will be able to develop a Bridge communications plan • By completing Activity 2 – Collaborative Problem Solving, learners will be able to know how to react to an action alarm	<u>Activity 1 – Individual Case Study Overview – Develop a Bridge Comms Plan (45 mins)</u> • Learners will respond to a bridge scenario description by examining the facts of the case, critically analyze them, develop a Bridge comms plan and then conduct a back brief to the rest of the class. This activity will facilitate learner to content, learner to learner, and learner to instructor interactions to enhance the learning experience. **Instructor Instructions:** 1. Present the activity instructions to the class and answer any questions from learners; 2. Hand out the case studies to learners and tell them when to start; 3. Monitor activity by responding to any learner questions, providing time warning(s) and observing whether all learners are ahead of time given; 4. Ask for volunteers to back brief and/or select students; 5. Project the corresponding scenario for each student back brief on the projector, so the rest of the students understand the scenario; 6. After the completion of a back brief, ask students to provide feedback and then provide instructor feedback, including DS solution 7. After the completion of all back briefs, conduct an activity debrief by asking students: "What are your tips and tricks to develop a Bridge comms plan?" **Student Instructions:** 1. Read the instructions; 2. Examine and analyze the facts and incidents of the case study; 3. Develop a bridge comms plan with the following information: a. How are you going to communicate to the Bridge team and take charge of the watch? b. Which networks will you punch into, where on the bridge, and why? c. What do you do if you lose communications on the bridge? 4. Conduct a 5 mins back brief back to the class. <u>Activity 2 –React to an Action Alarm (1x45 mins + afternoon on FFH) (Classroom first, linked with doing it on the bridge of a FFH)</u> • Learners will work in pairs to solve a problem based on an action scenario such as a sub-surface, surface or air threat. Teams will critically analyze the facts of the problem and come up with a solution and then conduct a back brief to the rest of the class. This activity will facilitate learner to content, learner to learner, and learner to	**To be developed**

Overview	Objective(s)	Activity Details		Lessons Identified
		instructor interactions to enhance the learning experience. (What are you thinking about as a NavO when you hear that action alarm – getting to the bridge, taking the watch) **Instructor Instructions:** 1. Present the activity instructions to the class and answer any questions from learners; 2. Pair learners into teams of two; 3. Hand out the scenario problems to each group and tell them when to start; 4. Monitor activity by responding to any learner questions, providing time warning(s) and observing whether groups are ahead of time given; 5. Ask for group volunteers to back brief and/or select groups; 6. Project the corresponding problem scenario for each group back brief on the projector, so the rest of the students understand the problem; 7. After the completion of a back brief, ask students to provide feedback and then provide instructor feedback, including DS solution; and 8. After the completion of all back briefs, conduct an activity debrief by asking students questions directly related to the problem itself. **Student Instructions:** • Read the instructions; • Examine and analyze the facts of the problem scenario; • Develop a solution to the problem; and • Conduct a 5 mins back brief back to the class.		
	Why	**Resources**		
	Preparation for EC 006.01 practical assessment Preparation for PC 001	**For student:** Activity instructions Case study Problem scenario	**For instructor:** PPT Lecture Activity instructions Case studies Problem scenarios Debrief questions DS solution **References:** A1, A6, A8	

Overview	Objective(s)	Activity Details		Lessons Identified

LO 006.01.02 – Tactics and Doctrine (FNO)

Overview	Objective(s)	Activity Details		Lessons Identified
Strategy: Interactive Instruction **Media**: DWAN and CSNI computers Projector Printed scenario Whiteboard or flipcharts **Methods**: Interactive PPT lecture Collaborative Problem Solving Interactive PPT lecture Brainstorming and Guided Discussion **Time**: 14 x 45 min (630min) **Environment**: Classroom - Secret	SWBAT 1. Respond to Threats; 2. React to Damage; 3. Execute Counter-Measures; 4. Enable fundamentals of Action Stations OOW 5. Provide support to operations **Activity Objectives** To become familiar with various references related to tactics and doctrine. To provide a brief overview of NavOs role in operations and support to operations.	<u>Activity 1: Tactics and Doctrine Introductory Lecture (1 x 45 min)</u> ORO qualified/FNO background instructor preferred - Introductory lecture covering the role of the NavO/importance of NavO role to operations - Exposure to available references, including where to find classified refs - Where to find operational planning information and how this relates to the NavO role <u>Activity 2: The Bridge at Action Stations - Tactics and Doctrine (6x45min)</u> Part 1 – Interactive lecture going through various topics regarding NavO role when the ship is at action stations. Topics to include: Patrolling sectors, Anti-ship Missile Defence, Defence against aircraft, TCMs, Naval Gunfire Support, Mine detection and transit, target motion analysis Part 2 – Classified pub crawl/self-study of references related to the topics covered in part 1 <u>Activity 3: NavO support to Operations(7x45min)</u> Part 1 – Interactive lecture going through various topics regarding NavO role in supporting operations Topics to include: message drafting, vessel interaction and hails, queries and warnings, GPS jamming/spoofing, use of AIS in warfare, surveillance and patrol, first visual chance of detection, NAVWAR, deceptive lighting, and RCN Risk Assessment Part 2 – Classified pub crawl/self-study of references related to the topics covered in part 1		**To be developed**
	Why	**Resources**		
	Preparation for EC 006.01 practical assessment Preparation for PC 001 Create a baseline of knowledge prior to skills development	**For student:**	**For instructor:** PPT Lecture Activity instructions Problem scenarios Debrief questions DS solution Post-deployment reports on C2D2E **References**: A1, A2, A4, A5, A6, A7, A8, B1, B3, B7, B9, B10, B11, B12, B13, B14, B15, B16, B17	

LO 006.01.03 – BRM during Action Stations (FNO)

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Strategy:
Experiential
Interactive

Media:
CMS 330 (actual or emulated);
Whiteboard
Flipchart

Method:
Interactive lecture
Practical

Time:
2 x 45 min
(90min)

Environment:
Combat Trainer and Classroom (SECRET)
Note that this could be delivered aboard a Halifax Class ship

TPts not listed due to classification</td><td>SWBAT
• Operate the CMS to include role selection and threat pages;
• Discuss the importance of the visual and navigation picture during action.

Activity Objectives:

By completing Activity 1, Operate CMS, students will have confidence in the set-up of the CMS .
By completing Activity 2, Open discussion, students will be able to apply BRM principles as Action Stations OOW.</td><td colspan="2"><u>Activity 1 - Operate CMS</u> (3x45min)
This activity is to follow Lo 006.01.02 A2, where they review the material prior to putting it into practice during this LO.

Students will go through set-up procedures in the Combat Trainer, with guidance from instructors.

FNI instructions:
Arrange Security brief to students regarding SEACOT (separate from N92a SDA)
Guide students into SEACOT
Orient the students to the operations room set-up
Students will be seated individually or in pairs at a CMS station
Talk the students through set-up and threat pages

<u>Activity 2 - BRM in action</u> (2x45) (unclass) concurrent 006.01.01 A2
Command, Control, Charge during Action
Liaison with EOOW, Ops, being the eyes of the ship
Inputs coming to the bridge (to inform 006.01.01 A1 bridge comm plan)

1x45min to take place under direct action stations watch 006.01.01 (unclass)
1x45min to take place during FFH tour on the bridge of FFH</td><td></td></tr>
<tr><td></td><td></td><td></td><td>To be developed</td></tr>
<tr><td></td><td>Why</td><td colspan="2">Resources</td><td rowspan="2"></td></tr>
<tr><td></td><td>Prep for EC 006.01</td><td>For student:
Set-up instructions</td><td>For instructor:
DS solution with security brief

References:
A1, A2, A4, A6, A7, A8</td></tr>
</table>

EO 006.02 – Provide Navigation Support to Operations

LO 006.02.01 – NavO Role in Operations other than Pilotage (MWVNO)

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Strategy:
Interactive learning

Media:

DWAN computer
WECDIS
Projector
Whiteboard
Flipcharts
Paper

Method:
Jigsaw

Time:
2 x 45 min
(90min)

Environment:

Classroom - UNCLAS</td><td>SWBAT

Describe navigation considerations pertaining to:
NEO/HADR;
- OGD Support incl FISHPAT/SARZONE;
SAR Response;
TTW/CZ;
MIO;
Small boat ops;
Force protection;
Info ops

Activity Objectives:
Students will be able to describe the considerations in navigation support as well as know the key references and OPIs.</td><td colspan="2"><u>Activity 1 – Guided Discussion – OGD Support (1x45min)</u>
- Discuss Role of NavO in Ops other than pilotage including: NEO/HADR/OGD support/TTW/MIO

<u>Activity 2 – Nav War Lecture and Guided Discussion (1x45min)</u>
- What is NavWar, how to conduct it, why it is important, discuss examples and refs
- FNO learning objectives regarding Nav War can be amplified during Ops Phase when in a secret discussion space as required (i.e. deceptive lighting)</td><td></td></tr>
<tr><td></td><td></td><td></td><td>To be developed</td></tr>
<tr><td></td><td>Why</td><td colspan="2">Resources</td><td rowspan="2"></td></tr>
<tr><td></td><td>Prep for PC 006
This is an mportant aspect of your job at sea.</td><td>For student:
Jigsaw card
Paper
Pen/pencil</td><td>For instructor:
Jigsaw Activity
DS Solution with key TPs for each aspect

References:
A1, A2, A3, A4, A5, A6, A7, A10, B1, B3, B4, B5, B6, B7, B15, B16, C11</td></tr>
</table>

LO 006.02.02 – Contributing to Intelligence Collection and Joint Fires (MWVNO)

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Strategy:
Direct/Interactive Lecture

Media:
CSNI Computer Projector

Method:
Lecture

Time:
2 x 45 min
(90min)

Environment:
Classroom – SECRET

TP not listed due to classification</td><td>SWBAT

• Describe expected interactions with intelligence collection, including regulations (where to find them) and resources.

• Describe navigation support to Joint Fires Planning.</td><td colspan="2"><u>Activity 1 - Lecture with Interactions (1x45 min) (MWVNO)</u>
Students will brainstorm what types of intelligence and what sorts of situations they may be asked as Navigators to collect or interact with intelligence. This will be followed by a guided discussion/lecture on regulations and the resources available. This will include data recording and transfer, IntReps, Port Intel reports and Trinity resources.

FNI Instructions:
35. Present the activity
36. Monitor the activity,

Student Instruction:
37. Do the activity!

<u>Activity 2 - Lecture (1x45 min) (MWVNO)</u>
Students will receive a lecture on Joint Fires processes and the expected interaction in the planning and execution stages of a Joint Fires tasking or mission. Particular attention was given to how spotting works and calling corrections for fall of round.</td><td></td></tr>
<tr><td></td><td></td><td></td><td>To be developed</td></tr>
<tr><td></td><td>Why</td><td colspan="2">Resources</td><td rowspan="2"></td></tr>
<tr><td></td><td>To prep for the role as FNO</td><td>For student:</td><td>For instructor:
Whiteboards or flip charts
Markers

References:
A1, A2, A6, A7, A8, B3, B4, B6, B7, B10</td></tr>
</table>

LO 006.02.03 – Creating Operational Navigation Products (MWVNO)

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="4">Strategy:
Interactive

Media:
DWAN computer with WECDIS
Projector
AML CD

Method:
EDI – Activity 1
Lecture – Activity 2
EDI – Activity 3

Time:
4 x 45 min
(180min)

Environment:
Classroom UNCLAS
Outside</td><td>SWBAT:

Create Waterspace management products in WECDIS

Teaching Points:
Creation of – moving havens, 4W grid, Op Areas, TTW, NAVWARNS, Screens, Operator Contacts, AMLs, WECDIS as CMS.

Survey teaching points as per BR 45</td><td colspan="2"><u>Activity 1 – Waterspace management production WECDIS Lead-through (2 x 45 min)</u>.

In class lead through to include:
- Moving havens
- 4W grid
- Op Areas
- TTW
- NAVWARNS
- Screens
- Operator Contacts
- AMLs
- WECDIS as a CMS

<u>Activity 2 – Intro to Hydrographic Survey (1 x 45 min)</u>
Lecture “Conducting a Minor Survey.” Intro to principals and practice

<u>Activity 3 – Practical Lead through (3 x 45 min)</u>
FNI leads students through a ‘minor survey’ on the WorkPoint Footprint (Land-based survey) in prep for EC 006.03. This adds in additional practice in the use of a marine sextant (practical).</td><td rowspan="4">To be developed</td></tr>
<tr><th>Why</th><th colspan="2">Resources</th></tr>
<tr><td>In prep of PC 006A
To support EC 006.04 (survey)</td><td>For student:
WECDIS Laptop
Sextant,
Norries Tables
Portable Echo- Sounder
Portable GPS</td><td>For instructor:
WECDIS laptop
AML CD

References:
A1, A4, A5, A6, A7, A15, A19, B1, B3, B6, B7, C1, C3, C4, C5, C18</td></tr>
</table>

EC 006.01 – Produce a Survey (MWVNO)

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Assessment Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="3">Media:
Paper and WECDIS

Method:
Practical

Time: 480min
240 min survey
240 min create a product

Environment:

Outdoor/Classroom</td><td rowspan="2">To confirm the student's ability to complete a minor survey and confirm proper use of a marine sextant.</td><td colspan="2"><u>Before the assessment:</u>
1. FNIs will give the students a discussion of the classification of the exam

<u>During the assessment:</u>
1. The student will conduct a minor survey as a group. The task will be framed by a scenario. A crusty FNI wants to win the Shadwell Prize and, while sipping their overly filled coffee, asks the students to demonstrate their skills so he can borrow their work. The students will survey a given area and selected objects, to include soundings, within a given footprint. Measurement will include VSA/HSA to confirm sextant proficiencies. The product will be a small (100x by 100x) chart correction on WECDIS, and will be conducted IAW BR45 (1) Ch 17 Minor Survey.

<u>Assessment feedback:</u>
1. After the student group completes the survey, the FNIs will review the chart correction. The chart correction will be submitted to HSO for printing and display</td><td rowspan="2">To be developed</td></tr>
<tr><th>Impact and Limitations</th><th colspan="2">Resources</th></tr>
<tr><td>Impact: Formative feedback will be given upon marking. No re-attempts are required.

Limitations/Remarks: The students will be expected to share the workload and to remain in a given footprint to complete a survey of the indicated area.</td><td>For student:
GPS datum/mark
Sextant
Blank paper
Stopwatch
Pen/pencil
Ruler
Lead Line or hand-held depth sounder
WECDIS laptop</td><td>For instructor:
DS Solution

References:
A1, A4, A5, A6, A7, A15, A19, B1, B3, B6, B7, C1, C3, C4, C5, C18</td></tr>
</table>

Part 3 - Assessment Checklist for PC 001A/B & CEC 001.05 (V4.1)

Assessment Checklist for PC 001A/B & CEC 001.05 (V4.1)

Navigator: Lt(N) A B C D E F	FNI: Lt(N)	Captain:	Control: NavO/CO
SOA: SR: Traffic Density:	From: To: Via:	Day/ Night Vis: U/R Restricted: x	
Date: Planning Time: hrs		Run:	Developmental/PCA/PCB/CEC **1 2 3 4** / 5 6 7

Planning	**Fail**	**Pass**	**Track**	°	°	°	°	°	°	°	°	°	°	°	⚓
Brief/Research			Track Assessment	LM PI EP CPA	LM PI EP CPA	LM PI EP CPA	LM PI EP CPA	LM PI EP CPA	LM PI EP CPA	LM PI EP CPA	LM PI EP CPA	LM PI EP CPA	LM PI EP CPA	LM PI EP CPA	10 ☐
LDL			Fixing	‘	‘	‘	‘	‘	‘	‘	‘	‘	‘	‘	9 ☐
Clearing Methods			Shoot Ups/OB/CB/Tx	SU OB CB TX	SU OB CB TX	SU OB CB TX	SU OB CB TX	SU OB CB TX	SU OB CB TX	SU OB CB TX	SU OB CB TX	SU OB CB TX	SU OB CB TX	SU OB CB TX	8 ☐
Route			Speed Management ETA: _____ Arrival: _____	TG SP	TG SP	TG SP	TG SP	TG SP	TG SP	TG SP	TG SP	TG SP	TG SP	TG SP	7 ☐ 6 ☐ 5 ☐ 4 ☐
Notebook			Safety Methods	CB SI CD CR	CB SI CD CR	CB SI CD CR	CB SI CD CR	CB SI CD CR	CB SI CD CR	CB SI CD R	CB SI CD CR	CB SI CD CR	CB SI CD CR	CB SI CD R	3 ☐ 2 ☐
Bridge Setup	**Dev**	**Sat**	Cable Countdown	5 3 1 RAI	5 3 1 RAI	5 3 1 RAI	5 3 1 RAI	5 3 1 RAI	5 3 1 RAI	5 3 1 RAI	5 3 1 RAI	5 3 1 RAI	5 3 1 RAI	5 3 1 RAI	1 ☐ Fix ☐
Plan/Brief Comments:			Turn Assessment	QB PI LM 1/2	QB PI LM 1/2	QB PI LM 1/2	QB PI LM 1/2	QB PI LM 1/2	QB PI LM 1/2	QB PI LM 1/2	QB PI LM 1/2	QB PI LM 1/2	QB PI LM 1/2	QB PI LM 1/2	____ x

Equipment Losses and Checks:

☐GPS
☐RIO
☐Radar
☐DSL
☐Gyro Error
☐Gyro
☐Radar
☐SDM
☐ Gyro Check
☐ Radar Azimuth Check
☐ Radar Index Check
☐_____
_____

Navigation Routine Comments:

Anchorage/Shiphandling	
Plan/Position Selection	
Countdown/Approach	
Speed Plan	
Track Assessment	
Final Position/Advice	

Collision Avoidance:						
Application of RoR						
Integration with Navigation						

Comments: *If an alongside, refer to separate PC 004 marking sheet. Alongsides/Departures are not evaluated during PCs or CECs.*	*Contact avoidance comments:* ☐ Multi-Contact ☐ Narrows

Evaluation	**Score** 0-4 (UNSAT, NI, Acceptable, Good, Very Good)	**Comments** *The candidate must achieve a minimum cumulative score of 18 in order for the run to be assessed as a pass at the FNO level and 14 at the MWVNO level. Max Score is 28. Any assessment of "UNSAT" results in a failure of the run for performance.*
Planning and Briefing		
Navigation Routine & Speed Management		
Contact Avoidance and Traffic Management		
Navigation Safety, Situational & Spatial Awareness		
Use of Safeties and technical skills		
Advice to Command		
Use of Bridge Team and BRM		
Automatic Safety Failure Criteria **Failure of any of these training objectives constitutes an immediate failure of the run.*		

☐ Unsafe planning (LDL, Safety Methods, Violation of Regulation or Sailing Directions, Unsafe Anchorage) ☐ Entered a safety method without briefing ☐ Violated ColRegs without brief ☐ Placed ship into danger or Collision/Grounding ☐ Presented risk to the ship unnecessarily ☐ Unable to assess ship's position (within 50x) or dangerously incorrect assessment without taking corrective action ☐ Recommendation that would have placed ship into danger (within 2min at sea) ☐ Failure to arrive within +/- 2min of updated ETA, (Max 8min from planned arrival time) ☐ Instructor determined failure:	**Overall Score** UNSAT Needs Improvement Acceptable Good Very Good

PASS/SAT ☐ Criteria For Evaluation not met	FAIL/DEV	☐ You are hereby warned that your performance as a Navigator has, thus far, been unsatisfactory and you must improve to meet the minimum standard. You are hereby issued the following formal caution.	Student: FNI: Snr Eval:

Annex L – TEL Plan

Application	COA 1 TEL Delivery	COA 2 Interim Delivery	Impact if TEL not implemented - Realized benefits - E/W Coast Delta	Priority
PIP	DLN Delivery – with key academic foundational knowledge. Assessments. DLN PIP, including experiential PIP similar to what already exists (more focused) with MITE as a prerequisite for loading on FNO course.	Google Classroom to make learning objects available to students & current PIP with Threshold Knowledge Exam	The current PIP is outdated and requires reworking for validity and valuable preparation for the course. Updated QSP does not provide time to refresh or review material delivered in NWO core training. Current PIP does not provide any academic refresher for students in advance of the course and will therefore result in an increased student failure rate.	High
PO 004	Planetarium/Celestial Navigation Simulator (NFS owned) – requires purchase Can use sextant effectively within	Field Trip to the Dominion Observatory (Planetarium) - off-site trips create costs and possible danger to students -cannot use sextant inside this planetarium	Unable to develop students' practical celestial navigation skills when not at sea. The Dominion Observatory is useful for concepts; however, sextants cannot be used. The planetarium simulator would provide for conceptual as well as practical training Sextant used in survey PO; however, without the simulator, students will not have the opportunity to take a celestial fix and therefore are not fully prepared for their role as a navigator at sea.	Med
PO 001 CEC 001.04	Duties of the Duty NavO – video clips? Or a day in the life of Duty NavO	Best practices discussion or handout	The TEL option would best provide students with full SA on the Duty NavO role.	Low

Application	COA 1 TEL Delivery	COA 2 Interim Delivery	Impact if TEL not implemented - Realized benefits - E/W Coast Delta	Priority
			The video clips could also be used effectively as recruiting tools.	
PO 001 PO 004	Six to eight small powerboats to be used for practical experience in shiphandling, berthing, slipping, tow approaches. Location: Esquimalt Power Boat Association	Simulator time and sea phase, which is costly.	Benefit: more hands-on practice. Simulator modelling is not accurate to reflect risk and more realistic training. Consequences are more immediate and impactful. Students learn more with hands-on experience with real water, actual environmental conditions and responsibility for lives and equipment. Would be used for initial shiphandling training for NWOs as well.	Low

Annex M – Target Population Analysis & Training Establishment Details

<u>Key Target Population Details</u> – Fleet Navigation Officer Candidates

"The learners' level of skill and knowledge with respect to the performance objectives (POs) is a factor which depends mainly on previous related training or education, and experience. It may also be affected by strong opinions or attitudes about the subject matter. Information should be obtained on the learners" -CFITES Vol 4

Characteristic	Target population details
General data	
Course capacity	10 max, 6 optimal, 4 min (even numbers ideal)
Expected annual throughput	10 annually (2 courses/year, amount of ships etc)
Trade(s)	Military, NWO, PRes (non-NOPQ qual), RegF, Submariners (non-BWK qual) Diverse experience, Foreign Military
location(s) of the intended learners	Work Point (CFB Esquimalt) - NFSP, Collier Bldg Not East Coast – weather, geography, coast does not lend itself well to pilotage, availability of training resources (Orca)
Access to computers (on the job)	Yes, critical resource, DWAN account/access, internet access, USB (protected A) WECDIS – installed on laptop
Average age	19 - 30ish, average age is about 24 over last 4 courses
Learning skills and preferences	
Competencies (general)	Dedication, hardworking, focused, perseverance, resilience, responsibility, honesty, integrity, team player, trustworthiness, decisive
Reading and writing level	Reading - high/advanced level, writing – generic military administration (officer – CAFJOD level) – English – references and practical execution of the job is in English - Standard
Preferences/attitudes towards a specific type of learning material and equipment (i.e. TEL, aversion to lectures)	Hands-on learning, simulator and at sea Technical – university level Inclination toward self-learning

Characteristic	Target population details
Educational level	University Education – bachelor level at min & ideal, due to amount of self-study and analysis High school minimum (Reservists mostly – no NOPQ), CEOTP
Previous military training experience	NOPQ ideal, BWK? MWV cap if no NOPQ?
Current subject-matter competence	
Experience with target performance	NWO II, III, IV for all FNO = NWO, NOPQ + 6mo BWK MWV = BWK ideal PIP sets stage with a level of exposure for both
Ability to perform any part of the POs	Between basic and intermediate Only a few topics are new; other topics are just higher level
Knowledge of the subject matter (and how it was acquired)	NWO II, III, IV for all Get the PIP coming in Experiential learning (would have seen a Navigator at work in their previous work environments)
Perception of the impact of achieving the POs on self, work or career	Not a career breaker, will delay your career (another D level course will need to be completed) – this is a major career milestone FNO failures, no ill-will, within the culture self-impact high
Job/career stage	Director level course (part of progression DP2 training -> ORO -> promotion to rank of LCdr
Affective Characteristics	
Interests	The sea! Job as a hobby, inherently interested in Naval History, Captains, Mariner Movies, Reading, Lean more toward academics, hardworking and dedicated, will put job before self, Mariner, Leader, (Warrior), a tendency to overwork
Attitude towards subject matter/ learning (academic self-concept)	Arrive very passionate about the subject matter, keen, respect for the importance, treated with the responsibility mindset it deserves (WIIFM), don't have to sell the importance of the job "the navy needs me."

Characteristic	Target population details
Anxiety level	High to medium (culture of excellence) Groomed for this, respect and responsibility
Social Characteristics	
Relationship to peers	Good relationship with peers, worked together before, very small community (NWOs and BWKs) Only Live-in vs. local resident of Victoria
Feelings toward authority	Healthy level of respect of navigator to instructors, captains and experience, usually instructors are same rank level
Tendencies toward cooperation or competition	Healthy level of cooperation & competition on course, and post-qualification higher level of competition
Other	
Fitness level	Average – not overly one way or the other
Health status	Fit Sea Mental health – healthy recognition of limits

<u>Describing the learning environment</u> – NFS (division)

1. What are the characteristics of the teachers/trainers who will be using these materials?
<u>Instructors on East Coast</u>: N/A <u>Instructors on West Coast</u>: *What are the interests and preferences of the instructors?* The sea! Job as a hobby, inherently interested in Naval History, Captains, Mariner Movies, Reading, Lean more toward academics, hardworking and dedicated, will put job before self, Mariner, Leader, (Warrior), a tendency to overwork *How do instructors feel about having instruction delivered via media or other non-traditional methods?* Openness to new and innovative training delivery, different media, AIT, future capabilities *Do instructors like to use technology-based instruction?* Yes. Games? *What experience have instructors had with technology-based instruction?* Simulator use, videos, 3-D animations, practical instruction on the ship (hands-on)
2. Are there existing courses/programs into which this piece of instruction must fit?
If so, what is the strategy used in these materials? Pre-Course: NWOiv (Performance Based), NOPQ (ATR), BWK - trust/culture Requirement for navigator on a ship Feeds into ORO (academic & performance-based – CBT - simulator, sea-phase), then direct feed to CDC (RegF – self-study, CBT - performance-based, simulator, sea-phase, competency-based), No effect on follow-on courses Requirement for Submariner along with Sonics trg (CanSubFor) - AILV? Submariner req'mt
3. What hardware is commonly available in the potential learning environments?
Are computer workstations available? DWAN Computers, Simulators (NABS), NPTT, Smart Boards, Projectors, WECDIS-enabled laptops, Sextants, Bridge Set-Up, Star Globes, All classes of ship
4. What are the characteristics of the classes and facilities that will use the new instruction?
Collier Bldg (dedicated FNO classroom), Simulators, Available Classes of Ship, Outdoor, K-Sim (planetarium (*Stellarium) capability) - indoor sextant use (NABS capability required)

Annex N – Pre-Instruction Package (PIP)

FLEET NAVIGATING OFFICER (FNO)
PRE-INSTRUCTION PACKAGE (PIP)

FNO Candidate's Name: _____

FNO PIP

Revisions

Version	Revision Date	Remarks
PIP Version 3.0	25 Jun 2020	Formatting & Reading List Update

Foreword

1. This package is intended to provide potential Navigating Officers with an introduction to selected topics as well as an outline of the necessary bridge experience prior to D-level navigation training. The material and subjects outlined in this package are aimed at consolidating navigation skills and providing students with the necessary exposure to ensure a successful course.
2. Completion of the PIP is one of the pre-requisites for the FNO course. Candidates must have a signed and completed PIP when they arrive at NFS(P).
3. Any questions or suggestions for amendments to this package may be directed to the Navigation Section (250-363-7335).

Introduction

4. In principle, this package is designed to provide direction to potential navigation students and COs prior to a student arriving at VENTURE Division. For the student, it is intended to list objectives that are key to success on the course. After early identification of a prospective student by the CO, the PIP also provides a rough plan for the necessary exposure to pilotage and shiphandling and other navigation-related topics prior to the FNO course. Some of the requirements in this package have been taught previously in NWO III or IV as well as in the NOPQ ATRs. It is essential that prospective students ensure that they are familiar with all of the items. Much of the content in this PIP will be examined either on the first day of the course as a Threshold Knowledge Test or later in a Performance Check.
5. This PIP is divided into two parts: Part One contains the list of Enabling Objectives with their perspective sign-off levels, and Part Two contains the Specifications for each of the Enabling Objectives. Part two should be used by both the student and sign-off authority to ensure that the subject has been reviewed and researched sufficiently.

Conduct

6. The Ship or Unit should issue this document at least 6 months prior to proceeding on course. Ideally, this package should be completed while posted to a ship; however, this does not preclude individuals from working on elements of the package at any time. It is recommended that Commanding Officers review students' progress at suitable intervals to ensure that they are progressing in their preparations for the course.
7. Commanding Officers are strongly recommended to allow prospective students the opportunity to conduct some consolidation and, in particular, navigation passages in either a frigate or destroyer. Statistics have shown a clear indicator that students with frigate or destroyer navigation opportunities have a much higher rate of success on the course.

Commanding Officer's Review

Date	Signature	Remarks

Reading List

A. CFCD 130 – Canadian Navigation Manual

B. ATP 1 Vol II Ch. 1/4/5/6/7/8

C. ATP 16 Ch. 3

D. BR45 (Vol 1) – Principles of Navigation

E. BR45 (Vol 2) – Astro-Navigation

F. BR45 (Vol 8) WECDIS, Charts and Publications

G. BR 67 – Admiralty Manual of Seamanship: Ch 2/4/5/6

H. Harbour Control Instructions

I. BWK Handbook

J. Your Ships Navigation Data Book

Part 1: Practical Training Requirement Checklist

PTR	Item	Remarks	Initials	Date DD/MMM/YY	OPI
PTR 001 - Use the Sextant					
001.01	Calculate the errors present in a marine Sextant				NavO
001.02	Measure the bearing and altitude of celestial bodies				NavO
PTR 002 - Navigation Equipment					
002.01	Describe bridge equipment				NavO
002.02	Operate Navigation Equipment				NavO
002.03	Operate GMDSS Equipment, including the use of DSC, EPIRB and SART				NavO
002.04	Describe RCN ECDIS policy				NavO
PTR 003 - Rules of the Road					
003.01	Review Collision Regulations				NavO
003.02	Review "A guide to the Collision Avoidance Rules" by Cockroft				NavO
003.03	Discuss implications of 1. Rule 17 2. Rule 19				CO
PTR 004 - Interpret and Report Meteorological Information					
004.01	Use basic meteorological information/Online (DL) Met Course	FNIs will distribute Online Met PIP Course ahead of MWVNO/FNO start to candidates identified for course loading. If a candidate wishes to complete this training ahead of time, they are to contact the FNI for access to the Met content			Online/ Submit certs to FNI
004.02	Provide a MET brief to the CO				Met Tech/ NavO/CO
PTR 005 - Navigation					
005.01	Understudy the NavO in the preparation of a Coastal Passage				NavO

PTR	Item	Remarks	Initials	Date DD/MMM/YY	OPI
005.02	Understudy the NavO in the preparation of a Pilotage Plan				NavO
005.03	Execute 5 Pilotage passages				NavO/CO
005.04	Discuss the Canadian Buoyage and Aids to Navigation System				NavO
005.05	Review Canadian and US Tides				NavO
005.06	Research SHINNADS				NavO/CSE
005.07	Act as ORO during Pilotage Passages				OpsO/NavO
PTR 006 - Shiphandling					
006.01	Understudy the NavO for the following Evolutions: (emphasis on Navigation Issues)				NavO
006.01a	Coming to a Buoy				
006.01b	Towing				
006.01c	Anchorages				
006.01d	Replenishment at Sea (RAS)				
006.02	Conduct as a minimum 1 of each of the following evolutions:				NavO/CO
006.02a	Berthing/Unberthing				NavO/CO
006.02b	Buoy Approach				NavO/CO
006.02c	Replenishment at Sea (RAS)				NavO/CO
PTR 007 - Navigation Admin					
007.01	Review Ship's Navigation Outfit				NavO/NAV YEO
007.02	Describe Maintenance of Navigation Records				NavO
007.03	Draft Routine Navigation Messages				NavO
007.04	Conduct Chart Corrections				NavO
PTR 008 - Commanding Officer's Forums					
008.01	In a Forum format discuss the following items:				CO

PTR	Item	Remarks	Initials	Date DD/MMM/YY	OPI
008.01a	Rules of the Road WRT Navigation				
008.01b	Shiphandling Characteristics				
008.01c	Berthing/Unberthing				
008.01d	Turning in a Confined Space				
008.01e	RAS Approaches				
008.01f	Towing				
008.01g	Acceleration Deceleration				
008.01h	Windage/Drift Rate				

Part 2: Practical Training Requirement Specifications

PTR 001 Use a Sextant

001.01 Calculate the Errors Present in a Sextant

1. Performance: Calculate the Errors Present in a Sextant
2. Conditions:
    a. Given:
        (1) Class A or B sextant
        (2) Assistance
    b. Denied: N/A
3. Standard: The officer shall demonstrate the ability to correctly use a marine sextant, including:
    a. Describe the errors a sextant can have; and
    b. Calculate and correct those errors that are correctable
4. Remark: Students can measure terrestrial or celestial angles
5. References: Admiralty Manual of Navigation – Vol 2

PTR 001 Use a Sextant

001.02 Measure the Bearing and Altitude of Celestial and Terrestrial Bodies

1. Performance: Measure the Bearing and Altitude of Celestial and Terrestrial Bodies
2. Conditions:
    a. Given:
        (1) Class A or B sextant
        (2) NavPac PC
        (3) Assistance
    b. Denied: N/A
3. Standard: The officer shall:
    a. measure the sextant altitude of a celestial body as well as its true bearing;
    b. use NavPac PC to identify the heavenly body taken; and
    c. measure the vertical and horizontal sextant angles of terrestrial objects
4. Remark: The NavO must show the officer how to utilize NavPac PC.
5. References: N/A

PTR 002 Navigation Equipment

002.01 Describe Bridge Equipment

1. Performance: Describe Bridge Equipment
2. Conditions:
    a. Given: Bridge equipment
    b. Denied: Assistance
3. Standard: The officer shall describe in detail all bridge navigation equipment, including:
    a. Principles of operation;
    b. Capabilities and limitations;
    c. Environmental effects on capability;
    d. Errors – how to calculate and correct for them;
    e. Maintenance
4. Remark: The officer should focus primarily on the class of vessel they are assigned to; however, any knowledge of other equipment (specifically FFH and DDH) will assist them on the FNO course.
5. References: N/A

PTR 002 Navigation Equipment

002.02 Operate Navigation Equipment

1. Performance: Operate Navigation Equipment
2. Conditions:
    a. Given: Navigation equipment
    b. Denied: Assistance
3. Standard: The officer shall demonstrate to a proficiency level 3 (the level of proficiency required to independently and correctly perform duties and tasks) the use of the following equipment:
    a. Radar;
    b. GPS/INS;
    c. Range finding Equipment (Weymouth Ross, Stuarts Distance Meters);
    d. Three Arm Protractor;
    e. Speed and Distance Log;
    f. Echo Sounder
4. Remark: The officer should comfortably use the above equipment without supervision.
5. References: Equipment operator manuals; CFCD 130; CFITES Vol 3

PTR **002** **Navigation Equipment**

002.03 Operate GMDSS Equipment

1. Performance: Operate GMDSS Equipment
2. Conditions:
    a. Given:
        (1) GMDSS equipment; and
        (2) Assistance
    b. Denied: N/A
3. Standard: The officer shall perform the following functions:
    a. Initiate and receive a DSC call to another vessel;
    b. Describe how to send an emergency call via DSC;
    c. Describe how to activate and test an EPIRB;
    d. Activate a SART; and
    e. Identify a SART signal on radar.
4. Remark: The officer should use the above equipment with minimal supervision.
5. References: Equipment manuals

PTR 002 Navigation Equipment

002.04 Describe CF ECDIS Policy

1. Performance: Describe CF ECDIS Policy
2. Conditions:
    a. Given:
        (1) SDM equipment;
        (2) References; and
        (3) Assistance
    b. Denied: N/A
3. Standard: The officer shall describe and demonstrate the application of CF policy with respect to the setup and use of SDM to include:
    a. Fixing;
    b. Security;
    c. Types and usage of electronic charts;
    d. Feature sets;
    e. Safety settings;
    f. System failure responses
4. Remark: The officer should use the above equipment with minimal supervision.
5. References: Equipment manuals

PTR 003 Rules of the Road

003.01 Review Collision Regulations

1. Performance: Review Collision Regulations
2. Conditions:
    a. Given:
        (1) References;
        (2) Assistance.
    b. Denied: N/A
3. Standard: The officer shall review the necessary references with the aim of being able to interpret the following:
    a. International Regulations for Preventing Collisions at sea;
    b. Canadian modifications;
    c. U.S. Inland rules;
    d. Allied naval vessels rules;
    e. local rules;
    f. Annual Notice to Mariners; and
    g. routing systems and VTMS regulations.
4. Remark: The officer may choose to use the Rules Master software to assist with studying. A Threshold Knowledge Test will be administered on the first day of the course. The officer is expected to pass this TKT in order to demonstrate their preparedness for the course.
5. References: Canada Shipping Act – Collision Regulations, Canada Shipping Act; Annual Notice to Mariners; ATP 1(D); US Coast Guard – Navigation Rules – International and Inland

PTR 003 Rules of the Road

003.02 Review "A Guide to the Collision Avoidance Rules" by Cockroft & Lameijer

1. Performance: Review "A Guide to the Collision Avoidance Rules" by Cockroft & Lameijer
2. Conditions:
    a. Given: References
    b. Denied: Assistance
3. Standard: The officer shall review the collision regulations and the court case examples stated in the reference and discuss them with the Navigating Officer.
4. Remark: A Threshold Knowledge Test will be administered on the first day of the course. The officer is expected to pass this TKT in order to demonstrate their preparedness for the course.
5. References: A Guide to the Collision Avoidance Rules – Cockroft & Lameijer

PTR 003 Rules of the Road

003.03 Discuss Implications of Rules 17 and 19

1. Performance: Discuss Implications of Rules 17 and 19
2. Conditions:
    a. Given: N/A
    b. Denied: Assistance
3. Standard: The officer shall discuss with the Commanding Officer the interpretation of Rule 17 – Action by Stand-on Vessel; and Rule 19 – Conduct of Vessels in Restricted Visibility.
4. Remark: The knowledge from this EO will be examined in the Threshold Knowledge Test on the first day of class.
5. References: Canada Shipping Act – Collision Regulations

PTR 004 Interpret and Report Meteorological Information

004.01 Use Basic Meteorological Information

1. Performance: Use Basic Meteorological Information
2. Conditions:
    a. Given: Online (DL) Met Course access
    b. Denied: Assistance
3. Standard: Understand the basic principles of meteorology as covered in the Online (DL) Met Course. The officer shall be capable of understanding and explaining the following:
    a. The characteristics of the atmosphere, to include:
        (1) Vertical distribution of temperature;
        (2) Pressure systems;
        (3) Atmospheric pressure;
        (4) Pressure terminology;
        (5) Characteristics of high- and low-pressure systems; and
        (6) Pressure gradient force;
    b. The characteristics of wind, to include:
        (1) Wind terminology
        (2) Wind scale
    c. Fronts and frontal development, to include:
        (1) Air masses;
        (2) Stability;
        (3) Movement;
        (4) Weather associated with the passage of a warm or cold front, to include:
            (a) Temperature;
            (b) Visibility;
            (c) Clouds;
            (d) Wind; and

(e) Pressure;

(5) Weather associated with the passage of a cold front;

(6) Frontal waves; and

(7) Occluded fronts;

d. Cloud types and precipitation to include:

(1) Stratiform, cumuliform, nubiform and cirriform;

(2) Dew point;

(3) Lifting process to include:

(a) Orographic;

(b) Frontal;

(c) Convection; and

(d) Convergence.

e. Types of fog and how they are formed, to include:

(1) Sea/Advection;

(2) Radiation;

(3) Upslope/Orographic;

(4) Precipitation/Frontal; and

(5) Arctic Sea Smoke/Steam;

f. The formation of tropical depressions/tropical storms and hurricanes;

g. Weather and wind warnings;

h. Hazards to shipping in the vicinity of a hurricane;

i. Define sea and swell;

j. Factors affecting wave height, to include:

(1) Wind speed;

(2) Duration; and

(3) Fetch.

k. Factors that lead to a decrease in swell, to include:

(1) Angular dispersion;

(2) Differing speeds; and

(3) Opposing winds.

l. Ice formation and the effects of freezing spray.

4. Remark: The above material will be examined throughout the online Met Course and all certificates are to be submitted to the FNIs. Further testing of Met material will take place as part of the Threshold Knowledge Test, given on day one of the FNO course, and throughout the course in support of other POs.

5. References:

a. Admiralty Manual of Navigation, Vol 2

b. NWO III Handbook, Chapter 5

c. Venture Div Meteorological Videos (NISP or from FNIs)

d. Online Met Course content

e. NP 100 Mariners Handbook

PTR **004** **Interpret and Report Meteorological Information**

004.02 Provide a Meteorological Brief to the CO

1. Performance: Provide a Meteorological Brief to the CO
2. Conditions:
    a. Given: Surface analysis and prognosis charts for a given area
    b. Denied: N/A
3. Standard:
    a. Interpret a surface analysis chart to include:
        (1) Frontal systems;
        (2) Winds speed and direction;
        (3) Precipitation; and
        (4) Temperatures.
    b. Interpret a surface prognosis chart to include:
        (1) Movement of frontal systems;
        (2) Forecasted winds;
        (3) Forecasted temperatures; and
        (4) Forecasted precipitation
    c. Provide a brief to the Commanding Officer and navigation team. The brief is to follow the current Sea Training format and is to include (as a minimum):
        (1) Synopsis
        (2) Forecast (12, 24 and 36 hours)
        (3) Current weather
            (a) Wind direction and velocity
            (b) Sea and swell
            (c) Cloud cover and type
            (d) Precipitation, if any
            (e) Sea surface temperature
            (f) Sea survival time for a person overboard

4. Remark: The knowledge from the above material will be examined as part of the Threshold Knowledge Test, given on day one of the FNO course. The practical portion will be used during both the academic and sea phases.

5. References:

   a. Admiralty Manual of Navigation, Vol 2

   b. NWO III Handbook

   c. Marine Weather Observing (MANMAR)

PTR 005 Navigation

005.01 Understudy the NavO in the preparation of a Coastal Passage

1. Performance: Understudy the NavO in the preparation of a Coastal Passage
2. Conditions:
    a. Given:
        (1) Assistance;
        (2) A chart outfit; and
        (3) Coastal passage data
    b. Denied: N/A
3. Standard: The officer shall :
    a. Assist in the planning of a coastal passage to include the following factors:
        (1) times of departure and arrival;
        (2) fog, low visibility and bad weather - the effects and implications on the plan;
        (3) ship's routing and traffic separation schemes;
        (4) narrow channels, choke points and traffic density;
        (5) operations or exercises;
        (6) speed, endurance and economical steaming;
        (7) clearance from the coast and under keel clearance;
        (8) territorial limits;
        (9) selection of largest scale, best scale to cover the route;
        (10) production of the passage graph;
        (11) preparation of the plan;
        (12) preparation of signalled reports; and
        (13) preparation of the navigation notebook.
    b. explain the factors involved in the execution of a coastal passage to a Command qualified NWO officer
4. Remark: N/A

5. References: CFCD 130 – Canadian Navigation Manual; Admiralty Manual of Navigation Vol 1; Ship's Standing Orders; NP 100 – Mariner's Handbook; Tide and Current Tables.

PTR 005 Navigation

005.02 Understudy the NavO in the preparation of a Pilotage Plan

1. Performance: Understudy the NavO in the preparation of a Pilotage Plan
2. Conditions:
    a. Given:
        (1) Chart outfit;
        (2) Ship characteristics;
        (3) References; and
        (4) Assistance.
    b. Denied: N/A
3. Standard: The officer shall assist in the planning of a pilotage passage to include the following factors:
    a. Research references;
    b. Choice of ETA;
    c. Limiting danger line;
    d. Track selection;
    e. Clearing bearings;
    f. Wheel over bearings;
    g. Other planning factors;
        (1) Shoot ups;
        (2) Gyro and equipment checks;
        (3) Point of no return;
        (4) Wind direction;
        (5) Echo sounder policy;
        (6) Min expected depths;
        (7) Emergency anchorage policy;
        (8) Special parties;
        (9) Machinery states;
        (10) Ceremonial;

(11) Calling in points; and

(12) Briefs;

h. Notebook;

i. Anchoring/mooring and buoy work;

j. Berthing/unberthing; and

k. Radar navigation.

4. Remark: N/A

5. References: Charts; Sailing directions; MARPIF; Fleet and Port Guides; Tide and current tables; List of lights; ALRS; Navigation data book; Nautical almanac

PTR 005 Navigation

005.03 Execute 5 Pilotage Passages

1. Performance: Execute 5 Pilotage Passages
2. Conditions:
    a. Given:
        (1) A DDH/FFH/MCDV/Orca;
        (2) References; and
        (3) Supervision.
    b. Denied: Assistance
3. Standard: The officer shall use the preparation techniques provided in EO 006.02 and execute a minimum of 5 pilotage passages. The passages should include:
    a. Passage termination at an anchorage (can be virtual) or a buoy;
    b. A mixture of visibility to observe the transition between states;
    c. At least 3-day passages and 2-night passages;
    d. A combination of SDM and paper;
        (1) SDM passages should include a loss of GPS input, requiring manual fix inputs;
    e. A loss of equipment (gyro, radar, GPS); and
    f. Use of radian rule/mental math.
4. Remark: The officer's chances of success on the course is greatly enhanced with increased numbers of passages.
5. References: As per EO 006.02

PTR 005 Navigation

005.04 Discuss the Canadian Buoyage and Aids to Navigation System

1. Performance: Discuss the Canadian Buoyage and Aids to Navigation System
2. Conditions:
    a. Given: N/A
    b. Denied: Assistance
3. Standard: The officer shall describe all aids to navigation used in the Canadian Aids to Navigation System, including their colour, flashing characteristics (if any), their numbering or lettering scheme and their purpose.
4. Remark: N/A
5. References: TP 968(E) – The Canadian Aids to Navigation System

PTR 005 Navigation

005.05 Review Canadian and US Tides

1. Performance: Review Canadian and US Tides
2. Conditions:
    a. Given:
        (1) References; and
        (2) Assistance
    b. Denied: N/A
3. Standard: The officer shall solve tide and tidal stream problems using Canadian and US tabular methods to include:
    a. Extract tidal information for reference stations to include:
        (1) Height of tide at a specific time;
        (2) Time at a specific height; and
        (3) Tidal stream information.
    b. Calculate tidal information for secondary ports to include:
        (1) Reference port;
        (2) Mean or large tides;
        (3) Application of secondary port corrections;
        (4) Change in Chart Datum in a river; and
        (5) Calculate the velocity of current.
    c. Determine the daily predicted tidal set using the Current Atlas part A;
    d. Represent daily tide and tidal stream information in graph format as follows:
        (1) Maximize the scale of the graph;
        (2) Properly label x and y-axis;
        (3) Title;
        (4) Hashing off pertinent timings; and
        (5) Use of the French curve.

e. Determine vertical clearance over a submerged object and under a fixed, elevated object;

f. Determine specific times that are safe to transit over or under an object; and

g. Calculation to an accuracy of:

(1) height +/- 0.5 meters;

(2) time +/- 5 minutes; and

(3) velocity +/- 0.5 knots.

4. Remark: The Canadian and US tides portion of the course is no longer instructed during the FNO course. This material is taught during NWO III/IV, and all FNO students will be examined on the subject as part of the Tides Performance Check. British Admiralty tides will continue to be taught in class.

5. References: Canadian and US Tide and Current Tables, Current Atlas

PTR 005 Navigation

005.06 Research SHINNADS

1. Performance: Research SHINNADS
2. Conditions:
    a. Given:
        (1) SDM system;
        (2) References; and
        (3) Assistance.
    b. Denied: N/A
3. Standard: The officer shall describe the following:
    a. SHINNADS Dual Military system, including the following:
        (1) The hardware components (core and peripheral);
        (2) Optional components;
        (3) Sensor inputs (types and selection);
        (4) Backup power sources;
    b. Electronic charts, including:
        (1) Raster versus vector;
        (2) Types of charts available and hierarchy IAW 130; and
        (3) IMO compliancy.
    c. System set up, including:
        (1) User accounts and security;
        (2) Approved feature sets;
        (3) Own ship data;
        (4) Voyage Data Recording settings;
        (5) Adjusting radar image overlay;
        (6) Safety limits, alarms and warnings; and
        (7) Working with contacts (AIS and ARPA)
4. Remark: N/A

5. References: CFCD 130 – Canadian Manual of Navigation; Admiralty Manual of Navigation – Vol 8.

PTR 005 Navigation

005.07 Act as ORO during Pilotage Passages

1. Performance: Act as ORO during Pilotage Passages
2. Conditions:
    a. Given:
        (1) Ship at sea;
        (2) SDM; and
        (3) Supervision.
    b. Denied: N/A
3. Standard: The officer shall conduct a minimum of 2 pilotage passages as ORO
4. Remark: The emphasis for this requirement is to become familiar with the expected level of support to the Navigating Officer from the Operations Room during a pilotage passage.
5. References: CFCD 130 – Canadian Manual of Navigation

PTR 006 Shiphandling

006.01 Understudy the NavO for Evolutions

1. Performance: Understudy the NavO for Evolutions
2. Conditions:
    a. Given:
        (1) A ship at sea;
        (2) Assistance; and
        (3) Supervision.
    b. Denied:
3. Standard: The officer shall understudy a minimum of one of each of the following evolutions, focusing on the navigation and shiphandling aspects:
    a. Coming to a buoy;
    b. Towing;
    c. Anchorage; and
    d. Replenishment at Sea
4. Remark: This PTR should be done in conjunction with PTR 008 – Commanding Officer's Forums
5. References: Admiralty Manual of Navigation – Vol 6

PTR 006 Shiphandling

006.02 Conduct Seamanship Evolutions

1. Performance: Conduct Seamanship Evolutions
2. Conditions:
    a. Given:
        (1) A ship at sea;
        (2) Assistance; and
        (3) Supervision
    b. Denied: N/A
3. Standard: The officer shall be provided with an opportunity to have control of the ship for a minimum of one of each of the following evolutions:
    a. Berthing/unberthing;
    b. Buoy approach; and
    c. Replenishment at Sea.
4. Remark: This PTR should be done in conjunction with PTR 008 – Commanding Officer's Forums
5. References: Admiralty Manual of Navigation – Vol 6

PTR 007 Navigation Administration

007.01 Review Ship's Navigation Outfit

1. Performance: Review Ship's Navigation Outfit
2. Conditions:
    a. Given:
        (1) References;
        (2) A ship's navigation outfit; and
        (3) Supervision.
    b. Denied: N/A
3. Standard: Complete the Navigation Administration Inspection Checklist in Annex A of CFCD 130.
4. Remark: N/A
5. References: CFCD 130 – Canadian Manual of Navigation

PTR 007 Navigation Administration

007.02 Describe the Maintenance of Navigation Records, Reports and Returns

1. Performance: Describe the Maintenance of Navigation Records, Reports and Returns
2. Conditions:
    a. Given: References
    b. Denied: N/A
3. Standard: The officer shall describe the SOPs for maintaining the following records;
    a. Navigation workbook;
    b. Navigation notebook;
    c. Navigation databooks;
    d. Officer of the Watch notebooks;
    e. Ship's Log;
    f. ECDIS Voyage Data Records;
    g. Echo sounder logs;
    h. Night Order books
    i. Ship Activity Reports (SHIPACT);
    j. Maritime Port Information Reports (MARPIFs);
    k. Unit Historical Reports;
    l. Hydrographic notices;
    m. Professional Notes (PRONOTES);
    n. Post Deployment Reports (PDR);
    o. Security of Navigation records.
4. Remark: N/A

5. References: CFCD 130 – Canadian Manual of Navigation; Navigator's Yeoman Handbook; Admiralty Manual of Navigation – Vol 4

PTR 007 Navigation Administration

007.03 Draft Routine Navigation Messages

1. Performance: Draft Routine Navigation Messages
2. Conditions:
    a. Given:
        (1) References;
        (2) Assistance.
    b. Denied: N/A
3. Standard: The officer shall complete a minimum of one of each of the following;
    a. Weather report;
    b. Daily OPSUM;
    c. Harbour Training Intentions;
    d. Exercise support request;
    e. OCS intentions;
    f. OOW designator list;
    g. MOVREP/NAV PIM message;
    h. Marine Mammal sighting report/message;
    i. Maritime Port Information Report (MARPIF), or an update to an existing one; and
    j. Hydrographic observation.
4. Remark: N/A
5. References: CFCD 130 – Canadian Manual of Navigation, Annex B

PTR 007 Navigation Administration

007.04 Conduct Chart Corrections

1. Performance: Conduct Chart Corrections
2. Conditions:
    a. Given:
        (1) Ship's navigation outfit;
        (2) References; and
        (3) Assistance.
    b. Denied: N/A
3. Standard: The officer shall correctly apply chart corrections to paper and electronic charts, including recording the correction.
4. Remark: Understudy the Navigator's Yeoman
5. References: CFCD 130 – Canadian Manual of Navigation; Navigator's Yeoman Handbook

PTR 008 Commanding Officer's Forums

008.01 Discuss Navigation and Shiphandling

1. Performance: Discuss Navigation and Shiphandling
2. Conditions:
    a. Given:
        (1) Forum with Commanding Officer;
    b. Denied: N/A
3. Standard: The officer shall meet with the Commanding Officer and discuss the following navigation and shiphandling areas:
    a. Rules of the Road (WRT navigation);
    b. Shiphandling characteristics;
    c. Berthing and unberthing;
    d. Turning in a confined space;
    e. RAS approaches;
    f. Towing;
    g. Acceleration & deceleration; and
    h. Windage and Drift rate.
4. Remark: As per PTR 006, the officer should be provided with an opportunity to conduct evolutions.
5. References: Admiralty Manual of Navigation Vols 1 and 6; CFCD 130 – Canadian Manual of Navigation